SAT Math — Teacher Copies

Explanations with worked examples · a labelled figure for every rule, every example and every question · Desmos questions · Grade 11 · Ellipse International School

Domain	Lessons	Worked examples	Questions	Figures
SAT Math 1 — Algebra	6	108	132	**220**
SAT Math 2 — Advanced Math	8	144	176	**303**
SAT Math 3 — Problem-Solving & Data Analysis	8	144	176	**288**
SAT Math 4 — Geometry & Trigonometry	6	108	132	**235**
All domains	**28**	**504**	**616**	**1046**

SAT Math 1 — Algebra

1. **1.1** Linear Equations in One Variable
2. **1.2** Linear Functions
3. **1.3** Linear Equations in Two Variables
4. **1.4** Systems of Linear Equations
5. **1.5** Linear Inequalities
6. **1.6** Absolute Value

SAT Math 2 — Advanced Math

1. **2.1** Equivalent Expressions
2. **2.2** Quadratic Equations
3. **2.3** Polynomials
4. **2.4** Rational & Radical Equations
5. **2.5** Exponential Functions
6. **2.6** Quadratic & Polynomial Graphs
7. **2.7** Function Notation & Transformations
8. **2.8** Nonlinear Systems

SAT Math 3 — Problem-Solving & Data Analysis

1. **3.1** Ratios, Rates, Proportion & Units
2. **3.2** Percentages
3. **3.3** One-Variable Data: Centre & Spread
4. **3.4** Data Displays
5. **3.5** Two-Variable Data & Models
6. **3.6** Probability
7. **3.7** Inference & Margin of Error
8. **3.8** Evaluating Statistical Claims

SAT Math 4 — Geometry & Trigonometry

1. **4.1** Lines, Angles & Triangles
2. **4.2** Similarity & Congruence
3. **4.3** Right Triangles & Trigonometry
4. **4.4** Circles
5. **4.5** Area & Volume
6. **4.6** Coordinate Geometry

Each domain is also supplied as its own file. Every numeric answer in this booklet is recomputed by the checkers in the source folder; every figure is drawn to the numbers in its own question.

SAT Math 1.1 — Linear Equations in One Variable

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Multi-step equations

undo the operations in reverse order | whatever you do to one side, do to the other

Work from the outside in: deal with addition and subtraction before multiplication and division. Substituting your answer back is a ten-second check that catches almost every slip.



then check: 5(8 − 2) = 30 ✓

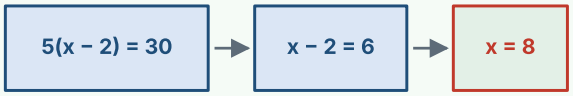
Each box undoes one operation, in the reverse of the order they were applied. The last box is the answer, and the check puts it straight back into the original.

Example 1. **Solve 5(x − 2) = 30.**

Solution.

Divide by 5: x − 2 = 6.

x = **8**. Check: 5(6) = 30.

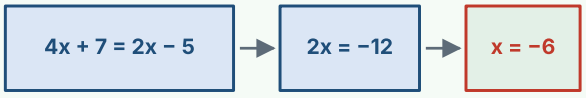


Example 2. **Solve 4x + 7 = 2x − 5.**

Solution.

Collect x on one side: 2x = −12.

x = **−6**.

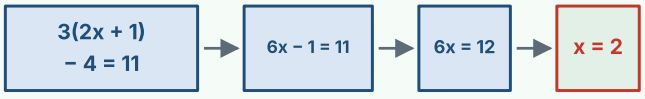


Example 3. **Solve 3(2x + 1) − 4 = 11.**

Solution.

6x + 3 − 4 = 11, so 6x = 12.

x = **2**.



2 Clearing fractions and decimals

multiply every term by the common denominator

Fractions make arithmetic slow and errors likely. Multiply the whole equation through by the lowest common denominator first, and the problem becomes an ordinary one.

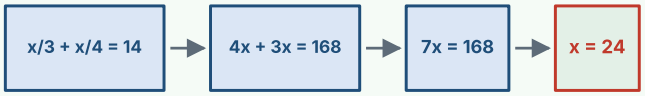


every term — including the one on the right

Twelve is the lowest number both 3 and 4 divide into, so one multiplication clears both fractions at once.

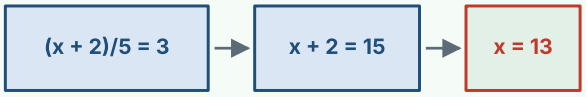
Example 1. **Solve x/3 + x/4 = 14.**

Solution.
Multiply everything by 12: 4x + 3x = 168.
7x = 168, so x = **24**.



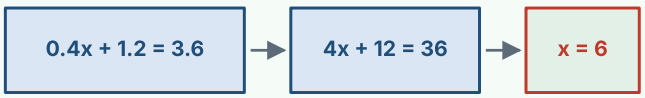
Example 2. **Solve (x + 2)/5 = 3.**

Solution.
Multiply by 5: x + 2 = 15.
x = **13**.



Example 3. **Solve 0.4x + 1.2 = 3.6.**

Solution.
Multiply by 10 to clear decimals: 4x + 12 = 36.
4x = 24, so x = **6**.



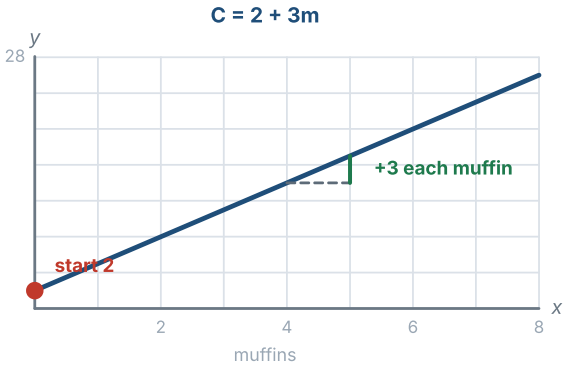
3 Translating a sentence

name the unknown | write one equation | answer the question that was asked

Most SAT algebra is a sentence in disguise. ‘A fixed charge plus so much each’ is always b + mx. Write the equation before touching any arithmetic.

the words map onto the symbols, phrase by phrase

a fixed charge plus so much each	b + mx
‘$2 for a box plus $3 per muffin’	2 + 3m
‘three more than twice a number’	2n + 3



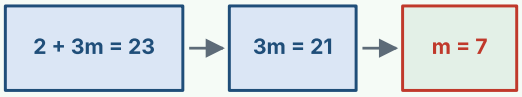
The fixed charge is where the line starts and the per-item charge is how fast it climbs. Paying $23 means asking which m puts you at that height.

Example 1. **A bakery charges $2 for a box plus $3 per muffin. A customer pays $23. How many muffins?**

Solution.

2 + 3m = 23, so 3m = 21.

m = **7 muffins**.



Example 2. **A fabric shop charges $4.50 per metre plus a $2 cutting fee. A customer pays $29. How many metres?**

Solution.

4.5x + 2 = 29, so 4.5x = 27.

x = **6 metres**.

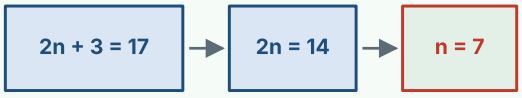


Example 3. **Three more than twice a number is 17. Find the number.**

Solution.

2n + 3 = 17, so 2n = 14.

n = **7**.



4 Literal equations

treat the other letters as numbers and isolate the one you want

Rearranging a formula uses exactly the same moves as solving for x. If the target letter appears twice, gather those terms together and factor it out.



the same three moves as solving for a number

when the letter appears twice, factor it out

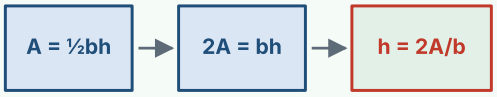
Top: isolate h exactly as you would isolate x. Bottom: with x on both sides you must collect first — dividing early is what loses the mark.

Example 1. **Solve A = ½bh for h.**

Solution.

Multiply by 2: 2A = bh.

h = **2A/b**.

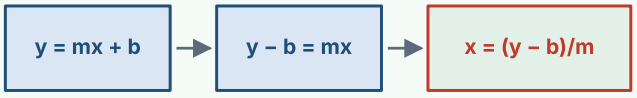


Example 2. **Solve y = mx + b for x.**

Solution.

y − b = mx.

x = **(y − b)/m**.

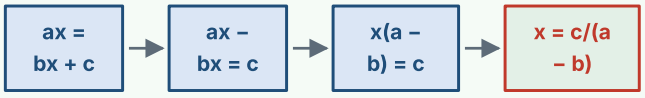


Example 3. **Solve ax = bx + c for x, where a ≠ b.**

Solution.

x appears twice, so gather and factor: ax − bx = c, giving x(a − b) = c.

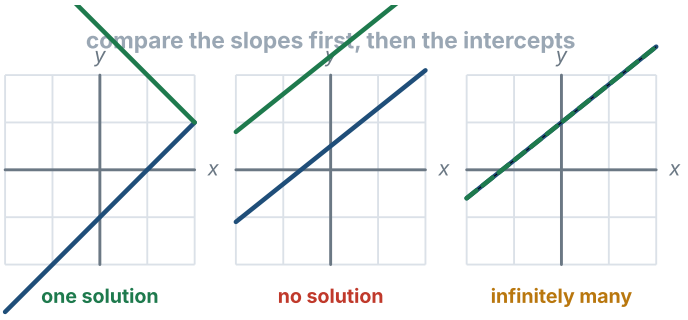
x = **c/(a − b)**.



5 No solution or infinitely many

variables cancel and the statement is false → no solution | true → infinitely many

Simplify both sides fully. If the x terms vanish, look at what is left: 5 = 7 is false, so nothing works; 5 = 5 is true, so everything works. For ax + b = cx + d this means a = c with b ≠ d, or a = c with b = d.



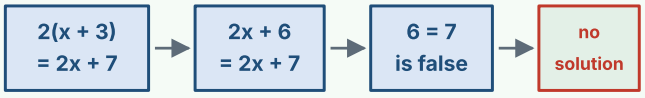
The same three cases drawn as lines: crossing once, running parallel with no crossing, or lying on top of each other so every point works.

Example 1. **Solve 2(x + 3) = 2x + 7.**

Solution.

2x + 6 = 2x + 7 gives 6 = 7, which is false.

No solution.

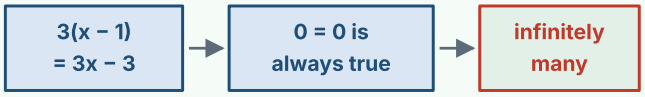


Example 2. **Solve 3(x − 1) = 3x − 3.**

Solution.

3x − 3 = 3x − 3 gives 0 = 0, which is always true.

Infinitely many solutions.



Example 3. **For what k does kx + 5 = 4x + 5 have infinitely many solutions?**

Solution.

The sides must match exactly.

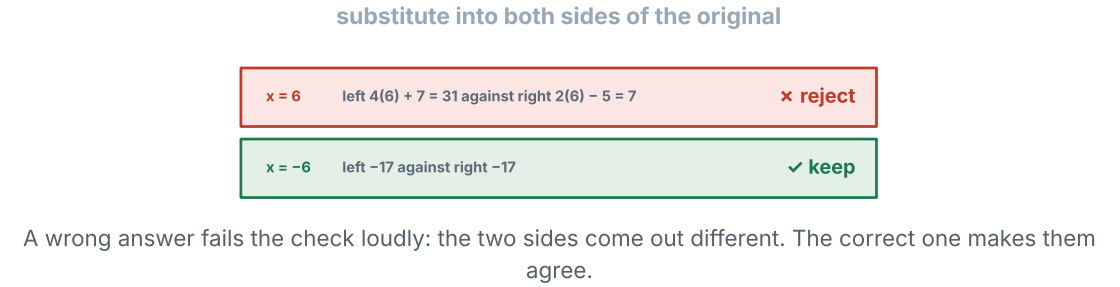
k = **4**.

kx + 5 = 4x + 5	the constants already match
so the x terms must too	k = 4

6 Checking and interpreting

substitute back | and make sure the answer is the quantity asked for

Two different habits, both worth marks. Substituting catches arithmetic errors. Re-reading the question catches the trap where you solve for x but the question wanted 2x, or the price rather than the quantity.

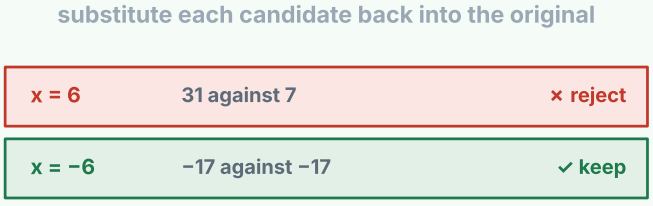


A wrong answer fails the check loudly: the two sides come out different. The correct one makes them agree.

Example 1. **A student solves 4x + 7 = 2x − 5 and gets x = 6. Check it.**

Solution.

Left: 31. Right: 7. They differ, so **x = 6 is wrong**; the correct answer is −6.

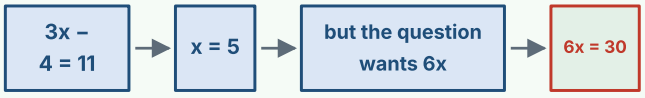


Example 2. **If 3x − 4 = 11, what is the value of 6x?**

Solution.

3x = 15, so x = 5.

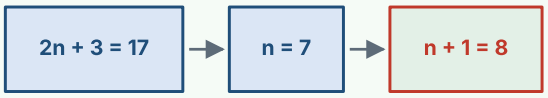
6x = **30**, not 5 — read what was asked.



Example 3. **If 2n + 3 = 17, what is n + 1?**

Solution.

n = 7, so n + 1 = **8**.



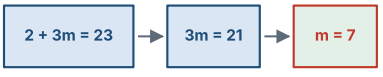
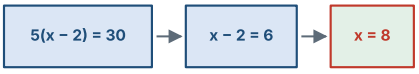
Part 1 — With Desmos (10 questions · every one is faster on the graph)

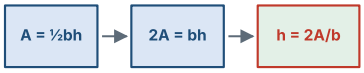
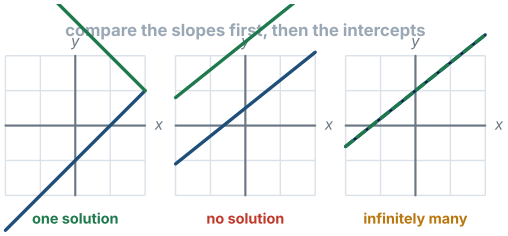
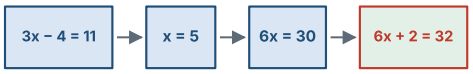
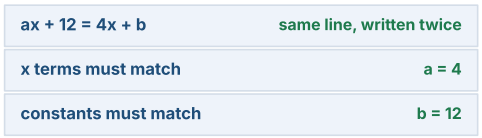
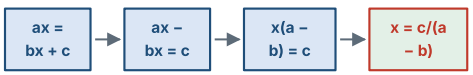
The SAT gives you Desmos on every maths question. Here it is **an equation solver**. Type the equation exactly as written and Desmos returns the value; the skill is turning the sentence into that equation.

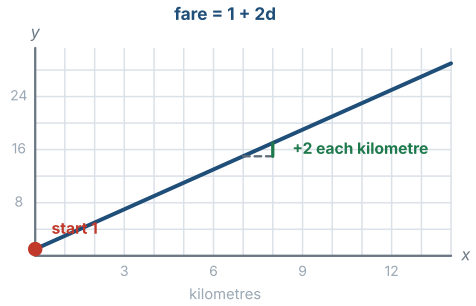
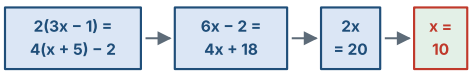
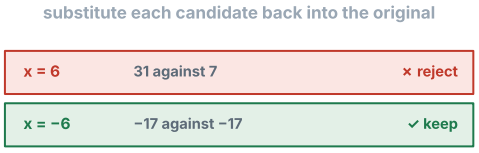
#	Question	Type into Desmos	Then	Answer
D1	If 5(x − 2) = 30, what is x?	`5(x-2)=30`	Type the equation; Desmos solves it.	8
D2	Solve 4x + 7 = 2x − 5.	`4x+7=2x-5`	Variables on both sides are no harder.	-6
D3	Solve x/3 + x/4 = 14.	`x/3+x/4=14`	Fractions are fine typed directly.	24
D4	A bakery charges $2 plus $3 per muffin and the total is $23. How many muffins?	`2+3m=23`	Write the model, then solve.	7
D5	A shop charges $4.50 per metre plus a $2 fee and the total is $29. How many metres?	`4.5x+2=29`	Fixed part plus rate times amount.	6
D6	Solve 0.4x + 1.2 = 3.6.	`0.4x+1.2=3.6`	Decimals need no clearing in Desmos.	6
D7	If 3x − 4 = 11, what is 6x?	`6*((11+4)/3)`	Solve, then answer what was asked.	30
D8	Solve 3(2x + 1) − 4 = 11.	`3(2x+1)-4=11`	Brackets are handled as typed.	2
D9	For what k does kx + 5 = 4x + 5 have infinitely many solutions?	`4`	The two sides must be identical.	4
D10	Solve (x + 2)/5 = 3.	`(x+2)/5=3`	One step after clearing the denominator.	13

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Several of these solve to a value that is not the answer — re-read the final sentence before writing anything down.

#	Level	Question	Answer	Solution
P1	Hard	A bakery charges $2 for a gift box plus $3 for each muffin placed in it. A customer pays $23 in total for one box of muffins. How many muffins are in the box? 	7	2 + 3m = 23, so 3m = 21. m = **7 muffins**. A common wrong answer comes from 23/3 rounded, ignoring the box charge.
P2	Hard	If 5(x − 2) = 30, what is the value of x? 	8	Divide both sides by 5 first: x − 2 = 6. x = **8**. A common wrong answer is x − 2, one step short of the answer.

#	Level	Question	Answer	Solution
P3	Hard	A fabric shop charges $4.50 per metre of linen plus a flat $2 cutting fee. A customer pays $29 in total. How many metres of linen did the customer buy? 	6	4.5x + 2 = 29 gives 4.5x = 27. x = **6 metres**.
P4	Hard	If x/3 + x/4 = 14, what is the value of x? 	24	Multiply every term by 12: 4x + 3x = 168, so 7x = 168. x = **24**.
P5	Hard	The equation A = ½bh gives the area of a triangle. Which expression gives h in terms of A and b? 	2A/b	Multiply both sides by 2, then divide by b. h = **2A/b**. A common wrong answer divides by 2 instead of multiplying.
P6	Hard	How many solutions does the equation 2(x + 3) = 2x + 7 have? 	Zero	Expanding gives 2x + 6 = 2x + 7, and the x terms cancel to leave 6 = 7. That is false, so there are **no solutions** — the two sides never meet.
P7	Very hard	If 3x − 4 = 11, what is the value of 6x + 2? 	32	3x = 15, so x = 5 and 6x = 30. 6x + 2 = **32**. A common wrong answer is x itself, and another is 6x — both are steps along the way, not the answer.
P8	Very hard	In the equation ax + 12 = 4x + b, a and b are constants. For which values does the equation have infinitely many solutions? 	a = 4 and b = 12	Infinitely many solutions means the two sides are the **same expression**, so the x coefficients match and the constants match. **a = 4 and b = 12.** A common wrong answer gives no solution; another gives exactly one.
P9	Very hard	Solve the equation ax = bx + c for x, where a ≠ b. 	c/(a − b)	x appears on both sides, so gather and factor: ax − bx = c gives x(a − b) = c. x = **c/(a − b)**. Dividing before factoring is what leads to the wrong answer.

#	Level	Question	Answer	Solution
P10	Very hard	A taxi charges a fixed fee plus a rate per kilometre. A 5 km ride costs \$11 and a 12 km ride costs \$25. What is the fixed fee, in dollars? 	1	The rate is (25 − 11)/(12 − 5) = 14/7 = \$2 per km. At 5 km: 11 = f + 2(5), so f = **\$1**.
P11	Very hard	If 2(3x − 1) = 4(x + 5) − 2, what is the value of x? 	10	6x − 2 = 4x + 20 − 2 = 4x + 18, so 2x = 20. x = **10**. Forgetting to distribute the 4 over both terms gives 8.
P12	Very hard	A student solves 4x + 7 = 2x − 5 and obtains x = 6. Substituting x = 6 gives 31 on the left and 7 on the right. What is the correct value of x? 	−6	Collecting correctly: 4x − 2x = −5 − 7, so 2x = −12. x = **−6**. Check: 4(−6) + 7 = −17 and 2(−6) − 5 = −17. The sign of the constant was the error.

The one habit to build. Substitute your answer back into the original equation. It takes ten seconds and catches almost every sign slip.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

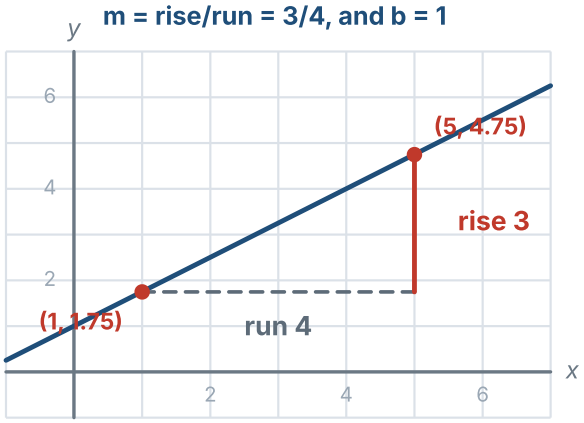
SAT Math 1.2 — Linear Functions

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Slope as a rate of change

$$m = (y_2 - y_1)/(x_2 - x_1) = \text{rise over run}$$

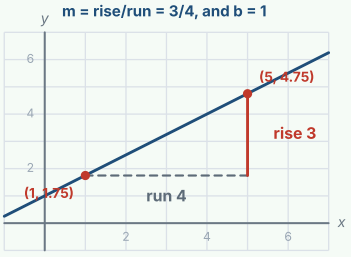
Slope is how much y changes for a one-unit rise in x. In a real context it always carries units: dollars per kilometre, degrees per hour. Keep the two points in the same order top and bottom.



From (1, 1.75) to (5, 4.75) the line climbs 3 while running 4, so the slope is 3/4. Swapping both differences gives −3/−4 — the same number.

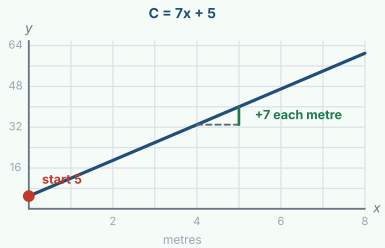
Example 1. **Find the slope through (1, 1.75) and (5, 4.75).**

Solution.
(4.75 − 1.75)/(5 − 1) = 3/4.
m = **0.75**.



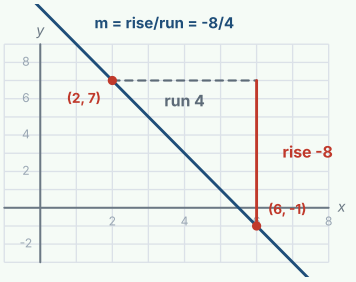
Example 2. **A tailor charges $19 for 2 m and $40 for 5 m. Find the rate per metre.**

Solution.
(40 − 19)/(5 − 2) = 21/3 = **$7 per metre**.



Example 3. **What is the slope of the line through (2, 7) and (6, −1)?**

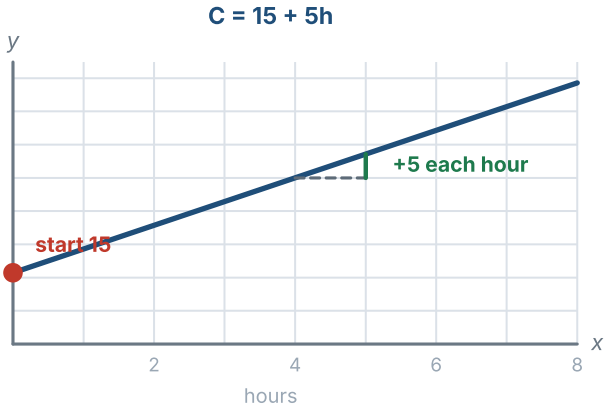
Solution.
(−1 − 7)/(6 − 2) = −8/4 = **−2**.



2 y = mx + b in context

b is the starting or fixed amount | m is the amount per unit

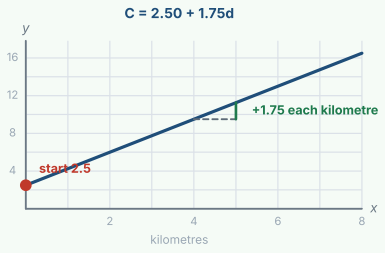
Interpretation questions are worth easy marks if you name the units. In C = 2.50 + 1.75d, the 2.50 is dollars and the 1.75 is dollars *per kilometre* — the 'per' is what makes it a rate.



The line starts at the fixed 15 and climbs 5 for every extra hour. Read b where the line meets the axis and m from one step across.

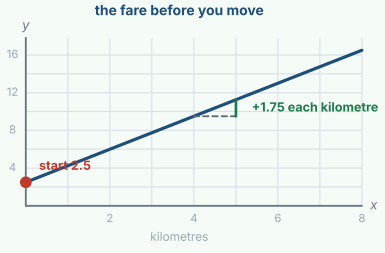
Example 1. **C = 2.50 + 1.75d gives a taxi fare for d kilometres. What does 1.75 mean?**

Solution.
The cost **per kilometre**, in dollars.



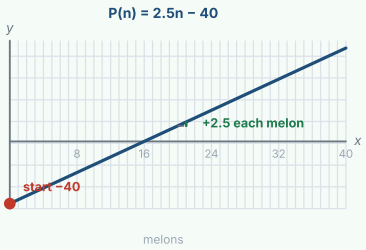
Example 2. **What does 2.50 mean in that model?**

Solution.
The **fixed starting fare** before any distance is travelled.



Example 3. **P(n) = 2.5n − 40 gives daily profit for n melons. What does 2.5 mean?**

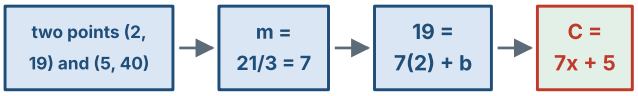
Solution.
The **increase in profit for each extra melon sold**, in dollars.



3 Writing a line from information

from two points: find m, then use y = mx + b with one point to get b

Three routes to the same line: two points, a point and a slope, or a table. Find the slope first in every case, then pin down b by substituting a known point.

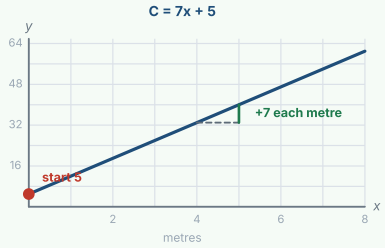


slope first, intercept second — always in that order

The rate comes from the two points; the fixed part comes from putting one of them back in. Doing it the other way round has nothing to substitute.

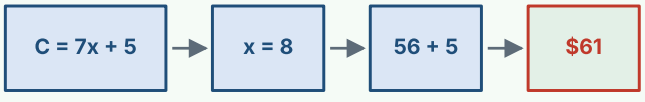
Example 1. **A tailor charges $19 for 2 m and $40 for 5 m. Write the model.**

Solution.
m = 7, and 19 = 7(2) + b gives b = 5.
C = 7x + 5.



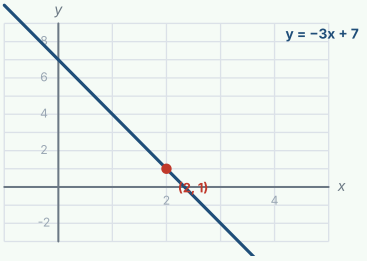
Example 2. **For that model, what is the charge for 8 metres?**

Solution.
7(8) + 5 = **$61**.



Example 3. **Write the equation of the line with slope −3 through (2, 1).**

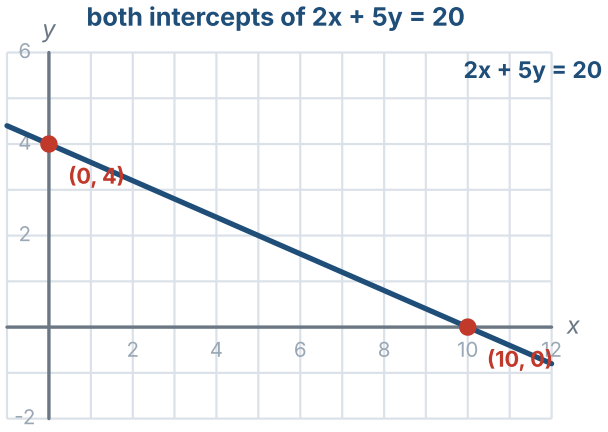
Solution.
1 = −3(2) + b gives b = 7.
y = −3x + 7.



4 Intercepts and what they mean

y-intercept: set x = 0 | x-intercept: set y = 0

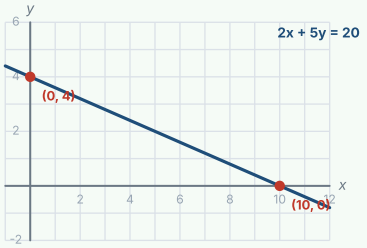
In context the y-intercept is the starting value and the x-intercept is when the quantity reaches zero — the break-even point, or when a tank empties.



Setting x = 0 gives the height where the line starts; setting y = 0 gives where it reaches the axis. Each is one substitution.

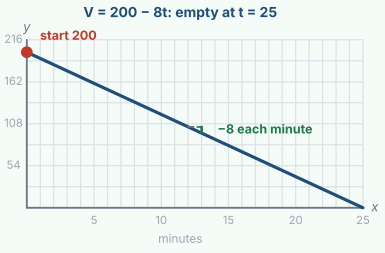
Example 1. **Find both intercepts of 2x + 5y = 20.**

Solution.
x = 0 gives y = 4; y = 0 gives x = 10.
(0, 4) and (10, 0).



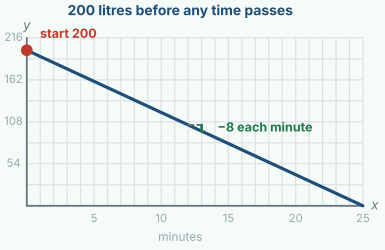
Example 2. **A tank model is V = 200 − 8t litres. When is it empty?**

Solution.
200 − 8t = 0, so t = **25 minutes**.



Example 3. **What does the 200 represent?**

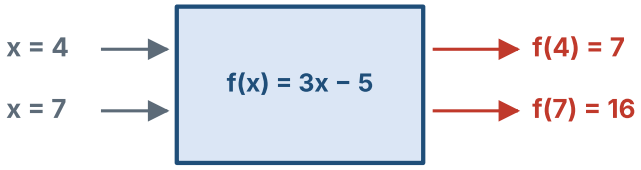
Solution.
The **starting volume**, in litres, at t = 0.



5 Function notation

f(3) means substitute 3 | f(x) = 3 means solve for x

The notation is compact but precise. Reading f(a) = b off a graph means the point (a, b) lies on it. Evaluating and solving are opposite directions, and the SAT tests both.

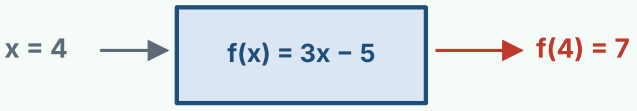


put the number in, read the number out

Put 4 in and 7 comes out, so f(4) = 7. Running it backwards from 16 asks which input produces that output.

Example 1. **f(x) = 3x − 5. Find f(4).**

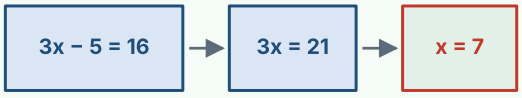
Solution.
3(4) − 5 = **7**.



put the number in, read the number out

Example 2. **For that f, solve f(x) = 16.**

Solution.
3x − 5 = 16, so 3x = 21.
x = **7**.



Example 3. **The graph of f passes through (2, 9). Write that in function notation.**

Solution.
f(2) = 9.

the point (2, 9)	is on the graph
in function notation	f(2) = 9

6 Linear models and tables

a constant difference in a table means the model is linear

Check the y-differences for equal steps. Once you know it is linear, the common difference is the slope and the value at x = 0 is b.

	x = 0	x = 1	x = 2	x = 3
	5	8	11	14
differences		+3	+3	+3
ratios		×1.6	×1.38	×1.27

equal differences → linear

Equal jumps of 3 for equal steps of 1: that is a slope of 3, and the value at x = 0 is the intercept.

Example 1. A table gives y = 5, 8, 11, 14 for x = 0, 1, 2, 3. Write the model.

Solution.
Differences are all 3, and y = 5 at x = 0.
y = 3x + 5.

	x = 0	x = 1	x = 2	x = 3
	5	8	11	14
differences		+3	+3	+3
ratios		×1.6	×1.38	×1.27

equal differences → linear

Example 2. A table gives y = 20, 17, 14 for x = 1, 2, 3. Write the model.

Solution.
Difference −3, and at x = 1, y = 20, so b = 23.
y = −3x + 23.

	x = 1	x = 2	x = 3
	20	17	14
differences		−3	−3
ratios		×0.85	×0.82

equal differences → linear

Example 3. **Is y = 2, 4, 8, 16 linear?**

Solution.

Differences 2, 4, 8 are not constant.

No — it is exponential, with ratio 2.

	x = 0	x = 1	x = 2	x = 3
	2	4	8	16
differences		+2	+4	+8
ratios		×2	×2	×2

equal ratios → exponential

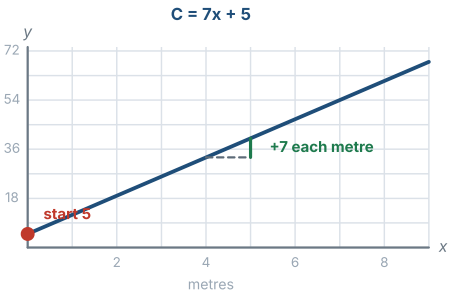
Part 1 — With Desmos (10 questions · every one is faster on the graph)

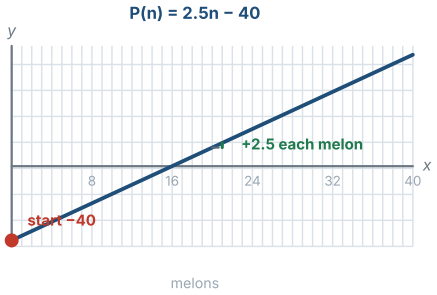
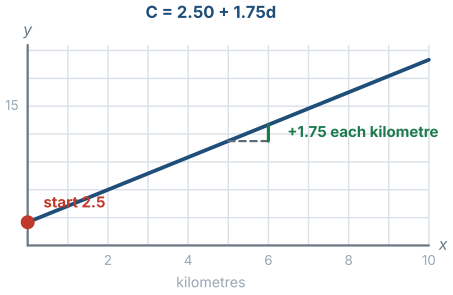
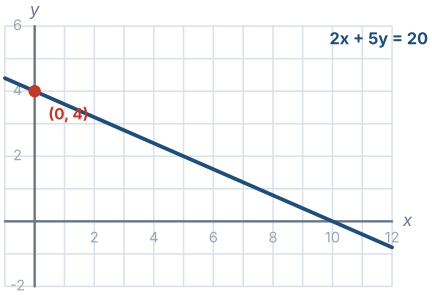
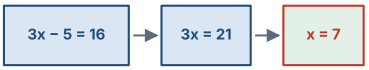
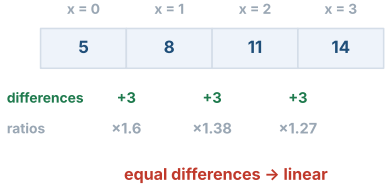
The SAT gives you Desmos on every maths question. Here it is **a line reader**. Plot two points, draw the line and Desmos gives the equation, the slope and both intercepts.

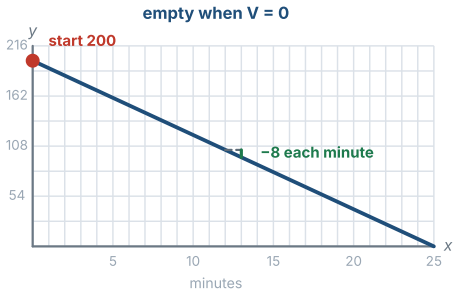
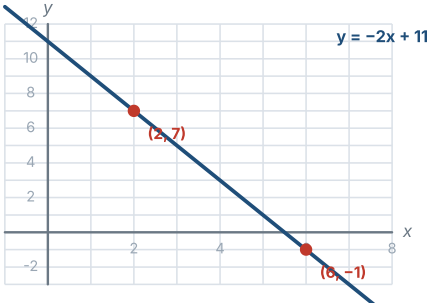
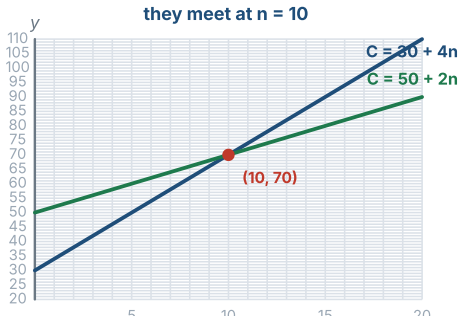
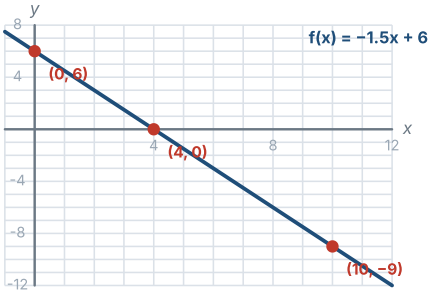
#	Question	Type into Desmos	Then	Answer
D1	What is the slope of the line through (1, 1.75) and (5, 4.75)?	`(4.75-1.75)/(5-1)`	Rise over run.	0.75
D2	A tailor charges $19 for 2 m and $40 for 5 m. What is the rate per metre?	`(40-19)/(5-2)`	The slope of the cost model.	7
D3	For that tailor, what is the charge for 8 metres?	`7*8+5`	Rate times amount plus the fixed part.	61
D4	What is the slope through (2, 7) and (6, −1)?	`(-1-7)/(6-2)`	Keep the order consistent.	-2
D5	Find the y-intercept of 2x + 5y = 20.	`20/5`	Set x = 0.	4
D6	Find the x-intercept of 2x + 5y = 20.	`20/2`	Set y = 0.	10
D7	V = 200 − 8t. At what t is V = 0?	`200/8`	The x-intercept in context.	25
D8	If f(x) = 3x − 5, what is f(4)?	`3*4-5`	Substitute.	7
D9	If f(x) = 3x − 5 and f(x) = 16, what is x?	`(16+5)/3`	Solve, do not substitute.	7
D10	A table gives y = 5, 8, 11, 14 at x = 0, 1, 2, 3. What is the slope?	`8-5`	Constant difference means linear.	3

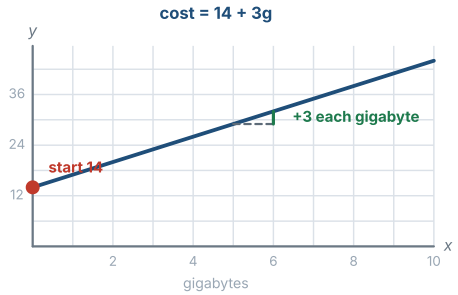
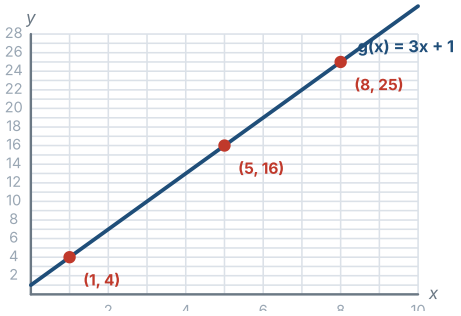
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Every interpretation question is answered by naming the **units**: dollars, or dollars per kilometre.

#	Level	Question	Answer	Solution
P1	Hard	A tailor's total charge is a linear function of the length of fabric used. The charge is $19 for 2 metres and $40 for 5 metres. What is the charge for 8 metres of fabric? 	$61	Rate = (40 − 19)/(5 − 2) = $7 per metre, and 19 = 7(2) + b gives b = $5. 7(8) + 5 = **$61**. A common wrong answer is 40 + 7×4, counting 5 to 8 as four steps instead of three; that answer drops the fixed $5. Only two of your thirteen got this on Test 5.

#	Level	Question	Answer	Solution
P2	Hard	The function P(n) = 2.5n − 40 gives the daily profit P, in dollars, of a fruit stall that sells n melons. What is the best interpretation of the number 2.5? 	**The increase in profit, in dollars, for each additional melon sold.**	The coefficient of n is a **rate**: profit per extra melon. A common wrong answer describes −40, the value when n = 0.
P3	Hard	The taxi fare, in dollars, for a ride of d kilometres is C = 2.50 + 1.75d. What is the best interpretation of 1.75? 	**The cost, in dollars, for each kilometre travelled.**	1.75 multiplies d, so it is the cost **per kilometre**. A common wrong answer is close but wrong: a 1 km ride costs 2.50 + 1.75 = $4.25.
P4	Hard	What is the y-coordinate of the y-intercept of the line 2x + 5y = 20? 	**4**	Set x = 0: 5y = 20. y = **4**.
P5	Hard	If f(x) = 3x − 5, for what value of x does f(x) = 16? 	**7**	3x − 5 = 16 gives 3x = 21. x = **7**. Substituting 16 instead would give 43, which answers a different question.
P6	Hard	A table shows y = 5, 8, 11 and 14 when x = 0, 1, 2 and 3. Write the equation that models the data. 	**y = 3x + 5**	The differences are all 3, so the slope is 3, and y = 5 when x = 0. **y = 3x + 5.** A common wrong answer swaps the slope and the intercept.

#	Level	Question	Answer	Solution
P7	Very hard	A water tank is modelled by V = 200 − 8t, where V is litres and t is minutes. After how many minutes is the tank empty? 	25	Set V = 0: 200 = 8t. t = **25 minutes**. This is the x-intercept read in context.
P8	Very hard	A line passes through (2, 7) and (6, −1). Write the equation of that line. 	y = −2x + 11	Slope = −8/4 = −2, and 7 = −2(2) + b gives b = 11. **y = −2x + 11.** A common wrong answer uses the y-value of the first point as the intercept.
P9	Very hard	Two linear models for a monthly cost are C = 30 + 4n and C = 50 + 2n, where n is the number of items. For what value of n do the two cost the same? 	10	30 + 4n = 50 + 2n gives 2n = 20. n = **10 items**, where both cost $70.
P10	Very hard	The graph of a linear function f passes through (0, 6) and (4, 0). What is the value of f(10)? 	−9	Slope = (0 − 6)/(4 − 0) = −1.5, and b = 6, so f(x) = −1.5x + 6. f(10) = −15 + 6 = **−9**. A common wrong answer assumes the line stops at zero.

#	Level	Question	Answer	Solution
P11	Very hard	A phone plan charges a fixed monthly fee plus a rate per gigabyte. Using 4 GB costs \$26 and using 9 GB costs \$41. What is the fixed monthly fee, in dollars? 	14	Rate = (41 − 26)/(9 − 4) = 15/5 = \$3 per GB. 26 = f + 3(4), so f = **\$14**. A common wrong answer is the rate, not the fee.
P12	Very hard	For a linear function g, g(1) = 4 and g(5) = 16. What is the value of g(8)? 	25	Slope = (16 − 4)/(5 − 1) = 3, and 4 = 3(1) + b gives b = 1. g(8) = 24 + 1 = **25**. A common wrong answer adds the slope once rather than three times.

The one habit to build. Find the rate first and the fixed part second. On Test 5 only two of thirteen got the tailor question, and the usual cause was counting the step from 5 to 8 as four instead of three.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

SAT Math 1.3 — Linear Equations in Two Variables

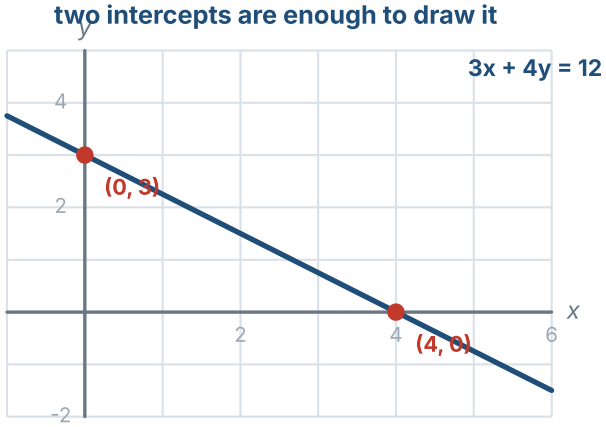
Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Graphing a line

plot both intercepts and join them | or start at b and step by the slope

From standard form, the intercepts are the fastest two points: set x = 0, then set y = 0. From slope-intercept form, plot b on the y-axis and step rise over run.

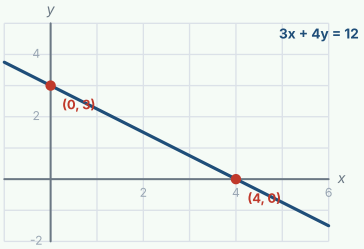
two intercepts are enough to draw it



Two points fix a line, and the intercepts are the two easiest to find — each costs one substitution.

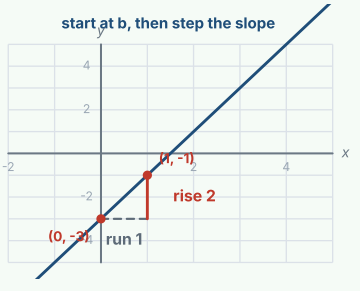
Example 1. **Find both intercepts of 3x + 4y = 12.**

Solution.
x = 0 gives y = 3; y = 0 gives x = 4.
(0, 3) and (4, 0).



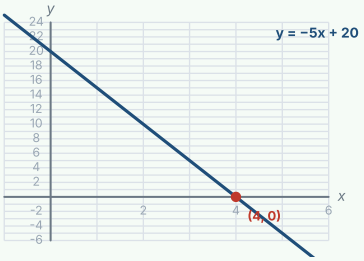
Example 2. **Describe how to graph y = 2x − 3 without a table.**

Solution.
Start at **(0, −3)**, then go up 2 and right 1 to reach (1, −1).



Example 3. **Find the x-intercept of y = −5x + 20.**

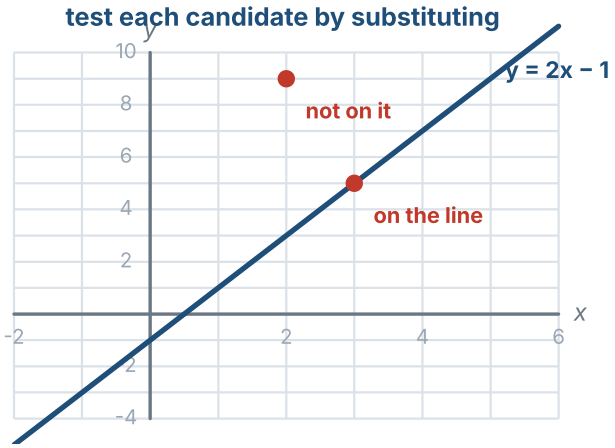
Solution.
0 = −5x + 20, so x = **4**.



2 Is this point on the line?

substitute both coordinates and see whether the equation holds

There is nothing clever here, and that is the point: a question asking which point lies on a line is answered by testing options until one works.



A point is on the line exactly when its coordinates satisfy the equation. One substitution settles it.

Example 1. **Does (3, 5) lie on y = 2x − 1?**

Solution.

2(3) − 1 = 5, which matches.

Yes.

substitute x and compare with y

(3, 5)	2(3) − 1 = 5 ✓	✓ keep

Example 2. **Does (2, 9) lie on 4x + y = 16?**

Solution.

8 + 9 = 17, not 16.

No.

substitute x and y into the equation

(2, 9)	8 + 9 = 17, not 16	× reject

Example 3. **The point (k, 7) lies on y = 3x + 1. Find k.**

Solution.

7 = 3k + 1, so 3k = 6.

k = **2**.

7 = 3k + 1 → 3k = 6 → k = 2

3 The same line in different clothes

slope-intercept $y = mx + b$ | standard $Ax + By = C$ | point-slope $y - y_1 = m(x - x_1)$

Two equations describe the same line when one is a multiple of the other. Rearranging to $y = mx + b$ is the quickest way to compare.

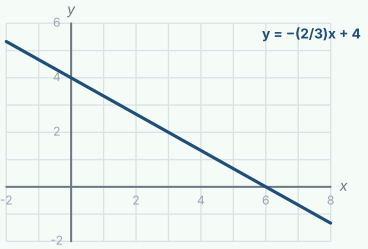
one line, three appearances

$2x + 3y = 12$	standard form
$y = -(2/3)x + 4$	the same line, rearranged
$4x + 6y = 24$	the same line again, doubled

Doubling every coefficient changes nothing about the line. Putting each into $y = mx + b$ makes that obvious at a glance.

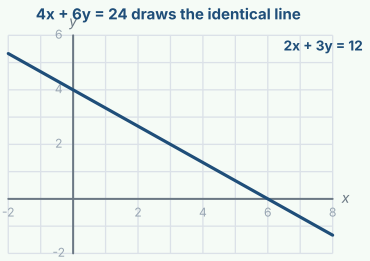
Example 1. **Write $2x + 3y = 12$ in slope-intercept form.**

Solution.
$3y = -2x + 12$.
$y = -(2/3)x + 4$.



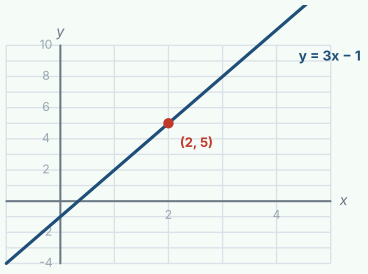
Example 2. **Are $2x + 3y = 12$ and $4x + 6y = 24$ the same line?**

Solution.
The second is exactly twice the first.
Yes — one line, written two ways.



Example 3. **Write the line through (2, 5) with slope 3 in point-slope form.**

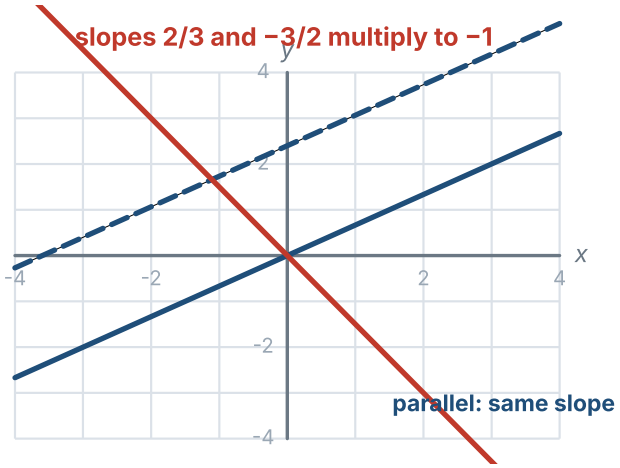
Solution.
$y - 5 = 3(x - 2)$, which rearranges to $y = 3x - 1$.



4 Parallel and perpendicular

parallel → same slope | perpendicular → slopes multiply to −1

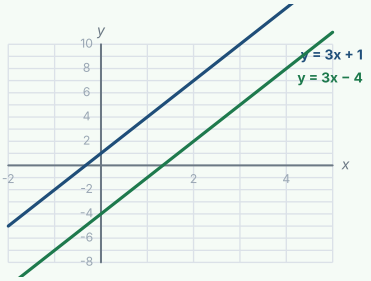
Rearrange both equations into y = mx + b before comparing. To get a perpendicular slope, flip the fraction and change the sign.



The two blue lines share a slope and never meet; the red one crosses at a right angle, with the slope flipped and negated.

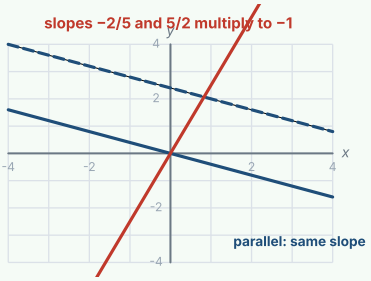
Example 1. **Is y = 3x + 1 parallel to 6x − 2y = 8?**

Solution.
The second gives y = 3x − 4, the same slope.
Yes, parallel.



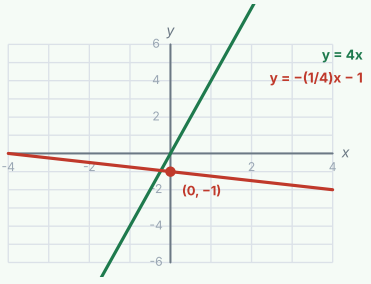
Example 2. **Find the slope perpendicular to y = −(2/5)x + 1.**

Solution.
Flip and negate.
5/2.



Example 3. **Write the line through (0, −1) perpendicular to y = 4x.**

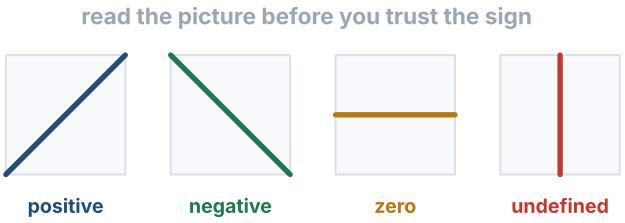
Solution.
Perpendicular slope −1/4.
y = −(1/4)x − 1.



5 Horizontal and vertical lines

y = c is horizontal, slope 0 | x = c is vertical, slope undefined

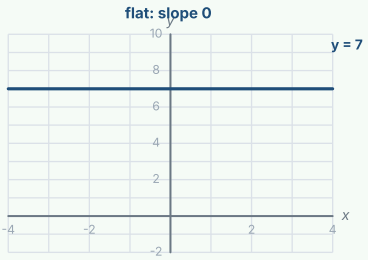
A horizontal line has zero slope because nothing rises. A vertical line has no slope at all, because the run is zero and you cannot divide by it — 'zero slope' and 'undefined slope' are different things.



Four lines, four slopes: positive, negative, zero (flat) and undefined (vertical). The last two are the pair that get confused.

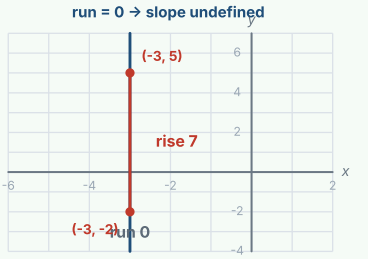
Example 1. **What is the slope of y = 7?**

Solution.
0 — it is horizontal.



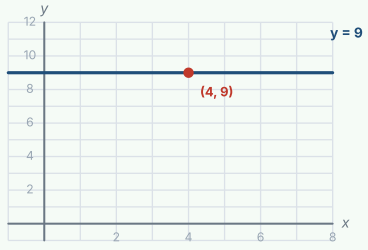
Example 2. **What is the slope of x = −3?**

Solution.
Undefined — it is vertical, and the run is zero.



Example 3. **Write the line through (4, 9) that is parallel to the x-axis.**

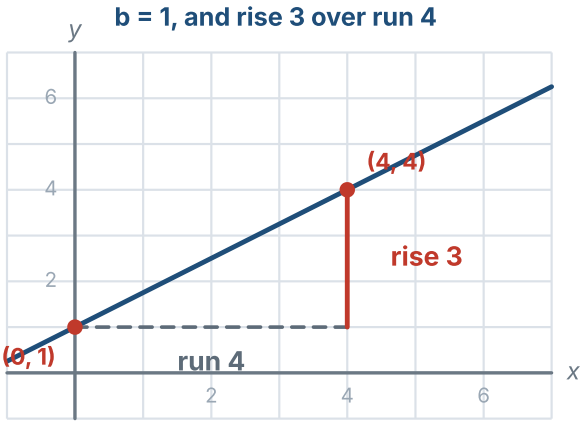
Solution.
Parallel to the x-axis means horizontal.
y = 9.



6 Reading a line off a graph

find b where it crosses the y-axis, then count rise over run between two grid points

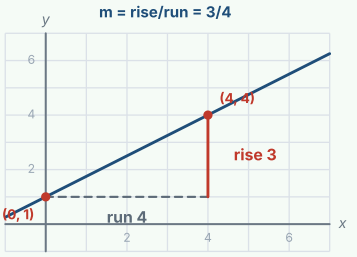
Pick two points where the line passes exactly through grid intersections, or the slope you count will be wrong. Then read b directly.



Both marked points sit on grid corners, so the count is exact: up 3, across 4, starting from a y-intercept of 1.

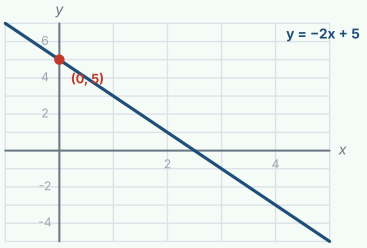
Example 1. **A line crosses the y-axis at 1 and passes through (4, 4). Find its equation.**

Solution.
m = (4 − 1)/4 = 3/4.
y = (3/4)x + 1.



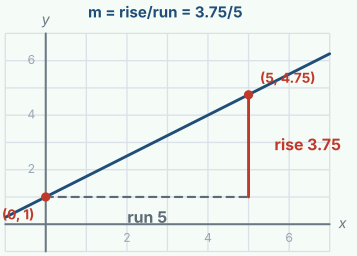
Example 2. **A line falls 2 for every 1 across and passes through (0, 5). Find its equation.**

Solution.
y = −2x + 5.



Example 3. **A graph shows a line through (0, 1) and (5, 4.75). Find its slope.**

Solution.
(4.75 − 1)/5 = **0.75**.



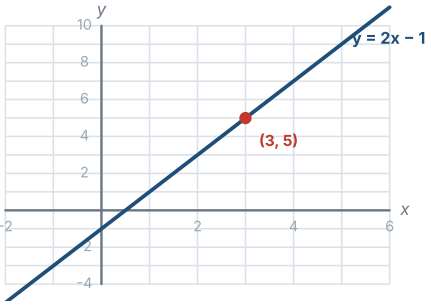
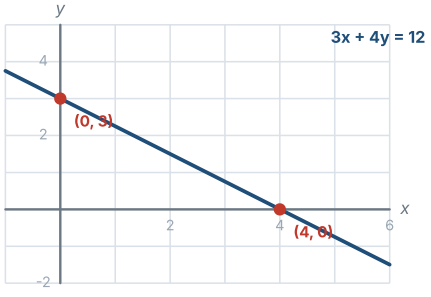
Part 1 — With Desmos (10 questions · every one is faster on the graph)

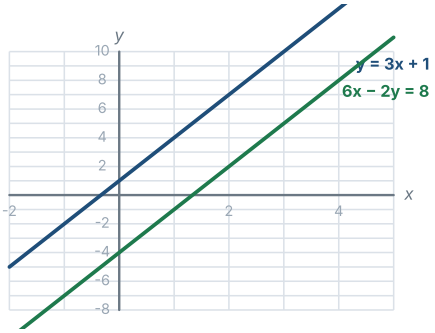
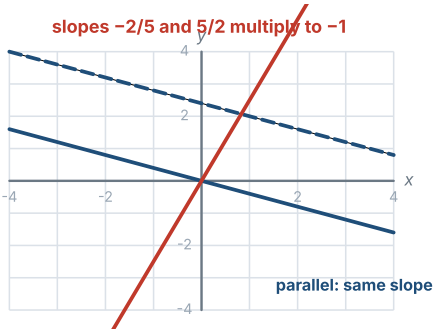
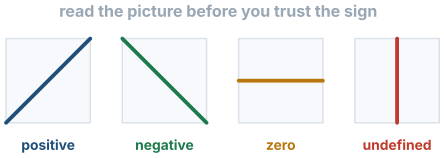
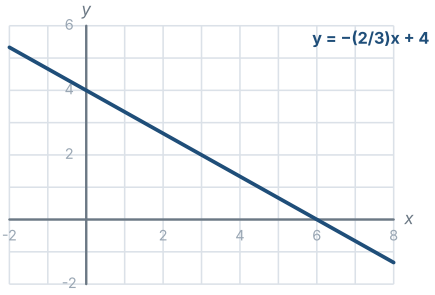
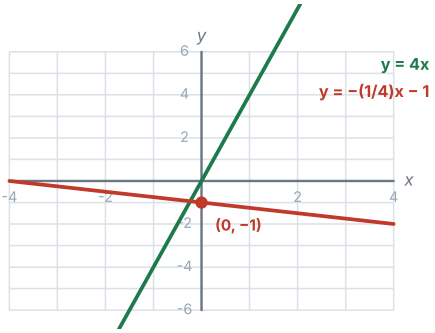
The SAT gives you Desmos on every maths question. Here it is **a grapher**. Type any form of the equation and read the slope and the intercepts straight off the picture.

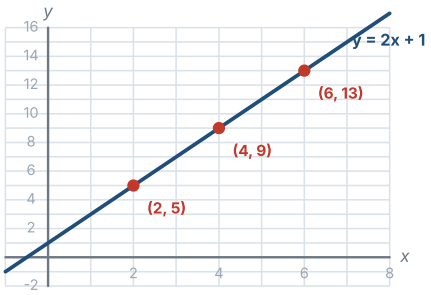
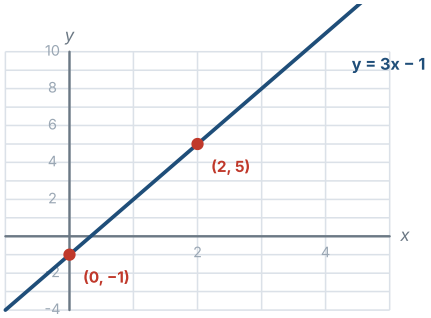
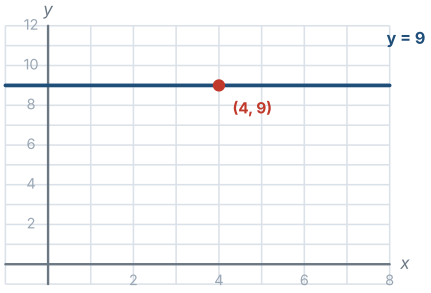
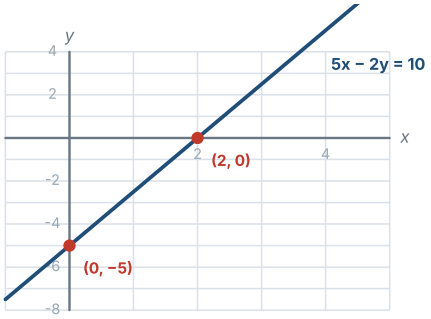
#	Question	Type into Desmos	Then	Answer
D1	What is the y-intercept of 3x + 4y = 12?	`12/4`	Set x = 0.	3
D2	What is the x-intercept of 3x + 4y = 12?	`12/3`	Set y = 0.	4
D3	Does the point (3, 5) lie on y = 2x − 1?	`2*3-1`	Compare with 5.	yes
D4	The point (k, 7) is on y = 3x + 1. What is k?	`(7-1)/3`	Solve for the input.	2
D5	Write 2x + 3y = 12 in the form y = mx + b. What is m?	`-2/3`	Rearranging is easier than it looks.	-2/3
D6	What slope is perpendicular to −2/5?	`-1/(-2/5)`	Flip and negate.	2.5
D7	Are y = 3x + 1 and 6x − 2y = 8 parallel?	`6/2`	Rearrange the second to compare.	yes, both slope 3
D8	What is the slope of the line through (0, 1) and (5, 4.75)?	`(4.75-1)/5`	Rise over run.	0.75
D9	What is the x-intercept of y = −5x + 20?	`20/5`	Set y = 0.	4
D10	A line has slope −2 and passes through (0, 5). What is y at x = 3?	`-2*3+5`	Substitute into y = mx + b.	-1

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Rearrange to y = mx + b before comparing two lines — two equations can look different and be the same line.

#	Level	Question	Answer	Solution
P1	Hard	Give a point that lies on the line y = 2x − 1. 	(3, 5)	Test each: at x = 3, y = 2(3) − 1 = **5**. A common wrong answer gives 2(2) − 1 = 3, not 5 — close enough to catch a careless reader.
P2	Hard	What is the x-intercept of the line 3x + 4y = 12? 	4	Set y = 0: 3x = 12. x = **4**. Setting x = 0 instead gives the y-intercept, 3.

#	Level	Question	Answer	Solution
P3	Hard	Write the equation of a line parallel to y = 3x + 1. 	6x − 2y = 8	Rearranging 6x − 2y = 8 gives y = 3x − 4, which has the same slope. **Parallel.** A common wrong answer negates the slope, and another inverts it — neither is parallel.
P4	Hard	A line has slope −2/5. What is the slope of a line perpendicular to it? 	5/2	Perpendicular slopes are negative reciprocals: flip and change the sign. **5/2.** A common wrong answer flips without negating.
P5	Hard	What is the slope of the line x = −3? 	undefined	A vertical line has a run of zero, and division by zero is undefined. **Undefined.** A common wrong answer is the slope of a *horizontal* line — the two are often confused.
P6	Hard	Write the equation that 2x + 3y = 12 becomes in slope-intercept form. 	y = −(2/3)x + 4	3y = −2x + 12, then divide by 3. **y = −(2/3)x + 4.** A common wrong answer inverts the slope.
P7	Very hard	Line k passes through (0, −1) and is perpendicular to the line y = 4x. What is the equation of line k? 	y = −(1/4)x − 1	The perpendicular slope to 4 is −1/4, and the y-intercept is given as −1. **y = −(1/4)x − 1.** A common wrong answer negates without flipping.

#	Level	Question	Answer	Solution
P8	Very hard	A line passes through (2, 5) and (6, 13). Give one other point with integer coordinates that lies on this line. 	(4, 9)	Slope = (13 − 5)/(6 − 2) = 2, and 5 = 2(2) + b gives b = 1, so y = 2x + 1. At x = 4: y = **9**, so (4, 9). A common wrong answer would need y = 21.
P9	Very hard	The equations 2x + 3y = 12 and 4x + 6y = k represent the same line. What is the value of k? 2x + 3y = 12 — double everything 4x + 6y = 24 — so k = 24	24	The second equation must be a multiple of the first; doubling 2x + 3y = 12 gives 4x + 6y = 24. k = **24**.
P10	Very hard	A line with slope 3 passes through the point (2, 5). What is its y-intercept? 	−1	5 = 3(2) + b gives b = 5 − 6. b = **−1**. A common wrong answer adds instead of subtracting.
P11	Very hard	Which is the equation of the line through (4, 9) that is parallel to the x-axis? 	y = 9	Parallel to the x-axis means **horizontal**, so y is constant at the given y-value. **y = 9.** A common wrong answer is the vertical line through that point.
P12	Very hard	In the xy-plane, line m has equation 5x − 2y = 10. What is the sum of its x-intercept and its y-intercept? 	−3	Set y = 0: x = 2. Set x = 0: −2y = 10, so y = −5. 2 + (−5) = **−3**.

The one habit to build. Zero slope and undefined slope are different things. A horizontal line has slope 0; a vertical line has none at all.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

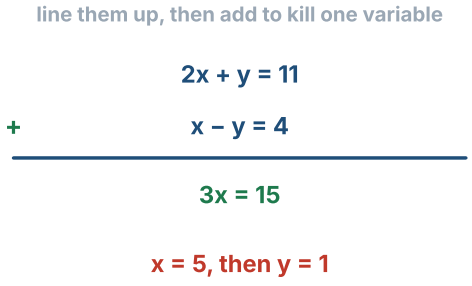
SAT Math 1.4 — Systems of Linear Equations

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Elimination

line the equations up and add or subtract to kill one variable

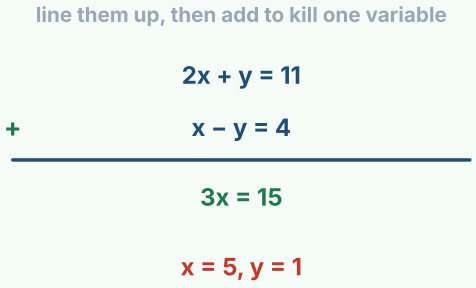
Elimination is usually faster than substitution on the SAT, because the coefficients are chosen to cooperate. If they do not match, multiply one equation through first.



The y terms are +y and −y, so adding the two equations removes y outright and leaves one equation in x.

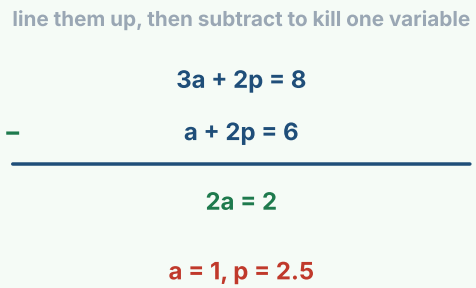
Example 1. **Solve 2x + y = 11 and x – y = 4.**

Solution.
Adding kills y: 3x = 15, so x = 5.
Then y = 1, giving **(5, 1)**.



Example 2. **Solve 3a + 2p = 8 and a + 2p = 6.**

Solution.
Subtracting kills p: 2a = 2, so a = 1.
Then p = 2.5, giving **a = 1, p = 2.5**.



Example 3. **Solve 3x + 2y = 57 and x + 4y = 39.**

Solution.
Multiply the second by 3: 3x + 12y = 117. Subtract: 10y = 60, so y = 6.
Then x = 15, giving **(15, 6)**.

line them up, then subtract to kill one variable

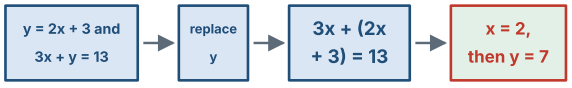
$$\begin{array}{rr} & 3x + 2y = 57 \\ - & 3x + 12y = 117 \\ \hline & 10y = 60 \end{array}$$

$$y = 6,\ x = 15$$

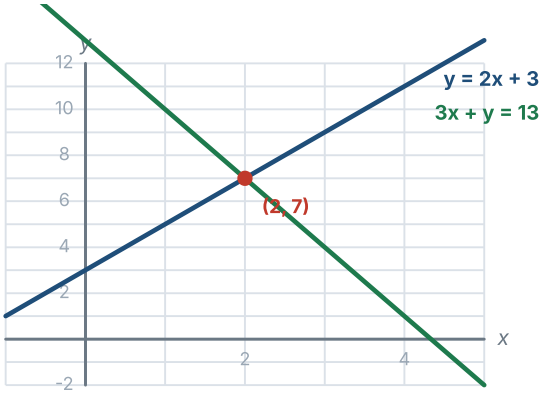
2 Substitution

when one variable is already alone, put that expression into the other equation

Substitution is the natural choice when an equation reads y = something. Replace, solve for the remaining variable, then go back for the other one.



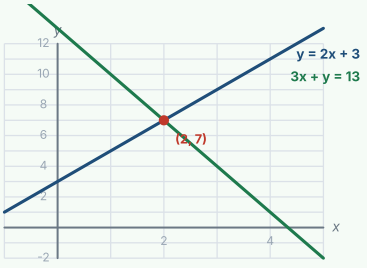
solve for one, then go back for the other



The algebra and the picture agree: the pair (2, 7) is the one point that satisfies both equations at once.

Example 1. **Solve y = 2x + 3 and 3x + y = 13.**

Solution.
3x + (2x + 3) = 13 gives 5x = 10, so x = 2.
Then y = 7: **(2, 7)**.

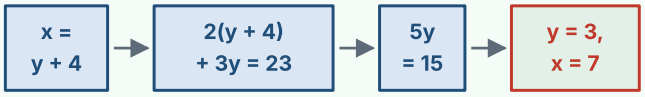


Example 2. **Solve x = y + 4 and 2x + 3y = 23.**

Solution.

2(y + 4) + 3y = 23 gives 5y = 15, so y = 3.

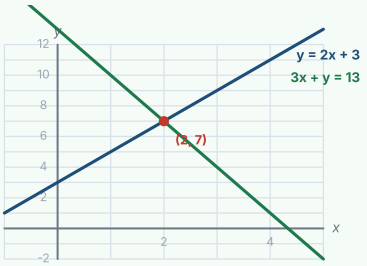
Then x = 7: **(7, 3)**.



Example 3. **Why finish by finding the second variable?**

Solution.

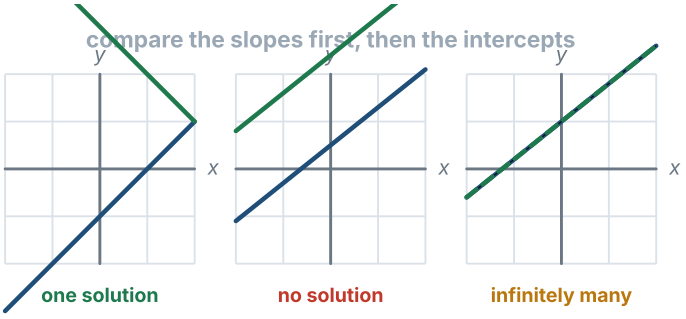
A system's solution is a **point**. Stopping at x answers only half the question — though do check whether the question wanted only one of them.



3 One, none, or infinitely many

different slopes → one | same slope, different intercept → none | identical → infinitely many

Two lines either cross once, run parallel forever, or sit on top of each other. For $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$, compare the ratios a_1/a_2, b_1/b_2 and c_1/c_2.



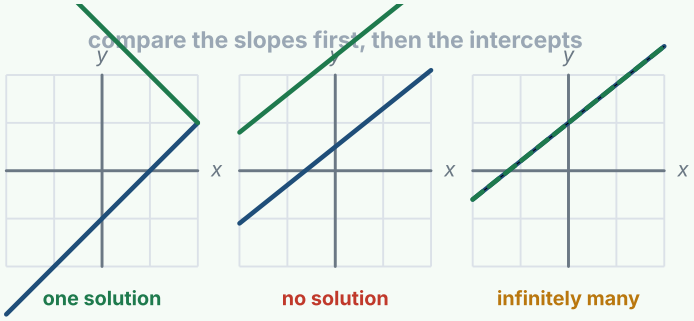
Crossing once, never meeting, or lying on top of one another — and the ratios of the coefficients tell you which before you solve anything.

Example 1. **How many solutions does 3x + 5y = 21 and 6x + 10y = 42 have?**

Solution.

The second is exactly twice the first.

Infinitely many — one line.

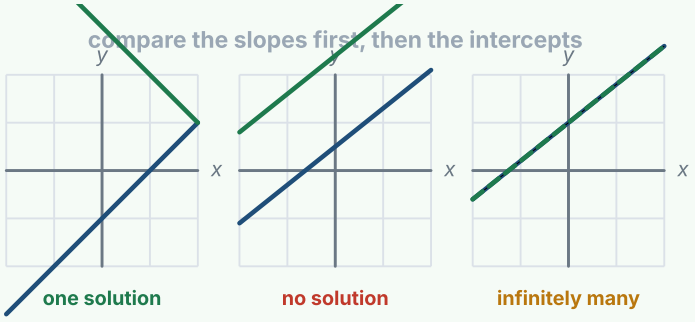


Example 2. **How many solutions does 3x + 5y = 21 and 6x + 10y = 35 have?**

Solution.

Same slope, different constant: parallel.

None.

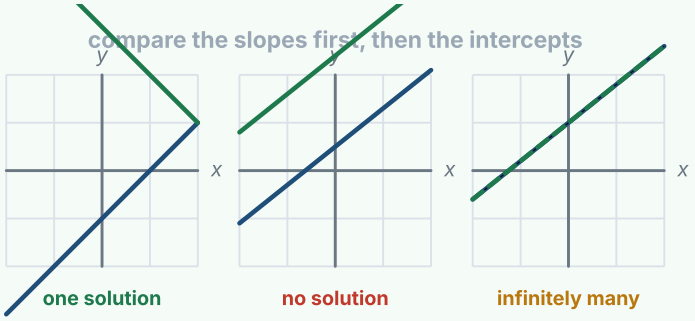


Example 3. **How many does 3x + 5y = 21 and 2x − y = 4 have?**

Solution.

The slopes differ.

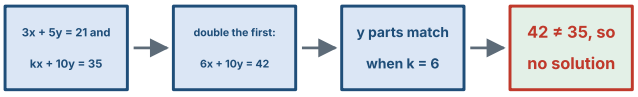
Exactly one.



4 Finding the parameter

no solution → match the coefficients but not the constants | infinitely many → match everything

These questions give you a letter in one coefficient and ask you to force a case. Scale one equation until the x and y parts line up, then compare the constants.

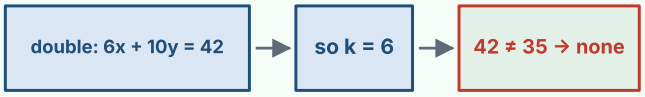


match the coefficients, then check the constants

Doubling makes the y terms agree, which fixes k. The constants then decide whether the answer is 'no solution' or 'infinitely many'.

Example 1. **For what k does 3x + 5y = 21 and kx + 10y = 35 have no solution?**

Solution.
Doubling the first gives 6x + 10y = 42, so k = **6**, and 42 ≠ 35 confirms no solution.



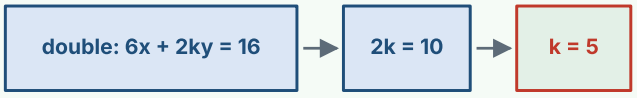
Example 2. **For what k does 2x + 6y = 10 and kx + 9y = 15 have infinitely many solutions?**

Solution.
Multiplying the first by 1.5 gives 3x + 9y = 15.
k = **3**.



Example 3. **For what value of k does 3x + ky = 8 and 6x + 10y = 9 have no solution?**

Solution.
Doubling the first gives 6x + 2ky = 16, so 2k = 10.
k = **5**.



5 Two-constraint word problems

one equation for the count, one for the value

Tickets, coins, mixtures and two-rate problems all share this shape: how many things, and how much they are worth. Define both variables before writing anything.

two sentences, two equations

how many: a + s = 45	the count equation
how much: 12a + 7s = 440	the value equation
substitute s = 45 − a	5a = 125, so a = 25

The first equation counts the tickets; the second prices them. Two unknowns need exactly these two facts.

Example 1. **Adult tickets cost $12 and student $7. 45 tickets sold for $440. How many adult tickets?**

Solution.
a + s = 45 and 12a + 7s = 440.
Substituting: 12a + 7(45 − a) = 440 gives 5a = 125, so a = **25**.

a + s = 45	count
12a + 7s = 440	value
5a = 125	a = 25

Example 2. **3 apples and 2 pears cost $8.00; 1 apple and 2 pears cost $6.00. Find the price of a pear.**

Solution.
Subtracting: 2 apples cost $2, so an apple is $1.
Then 2 pears cost $5: a pear is **$2.50**.

line them up, then subtract to kill one variable

$$\begin{array}{rr} & 3a + 2p = 8 \\ - & a + 2p = 6 \\ \hline & 2a = 2 \end{array}$$

a = 1, so p = 2.50

Example 3. **3 crates and 2 baskets cost 57; 1 crate and 4 baskets cost 39. Find the crate price.**

Solution.
Multiply the second by 3 and subtract: 10 baskets = 60, so a basket is 6.
A crate is (57 − 12)/3 = **15**.

line them up, then subtract to kill one variable

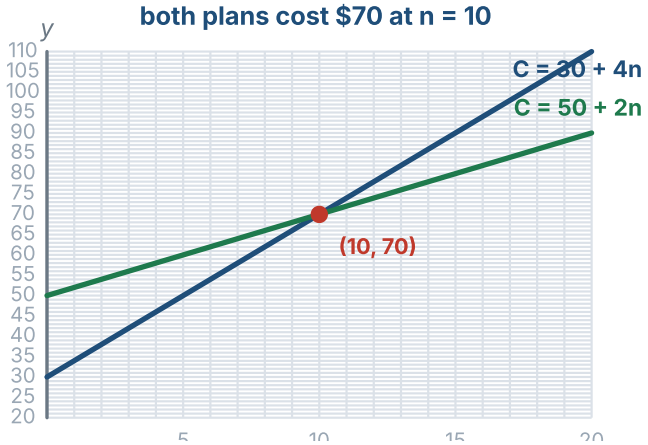
$$\begin{array}{rr} & 3c + 2b = 57 \\ - & 3c + 12b = 117 \\ \hline & 10b = 60 \end{array}$$

$$b = 6,\ c = 15$$

6 Interpreting the solution

the solution is the point where both conditions hold at once

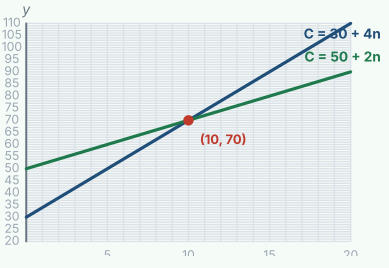
In context the intersection is the break-even point, the moment two plans cost the same, or the mix that satisfies both requirements. Check your answer makes sense: you cannot sell 2.5 tickets.



Before n = 10 one plan is cheaper; after it, the other. The crossing point is the only place they agree.

Example 1. **Two plans cost C = 30 + 4n and C = 50 + 2n. What does their intersection mean?**

Solution.
The number of items at which **both plans cost the same**: n = 10, at $70.



Example 2. **A system gives a = 25 and s = 20 for tickets. Check it against 45 tickets for $440.**

Solution.

25 + 20 = 45 ✓ and 12(25) + 7(20) = 300 + 140 = 440 ✓.

Both hold.

a solution must satisfy both equations

count: 25 + 20	= 45 ✓	✓ keep
value: 300 + 140	= 440 ✓	✓ keep

Example 3. **A ticket system solves to a = 22.5. What does that tell you?**

Solution.

That something is wrong — you cannot sell **half a ticket**. Recheck the setup.

does the answer make sense in context?

a = 22.5	tickets come in whole numbers	× reject

Part 1 — With Desmos (10 questions · every one is faster on the graph)

The SAT gives you Desmos on every maths question. Here it is **an intersection finder**. Enter both equations and click where the lines cross; Desmos gives the point exactly.

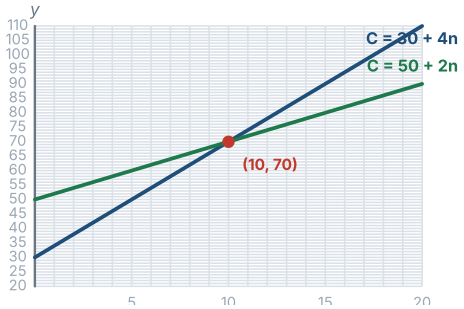
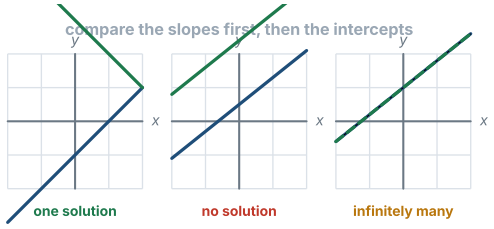
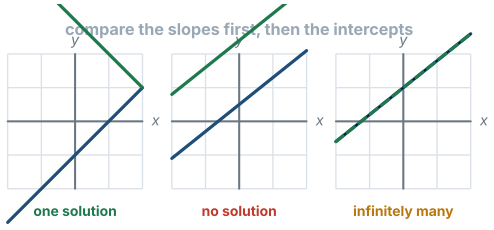
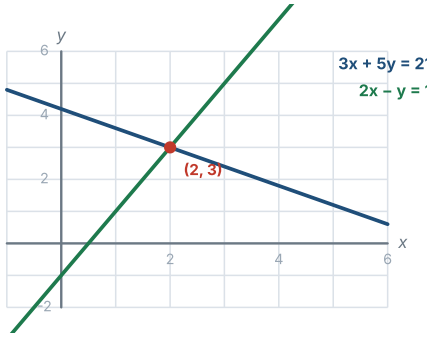
#	Question	Type into Desmos	Then	Answer
D1	Solve 2x + y = 11 and x − y = 4. What is x?	`2x+y=11, x-y=4`	Enter both equations and click the intersection.	5
D2	For that system, what is x + y?	`5+1`	Find both, then combine.	6
D3	Solve y = 2x + 3 and 3x + y = 13. What is x?	`y=2x+3, 3x+y=13`	Desmos marks the crossing point.	2
D4	3 apples and 2 pears cost $8.00; 1 apple and 2 pears cost $6.00. What does a pear cost?	`3a+2p=8, a+2p=6`	Two equations, two unknowns.	2.5
D5	Adult tickets cost $12 and student $7; 45 tickets for $440. How many adult tickets?	`a+s=45, 12a+7s=440`	Count equation and value equation.	25
D6	3 crates and 2 baskets cost 57; 1 crate and 4 baskets cost 39. What does a crate cost?	`3c+2b=57, c+4b=39`	Same structure again.	15
D7	For what k do 3x + ky = 8 and 6x + 10y = 9 have no solution?	`10/2`	Match the coefficients.	5
D8	For what k do 2x + 6y = 10 and kx + 9y = 15 have infinitely many solutions?	`2*1.5`	Scale the whole first equation.	3
D9	Two plans cost 30 + 4n and 50 + 2n. At what n are they equal?	`30+4n=50+2n`	Set them equal.	10
D10	Solve 3x + 2y = 57 and x + 4y = 39. What is y?	`3x+2y=57, x+4y=39`	Eliminate x first if you work by hand.	6

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. For word problems write one equation for the **count** and one for the **value** before solving anything.

#	Level	Question	Answer	Solution
P1	Hard	The system 2x + y = 11 and x − y = 4 has solution (x, y). What is the value of x + y? line them up, then add to kill one variable 2x + y = 11 + x − y = 4 3x = 15 x = 5, y = 1, so x + y = 6	6	Adding the equations gives 3x = 15, so x = 5, and then y = 1. x + y = **6**. A common wrong answer is x + 4, and another is the first constant — both are numbers already in the problem.
P2	Hard	At a farmers market, 3 apples and 2 pears cost $8.00, and 1 apple and 2 pears cost $6.00. What is the price of one pear? line them up, then subtract to kill one variable 3a + 2p = 8 − a + 2p = 6 2a = 2 a = $1, so p = $2.50	$2.50	Subtracting the equations removes the pears: 2 apples cost $2, so an apple is $1. Then 2 pears cost $5, so a pear is **$2.50**. A common wrong answer is the apple price.

#	Level	Question	Answer	Solution
P3	Hard	At a basketball tournament, adult tickets cost $12 each and student tickets cost $7 each. A total of 45 tickets were sold for $440. How many adult tickets were sold? a + s = 45 — the count 12a + 7s = 440 — the value 5a = 125 — a = 25 adult tickets	25	a + s = 45 and 12a + 7s = 440. Substituting s = 45 − a gives 12a + 315 − 7a = 440. 5a = 125, so a = **25**.
P4	Hard	In the system 3x + ky = 8 and 6x + 10y = 9, k is a constant. For which value of k does the system have no solution? double: 6x + 2ky = 16 → 2k = 10 → k = 5 → 16 ≠ 9 → none	5	No solution means the lines are parallel, so doubling the first must match the second's x and y parts: 6x + 2ky = 16 needs 2k = 10. k = **5**, and 16 ≠ 9 confirms they never meet.
P5	Hard	At a market, 3 crates of oranges and 2 baskets of figs cost 57 talens, while 1 crate and 4 baskets cost 39 talens. What is the price, in talens, of one crate of oranges? line them up, then subtract to kill one variable 3c + 2b = 57 − 3c + 12b = 117 10b = 60 b = 6, so c = 15	15	Multiply the second equation by 3 and subtract: 10 baskets cost 60, so a basket is 6. Then 3 crates cost 57 − 12 = 45, so a crate is **15**. A common wrong answer is the basket price.
P6	Hard	How many solutions does the system 3x + 5y = 21 and 6x + 10y = 35 have? compare the slopes first, then the intercepts one solution — no solution — infinitely many	Zero	Doubling the first gives 6x + 10y = 42, which contradicts 35. Same slope, different intercept means the lines are **parallel** and never meet.
P7	Very hard	A weaver's pricing system is modelled by 2x + 6y = 10 and kx + 9y = 15, where k is a constant. If the system has infinitely many solutions, what is the value of k? 9/6 = 1.5 → 15/10 = 1.5 ✓ → k = 2 × 1.5 → k = 3	3	Infinitely many solutions means one equation is a multiple of the other. The y terms give the factor: 9/6 = 1.5, and 15/10 = 1.5 agrees. k = 2 × 1.5 = **3**.

#	Level	Question	Answer	Solution
P8	Very hard	Two mobile plans charge C = 30 + 4n and C = 50 + 2n dollars for n items. At the number of items where the costs are equal, what is that cost, in dollars? 	70	30 + 4n = 50 + 2n gives 2n = 20, so n = 10. C = 30 + 40 = **$70**. Answering 10 gives the number of items, not the cost.
P9	Very hard	A chemist mixes a 20% solution with a 50% solution to make 30 litres of a 40% solution. How many litres of the 20% solution are used? x + y = 30 — the volume equation 0.2x + 0.5y = 12 — the concentration equation −0.3x = −3 — x = 10 litres	10	Let x be the 20% volume: x + y = 30 and 0.2x + 0.5y = 12. Substituting: 0.2x + 0.5(30 − x) = 12 gives −0.3x = −3, so x = **10 litres**.
P10	Very hard	The system y = 2x + 3 and y = ax + 7 has no solution. What is the value of a? 	2	No solution means the lines are parallel, so the slopes match. a = **2**, and because 3 ≠ 7 the lines really are distinct.
P11	Very hard	A system of two linear equations is graphed, and the two lines lie exactly on top of one another. State what is true here. 	The system has infinitely many solutions.	Every point on the shared line satisfies both equations. **Infinitely many solutions.** A common wrong answer describes parallel lines, which look similar on a small sketch but are never identical.
P12	Very hard	In the system 3x + 5y = 21 and 2x − y = 1, what is the value of y? 	3	From the second equation, y = 2x − 1. Substituting: 3x + 5(2x − 1) = 21 gives 13x = 26, so x = 2. Then y = 2(2) − 1 = **3**. A common wrong answer is x — check which variable the question asked for.

The one habit to build. Finish the problem. A system’s solution is a point, and the question may want x, y, their sum, or the cost at the intersection — not the first number you find.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

SAT Math 1.5 — Linear Inequalities

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Solving an inequality

solve exactly as for an equation — but flip the sign when you multiply or divide by a negative

Every move is the same as for an equation except one. Multiplying or dividing both sides by a negative number reverses the direction, and forgetting that is the classic error.



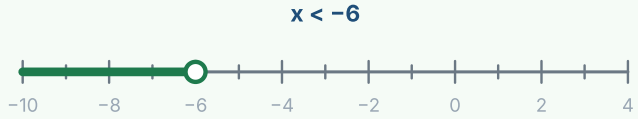
The flip is the whole difference from an equation. The open circle at −6 shows that the boundary itself is excluded by the strict inequality.

Example 1. **Solve 5 − 2x > 17.**

Solution.

−2x > 12, and dividing by −2 flips the sign.

x < **−6**.

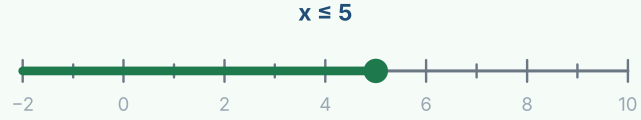


Example 2. **Solve 3x + 4 ≤ 19.**

Solution.

3x ≤ 15.

x ≤ **5**. No flip — we divided by a positive.

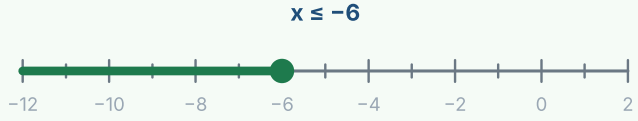


Example 3. **Solve −x/3 ≥ 2.**

Solution.

Multiply by −3 and flip.

x ≤ **−6**.



2 Translating the words

‘at least’ → ≥ | ‘at most’ / ‘no more than’ → ≤ | ‘more than’ → >

The wording decides whether the boundary counts. ‘Greater than $50’ excludes $50 exactly, so the least whole number of sales is one past the boundary.

	does the boundary value itself count?
at least 20	m ≥ 20 — 20 counts
at most 600 kg	L ≤ 600 — 600 counts
more than $50	p > 50 — 50 does not count

Two of these include their boundary and one does not. That single word is what separates 50 from 51 in the answer.

Example 1. **A vendor spends $75 and sells at $2.50 a cup. How many cups for profit over $50?**

Solution.

2.5n − 75 > 50 gives n > 50.

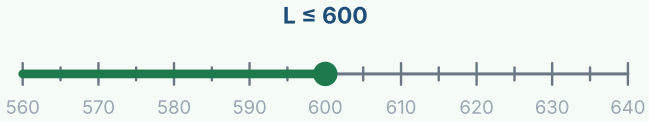
The least whole number is **51**, not 50.



Example 2. **A lift holds at most 600 kg. Write this as an inequality for the load L.**

Solution.

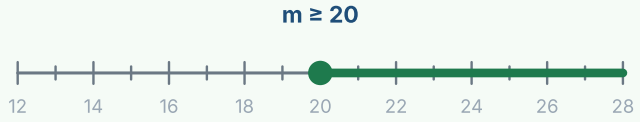
L ≤ **600** — 600 itself is allowed.



Example 3. **A club needs at least 20 members. Write it for membership m.**

Solution.

m ≥ **20**.



3 Interpreting a constant in context

ask what the units of each side are

In 3a + 7b ≤ 84, the left side is a cost in talens, so 84 is the **most** that can be spent — not a count of items. Checking units settles these questions immediately.

follow the units on each side

3a + 7b	3 and 7 are prices, so this is money
≤ 84	so 84 is a maximum amount of money
not a count	the units do not match items

Because the coefficients are prices, everything on that side is money. The number on the right has to be money too.

Example 1. **A stallholder buys a bags at 3 each and b jars at 7 each, with 3a + 7b ≤ 84. What is 84?**

Solution.
Both sides are costs, so 84 is the **maximum amount that can be spent**.

3a + 7b	a cost in talens
84	the most that can be spent

Example 2. **In 5x + 2y ≥ 40 where x is hours worked, what does 40 represent?**

Solution.
The **minimum** total the quantities must reach.

≥ 40	a floor, not a ceiling
the total must reach it	at least 40

Example 3. **Why is ‘the total number of items’ a wrong reading of 84?**

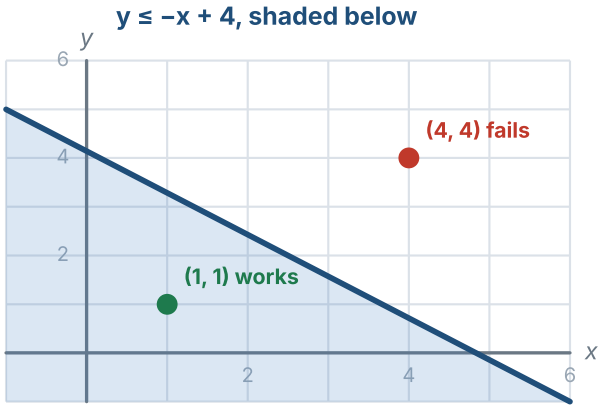
Solution.
The coefficients 3 and 7 are prices, so the product is money, not a count. **The units do not match.**

3 × bags	talens, not bags
a count would need	coefficients of 1

4 Graphing an inequality

solid line for ≤ or ≥ | dashed for < or > | test a point to decide which side

Draw the boundary line first, then pick any point not on it — (0, 0) is easiest — and see whether it satisfies the inequality. If it does, shade its side.



The line itself is included, so it is solid. (1, 1) satisfies the inequality and sits in the shaded region; (4, 4) fails and sits outside it.

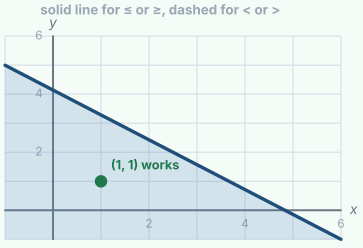
Example 1. **Describe the graph of y ≤ −x + 4.**

Solution.
A **solid** line through (0, 4) and (4, 0), with the region **below** shaded.



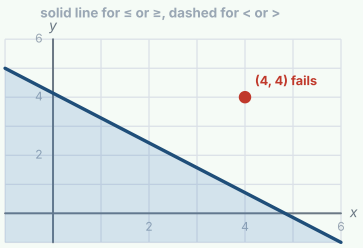
Example 2. **Does (1, 1) satisfy y ≤ −x + 4?**

Solution.
1 ≤ 3.
Yes, so it is in the shaded region.



Example 3. **Does (4, 4) satisfy it?**

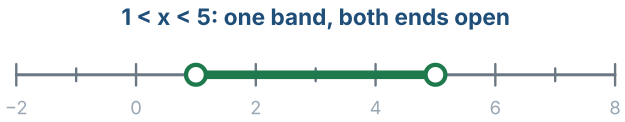
Solution.
4 ≤ 0 is false.
No — it lies above the line.



5 Compound inequalities

‘and’ → the overlap | ‘or’ → either region

Solve a three-part inequality by doing the same thing to all three parts at once. Keep the variable in the middle and remember that flipping applies to the whole chain.



An ‘and’ solution is a single stretch between two boundaries. Only the integers inside it — 2, 3 and 4 — qualify.

Example 1. **Solve $3 < 2x + 1 < 11$.**

Solution.

Subtract 1 throughout: $2 < 2x < 10$.

$1 < x < 5$.

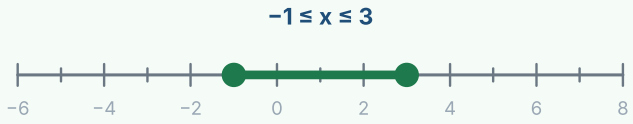


Example 2. **Solve $-4 \le 5 - 3x \le 8$.**

Solution.

Subtract 5: $-9 \le -3x \le 3$. Divide by -3 and flip both signs.

$-1 \le x \le 3$.



Example 3. **How many integers satisfy $1 < x < 5$?**

Solution.

$x = 2, 3, 4$.

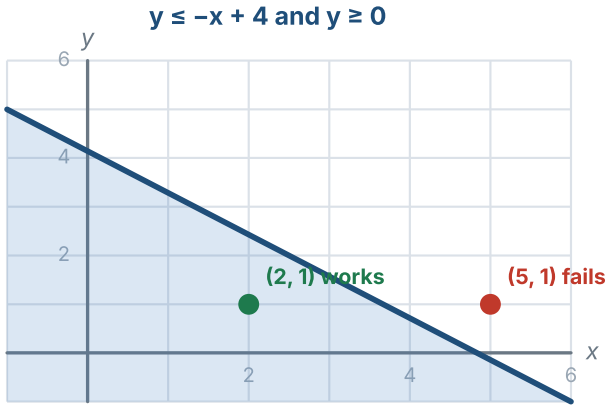
Three.



6 Systems of inequalities

the solution is the region satisfying every inequality at once

Shade each inequality and look for the overlap. To test whether a point is a solution, check it in every inequality — it only takes one failure to rule a point out.



(2, 1) satisfies both conditions; (5, 1) is above the line, so one failure is enough to exclude it.

Example 1. **Does (2, 1) satisfy y ≤ −x + 4 and y ≥ 0?**

Solution.

1 ≤ 2 ✓ and 1 ≥ 0 ✓.

Yes, both hold.

a solution must pass every inequality

y ≤ −x + 4	1 ≤ 2 ✓	✓ keep
y ≥ 0	1 ≥ 0 ✓	✓ keep

Example 2. **Does (5, 1) satisfy that system?**

Solution.

1 ≤ −1 is false.

No — one failure is enough.

one failure rules the point out

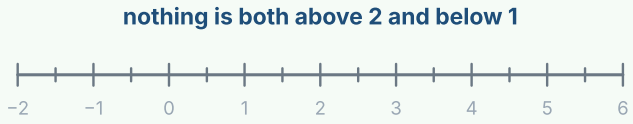
y ≤ −x + 4	1 ≤ −1 ×	× reject
y ≥ 0	1 ≥ 0 ✓	✓ keep

Example 3. **A system is y > 2 and y < 1. What is the solution region?**

Solution.

No value of y is both above 2 and below 1.

Empty — there is no solution.



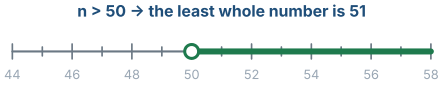
Part 1 — With Desmos (10 questions · every one is faster on the graph)

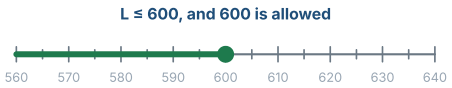
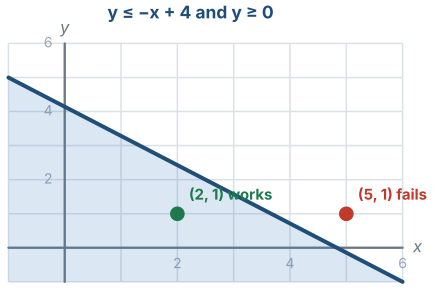
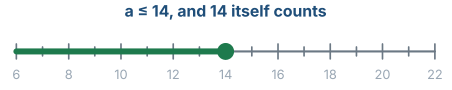
The SAT gives you Desmos on every maths question. Here it is **a region shader**. Type the inequality and Desmos shades the solution, boundary style included.

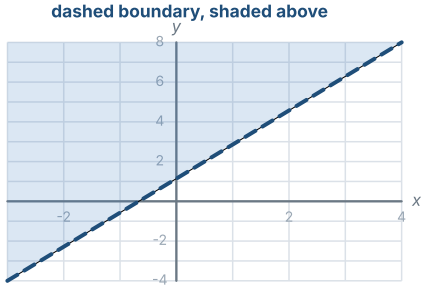
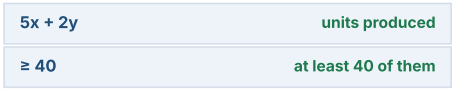
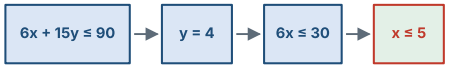
#	Question	Type into Desmos	Then	Answer
D1	Solve 5 − 2x > 17. What is the boundary value of x?	`5-2x=17`	Solve the equation, then decide the direction.	-6
D2	A vendor spends $75 and sells at $2.50 a cup. What is the break-even-plus-$50 cup count?	`(50+75)/2.5`	The boundary; the answer is one more.	50
D3	Solve 3x + 4 ≤ 19. What is the largest x?	`(19-4)/3`	Dividing by a positive keeps the sign.	5
D4	Solve 3 < 2x + 1 < 11. What are the two boundary values of x?	`2x+1=3`	Do each end separately.	1 and 5
D5	Solve −4 ≤ 5 − 3x. What is the boundary value of x?	`(5+4)/3`	Flip the sign when dividing by −3.	3
D6	Does (1, 1) satisfy y ≤ −x + 4?	`-1+4`	Compare the result with 1.	yes
D7	Does (4, 4) satisfy y ≤ −x + 4?	`-4+4`	4 is not less than or equal to 0.	no
D8	How many integers satisfy 1 < x < 5?	`4-1-0`	Count 2, 3 and 4.	3
D9	Graph y ≤ −x + 4 and shade. Which side contains the origin?	`y<=-x+4`	Desmos shades automatically.	below
D10	If 3a + 7b ≤ 84 and b = 6, what is the greatest whole value of a?	`(84-42)/3`	Substitute, then solve.	14

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Decide whether the boundary value itself counts: 'more than 50' excludes 50, so the least whole number is 51.

#	Level	Question	Answer	Solution
P1	Hard	Which describes all values of x that satisfy 5 − 2x > 17? 	x < −6	−2x > 12, and dividing by −2 **reverses** the inequality. x < **−6**. A common wrong answer is what you get by forgetting to flip — the single most common error in this topic.
P2	Hard	A lemonade vendor spends $75 on supplies each morning and sells lemonade for $2.50 per cup. What is the least whole number of cups the vendor must sell so that the day's profit is greater than $50? 	51	2.5n − 75 > 50 gives 2.5n > 125, so n > 50. The least whole number strictly greater than 50 is **51**. A common wrong answer is the boundary itself, which 'greater than' excludes.

#	Level	Question	Answer	Solution
P3	Hard	A stallholder buys a bags of rice at 3 talens each and b jars of honey at 7 talens each, where 3a + 7b ≤ 84. What is the best interpretation of 84? 3a + 7b — prices × quantities = money ≤ 84 — a maximum amount of money not a count of items — the units would not match	**The maximum number of talens the stallholder can spend.**	Both sides are amounts of money, and ≤ caps the left side. **The most that can be spent.** The trap five of your class took: 84 cannot be a count, because 3 and 7 are prices.
P4	Hard	Which inequality represents 'a lift holds at most 600 kilograms', where L is the load? 	**L ≤ 600**	'At most' means it may reach 600 but not exceed it. **L ≤ 600.** A common wrong answer would forbid a load of exactly 600 kg.
P5	Hard	If 3 < 2x + 1 < 11, how many integer values can x take? 	**3**	Subtracting 1 throughout gives 2 < 2x < 10, so 1 < x < 5. The integers are 2, 3 and 4, so there are **3** of them. The endpoints are excluded.
P6	Hard	Which point satisfies the system y ≤ −x + 4 and y ≥ 0? 	**(2, 1)**	Test each: at (2, 1), 1 ≤ 2 ✓ and 1 ≥ 0 ✓. **(2, 1) works.** (5, 1) fails the first; (1, −2) fails the second.
P7	Very hard	If −4 ≤ 5 − 3x ≤ 8, which describes all possible values of x? 	**−1 ≤ x ≤ 3**	Subtract 5 throughout: −9 ≤ −3x ≤ 3. Dividing by −3 flips both signs and swaps the ends. **−1 ≤ x ≤ 3.** A common wrong answer is what you get without flipping.
P8	Very hard	In the inequality 3a + 7b ≤ 84, b = 6. What is the greatest possible whole-number value of a? 	**14**	3a + 42 ≤ 84 gives 3a ≤ 42. a ≤ 14, so the greatest whole value is **14** — the boundary is allowed because the sign is ≤.

#	Level	Question	Answer	Solution
P9	Very hard	A student must score at least 80 as an average over four tests. After three tests the scores are 72, 85 and 79. What is the minimum score needed on the fourth test? 	84	The total must be at least 4 × 80 = 320, and the first three sum to 236. 320 − 236 = **84**.
P10	Very hard	The graph of a linear inequality shows a **dashed** boundary line with the region above it shaded. Which inequality could it represent? 	y > 2x + 1	Dashed means the boundary is **not** included, so the sign is strict; shading above means y is greater. **y > 2x + 1.** A common wrong answer would need a solid line.
P11	Very hard	A company needs at least 40 units produced, using machine A at 5 units per hour and machine B at 2 units per hour. If x and y are the hours each runs, which inequality models this? 	5x + 2y ≥ 40	'At least 40 units' means the total production must reach 40 or more. **5x + 2y ≥ 40.** A common wrong answer counts hours rather than units.
P12	Very hard	A caterer has at most 90 talens to spend on x trays of bread at 6 talens each and y trays of cheese at 15 talens each. If 4 trays of cheese are bought, what is the greatest whole number of bread trays the caterer can buy? 	5	6x + 15y ≤ 90 with y = 4 gives 6x + 60 ≤ 90, so 6x ≤ 30. x ≤ 5, and the greatest whole number is **5**.

The one habit to build. Flip the inequality sign whenever you multiply or divide by a negative. It is the only rule that differs from solving an equation, and it is where the marks go.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

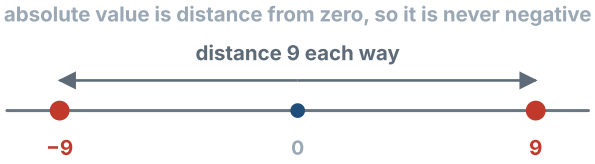
SAT Math 1.6 — Absolute Value

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 What absolute value means

|a| is the distance from zero, so it is never negative

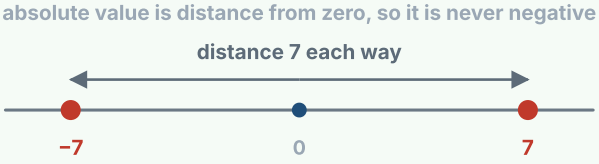
Distance has no direction, which is why |−7| = 7. An equation |x| = c with c negative has no solution at all, because no distance is negative.



Both −9 and 9 sit nine units from zero, so both satisfy |x| = 9. Nothing sits a negative distance away, which is why |x| = −4 has no solution.

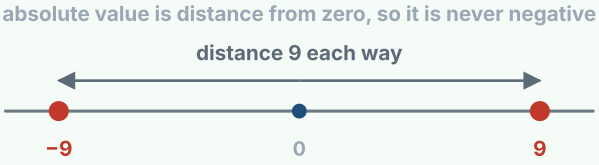
Example 1. **Evaluate |−7| + |3|.**

Solution.
7 + 3 = **10**.



Example 2. **Solve |x| = 9.**

Solution.
x is 9 away from zero in either direction.
x = **9 or −9**.



Example 3. **Solve |x| = −4.**

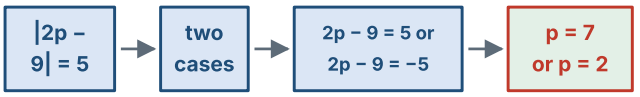
Solution.
No solution — an absolute value can never be negative.

|x| = −4	**a distance cannot be negative**
nothing to split	**no solution**

2 Solving |ax + b| = c

split into two equations: ax + b = c and ax + b = −c

The bars hide a plus-or-minus. Write both cases before solving either, and check c is not negative first.

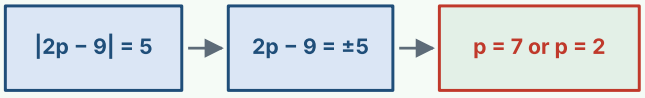


both cases are real solutions — neither is a spare

One set of bars, two equations. Taking only the positive case is what loses half the answer.

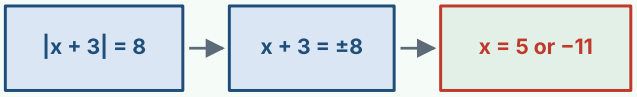
Example 1. **Solve |2p − 9| = 5.**

Solution.
2p − 9 = 5 gives p = 7; 2p − 9 = −5 gives p = 2.
p = 7 or 2, and their sum is 9.



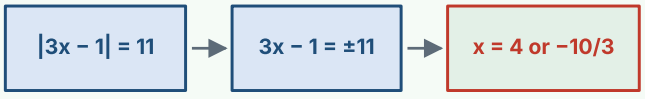
Example 2. **Solve |x + 3| = 8.**

Solution.
x + 3 = 8 or x + 3 = −8.
x = **5 or −11**.



Example 3. **Solve |3x − 1| = 11.**

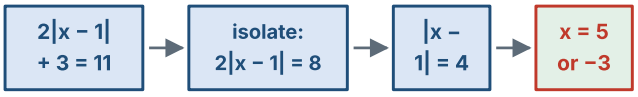
Solution.
3x − 1 = 11 gives x = 4; 3x − 1 = −11 gives x = −10/3.
x = **4 or −10/3**.



3 Isolate the bars first

get |...| alone on one side before splitting

If something is added to or multiplying the absolute value, undo that first. Splitting too early produces two wrong equations instead of one right pair.

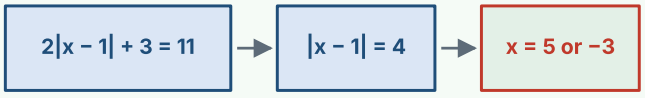


the +3 and the 2 apply to the whole distance

The 2 and the +3 sit outside the bars, so they act on the whole distance. Only once the bars stand alone can the split be made.

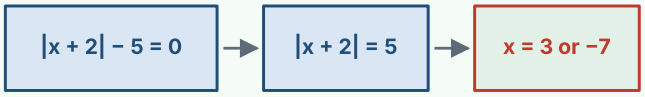
Example 1. **Solve $2|x - 1| + 3 = 11$.**

Solution.
Isolate: $2|x - 1| = 8$, so $|x - 1| = 4$.
$x - 1 = \pm 4$, giving $x =$ **5 or −3**.



Example 2. **Solve $|x + 2| - 5 = 0$.**

Solution.
$|x + 2| = 5$.
$x =$ **3 or −7**.



Example 3. **Why not split $2|x - 1| + 3 = 11$ straight away?**

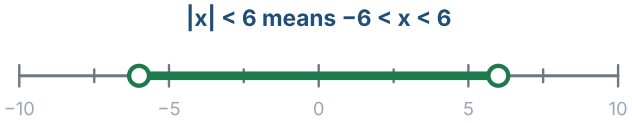
Solution.
The +3 and the 2 are outside the bars, so they apply to the whole distance. Splitting first would attach them to only one side and **both answers would be wrong**.

$2\|x - 1\| + 3 = 11$	the 2 and 3 are outside
isolate first	$\|x - 1\| = 4$
then split	$x - 1 = \pm 4$

4 Absolute value inequalities: the ‘and’ case

$|x| < c$ means $-c < x < c$ — one interval, between

Less than means close to zero, so the solution is a single band around the centre. Write it as a compound inequality and solve all three parts at once.



Everything within six units of zero, in either direction: one unbroken band, with both ends open because the inequality is strict.

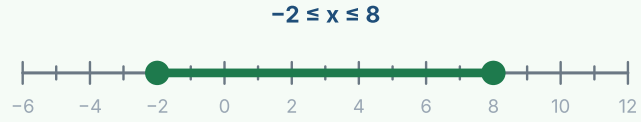
Example 1. **Solve $|x| < 6$.**

Solution.
$-6 < x < 6$.



Example 2. **Solve $|x - 3| \le 5$.**

Solution.
$-5 \le x - 3 \le 5$, so add 3 throughout.
$-2 \le x \le 8$.



Example 3. **Solve $|2x + 1| < 7$.**

Solution.
$-7 < 2x + 1 < 7$ gives $-8 < 2x < 6$.
$-4 < x < 3$.



5 Absolute value inequalities: the ‘or’ case

$|x| > c$ means $x < -c$ or $x > c$ — two separate pieces

Greater than means far from zero, so the solution splits into two rays with a gap in the middle. It can never be written as a single interval.

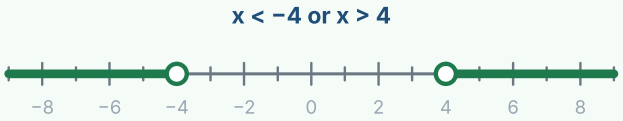


Two pieces with a hole between them. Writing this as one interval is impossible — the middle values fail the inequality.

Example 1. **Solve $|x| > 4$.**

Solution.

$x < -4$ or $x > 4$.

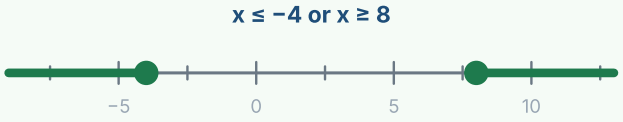


Example 2. **Solve $|x - 2| \geq 6$.**

Solution.

$x - 2 \geq 6$ or $x - 2 \leq -6$.

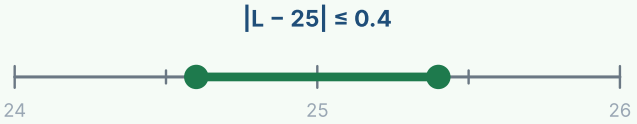
$x \geq 8$ or $x \leq -4$.



Example 3. **A machine part must be within 0.4 mm of 25 mm. Write that as an inequality.**

Solution.

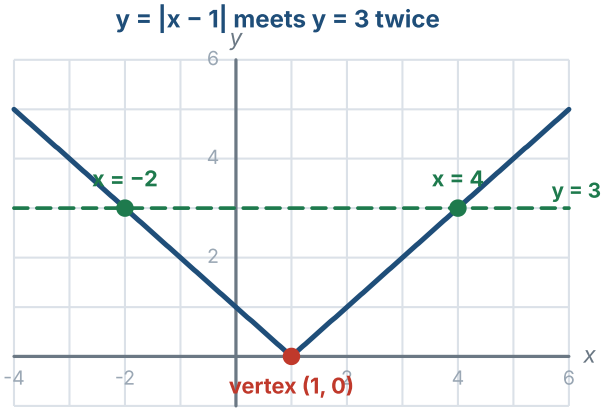
$|L - 25| \leq 0.4$, which means $24.6 \leq L \leq 25.4$.



6 The V-shaped graph

$y = |x - h| + k$ is a V with vertex (h, k)

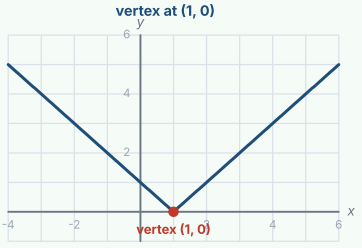
The graph folds upward at the vertex. A horizontal line cuts it twice above the vertex, once at the vertex, and not at all below — which is how the number of solutions is decided.



The line $y = 3$ sits above the vertex, so it crosses both arms, at $x = -2$ and $x = 4$. Lowering it to the vertex would give one crossing; below, none.

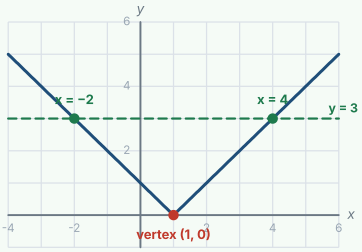
Example 1. **Find the vertex of $y = |x - 1|$.**

Solution.
(1, 0).



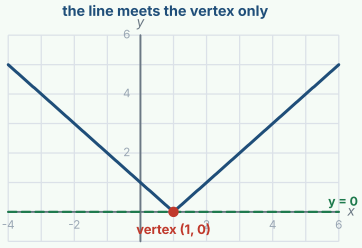
Example 2. **How many solutions does $|x - 1| = 3$ have?**

Solution.
The line $y = 3$ is above the vertex, so it cuts the V twice.
Two: $x = 4$ and $x = -2$.



Example 3. **How many solutions does $|x - 1| = 0$ have?**

Solution.
The line passes exactly through the vertex.
One: $x = 1$.



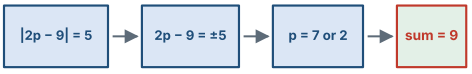
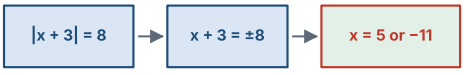
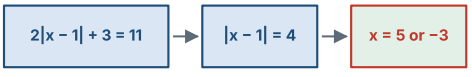
Part 1 — With Desmos (10 questions · every one is faster on the graph)

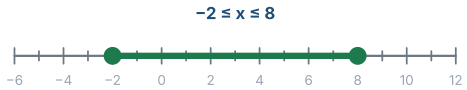
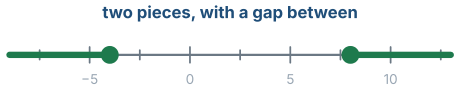
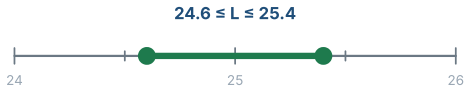
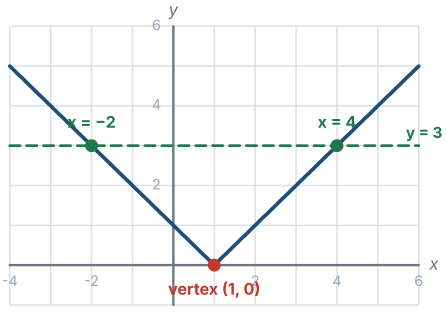
The SAT gives you Desmos on every maths question. Here it is **a V-grapher**. Type abs(x − 1) alongside a horizontal line and count the crossings to see how many solutions there are.

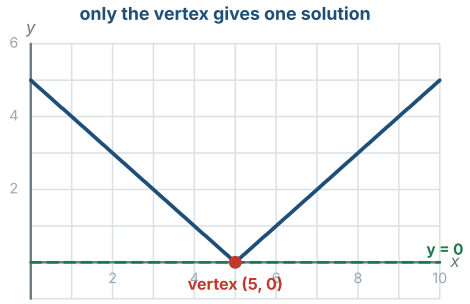
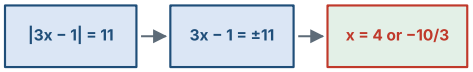
#	Question	Type into Desmos	Then	Answer
D1	What is $\|-7\| + \|3\|$?	`abs(-7)+abs(3)`	Use abs() in Desmos.	10
D2	Solve $\|2p - 9\| = 5$. What is the sum of the solutions?	`7+2`	Graph both sides and read the two crossings.	9
D3	Solve $\|x + 3\| = 8$. What is the smaller solution?	`-8-3`	Two cases, two answers.	-11
D4	Solve $2\|x - 1\| + 3 = 11$. What is the larger solution?	`4+1`	Isolate the bars before splitting.	5
D5	Solve $\|x - 3\| \le 5$. What is the upper bound of x?	`5+3`	The 'and' case.	8
D6	Solve $\|2x + 1\| < 7$. What is the lower bound of x?	`(-7-1)/2`	Three parts at once.	-4
D7	Solve $\|x - 2\| \ge 6$. What is the smaller boundary?	`-6+2`	The 'or' case.	-4
D8	A part must be within 0.4 mm of 25 mm. What is the smallest allowed length?	`25-0.4`	Centre minus the tolerance.	24.6
D9	Graph $y = \|x - 1\|$. What is the x-coordinate of the vertex?	`y=abs(x-1)`	The V folds at the vertex.	1
D10	How many solutions does $\|x - 1\| = 3$ have?	`y=abs(x-1) and y=3`	Count the intersections.	2

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Decide first whether the inequality is the 'and' case (one band) or the 'or' case (two pieces).

#	Level	Question	Answer	Solution
P1	Hard	The price p, in talens, of a lantern satisfies $\|2p - 9\| = 5$. What is the sum of all possible values of p? 	9	$2p - 9 = 5$ gives $p = 7$, and $2p - 9 = -5$ gives $p = 2$. $7 + 2 = \mathbf{9}$. Taking only the positive case and answering 7 is the usual error.
P2	Hard	Solve $\|x + 3\| = 8$. Give both solutions. 	x = 5 and x = −11	$x + 3 = 8$ gives $x = 5$; $x + 3 = -8$ gives $x = -11$. Both are valid: the bars hide a plus-or-minus.
P3	Hard	Solve $2\|x - 1\| + 3 = 11$. Give both solutions. 	x = 5 and x = −3	Isolate first: $2\|x - 1\| = 8$, so $\|x - 1\| = 4$. $x - 1 = \pm 4$ gives **5 or −3**. Splitting before isolating gives the wrong pair.

#	Level	Question	Answer	Solution
P4	Hard	Solve the inequality $\|x - 3\| \le 5$. Write your answer as a compound inequality. 	$-2 \le x \le 8$	Less than means a single band: $-5 \le x - 3 \le 5$. Adding 3 throughout gives **$-2 \le x \le 8$**.
P5	Hard	Solve the inequality $\|x - 2\| \ge 6$. Describe all values of x. 	$x \le -4$ or $x \ge 8$	Greater than means two separate pieces: $x - 2 \ge 6$ or $x - 2 \le -6$. **$x \ge 8$ or $x \le -4$.** This cannot be written as one interval.
P6	Hard	A machine part must be within 0.4 millimetres of a target length of 25 millimetres. Write an absolute-value inequality for the acceptable length L. 	$\|L - 25\| \le 0.4$	‘Within 0.4 of 25’ is a distance statement. **$\|L - 25\| \le 0.4$**, which unpacks to $24.6 \le L \le 25.4$.
P7	Very hard	How many solutions does the equation $\|3x - 6\| = -2$ have? Explain briefly. $\|3x - 6\| = -2$ — a distance cannot be negative check c first — no solutions, no work needed	none	An absolute value is a distance and can never be negative, so no value of x can satisfy it. **No solutions.** There is nothing to split — checking the right-hand side first saves the whole calculation.
P8	Very hard	Solve $\|2x + 1\| < 7$. Write your answer as a compound inequality. 	$-4 < x < 3$	$-7 < 2x + 1 < 7$ gives $-8 < 2x < 6$. Dividing by 2 gives **$-4 < x < 3$**.
P9	Very hard	The graph of $y = \|x - 1\|$ is drawn together with the horizontal line $y = 3$. At how many points do they intersect, and at which x-values? 	two, at $x = 4$ and $x = -2$	y = 3 sits above the vertex (1, 0), so it cuts both arms of the V. $\|x - 1\| = 3$ gives $x - 1 = \pm 3$, so x = **4 or −2**.

#	Level	Question	Answer	Solution
P10	Very hard	For what value of c does the equation $\|x - 5\| = c$ have exactly one solution? 	0	A horizontal line meets the V twice above the vertex and not at all below it. Exactly one intersection happens only at the vertex itself, so c = **0**.
P11	Very hard	Solve $\|3x - 1\| = 11$. Give both solutions. 	x = 4 and x = −10/3	$3x - 1 = 11$ gives $3x = 12$, so $x = 4$. $3x - 1 = -11$ gives $3x = -10$, so $x = $ **−10/3**.
P12	Very hard	A thermostat keeps a room within 1.5°C of 21°C. Write an absolute-value inequality for the temperature T, and state the range of acceptable temperatures. 	$\|T - 21\| \le 1.5$, so $19.5 \le T \le 22.5$	'Within 1.5 of 21' gives $\|T - 21\| \le 1.5$. Unpacking: $-1.5 \le T - 21 \le 1.5$, so **$19.5 \le T \le 22.5$**.

The one habit to build. Isolate the bars before splitting into two cases, and check the right-hand side is not negative — if it is, there is no solution and no work to do.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

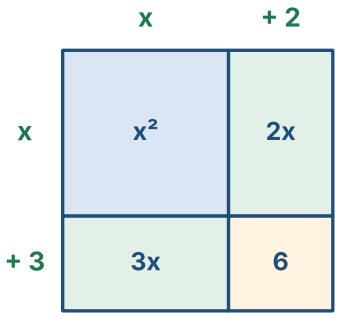
SAT Math 2.1 — Equivalent Expressions

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Multiplying and collecting

multiply every term by every term | then collect only like terms

Like terms share the same variable to the same power: $3x^2$ and $5x^2$ combine, $3x^2$ and $5x$ do not. The area model keeps every product in view and is worth drawing whenever a minus sign is involved.

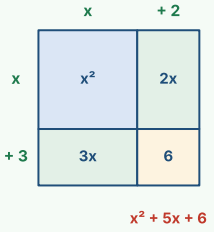


$$(x + 2)(x + 3) = x^2 + 5x + 6$$

The rectangle is (x + 2) wide and (x + 3) tall, so its area is the product. Each piece is one of the four multiplications, and the two green pieces are the like terms that collect into 5x.

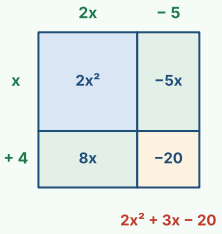
Example 1. **Expand (x + 2)(x + 3).**

Solution.
x·x + 3x + 2x + 6.
= **$x^2 + 5x + 6$**.



Example 2. **Expand (2x − 5)(x + 4).**

Solution.
$2x^2 + 8x - 5x - 20$.
= **$2x^2 + 3x - 20$**.



Example 3. **Simplify $(3x^2 + 2x - 1) - (x^2 - 4x + 6)$.**

Solution.
Distribute the minus to *every* term: $3x^2 + 2x - 1 - x^2 + 4x - 6$.
$= \mathbf{2x^2 + 6x - 7}$.

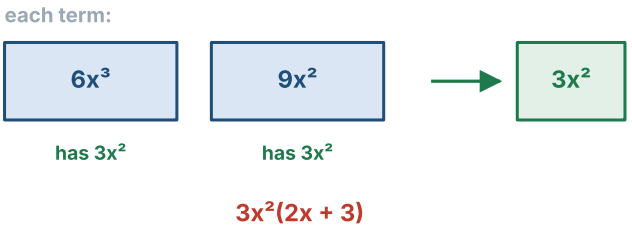
the minus belongs to every term in the bracket

$-(x^2 - 4x + 6)$	$= -x^2 + 4x - 6$
every sign inside flips	all three, not just the first

2 Common factor first, always

take out the greatest common factor before anything else

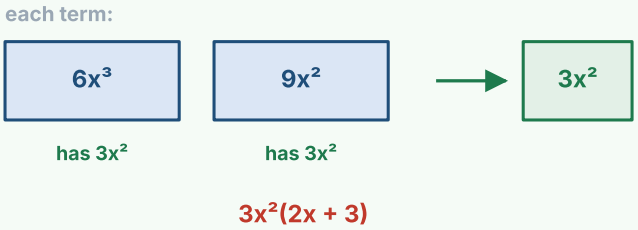
Pulling out the GCF turns an awkward expression into a familiar one and is the step most often skipped. Check the numbers and every variable: the GCF of $6x^3$ and $9x^2$ is $3x^2$, not 3.



Both terms carry three lots of x^2, so $3x^2$ comes out in front and what is left goes inside the bracket. Take the numbers *and* the letters — 3 alone would not be greatest.

Example 1. **Factor $6x^3 + 9x^2$ completely.**

Solution.
GCF is $3x^2$.
$= \mathbf{3x^2(2x + 3)}$.

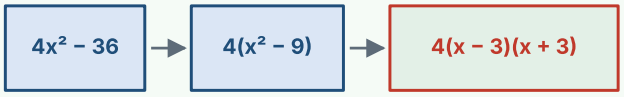


Example 2. **Factor $4x^2 - 36$ completely.**

Solution.

GCF 4 first: $4(x^2 - 9)$, and $x^2 - 9$ is a difference of squares.

= **$4(x - 3)(x + 3)$**. Stopping at $4(x^2 - 9)$ is not *completely* factored.



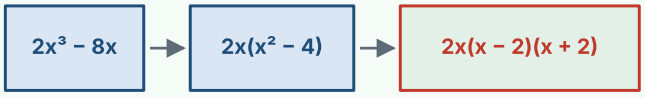
the GCF first, then the pattern underneath it

Example 3. **Factor $2x^3 - 8x$.**

Solution.

GCF $2x$: $2x(x^2 - 4)$.

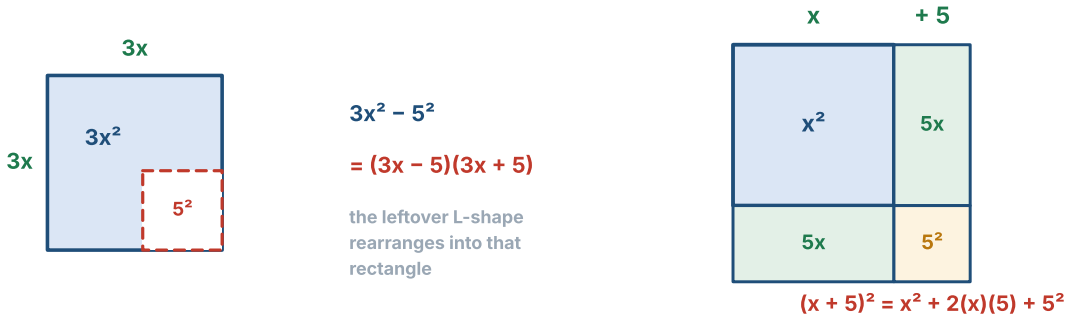
= **$2x(x - 2)(x + 2)$**.



3 Difference of squares and perfect squares

$$a^2 - b^2 = (a - b)(a + b) \;\big|\; a^2 \pm 2ab + b^2 = (a \pm b)^2$$

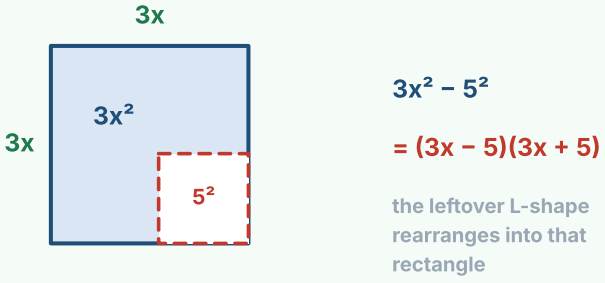
These two patterns appear constantly, often disguised. A sum of squares, $x^2 + 9$, does *not* factor over the real numbers — recognising that saves time on a question designed to waste it.



Left: cut a small square out of a big one and the L-shape left over rearranges into the rectangle (a − b)(a + b). Right: a perfect square has two identical middle pieces, which is where the 2ab comes from.

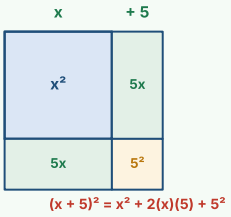
Example 1. **Factor $9x^2 - 25$.**

Solution.
This is $(3x)^2 - 5^2$.
$= \mathbf{(3x - 5)(3x + 5)}$.



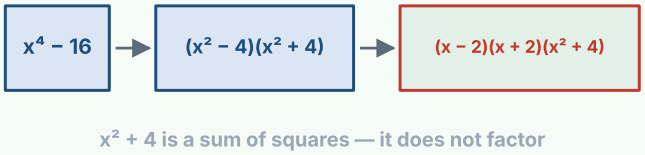
Example 2. **Factor $x^2 + 10x + 25$.**

Solution.
$25 = 5^2$ and $10x = 2(x)(5)$.
$= \mathbf{(x + 5)^2}$.



Example 3. **Factor $x^4 - 16$.**

Solution.
$(x^2)^2 - 4^2 = (x^2 - 4)(x^2 + 4)$, and the first bracket factors again.
$= \mathbf{(x - 2)(x + 2)(x^2 + 4)}$. The sum of squares stops here.

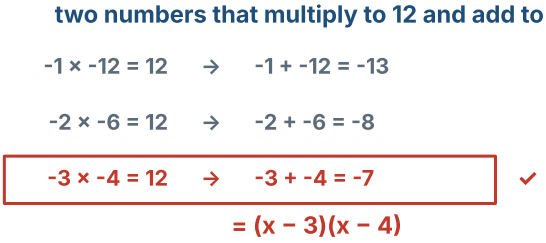


$x^2 + 4$ is a sum of squares — it does not factor

4 Factoring trinomials

$x^2 + bx + c$: find two numbers that multiply to c and add to b

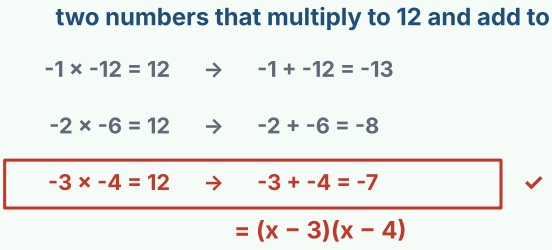
When the leading coefficient is not 1, multiply a by c, split the middle term using two numbers that multiply to ac and add to b, then factor by grouping. Writing out the factor pairs of ac beats guessing.



List the factor pairs and test the sum of each. Only one pair does both jobs, and that pair is the pair of brackets.

Example 1. **Factor $x^2 - 7x + 12$.**

Solution.
Two numbers multiplying to 12 and adding to −7: −3 and −4.
$= \mathbf{(x - 3)(x - 4)}$.



Example 2. **Factor $x^2 + 2x - 15$.**

Solution.

Multiply to −15, add to 2: 5 and −3.

= **(x + 5)(x − 3)**.

two numbers that multiply to −15 and add to 2

-1 × 15 = -15 → -1 + 15 = 14

1 × -15 = -15 → 1 + -15 = -14

3 × -5 = -15 → 3 + -5 = -2

5 × -3 = -15 → 5 + -3 = 2 ✓

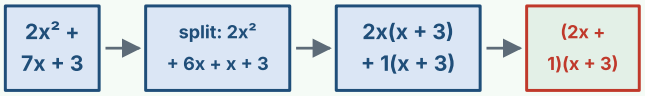
= (x + 5)(x − 3)

Example 3. **Factor $2x^2 + 7x + 3$.**

Solution.

ac = 6; two numbers multiplying to 6 and adding to 7 are 6 and 1.

$2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3) =$ **(2x + 1)(x + 3)**.



ac = 6, and 6 + 1 = 7

5 Exponent rules

$$x^m \cdot x^n = x^{m+n} \mid x^m / x^n = x^{m-n} \mid (x^m)^n = x^{mn} \mid x^{-n} = 1/x^n \mid x^0 = 1$$

Add exponents when multiplying, subtract when dividing, multiply when raising a power to a power. A negative exponent is a reciprocal, not a negative number: $2^{-3} = 1/8$, not -8.

five rules, and one worked instance of each

$x^m \cdot x^n = x^{m+n}$	$(3x^2y)(4x^3y^5) = 12x^5y^6$
$x^m / x^n = x^{m-n}$	$12^5/12^3 = 12^2 = 144$
$(x^m)^n = x^{mn}$	$(2x^3)^4 = 16x^{12}$
$x^{-n} = 1/x^n$	$5^{-2} = 1/25 = 0.04$
$x^0 = 1$	$7^0 = 1$

Read each rule with its example beside it. The fourth is the one that costs marks: a negative exponent flips the base, it never makes the value negative.

Example 1. **Simplify $(3x^2y)(4x^3y^5)$.**

Solution.
Multiply coefficients, add exponents.
= **$12x^5y^6$**.

coefficients: 3 × 4	= 12
x: 2 + 3	= x^5
y: 1 + 5	= y^6

Example 2. **Simplify $(2x^3)^4$.**

Solution.
Raise both parts to the fourth: $2^4 = 16$ and x^{12}.
= **$16x^{12}$**.

the 2 is inside the bracket too	$2^4 = 16$
exponents multiply	$(x^3)^4 = x^{12}$

Example 3. **Write 5^{-2} as a fraction.**

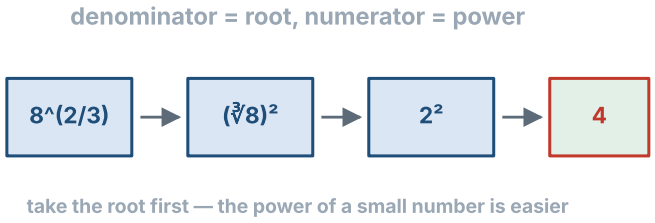
Solution.
1/25. A negative exponent flips the base; it never makes the value negative.

5^{-2}	$= 1/5^2 = 1/25 = 0.04$
not −25	the sign of the base never changes

6 Fractional exponents and radicals

$$x^{1/n} = \sqrt[n]{x} \mid x^{m/n} = \sqrt[n]{(x^m)}$$

The denominator of the exponent is the root; the numerator is the power. This conversion is the whole skill — once an expression is written with exponents, the ordinary exponent rules apply and radicals stop being special.



Denominator 3 means a cube root, numerator 2 means squared. Take the root first and you are squaring 2 instead of cube-rooting 64.

Example 1. **Write √x as a power of x.**

Solution.
$x^{1/2}$.

√x	= x^(1/2)
∛x	= x^(1/3)
ⁿ√x	= x^(1/n)

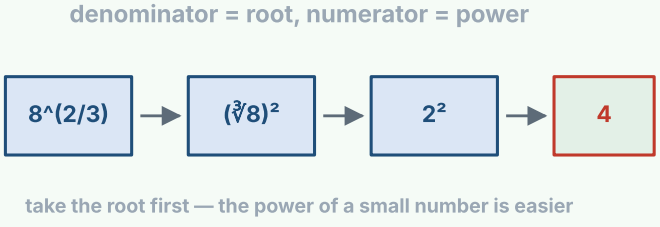
Example 2. **Simplify $x^{1/2} \cdot x^{1/3}$.**

Solution.
Add the exponents: 1/2 + 1/3 = 5/6.
= $x^{5/6}$.

same base → add the exponents	1/2 + 1/3 = 5/6
answer	x^(5/6)

Example 3. **Evaluate $8^{2/3}$.**

Solution.
Cube root first, then square: ∛8 = 2, and 2^2 = **4**. Taking the root first keeps the numbers small.



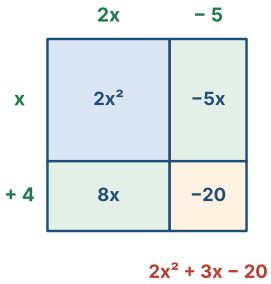
Part 1 — With Desmos (10 questions · every one is faster on the graph)

The SAT gives you Desmos on every maths question. For this topic it is **a checker, not a solver**. Desmos will not factor for you, but graphing your answer against the original shows instantly whether they are the same expression.

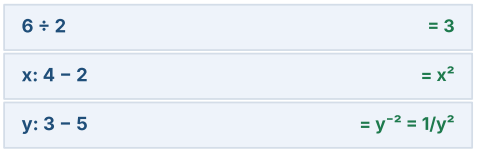
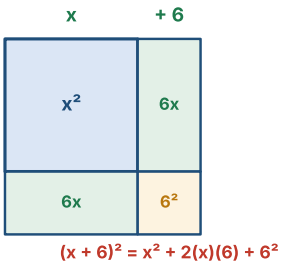
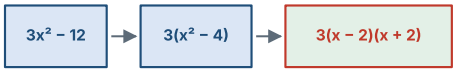
#	Question	Type into Desmos	Then	Answer
D1	Expand $(x + 2)(x + 3)$ and state the coefficient of x.	`(x+2)(x+3)`	Desmos will not expand symbolically; graph both forms to confirm they agree.	5
D2	What is the value of $8^{2/3}$?	`8^(2/3)`	Use the caret for exponents.	4
D3	What is the value of 5^{-2} as a decimal?	`5^(-2)`	A negative exponent gives a reciprocal.	0.04
D4	Simplify $12^5/12^3$.	`12^5/12^3`	Subtracting exponents gives 12^2.	144
D5	Evaluate $(2x^3)^4$ at x = 1.	`(2*1^3)^4`	Checks the coefficient $2^4 = 16$.	16
D6	At x = 5, what is the value of $x^2 - 7x + 12$?	`5^2-7*5+12`	Substitute to test a factorisation.	2
D7	At x = 3, does $(x - 3)(x - 4)$ equal zero?	`(3-3)*(3-4)`	A factor of zero makes the product zero.	0
D8	Evaluate $9x^2 - 25$ at x = 2.	`9*2^2-25`	Compare with $(3x - 5)(3x + 5)$ at x = 2.	11
D9	What is the value of $x^{1/2} \cdot x^{1/3}$ at x = 64?	`64^(1/2)*64^(1/3)`	Equals $64^{5/6}$.	32
D10	Evaluate $2x^3 - 8x$ at x = 2.	`2*2^3-8*2`	x = 2 is a zero, which confirms $(x - 2)$ is a factor.	0

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Read the word **completely** when it appears: more than one option may be equal in value, and only one is fully factored.

#	Level	Question	Answer	Solution
P1	Hard	Write an equivalent expression for $(2x - 5)(x + 4)$. 	$2x^2 + 3x - 20$	$8x - 5x = +3x$. $= 2x^2 + 3x - 20$. A common wrong answer multiplies only the first and last terms; C reverses the sign of the middle term.
P2	Hard	Write an equivalent expression for $(3x^2 + 2x - 1) - (x^2 - 4x + 6)$. the minus belongs to every term in the bracket $-(x^2 - 4x + 6)$ → $= -x^2 + 4x - 6$ every sign inside flips — all three, not just the first	$2x^2 + 6x - 7$	The minus sign applies to all three terms of the second bracket. $= 2x^2 + 6x - 7$. Two common wrong answers subtract only the first term.

#	Level	Question	Answer	Solution
P3	Hard	Factor completely: $4x^2 - 36$. $4x^2 - 36$ → $4(x^2 - 9)$ → $4(x - 3)(x + 3)$	$4(x - 3)(x + 3)$	Take the GCF 4 first, then the difference of squares. = **$4(x - 3)(x + 3)$**. Two common wrong answers equal in value but not *completely* factored — read the word 'complete' in the stem.
P4	Hard	What is the complete factorisation of $2x^2 + 7x + 3$? $2x^2 + 7x + 3$ → $2x^2 + 6x + x + 3$ → $2x(x + 3) + 1(x + 3)$ → $(2x + 1)(x + 3)$	$(2x + 1)(x + 3)$	ac = 6, and 6 + 1 = 7, so split the middle term. $2x(x + 3) + 1(x + 3)$ = **$(2x + 1)(x + 3)$**. Expanding that answer gives $2x^2 + 7x + 3$.
P5	Hard	Write an equivalent expression for $(2x^3y^2)^3$. the coefficient is cubed too — $2^3 = 8$ x: 3 × 3 — x^9 y: 2 × 3 — y^6	$8x^9y^6$	Every factor is cubed: $2^3 = 8$, and the exponents multiply by 3. = **$8x^9y^6$**. A common wrong answer multiplies the coefficient by 3 instead of cubing it.
P6	Hard	What is the value of $8^{2/3}$? denominator = root, numerator = power 8^(2/3) → $(\sqrt[3]{8})^2$ → 2^2 → 4 take the root first — the power of a small number is easier	4	The denominator is the root and the numerator is the power. $(\sqrt[3]{8})^2 = 2^2$ = **4**.
P7	Very hard	Write an equivalent expression for $x^4 - 16$, factored completely over the real numbers. $x^4 - 16$ → $(x^2 - 4)(x^2 + 4)$ → $(x - 2)(x + 2)(x^2 + 4)$	$(x - 2)(x + 2)(x^2 + 4)$	$x^4 - 16 = (x^2 - 4)(x^2 + 4)$, and $x^2 - 4$ factors again. = **$(x - 2)(x + 2)(x^2 + 4)$**. A sum of squares does not factor over the reals, so that answer is as far as it goes; A stops one step early.
P8	Very hard	If $x^{-2} = 1/49$ and $x > 0$, what is the value of x? $x^{-2} = 1/49$ — so $x^2 = 49$ $x > 0$ — $x = 7$, not -7	7	$x^{-2} = 1/x^2$, so $x^2 = 49$. With $x > 0$, x = **7**.
P9	Very hard	Write an equivalent expression for $(x^{1/2})(x^{1/3})$ for $x > 0$. same base → add — 1/2 + 1/3 = 5/6 multiplying them gives 1/6 — which is a different expression	$x^{5/6}$	Multiplying powers of the same base adds the exponents. 1/2 + 1/3 = 5/6, so the answer is **$x^{5/6}$**. A common wrong answer multiplies the exponents.

#	Level	Question	Answer	Solution
P10	Very hard	The expression $(6x^4y^3)/(2x^2y^5)$ is equivalent to which of the following, for x and y nonzero? 	$3x^2/y^2$	Divide the coefficients and subtract the exponents: 6/2 = 3, $x^{4-2} = x^2$, $y^{3-5} = y^{-2}$. = **$3x^2/y^2$**. A common wrong answer leaves the negative exponent in the numerator as a positive one.
P11	Very hard	For what value of k is $x^2 + kx + 36$ a perfect square trinomial, given k > 0? 	12	A perfect square $(x + a)^2$ needs $a^2 = 36$, so a = 6, and the middle term is 2a. k = 2(6) = **12**.
P12	Very hard	The expression $3x^2 − 12$ can be written as $3(x − a)(x + a)$ for a positive constant a. What is the value of a? 	2	$3x^2 − 12 = 3(x^2 − 4) = 3(x − 2)(x + 2)$. So a = **2**. A common wrong answer takes a to be the 4 inside the bracket rather than its square root.

The one habit to build. Take the common factor out first, every single time. Half the ‘hard’ questions in this topic become one-liners once the GCF is gone.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

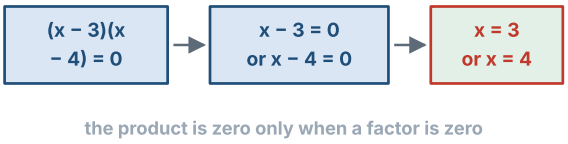
SAT Math 2.2 — Quadratic Equations

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

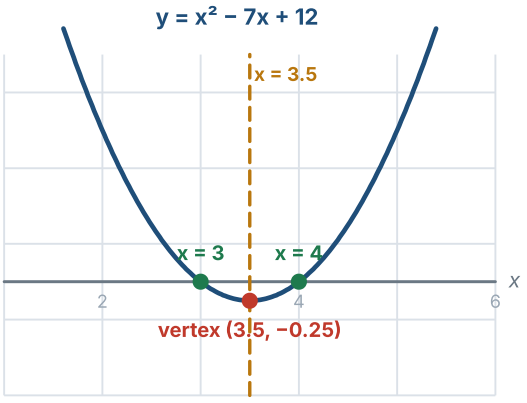
1 Solving by factoring

get = 0 first | factor | set each factor to zero

The zero-product property only works against zero, so move everything to one side before you factor. Dividing both sides by a variable loses a solution — factor it out instead.



the product is zero only when a factor is zero



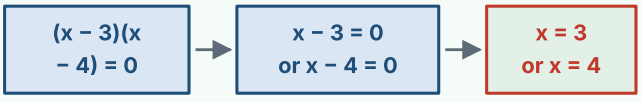
The factored form names the roots directly, and the graph shows them as the two places the curve meets the axis. Both brackets are zero nowhere else, which is why the method works.

Example 1. **Solve $x^2 - 7x + 12 = 0$.**

Solution.

$(x - 3)(x - 4) = 0$.

$x =$ **3 or 4**.



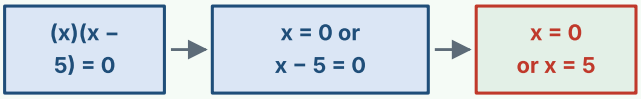
the product is zero only when a factor is zero

Example 2. **Solve $x^2 = 5x$.**

Solution.

$x^2 - 5x = 0$, so $x(x - 5) = 0$.

$x =$ **0 or 5**. Dividing by x at the start would have lost $x = 0$.



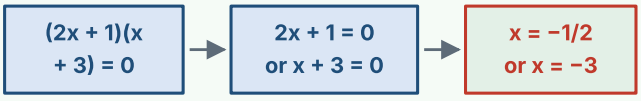
the product is zero only when a factor is zero

Example 3. **Solve $2x^2 + 7x + 3 = 0$.**

Solution.

$(2x + 1)(x + 3) = 0$.

$x =$ **$-1/2$ or -3**.



the product is zero only when a factor is zero

2 The quadratic formula

$x = [-b \pm \sqrt{(b^2 - 4ac)}] \div 2a$ — not on the reference sheet

Learn it: the SAT does not give it to you. Write a, b and c down before substituting, and keep the signs of b and c with their numbers. It works on every quadratic, factorable or not.

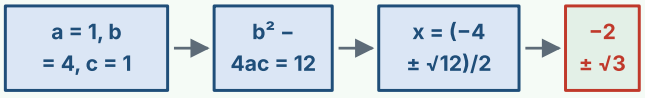
the formula works on every quadratic, factorable or not

$x = [-b \pm \sqrt{(b^2 - 4ac)}] \div 2a$	not on the reference sheet
write a, b and c down first	with their signs
$b^2 - 4ac$ is the discriminant	it decides how many roots

Three lines worth memorising. The sign of b travels with it: for $x^2 - 6x + 5$ the value of b is −6, so −b is +6.

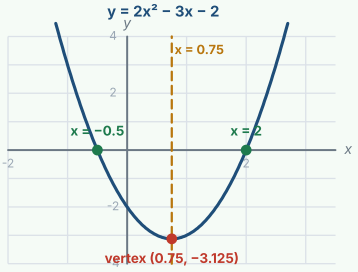
Example 1. **Solve $x^2 + 4x + 1 = 0$.**

Solution.
a = 1, b = 4, c = 1, so the discriminant is 16 − 4 = 12.
$x = (-4 \pm \sqrt{12})/2 =$ **$-2 \pm \sqrt{3}$**.



Example 2. **Solve $2x^2 - 3x - 2 = 0$.**

Solution.
Discriminant = 9 + 16 = 25, and √25 = 5.
x = (3 ± 5)/4, so x = **2 or −1/2**.



Example 3. **Why keep the sign with b?**

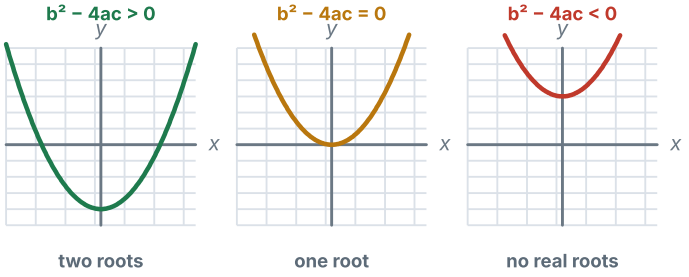
Solution.
For $x^2 - 6x + 5$, b = −6, so −b = +6. Writing b = 6 flips both roots — the single most common slip with the formula.

$x^2 - 6x + 5$	b = −6, so −b = +6
writing b = 6	gives −5 and −1 — both wrong

3 The discriminant

$b^2 - 4ac > 0$ → two real solutions | $= 0$ → one | < 0 → none

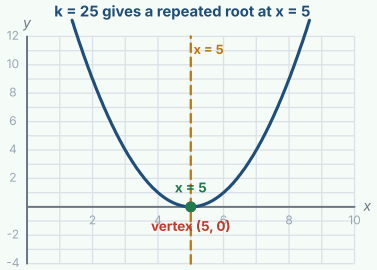
Whenever a question says ‘exactly one solution’, ‘a repeated root’, ‘touches the x-axis once’ or ‘the line is tangent’, it is asking you to set $b^2 - 4ac = 0$. One idea, many disguises.



The discriminant is the quantity under the root, so it counts the crossings: positive gives two, zero gives a touch, negative gives none because the root of a negative is not real.

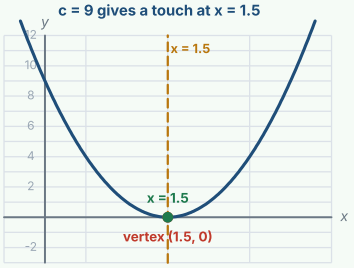
Example 1. **For what value of k does $x^2 - 10x + k = 0$ have exactly one solution?**

Solution.
$100 - 4k = 0$.
$k = \mathbf{25}$.



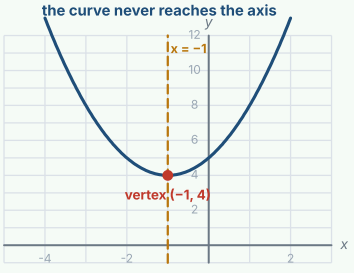
Example 2. **For what value of c does $4x^2 - 12x + c = 0$ have exactly one real solution?**

Solution.
$144 - 16c = 0$.
$c = \mathbf{9}$.



Example 3. **How many real solutions does $x^2 + 2x + 5 = 0$ have?**

Solution.
$4 - 20 = -16$, which is negative.
None.



4 Sum and product of the roots

sum = −b/a | product = c/a

When a question asks for the sum or product of the solutions, you never need the solutions themselves. Watch the sign: for $x^2 - 7x + 12$ the sum is +7, not −7.

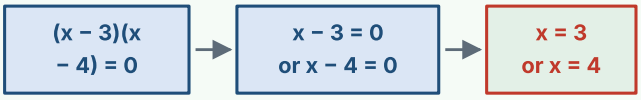
read them straight off the coefficients

sum of the roots	−b/a
product of the roots	c/a
$x^2 - 7x + 12$	sum −(−7)/1 = 7, product 12/1 = 12

The roots of $x^2 - 7x + 12$ are 3 and 4: they add to 7 and multiply to 12, exactly as −b/a and c/a predict — without solving anything.

Example 1. **What is the sum of the solutions of $x^2 - 7x + 12 = 0$?**

Solution.
−b/a = −(−7)/1 = **7**. Checking: 3 + 4 = 7.



the product is zero only when a factor is zero

Example 2. **What is the product of the solutions of $2x^2 + 7x + 3 = 0$?**

Solution.
c/a = 3/2 = **1.5**. Checking: (−1/2)(−3) = 1.5.

$2x^2 + 7x + 3$	a = 2, c = 3
product = c/a	3/2 = 1.5

Example 3. **The solutions of $x^2 + bx + 10 = 0$ sum to 7. What is b?**

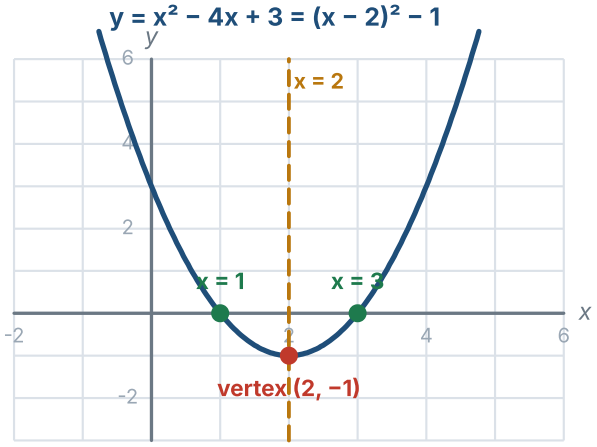
Solution.
−b = 7, so b = **−7**.

sum = −b/a = −b	and the sum is 7
so −b = 7	b = −7

5 Completing the square and vertex form

$x^2 + bx = (x + b/2)^2 - (b/2)^2$ | vertex form $y = a(x - h)^2 + k$ has vertex (h, k)

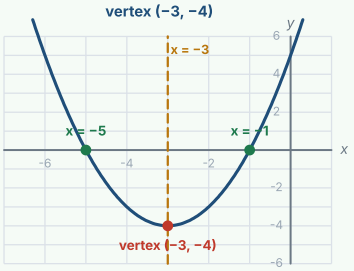
Halve the coefficient of x, square it, add and subtract it. Vertex form hands you the turning point directly, which is why the SAT asks you to produce it.



Both forms draw the same curve. Standard form shows the y-intercept 3; vertex form shows the turning point (2, −1) without any further work.

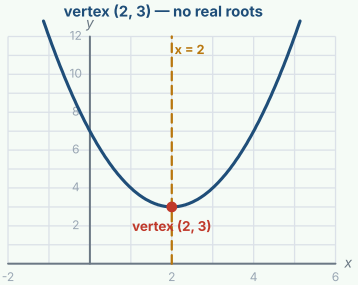
Example 1. **Write $x^2 + 6x + 5$ in vertex form.**

Solution.
Half of 6 is 3, and $3^2 = 9$.
$= (x + 3)^2 - 9 + 5 =$ **$(x + 3)^2 - 4$**, vertex $(-3, -4)$.



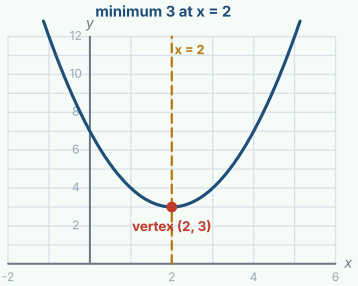
Example 2. **Write $x^2 - 4x + 7$ in vertex form.**

Solution.
Half of −4 is −2, squared is 4.
$=$ **$(x - 2)^2 + 3$**, vertex $(2, 3)$.



Example 3. **What is the minimum value of $y = x^2 - 4x + 7$?**

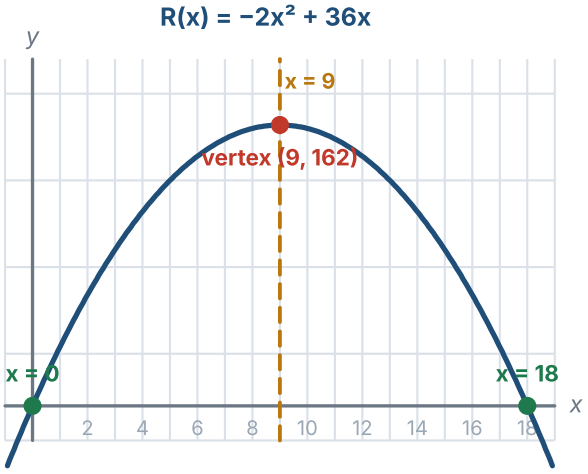
Solution.
From the vertex form, the smallest $(x - 2)^2$ can be is 0.
The minimum is **3**, reached at $x = 2$.



6 Quadratic word problems

maximum or minimum → the vertex | hits the ground / breaks even → a root

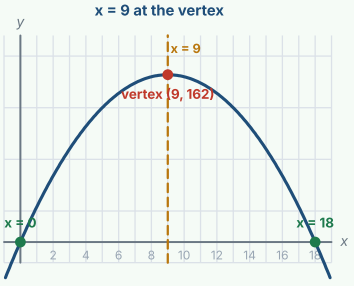
Translate the question into one of two requests. ‘What price gives the greatest revenue?’ wants the vertex’s x. ‘What is the greatest revenue?’ wants its y. Confusing those two costs marks every time.



The vertex (9, 162) answers both questions at once: x = 9 is the price that maximises revenue, y = 162 is the revenue itself. The roots are where revenue falls to zero.

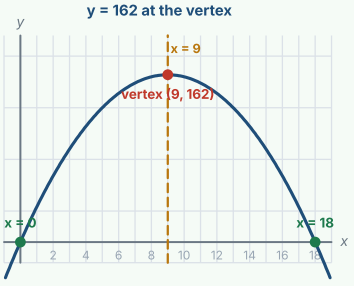
Example 1. **R(x) = −2x² + 36x. What price x gives the maximum revenue?**

Solution.
Vertex at x = −b/2a = −36/(−4) = **$9**.



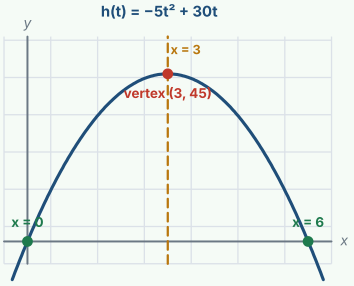
Example 2. **For that same R, what is the maximum revenue?**

Solution.
R(9) = −2(81) + 324 = **$162**. Note this answers a different question from the one above.



Example 3. **A ball’s height is h(t) = −5t² + 30t. When does it land?**

Solution.
−5t(t − 6) = 0, so t = 0 or 6.
It lands at **t = 6 seconds**; t = 0 is the launch.



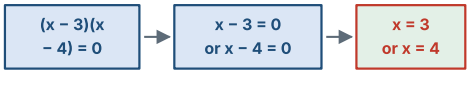
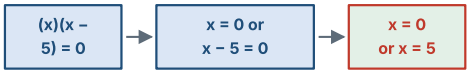
Part 1 — With Desmos (10 questions · every one is faster on the graph)

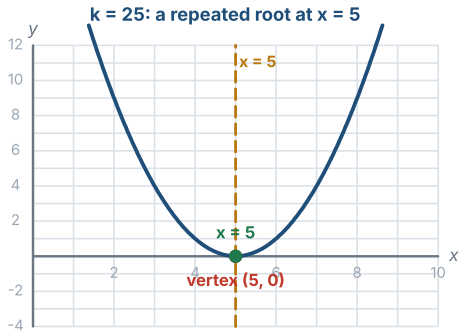
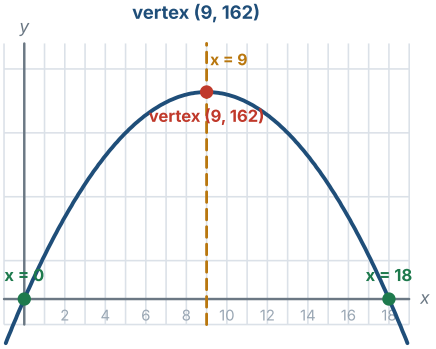
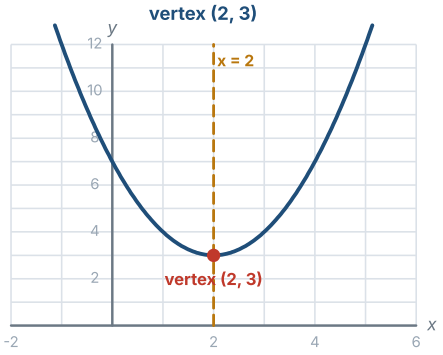
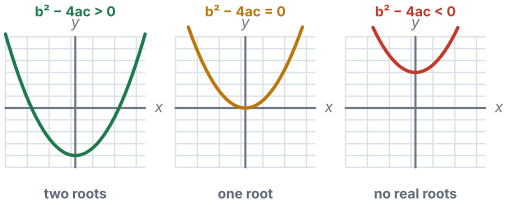
The SAT gives you Desmos on every maths question. Here it is **a root finder and a vertex finder**. Type the equation and read the x-intercepts, or type the function and click the turning point.

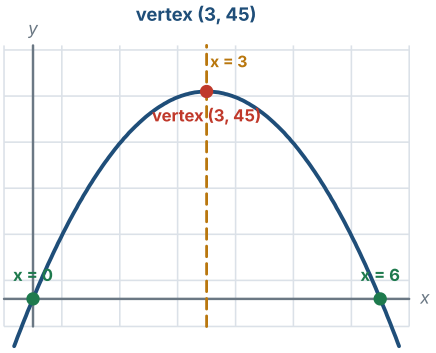
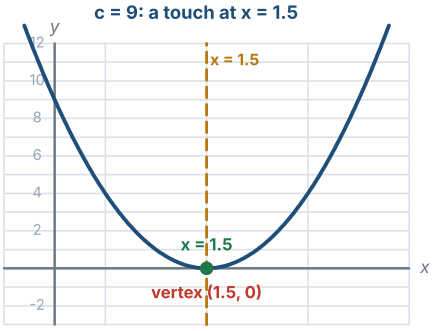
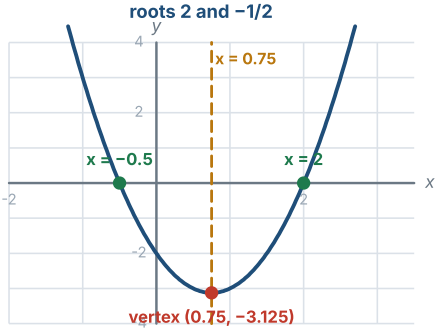
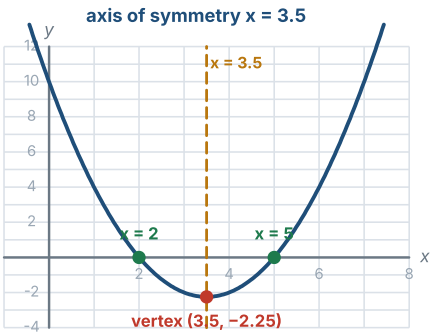
#	Question	Type into Desmos	Then	Answer
D1	What are the solutions of $x^2 - 7x + 12 = 0$?	`x^2-7x+12=0`	Type the equation; Desmos marks the roots on the x-axis.	3 and 4
D2	What is the sum of the solutions of $x^2 - 7x + 12 = 0$?	`3+4`	Or use −b/a directly.	7
D3	For what value of k does $x^2 - 10x + k = 0$ have exactly one solution?	`10^2/4`	Set the discriminant to zero: $k = b^2/4a$.	25
D4	What is the discriminant of $x^2 + 2x + 5 = 0$?	`2^2-4*1*5`	Negative means no real solutions.	-16
D5	Solve $2x^2 - 3x - 2 = 0$.	`2x^2-3x-2=0`	Graph it and read both x-intercepts.	2 and -0.5
D6	What is the minimum value of $y = x^2 - 4x + 7$?	`y=x^2-4x+7`	Graph it and click the turning point.	3
D7	$R(x) = -2x^2 + 36x$. What value of x gives the maximum?	`y=-2x^2+36x`	Click the vertex; Desmos labels it.	9
D8	For that R, what is the maximum value?	`-2*9^2+36*9`	The y of the vertex, not the x.	162
D9	At what positive time does $h(t) = -5t^2 + 30t$ return to zero?	`-5t^2+30t=0`	Read the right-hand root.	6
D10	What is the value of the discriminant of $4x^2 - 12x + 9$?	`12^2-4*4*9`	Zero means one repeated root.	0

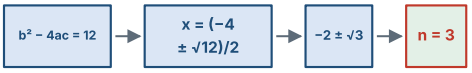
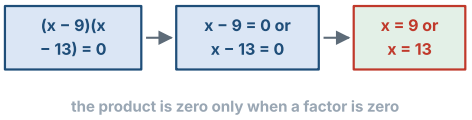
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Two questions decide most of these: is it asking **where** or **how much**, and does 'exactly one solution' appear anywhere in the stem?

#	Level	Question	Answer	Solution
P1	Hard	What is the sum of the solutions of the equation $x^2 - 7x + 12 = 0$?  the product is zero only when a factor is zero	7	Sum of roots = −b/a = −(−7)/1 = **7**, and indeed 3 + 4 = 7. A common wrong answer applies the formula without the sign of b — the error five of your class made on Test 5.
P2	Hard	If $x^2 = 5x$ and $x \neq 0$, what is the value of x?  the product is zero only when a factor is zero	5	$x^2 - 5x = 0$ gives $x(x - 5) = 0$, so x = 0 or 5. The condition $x \neq 0$ leaves **5**. Dividing by x at the start would also reach 5, but it silently discards a solution, which matters when the question does not rule it out.

#	Level	Question	Answer	Solution
P3	Hard	For what value of k does the equation $x^2 - 10x + k = 0$ have exactly one real solution? 	25	Exactly one solution means $b^2 - 4ac = 0$. $100 - 4k = 0$, so k = **25**. A common wrong answer is b/2 and B is b itself, both read off the equation rather than derived.
P4	Hard	A candle shop's daily revenue, in dollars, is $R(x) = -2x^2 + 36x$, where x is the price of one candle. What price gives the maximum daily revenue? 	$9	The vertex is at $x = -b/2a = -36/(-4) =$ **$9**. A common wrong answer is the maximum revenue R(9) = 162, which answers a different question; C is the coefficient read straight off the equation.
P5	Hard	Write an equivalent form of $x^2 - 4x + 7$. 	$(x - 2)^2 + 3$	Half of −4 is −2 and $(-2)^2 = 4$, so $x^2 - 4x = (x - 2)^2 - 4$. $=$ **$(x - 2)^2 + 3$**. A common wrong answer forgets to subtract the 4 back out.
P6	Hard	How many distinct real solutions does $x^2 + 2x + 5 = 0$ have? 	Zero	$b^2 - 4ac = 4 - 20 = -16 < 0$. A negative discriminant means **no real solutions** — the parabola never reaches the x-axis.

#	Level	Question	Answer	Solution
P7	Very hard	A ball is thrown upward and its height, in metres, after t seconds is $h(t) = -5t^2 + 30t$. What is the maximum height reached, in metres? 	45	The vertex is at $t = -30/(-10) = 3$ seconds. $h(3) = -45 + 90 =$ **45 metres**. Answering 3 gives the time, not the height.
P8	Very hard	In the equation $4x^2 - 12x + c = 0$, c is a constant. If the equation has exactly one real solution, what is the value of c? 	9	$144 - 16c = 0$, so $c = 144/16 =$ **9**. A common wrong answer is 12/4, read off the equation rather than derived.
P9	Very hard	The solutions of $2x^2 - 3x - 2 = 0$ are p and q. What is the value of pq? 	−1	Product of roots $= c/a = -2/2 =$ **−1**. Checking: the roots are 2 and −1/2, and $2 \times (-1/2) = -1$. A common wrong answer uses c/a without the sign of c.
P10	Very hard	The graph of $y = x^2 + bx + 10$ has its vertex on the line x = 3.5. What is the value of b? 	−7	The vertex sits at $x = -b/2a = -b/2$. $-b/2 = 3.5$ gives $b =$ **−7**. A common wrong answer drops the minus sign.

#	Level	Question	Answer	Solution
P11	Very hard	The equation $x^2 + 4x + 1 = 0$ has solutions of the form $-2 \pm \sqrt{n}$, where n is a constant. What is the value of n? 	3	Discriminant = 16 − 4 = 12, so $x = (-4 \pm \sqrt{12})/2 = -2 \pm \sqrt{12}/2$. $\sqrt{12}/2 = 2\sqrt{3}/2 = \sqrt{3}$, so n = **3**.
P12	Very hard	A rectangular garden has a perimeter of 44 metres and an area of 117 square metres. What is the length of its longer side, in metres? 	13	Half the perimeter is 22, so the sides are x and 22 − x with $x(22 - x) = 117$. $x^2 - 22x + 117 = 0$ factors as $(x - 9)(x - 13) = 0$, giving sides 9 and **13**. A common wrong answer is the shorter side.

The one habit to build. The quadratic formula is **not** on the reference sheet. Write a, b and c down with their signs before you substitute — losing the sign of b is the most expensive habit in this topic.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

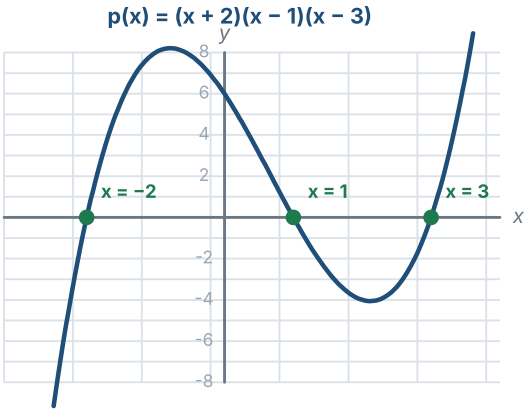
SAT Math 2.3 — Polynomials

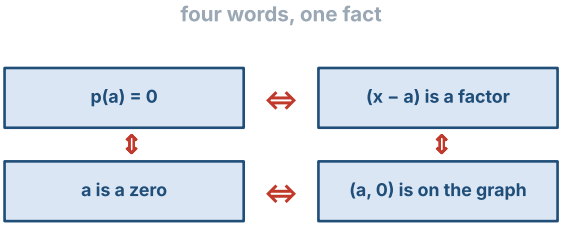
Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Zeros, roots, factors and intercepts

all four words mean the same thing: p(a) = 0 ⇔ (x − a) is a factor ⇔ (a, 0) is on the graph

The SAT swaps these words freely inside one question. Translate whichever you are given into whichever you need. A zero at x = 3 means the factor is (x − 3) — note the sign flip.

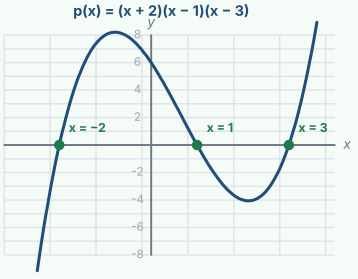




Each bracket switches off at one value of x, and that is where the curve meets the axis. The four statements on the right are the same fact in four vocabularies.

Example 1. **p(x) = (x + 2)(x − 1)(x − 3). What are its zeros?**

Solution.
Set each factor to zero.
x = **−2, 1 and 3**. Each sign flips.



Example 2. **A polynomial has zeros at x = −4 and x = 5. Give two of its factors.**

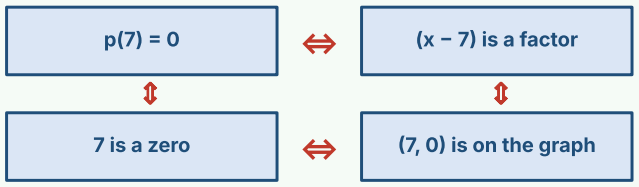
Solution.
(x + 4) and **(x − 5)**.

zero at x = −4	factor (x + 4)
zero at x = 5	factor (x − 5)
the sign always flips	x = a → (x − a)

Example 3. **The graph of p crosses the x-axis at (7, 0). What does that tell you about p?**

Solution.
p(7) = 0, and (x − 7) is a factor of p(x).

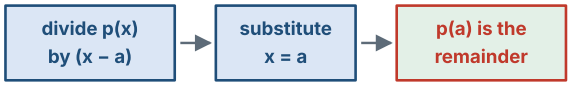
four words, one fact



2 The remainder and factor theorems

remainder on dividing by (x − a) is p(a) | remainder 0 ⇔ (x − a) is a factor

You never have to do the long division. To find the remainder, substitute. Watch the sign: dividing by (x + 2) means a = −2, because x + 2 = x − (−2).



and if p(a) = 0, the divisor is a factor

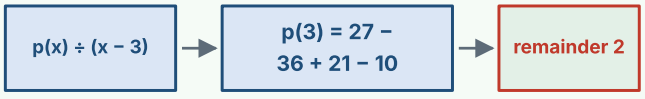
the sign flips on the way in

divide by (x − 3)	substitute x = +3
divide by (x + 2)	substitute x = −2

No division at all: put the number that switches the divisor off into p, and what comes out is the remainder. Zero means the divisor divides exactly.

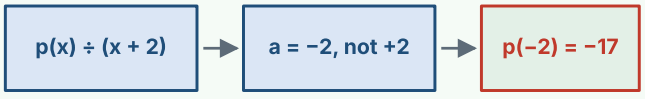
Example 1. **What is the remainder when $p(x) = x^3 - 4x^2 + 7x - 10$ is divided by x − 3?**

Solution.
p(3) = 27 − 36 + 21 − 10 = **2**.



Example 2. **What is the remainder when $p(x) = x^3 + 2x - 5$ is divided by x + 2?**

Solution.
Here a = −2.
p(−2) = −8 − 4 − 5 = **−17**.



Example 3. **Is $(x - 2)$ a factor of $x^3 - 3x^2 + 4$?**

Solution.

$p(2) = 8 - 12 + 4 = 0$.

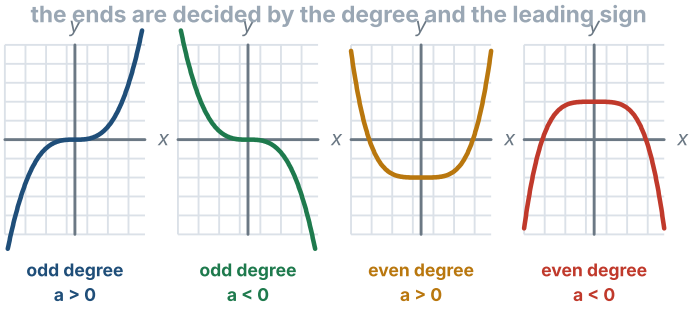
Yes — a remainder of zero means it is a factor.



3 Degree, end behaviour and turning points

degree n has at most n zeros and at most n − 1 turning points

The degree is the highest power. An odd-degree graph runs from bottom-left to top-right (or the reverse if the leading coefficient is negative); an even-degree graph has both ends pointing the same way.

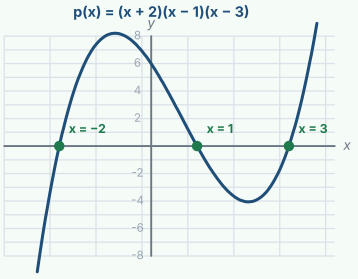


Four pictures cover every case. Odd degree sends the ends opposite ways; even degree sends them the same way; the leading sign says which way up.

Example 1. **How many turning points can a cubic have at most?**

Solution.

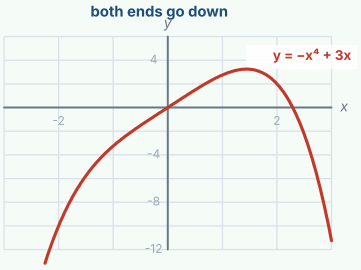
Degree 3, so at most **2** turning points.



Example 2. **Describe the end behaviour of $p(x) = -x^4 + 3x$.**

Solution.

Even degree with a negative leading coefficient, so **both ends go down**.



Example 3. **A polynomial has degree 5. What is the greatest number of x-intercepts it can have?**

Solution.

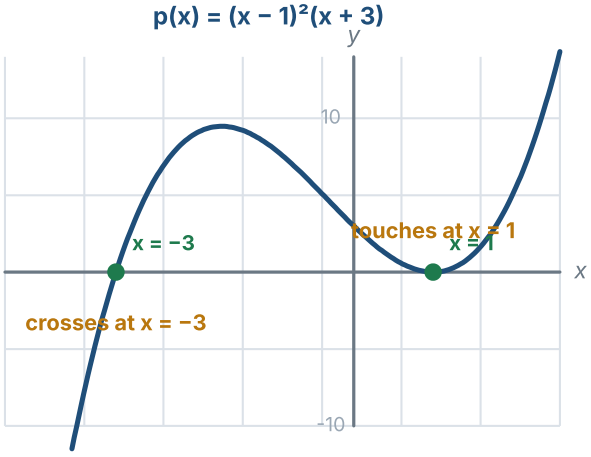
5. It could have fewer, but never more than its degree.

degree n	at most n zeros
degree n	at most n − 1 turning points
degree 5	5 zeros, 4 turning points at most

4 Multiplicity: crossing or touching

odd multiplicity → the graph crosses | even multiplicity → it touches and turns back

A repeated factor changes the shape at that zero. $(x - 2)^2$ makes the curve bounce off the axis at $x = 2$; $(x - 2)^3$ flattens but still passes through.

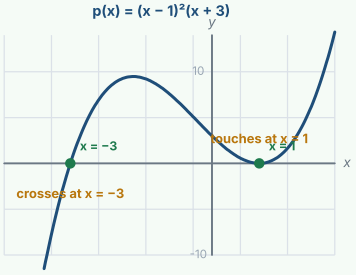


The squared factor at $x = 1$ turns the curve back without letting it through; the single factor at $x = -3$ lets it cross. Count the power on the bracket and you know which happens.

Example 1. **$p(x) = (x - 1)^2(x + 3)$. Describe the graph at each zero.**

Solution.

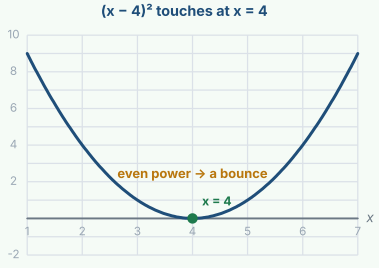
At $x = 1$ the multiplicity is 2, so it **touches and turns**. At $x = -3$ it is 1, so it **crosses**.



Example 2. **A graph touches the x-axis at $x = 4$ without crossing. What does that say about the factor?**

Solution.

$(x - 4)$ appears an **even** number of times — at least squared.



Example 3. **What is the degree of $p(x) = (x - 1)^2(x + 3)$?**

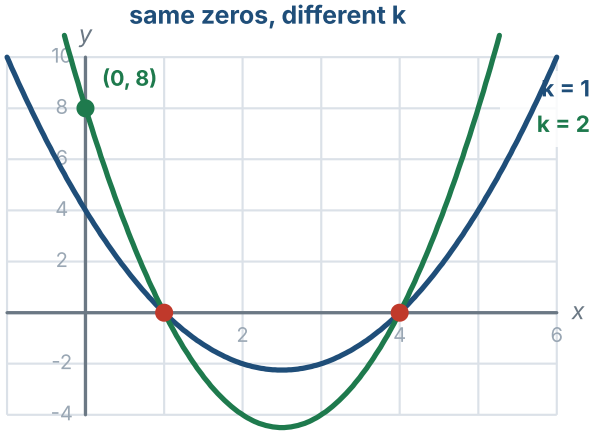
Solution.
Add the multiplicities: 2 + 1 = **3**.

$(x - 1)^2$	contributes 2
$(x + 3)$	contributes 1
degree	2 + 1 = 3

5 Building a polynomial from its zeros

zeros a, b, c → $p(x) = k(x - a)(x - b)(x - c)$

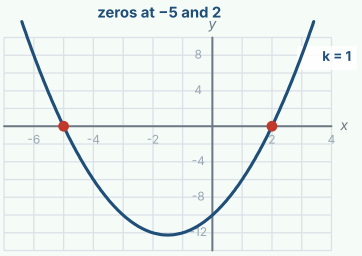
The zeros fix the shape but not the vertical scale, so a constant k remains. One extra point — usually the y-intercept — pins k down.



Both curves have zeros at 1 and 4; only the green one passes through (0, 8). That extra point is what fixes k = 2.

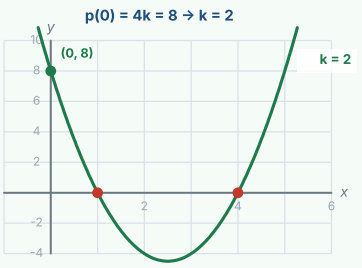
Example 1. **Write a quadratic with zeros 2 and −5.**

Solution.
$k(x - 2)(x + 5)$ for any nonzero k; with k = 1, $x^2 + 3x - 10$.



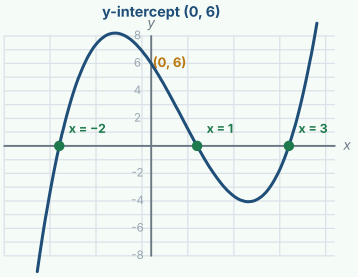
Example 2. **A quadratic has zeros 1 and 4 and passes through (0, 8). Find it.**

Solution.
$p(x) = k(x - 1)(x - 4)$, and $p(0) = 4k = 8$, so k = 2.
$p(x) = 2(x - 1)(x - 4)$.



Example 3. **What is the y-intercept of $p(x) = (x + 2)(x - 1)(x - 3)$?**

Solution.
$p(0) = (2)(-1)(-3) = \mathbf{6}$.



6 Dividing polynomials

$p(x) = (\text{divisor})(\text{quotient}) + \text{remainder}$

If a question gives you a quotient and a remainder, reassemble the original with this identity rather than dividing again. When the remainder is zero the divisor and quotient are factors.

reassemble instead of dividing again

$p(x) =$ [$x - 1$] [$x + 4$] + [3]

$= x^2 + 3x - 1$

Multiply the divisor by the quotient and add the remainder: the original polynomial reappears, with no long division anywhere.

Example 1. **$x^2 + 5x + 6$ divided by $(x + 2)$ gives what quotient?**

Solution.
$x^2 + 5x + 6 = (x + 2)(x + 3)$, so the quotient is $\mathbf{x + 3}$ with remainder 0.

reassemble instead of dividing again

$p(x) =$ [$x + 2$] [$x + 3$] + [0]

$= x^2 + 5x + 6$

Example 2. **$p(x)$ divided by $(x - 1)$ gives quotient $x + 4$ and remainder 3. Find $p(x)$.**

Solution.
$p(x) = (x - 1)(x + 4) + 3 = x^2 + 3x - 4 + 3$.
$= \mathbf{x^2 + 3x - 1}$.

reassemble instead of dividing again

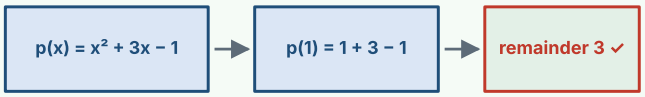
$p(x) =$ [$x - 1$] [$x + 4$] + [3]

$= x^2 + 3x - 1$

Example 3. **Check that result using the remainder theorem.**

Solution.

p(1) = 1 + 3 − 1 = **3**, which is the stated remainder. The identity holds.



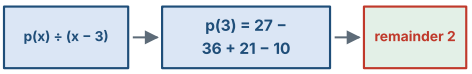
Part 1 — With Desmos (10 questions · every one is faster on the graph)

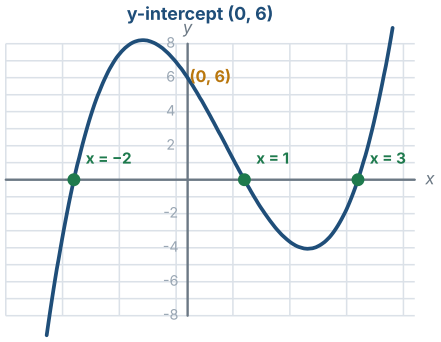
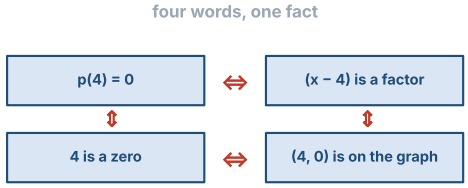
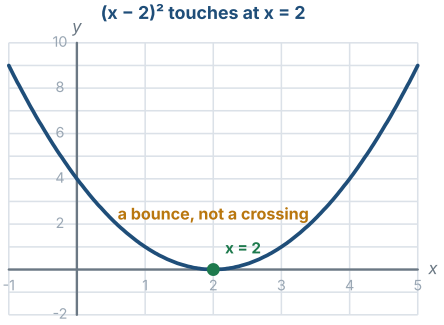
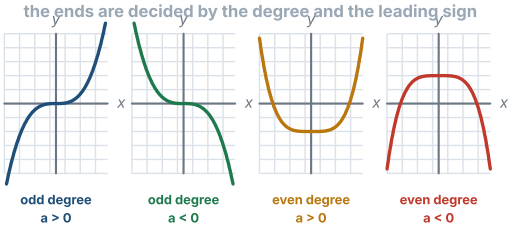
The SAT gives you Desmos on every maths question. For polynomials it is **a zero finder**. Graph p and read where it meets the axis; substitute a value to test a factor without dividing.

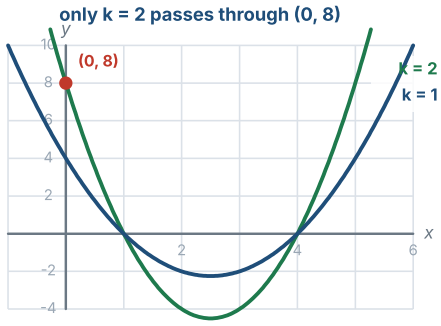
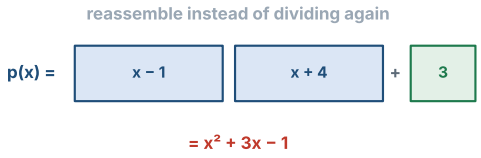
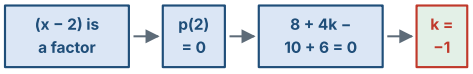
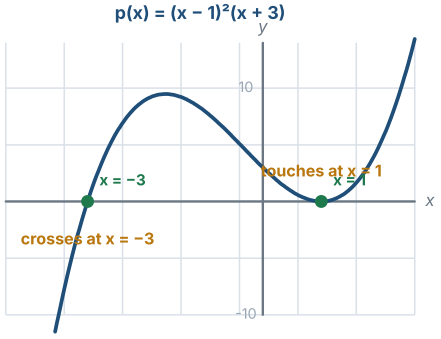
#	Question	Type into Desmos	Then	Answer
D1	What is the value of $x^3 - 4x^2 + 7x - 10$ at $x = 3$?	`3^3-4*3^2+7*3-10`	The remainder theorem in one line.	2
D2	What is the value of $x^3 + 2x - 5$ at $x = -2$?	`(-2)^3+2*(-2)-5`	Dividing by $x + 2$ means substituting -2.	-17
D3	Is $x = 2$ a zero of $x^3 - 3x^2 + 4$?	`2^3-3*2^2+4`	A value of 0 confirms the factor.	0 (yes)
D4	What is the y-intercept of $y = (x + 2)(x - 1)(x - 3)$?	`(0+2)*(0-1)*(0-3)`	Substitute $x = 0$.	6
D5	Graph $y = (x + 2)(x - 1)(x - 3)$. How many x-intercepts does it have?	`y=(x+2)(x-1)(x-3)`	Count where the curve meets the axis.	3
D6	For $p(x) = k(x - 1)(x - 4)$ with $p(0) = 8$, what is k?	`8/((0-1)*(0-4))`	Solve $4k = 8$.	2
D7	Expand $(x - 1)(x + 4) + 3$ at $x = 5$ to check a division.	`(5-1)*(5+4)+3`	Compare with $p(5)$ for $p(x) = x^2 + 3x - 1$.	39
D8	What is $x^2 + 3x - 1$ at $x = 5$?	`5^2+3*5-1`	Matches the line above, confirming the identity.	39
D9	Graph $y = (x - 1)^2(x + 3)$. At which zero does the curve only touch the axis?	`y=(x-1)^2(x+3)`	The squared factor makes it bounce.	x = 1
D10	What is the remainder when $x^4 - 1$ is divided by $x - 1$?	`1^4-1`	Zero means $(x - 1)$ is a factor.	0

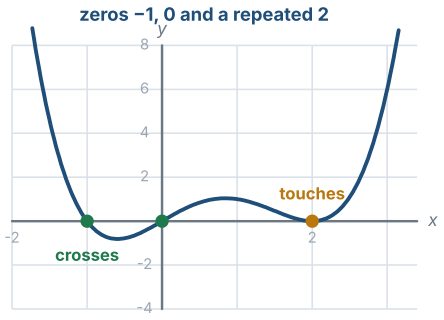
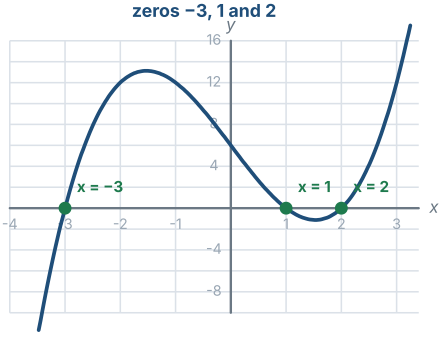
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Nearly every question here is the same idea in different words: **p(a) = 0** means $(x - a)$ is a factor means the graph crosses at $(a, 0)$.

#	Level	Question	Answer	Solution
P1	Hard	A shop's weekly profit index is modelled by $p(x) = x^3 - 4x^2 + 7x - 10$. What is the remainder when $p(x)$ is divided by $x - 3$? 	2	By the remainder theorem the answer is $p(3)$. $27 - 36 + 21 - 10 =$ **2**. No division is needed.

#	Level	Question	Answer	Solution
P2	Hard	The polynomial p is defined by p(x) = (x + 2)(x − 1)(x − 3). What is the y-intercept of the graph of y = p(x)? 	6	The y-intercept is p(0) = (2)(−1)(−3) = **6**. A common wrong answer loses one of the two negative signs.
P3	Hard	If (x − 4) is a factor of the polynomial p, which of the following must be true? 	p(4) = 0	A factor (x − a) corresponds to a zero at x = a, so **p(4) = 0**. A common wrong answer flips the sign, which is the single most common error in this topic.
P4	Hard	The graph of a polynomial p touches the x-axis at x = 2 without crossing it. Which could be a factor of p(x)? 	(x − 2)²	Touching without crossing means **even multiplicity**, so (x − 2)². Two common wrong answers use odd multiplicities, and both would cross.
P5	Hard	What is the greatest possible number of turning points on the graph of a polynomial of degree 5? 	4	A polynomial of degree n has at most n − 1 turning points. 5 − 1 = **4**.

#	Level	Question	Answer	Solution
P6	Hard	A quadratic function has zeros at x = 1 and x = 4 and passes through the point (0, 8). What is the value of the leading coefficient? 	2	p(x) = k(x − 1)(x − 4), so p(0) = k(−1)(−4) = 4k. 4k = 8 gives k = **2**.
P7	Very hard	When the polynomial p is divided by x − 1, the quotient is x + 4 and the remainder is 3. What is p(x)? 	$x^2 + 3x - 1$	p(x) = (x − 1)(x + 4) + 3 = x² + 3x − 4 + 3. = **x² + 3x − 1**. A common wrong answer forgets to add the remainder.
P8	Very hard	The polynomial p(x) = x³ + kx² − 5x + 6 has a factor of (x − 2). What is the value of k? 	−1	A factor (x − 2) means p(2) = 0. 8 + 4k − 10 + 6 = 0, so 4k + 4 = 0 and k = **−1**. Check: 8 − 4 − 10 + 6 = 0.
P9	Very hard	The graph of y = p(x) crosses the x-axis at x = −3, touches it at x = 1, and p has degree 3. Which could be p(x)? 	$(x + 3)(x - 1)^2$	Crossing at −3 needs (x + 3) to odd power; touching at 1 needs (x − 1) to even power; the degrees must total 3. = **(x + 3)(x − 1)²**. A common wrong answer swaps which zero is repeated; C has both signs wrong.
P10	Very hard	For p(x) = x³ + 2x − 5, what is the remainder when p(x) is divided by x + 2? 	−17	x + 2 = x − (−2), so substitute −2. −8 − 4 − 5 = **−17**. A common wrong answer substitutes +2 instead, which is the sign trap this question is built on.

#	Level	Question	Answer	Solution
P11	Very hard	A polynomial p of degree 4 has zeros at x = −1, 0, 2 and 2. State what is true here. 	**The graph crosses at x = −1 and x = 0 and touches at x = 2.**	x = 2 appears twice, so it has **even multiplicity** and the curve turns back there; the other two zeros are simple and the curve passes through. A common wrong answer confuses the degree with the number of distinct intercepts.
P12	Very hard	The polynomial p(x) = x^3 − 7x + 6 has x = 1 as a zero. What is the sum of its other two zeros? 	**−1**	Dividing by (x − 1) gives x^2 + x − 6, whose roots sum to −b/a = **−1** (they are 2 and −3). Alternatively all three zeros sum to 0 (no x^2 term), so the other two sum to 0 − 1 = −1.

The one habit to build. Watch the sign when you convert. A factor of (x + 2) means a zero at x = −2, and dividing by (x + 2) means substituting −2. Your class lost more marks to that flip than to any calculation.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

SAT Math 2.4 — Rational & Radical Equations

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Solving rational equations

multiply both sides by the denominator | then check nothing was made zero

Clearing the fraction turns the problem into something familiar. The one extra step is checking: any value that makes an original denominator zero must be thrown out, however correctly it was derived.



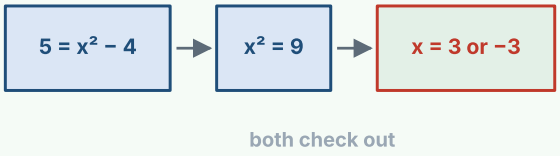
Clearing the denominator is a legal move, but it quietly forgets that x = 2 was never allowed. The number line keeps that exclusion in view while you solve.

Example 1. **Solve 5/(x − 2) = x + 2.**

Solution.

Multiply by (x − 2): 5 = (x + 2)(x − 2) = $x^2 - 4$.

$x^2 = 9$, so x = **3 or −3**. Neither makes x − 2 zero.

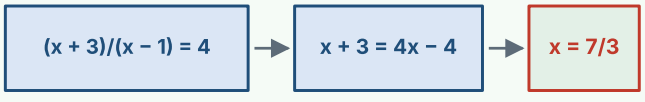


Example 2. **Solve (x + 3)/(x − 1) = 4.**

Solution.

x + 3 = 4(x − 1) = 4x − 4, so 7 = 3x.

x = **7/3**.



Example 3. **Solve 1/x + 1/(2x) = 3.**

Solution.

Multiply by 2x: 2 + 1 = 6x.

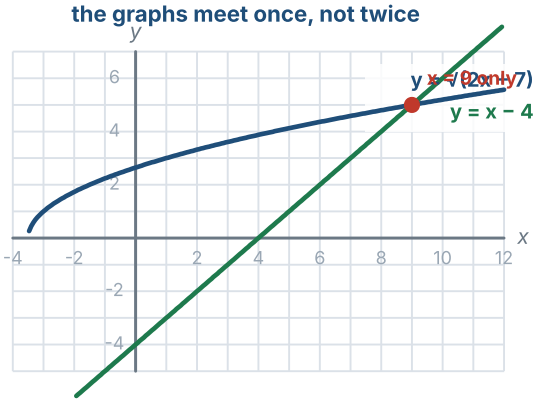
x = **1/2**. Check x ≠ 0, which it is not.



2 Extraneous solutions

squaring or clearing a denominator can invent solutions — always substitute back

This is the highest-value habit in the topic. Both squaring and multiplying out can produce values that satisfy the new equation but not the original. The check is part of the method, not an optional tidy-up.



substitute each candidate back into the original

x = 1	√9 = 3 against 1 − 4 = −3	× reject
x = 9	√25 = 5 against 9 − 4 = 5	✓ keep

Squaring turned one equation into another with an extra root. The graphs cross only at x = 9; x = 1 solves the squared equation and nothing else.

Example 1. **Solve √(2x + 7) = x − 4.**

Solution.

Squaring: 2x + 7 = x² − 8x + 16, so x² − 10x + 9 = 0 and x = 1 or 9.

Check x = 1: √9 = 3 but 1 − 4 = −3, so it fails. The solution set is **{9}**.

substitute each candidate back into the original

x = 1	√9 = 3 against 1 − 4 = −3	× reject
x = 9	√25 = 5 against 9 − 4 = 5	✓ keep

Example 2. **Why did x = 1 appear at all?**

Solution.
Squaring turns 3 = −3 into 9 = 9. The step is not reversible, so it can create solutions that were never there.

3 = −3	is false
square both sides	9 = 9 is true
so squaring	can invent a solution

Example 3. **Solve √(x + 5) = x − 1.**

Solution.
$x + 5 = x^2 - 2x + 1$ gives $x^2 - 3x - 4 = 0$, so x = 4 or −1.
x = −1 gives √4 = 2 but −2, so it fails. Answer **x = 4**.

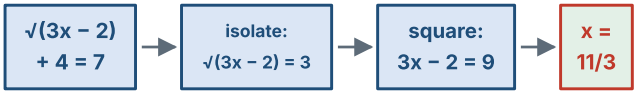
substitute each candidate back into the original

x = −1	√4 = 2 against −2	× reject
x = 4	√9 = 3 against 3	✓ keep

3 Radical equations

isolate the radical first, then square once

Get the root alone on one side before squaring, or you will square a sum and create cross terms. If two radicals appear, isolate one, square, then isolate and square again.

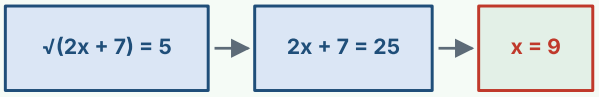


squaring before isolating leaves a radical behind

Four steps in order. Squaring the left side as it stands would give 3x − 2 + 8√(3x − 2) + 16 — still a radical, and worse than where you started.

Example 1. **Solve √(2x + 7) = 5.**

Solution.
Square both sides: 2x + 7 = 25.
x = **9**. Check: √25 = 5.

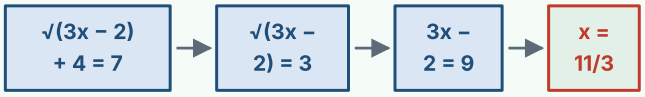


Example 2. **Solve √(3x − 2) + 4 = 7.**

Solution.

Isolate first: √(3x − 2) = 3, so 3x − 2 = 9.

x = **11/3**.



Example 3. **Why not square √(3x − 2) + 4 = 7 immediately?**

Solution.

The left side would become 3x − 2 + 8√(3x − 2) + 16, which still has a radical in it — **worse than where you started**.

$(\sqrt{a} + 4)^2$	$= a + 8\sqrt{a} + 16$
the radical survives	so isolate it first

4 Domain restrictions

a denominator cannot be zero | an even root cannot take a negative

Questions often ask which values are excluded rather than asking you to solve anything. Set the denominator to zero and solve; for a square root, set the inside greater than or equal to zero.



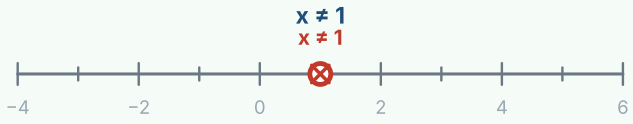
A hollow crossed circle marks a value x may not take; a filled dot with a ray marks where a square root starts being defined. At x = 3 the root is √0 = 0, which is allowed.

Example 1. **For what value of x is (x + 3)/(x − 1) undefined?**

Solution.

x − 1 = 0.

x = 1.



Example 2. **What is the domain of √(2x − 6)?**

Solution.

$2x - 6 \geq 0$, so $2x \geq 6$.

$x \geq 3$.



Example 3. **For what values of x is $1/(x^2 - 9)$ undefined?**

Solution.

$x^2 - 9 = 0$ gives $x = \pm 3$.

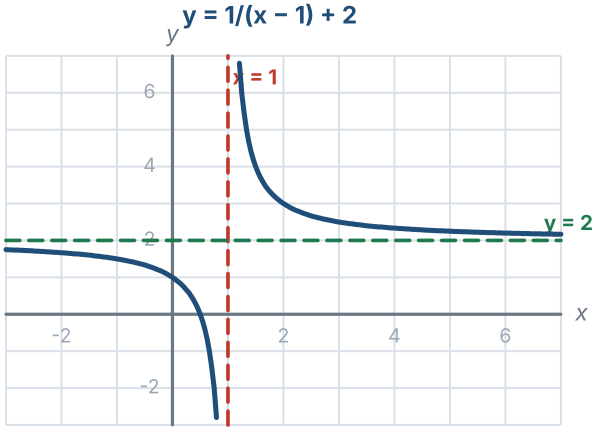
Undefined at **$x = 3$ and $x = -3$**.



5 Asymptotes of a rational function

vertical asymptote where the denominator is zero | horizontal asymptote from comparing degrees

A vertical asymptote is a value x cannot take. For the horizontal one: if the top's degree is smaller the asymptote is y = 0; if the degrees match it is the ratio of leading coefficients.

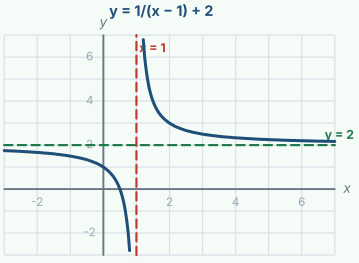


The curve runs away to infinity at x = 1, where the denominator vanishes, and flattens towards y = 2, the constant that was added on. Neither line is ever reached.

Example 1. **Find the vertical asymptote of $y = 1/(x - 1) + 2$.**

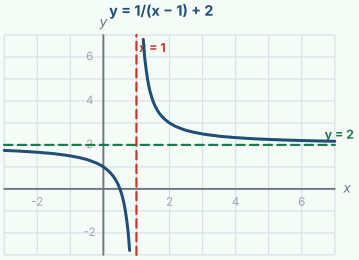
Solution.

The denominator vanishes at **$x = 1$**.



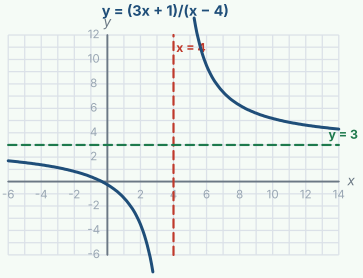
Example 2. **Find the horizontal asymptote of that same function.**

Solution.
As x grows, 1/(x − 1) tends to 0, so y tends to **2**.



Example 3. **Find the horizontal asymptote of y = (3x + 1)/(x − 4).**

Solution.
Degrees match, so take the ratio of leading coefficients.
y = 3.



6 Simplifying rational expressions

factor top and bottom, then cancel common factors — never cancel across a + sign

You may cancel a factor, never a term. $(x^2 + 3x)/x$ cancels to x + 3 because x is a factor of both terms on top; (x + 3)/x cancels to nothing at all.

cancel a factor, never a term

$$\frac{(x-3)\cancel{(x+3)}}{\cancel{(x+3)}} = x - 3 \quad \text{valid for } x \neq -3$$

The bracket (x + 3) multiplies everything above and below it, so it cancels. A loose x added to something else does not multiply the whole line and cannot.

Example 1. **Simplify $(x^2 - 9)/(x + 3)$.**

Solution.
Factor the top: (x − 3)(x + 3)/(x + 3).
= **x − 3**, for x ≠ −3.

cancel a factor, never a term

$$\frac{(x-3)\cancel{(x+3)}}{\cancel{(x+3)}} = x - 3 \quad x \neq -3$$

Example 2. **Simplify $(x^2 + 5x + 6)/(x^2 - 4)$.**

Solution.

$= (x + 2)(x + 3) / [(x - 2)(x + 2)].$

$=$ **$(x + 3)/(x - 2)$**, for $x \neq -2$.

cancel a factor, never a term

$$\frac{\cancel{(x+2)}\,(x+3)}{\cancel{(x+2)}} = (x+3)/(x-2), \quad x \neq -2$$

Example 3. **Can you cancel the x in (x + 4)/x?**

Solution.

No. On top x is a term, not a factor. The expression is already simplest.

$(x^2 + 3x)/x$	$= x + 3$ ✓ x is a factor
$(x + 4)/x$	cannot cancel — x is a term

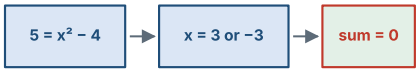
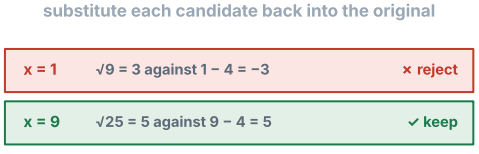
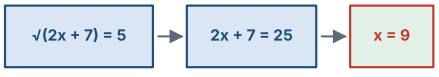
Part 1 — With Desmos (10 questions · every one is faster on the graph)

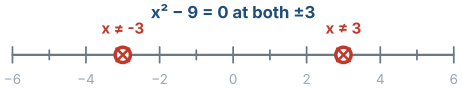
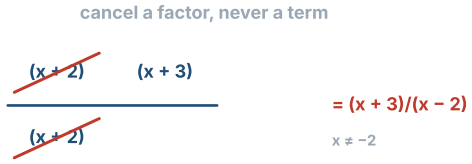
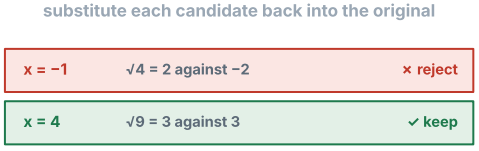
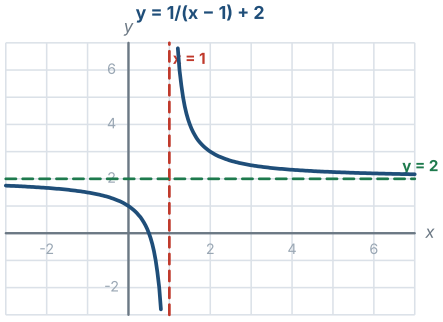
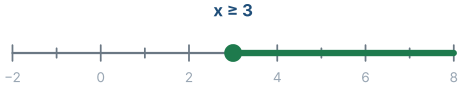
The SAT gives you Desmos on every maths question. Here it is **an extraneous-root filter**. Graph both sides: Desmos shows only the real intersections, so a root it does not show is one your algebra invented.

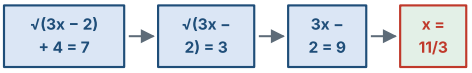
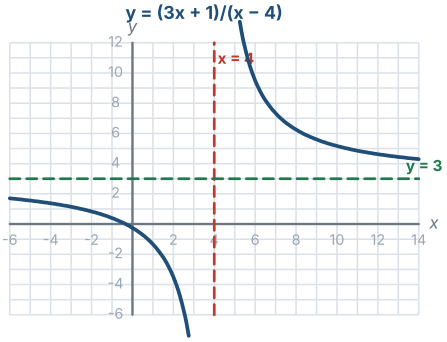
#	Question	Type into Desmos	Then	Answer
D1	Solve 5/(x − 2) = x + 2. What is the sum of the solutions?	`5/(x-2)=x+2`	Graph both sides; the roots are 3 and −3.	0
D2	Solve √(2x + 7) = x − 4.	`sqrt(2x+7)=x-4`	Desmos shows the genuine intersection, not the extraneous root.	9
D3	Solve √(2x + 7) = 5.	`sqrt(2x+7)=5`	Square both sides mentally to check.	9
D4	Solve (x + 3)/(x − 1) = 4.	`(x+3)/(x-1)=4`	Desmos gives the exact fraction.	7/3
D5	For what value of x is (x + 3)/(x − 1) undefined?	`x-1=0`	Graph it and look for the vertical asymptote.	1
D6	What is the horizontal asymptote of y = (3x + 1)/(x − 4)?	`y=(3x+1)/(x-4)`	Zoom out until the curve flattens.	3
D7	Solve 1/x + 1/(2x) = 3.	`1/x+1/(2x)=3`	Clear the denominators first if you prefer.	0.5
D8	Solve √(3x − 2) + 4 = 7.	`sqrt(3x-2)+4=7`	Isolate the radical first.	11/3
D9	Evaluate (x² − 9)/(x + 3) at x = 10.	`(10^2-9)/(10+3)`	Compare with x − 3 at x = 10.	7
D10	Solve √(x + 5) = x − 1.	`sqrt(x+5)=x-1`	Only one intersection appears; −1 is extraneous.	4

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Every radical equation on this sheet needs a check, and every rational one needs you to ask which values the denominator forbids.

#	Level	Question	Answer	Solution
P1	Hard	What is the sum of all solutions to the equation 5/(x − 2) = x + 2? 	0	Multiplying by (x − 2) gives 5 = x² − 4, so x² = 9 and x = ±3. 3 + (−3) = **0**. A common wrong answer is the positive root alone — the word *sum* means there is more than one.
P2	Hard	What is the solution set of the equation √(2x + 7) = x − 4? 	{9}	Squaring gives x² − 10x + 9 = 0, so x = 1 or 9. But x = 1 gives √9 = 3 against 1 − 4 = −3, which is false. Only **x = 9** survives. A common wrong answer is what you get by never checking — six of your class chose it on Test 5.
P3	Hard	If √(2x + 7) = 5, what is the value of x? 	9	Square both sides: 2x + 7 = 25, so 2x = 18 and x = **9**.

#	Level	Question	Answer	Solution
P4	Hard	The function f is defined by $f(x) = (x + 3)/(x - 1)$. If $f(a) = 4$, what is the value of a? 	7/3	$a + 3 = 4(a - 1) = 4a - 4$, so $7 = 3a$. $a =$ **7/3**.
P5	Hard	For what value of x is the expression $(x + 5)/(x^2 - 9)$ undefined, given $x > 0$? 	3	The expression is undefined where the denominator is zero: $x^2 = 9$ gives $x = \pm 3$. With $x > 0$, $x =$ **3**.
P6	Hard	Write an equivalent expression for $(x^2 + 5x + 6)/(x^2 - 4)$ for $x \neq \pm 2$. 	(x + 3)/(x − 2)	Factor both: $(x + 2)(x + 3)$ over $(x - 2)(x + 2)$, and cancel $(x + 2)$. = **(x + 3)/(x − 2)**.
P7	Very hard	What is the solution to $\sqrt{(x + 5)} = x - 1$? 	4	Squaring gives $x^2 - 3x - 4 = 0$, so $x = 4$ or -1. Testing $x = -1$: $\sqrt{4} = 2$ but $-1 - 1 = -2$, so it fails. Only **x = 4** works. A common wrong answer keeps the extraneous root.
P8	Very hard	The graph of $y = 1/(x - 1) + 2$ has a vertical asymptote and a horizontal asymptote. What are they? 	x = 1 and y = 2	The denominator vanishes at $x = 1$; as x grows, $1/(x - 1) \to 0$, leaving $y \to 2$. **x = 1 and y = 2.** A common wrong answer gives the asymptotes of $y = 1/(x - 1)$ before the shift.
P9	Very hard	State the complete domain of $f(x) = \sqrt{(2x - 6)}$. 	x ≥ 3	An even root needs a non-negative inside: $2x - 6 \geq 0$. **x ≥ 3.** The equals sign belongs: at $x = 3$ the value is $\sqrt{0} = 0$, which is defined.
P10	Very hard	If $1/x + 1/(2x) = 3$ and $x \neq 0$, what is the value of x? 	1/2	Multiply through by 2x: $2 + 1 = 6x$, so $6x = 3$. $x =$ **1/2**.

#	Level	Question	Answer	Solution
P11	Very hard	A student solves √(3x − 2) + 4 = 7 by squaring both sides immediately, obtaining 3x − 2 + 16 = 49. What is wrong, and what is the correct value of x? 	**The radical must be isolated first; x = 11/3.**	Squaring a sum is not squaring each piece: $(\sqrt{a} + 4)^2 = a + 8\sqrt{a} + 16$, so the radical survives. Isolating first gives √(3x − 2) = 3, then 3x − 2 = 9 and x = **11/3**.
P12	Very hard	What is the horizontal asymptote of the graph of y = (3x + 1)/(x − 4)? 	**y = 3**	The numerator and denominator have the same degree, so the asymptote is the ratio of leading coefficients. 3/1 = **y = 3**. A common wrong answer is the rule for when the top's degree is smaller.

The one habit to build. Substitute your answer back into the **original** equation. Squaring and clearing denominators are not reversible steps, and six of your class lost M2 Q25 on Test 5 by skipping the check.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

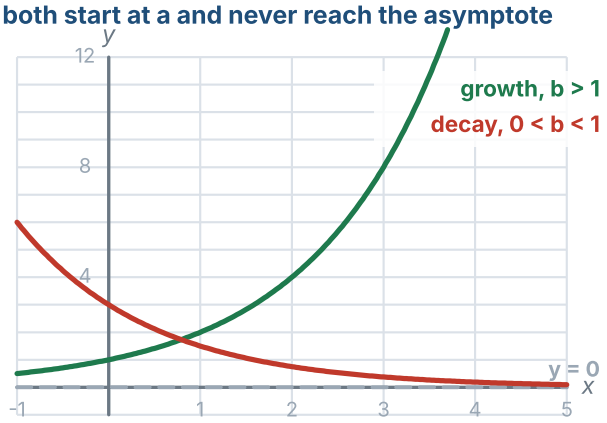
SAT Math 2.5 — Exponential Functions

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 The exponential model

$y = a \cdot b^x$ | a is the starting value, b is the growth factor per step

Read a at x = 0 and b as the multiplier for one step. Exponential means repeated multiplication, which is what separates it from a linear model's repeated addition.



Both curves cut the y-axis at their starting value a. Above b = 1 the curve climbs ever faster; below it the curve falls towards y = 0 without ever arriving.

Example 1. **A shop opens with 150 members and the number doubles each year. Write the model.**

Solution.
a = 150 and b = 2.
$M(t) = 150(2)^t$.



Example 2. **How many members after 3 years?**

Solution.
$150(2)^3 = 150 \times 8 =$ **1200**. Listing 150 → 300 → 600 → 1200 stops you counting one doubling short.

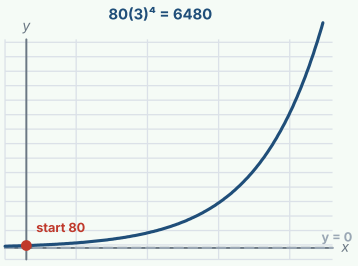
x = 0	x = 1	x = 2	x = 3
150	300	600	1200

differences	+150	+300	+600
ratios	×2	×2	×2

equal ratios → exponential

Example 3. **A culture starts at 80 and triples every hour. How many after 4 hours?**

Solution.
$80(3)^4 = 80 \times 81 = \mathbf{6480}$.



2 Growth, decay and the percentage

rise of r% → b = 1 + r | fall of r% → b = 1 − r

Convert between the base and the percentage in both directions. A base of 0.88 is a fall of 12%, not a fall of 88% and not 'keeps 12%'. A base of 1.07 is a rise of 7%.

the percentage is the distance from 1, never the base

rise of 15% →	b = 1 + 0.15 = 1.15
rise of 7% →	b = 1 + 0.07 = 1.07
fall of 8% →	b = 1 − 0.08 = 0.92
fall of 12% →	b = 1 − 0.12 = 0.88

The percentage is the distance from 1, never the base itself. 0.88 keeps 88% and therefore loses 12%.

Example 1. **A population falls 8% a year. What is the growth factor?**

Solution.
$1 - 0.08 = \mathbf{0.92}$.

the percentage is the distance from 1, never the base

fall of 8% → b = 1 − 0.08 = 0.92

Example 2. **V(t) = 4200(0.88)ᵗ. What does 0.88 mean?**

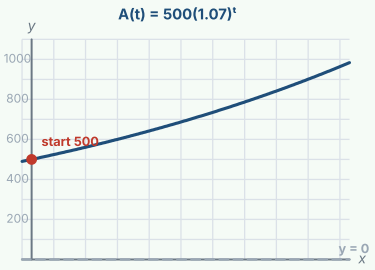
Solution.
The value **falls by 12% each year**, because 1 − 0.88 = 0.12. It keeps 88%, not 12%.

the percentage is the distance from 1, never the base

fall of 12% → b = 1 − 0.12 = 0.88

Example 3. **A deposit grows 7% a year. Write the model for $500.**

Solution.
$A(t) = 500(1.07)^t$.



3 Changing the time unit

$b^t = (b^{1/n})^{nt}$ — take the n-th root of the base, multiply the exponent by n

This is the subtopic your class scored lowest on. Changing from years to months never changes the base alone and never the exponent alone — it does both, in opposite directions. The tell in the question is a word like *monthly* or *quarterly*.

one year = 12 months

$V = a\cdot(0.88)^t$ → $a\cdot((0.88)^{(1/12)})^{12t}$

per year — per month

base → its 12-th root; exponent → ×12, both at once

$(0.88)^{(1/12)} \approx 0.9894$

Twelve monthly steps must come to one yearly step, so the monthly base is the twelfth root and the exponent is multiplied by twelve. Do one without the other and the model no longer matches.

Example 1. **$V(t) = 84{,}000(0.88)^t$ with t in years. Rewrite it with a monthly factor.**

Solution.
One year is 12 months, so t years is 12t months.
$84{,}000((0.88)^{1/12})^{12t}$, and $(0.88)^{1/12} \approx 0.9894$.

one year = 12 months

$V = a\cdot(0.88)^t$ → $a\cdot((0.88)^{(1/12)})^{12t}$

per year — per month

base → its 12-th root; exponent → ×12, both at once

$(0.88)^{(1/12)} \approx 0.9894$

Example 2. **Why is $84{,}000(0.88)^{12t}$ wrong?**

Solution.
That multiplies by 0.88 twelve times a *year* — a fall of about 78% annually instead of 12%.

$(0.88)^{12}$	≈ 0.216 — a 78% fall
$(0.88^{1/12})^{12}$	= 0.88 ✓ the same 12% fall

Example 3. **P = 200(1.6)^t with t in years. Write it with a quarterly factor.**

Solution.

Four quarters in a year.

200((1.6)$^{1/4}$)^{4t}.

one year = 4 quarters

$V = a\cdot(1.6)^t$	→	$a\cdot((1.6)^{(1/4)})^{4t}$
per year		per quarter

base → its 4-th root; exponent → ×4, both at once

(1.6)^(1/4) ≈ 1.1247

4 Exponential against linear

constant difference → linear | constant ratio → exponential

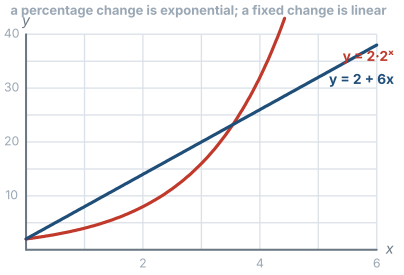
Test a table both ways. Subtract consecutive values: equal differences mean linear. Divide them: equal ratios mean exponential. Exponential growth eventually overtakes any linear growth, however slow it starts.

	x = 0	x = 1	x = 2	x = 3
	5	8	11	14
differences		+3	+3	+3
ratios		×1.6	×1.38	×1.27

equal differences → linear

	x = 0	x = 1	x = 2	x = 3
	5	10	20	40
differences		+5	+10	+20
ratios		×2	×2	×2

equal ratios → exponential



Same first value, two different engines. Adding a fixed amount draws a straight line; multiplying by a fixed amount bends upward and always wins in the end.

Example 1. **y = 5, 8, 11, 14. Which model?**

Solution.

Differences 3, 3, 3.

Linear, y = 3x + 2.

	x = 0	x = 1	x = 2	x = 3
	5	8	11	14
differences		+3	+3	+3
ratios		×1.6	×1.38	×1.27

equal differences → linear

Example 2. **y = 5, 10, 20, 40. Which model?**

Solution.

Ratios 2, 2, 2.

Exponential, $y = 5(2)^x$ counting from $x = 0$.

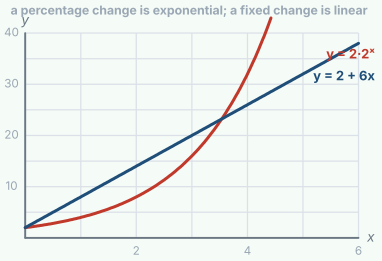
	x = 0	x = 1	x = 2	x = 3
	5	10	20	40
differences		+5	+10	+20
ratios		×2	×2	×2

equal ratios → exponential

Example 3. **A saver adds $50 a month; an investment grows 3% a month. Which is exponential?**

Solution.

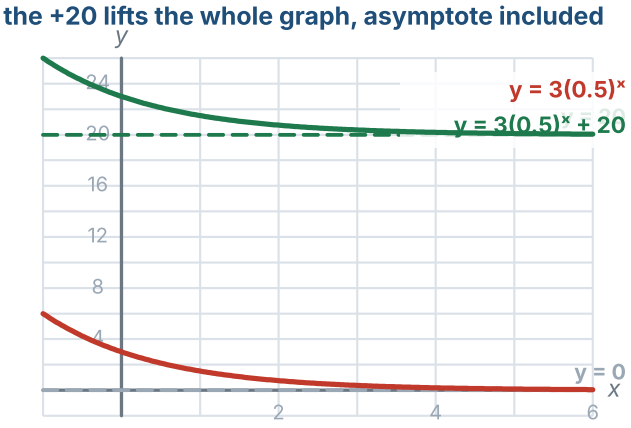
The **investment** — a percentage of a changing amount. Adding a fixed $50 is linear.



5 The horizontal asymptote

$y = a \cdot b^x$ approaches $y = 0$ | adding c shifts the asymptote to $y = c$

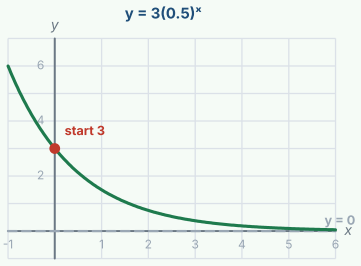
A decaying exponential gets arbitrarily close to its asymptote but never reaches it, which is why a model like 'cools towards room temperature' is written with a constant added on.



Both curves have the same shape; the green one sits 20 higher, so it levels off at 20 instead of 0. Whatever is added on is the value the model approaches.

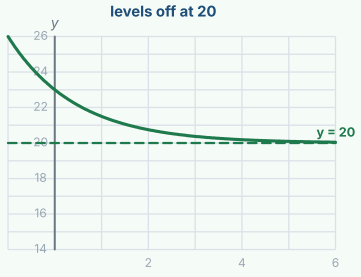
Example 1. **What is the horizontal asymptote of $y = 3(0.5)^x$?**

Solution.
$y = 0$. The curve falls towards the x-axis without touching it.



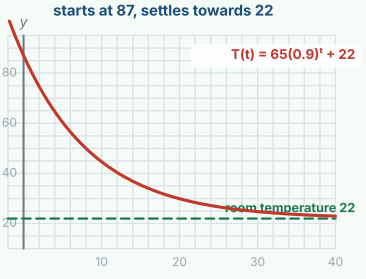
Example 2. **What is it for $y = 3(0.5)^x + 20$?**

Solution.
$y = 20$ — the whole graph is lifted 20.



Example 3. **A drink cools according to $T(t) = 65(0.9)^t + 22$. What does 22 represent?**

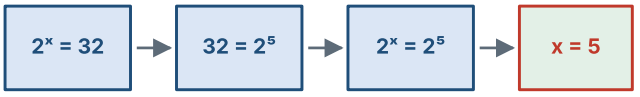
Solution.
The **room temperature**, 22 degrees — the value the drink approaches but never quite reaches.



6 Solving exponential equations

same base → equate the exponents | otherwise graph it in Desmos

Rewrite both sides as powers of the same number whenever you can. When the bases cannot be matched, the SAT expects you to read the intersection off a graph rather than use logarithms.

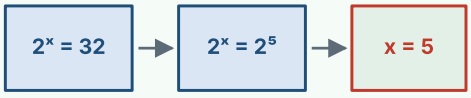


matching bases turns it into an ordinary equation

Once both sides are powers of 2, the exponents must agree. That is the whole method — no logarithms required.

Example 1. **Solve $2^x = 32$.**

Solution.
$32 = 2^5$.
$x = \mathbf{5}$.



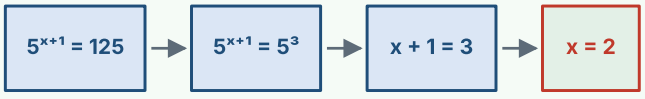
Example 2. **Solve $3^{2x} = 81$.**

Solution.
$81 = 3^4$, so $2x = 4$.
$x = \mathbf{2}$.



Example 3. **Solve $5^{x+1} = 125$.**

Solution.
$125 = 5^3$, so $x + 1 = 3$.
$x = \mathbf{2}$.



Part 1 — With Desmos (10 questions · every one is faster on the graph)

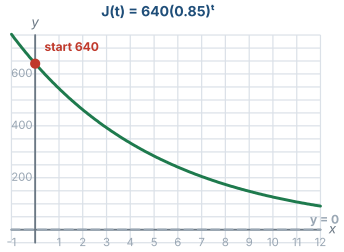
The SAT gives you Desmos on every maths question. Here it is **a growth calculator**. Type the whole model and evaluate it, or graph both sides of an exponential equation and read the crossing.

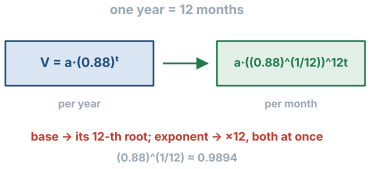
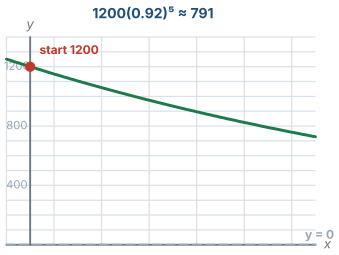
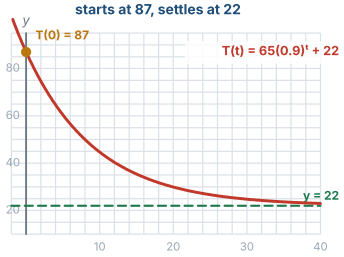
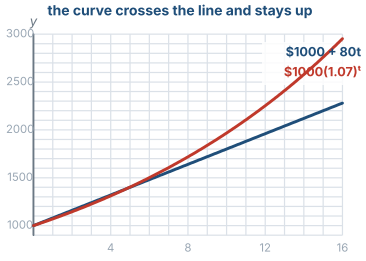
#	Question	Type into Desmos	Then	Answer
D1	A shop opens with 150 members and the number doubles each year. How many after 3 years?	`150*2^3`	Count the doublings carefully.	1200
D2	A town of 1200 falls 8% a year. What is the population after 5 years, to the nearest whole number?	`1200*0.92^5`	A fall of 8% means a factor of 0.92.	791
D3	What is $(0.88)^{1/12}$, to four decimal places?	`0.88^(1/12)`	The monthly factor behind an annual 0.88.	0.9894
D4	$500 grows at 7% a year. What is it worth after 3 years, to the nearest cent?	`500*1.07^3`	Repeated multiplication, not 500 × 1.21.	612.52
D5	Solve $2^x = 32$.	`2^x=32`	Graph both sides and read the intersection.	5
D6	A culture starts at 80 and triples hourly. How many after 4 hours?	`80*3^4`	Powers grow fast; let Desmos do it.	6480
D7	What is the growth factor for a rise of 15% per period?	`1+0.15`	A rise adds to 1.	1.15
D8	250 subscribers grow 20% monthly. How many after 2 months?	`250*1.2^2`	1.2 squared is 1.44.	360
D9	Solve $3^{2x} = 81$.	`3^(2x)=81`	Both sides are powers of 3.	2
D10	What does $65(0.9)^t + 22$ approach as t grows large?	`65*0.9^100+22`	Try a large t to see the asymptote.	22

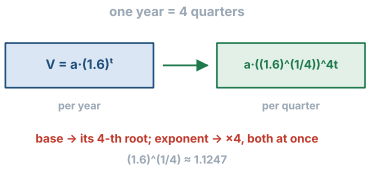
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Read the base carefully every time: it tells you growth or decay, and the percentage is its distance from 1 — not the base itself.

#	Level	Question	Answer	Solution
P1	Hard	When a tea shop opened it had 150 loyalty members, and the number has doubled every year since. How many members does the shop have 3 years after opening? 	1200	$150 \times 2^3 = 150 \times 8 =$ **1200**. A common wrong answer counts only two doublings — four of your class chose it on Test 5.

#	Level	Question	Answer	Solution
P2	Hard	The function $V(t) = 4200(0.88)^t$ gives the value, in talens, of an inventory t years after purchase. What is the best interpretation of 0.88? the percentage is the distance from 1, never the base fall of 12% → b = 1 − 0.12 = 0.88	**The value decreases by 12% each year.**	A base below 1 is decay, and 1 − 0.88 = 0.12. The value **falls 12% a year** and keeps 88%. A common wrong answer reverses which figure is kept.
P3	Hard	A spice stall opens with 640 jars and the number of unsold jars decreases by 15% each week. Which function gives the number of jars after t weeks? 	**$J(t) = 640(0.85)^t$**	A fall of 15% leaves 85%, so the base is 1 − 0.15 = 0.85. **$J(t) = 640(0.85)^t$.** A common wrong answer uses the rate as the base; C is linear.
P4	Hard	A subscription service has 250 subscribers and grows by 20% each month. How many subscribers will it have after 2 months? x = 0 | x = 1 | x = 2 250 | 300 | 360 differences +50 +60 ratios ×1.2 ×1.2 equal ratios → exponential	**360**	$250(1.2)^2 = 250 \times 1.44 =$ **360**. Adding 20% of 250 twice would give 350, which is the common error.
P5	Hard	If $3^{2x} = 81$, what is the value of x? $3^{2x} = 81$ → $3^{2x} = 3^4$ → $2x = 4$ → $x = 2$	**2**	$81 = 3^4$, so $2x = 4$ and $x =$ **2**. A common wrong answer stops at $2x = 4$ without halving.
P6	Hard	A table shows y = 5, 10, 20, 40 for x = 0, 1, 2, 3. Which function models the data? x = 0 | x = 1 | x = 2 | x = 3 5 | 10 | 20 | 40 differences +5 +10 +20 ratios ×2 ×2 ×2 equal ratios → exponential	**$y = 5(2)^x$**	The ratios are all 2, so the model is exponential with a = 5 and b = 2. **$y = 5(2)^x$.** A common wrong answer swaps the starting value and the base; at x = 0 it gives 2, not 5.

#	Level	Question	Answer	Solution
P7	Very hard	The value V(t), in dollars, of a tour bus t years after purchase is $V(t) = 84{,}000(0.88)^t$. Which expression is equivalent and uses the approximate monthly decay factor? 	$84{,}000(0.88^{1/12})^{12t}$	One year is 12 months, so take the twelfth root of the base and multiply the exponent by 12. **$84{,}000(0.88^{1/12})^{12t}$**, where $0.88^{1/12} \approx 0.9894$. Two common wrong answers decay twelve times too fast; C treats 0.12 as a factor rather than a rate.
P8	Very hard	A population of 1,200 decreases by 8% each year. To the nearest whole number, what is the population after 5 years? 	791	$1200(0.92)^5 = 790.89\ldots \approx$ **791**. Subtracting 8% of 1,200 five times would give 720 — that treats the decrease as linear.
P9	Very hard	A hot drink cools according to $T(t) = 65(0.9)^t + 22$, where T is in degrees Celsius and t in minutes. What does the 22 represent? 	The room temperature the drink approaches.	As t grows, $65(0.9)^t \to 0$, so T → **22**, the horizontal asymptote. The drink's initial temperature is T(0) = 87, which is that answer's trap.
P10	Very hard	Two investments start at \$1,000. The first adds \$80 each year; the second grows by 7% each year. State what is true here. 	The first leads at first, but the second eventually overtakes it.	After one year: \$1,080 against \$1,070. But 7% of a growing balance eventually exceeds a fixed \$80. **Exponential growth always overtakes linear growth in the end**, whatever the head start.
P11	Very hard	If $5^{x+1} = 125$, what is the value of x? 	2	$125 = 5^3$, so x + 1 = 3. x = **2**.

#	Level	Question	Answer	Solution
P12	Very hard	A quantity P is modelled by $P = 200(1.6)^t$, where t is in years. Which expression gives P using a quarterly growth factor? 	$200(1.6^{1/4})^{4t}$	A year holds 4 quarters, so take the fourth root of the base and multiply the exponent by 4. **$200(1.6^{1/4})^{4t}$.** A common wrong answer is algebraically equal to the original but uses an *annual* factor raised to a quarter power — it is not the quarterly factor the question asks for.

The one habit to build. Changing the time unit changes **both** the base and the exponent, in opposite directions. Your class scored 23% on that one subtopic, the second worst on the whole paper, so drill it until the shape of the answer is automatic.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

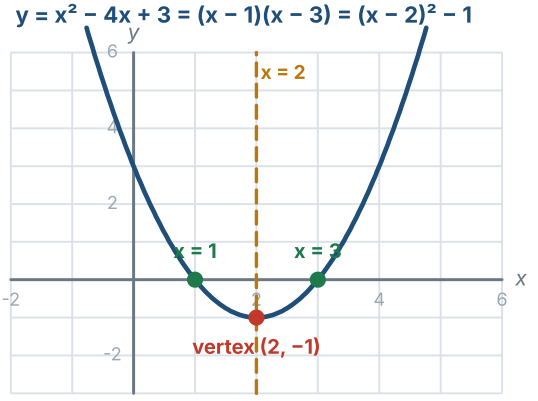
SAT Math 2.6 — Quadratic & Polynomial Graphs

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Three forms, three facts

standard ax^2+bx+c → y-intercept c | factored → x-intercepts | vertex $a(x-h)^2+k$ → vertex (h, k)

Each form hands you one fact for free. The question tells you which fact it wants, and that tells you which form to produce. Converting between them is the work; knowing which to aim for is the skill.



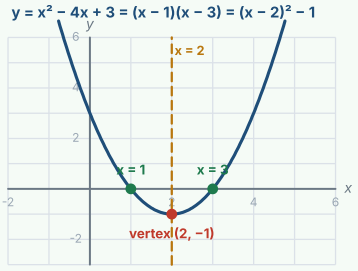
one curve, three ways of writing it

standard: $x^2 - 4x + 3$	y-intercept (0, 3)
factored: $(x - 1)(x - 3)$	x-intercepts 1 and 3
vertex: $(x - 2)^2 - 1$	vertex (2, −1)

All three expressions describe the identical parabola. Which one you write depends only on which number the question is asking for.

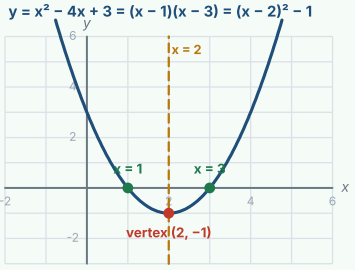
Example 1. **$y = x^2 - 4x + 3$. What is the y-intercept?**

Solution.
Standard form gives it immediately: c = **3**, so (0, 3).



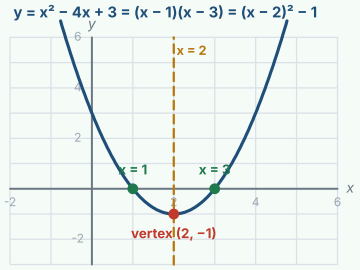
Example 2. **Write the same function in factored form and give the x-intercepts.**

Solution.
$(x - 1)(x - 3)$, so the x-intercepts are **(1, 0) and (3, 0)**.



Example 3. **Write it in vertex form and give the vertex.**

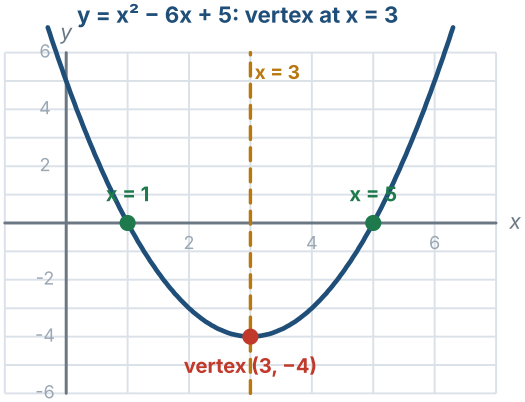
Solution.
$(x - 2)^2 - 1$, so the vertex is **(2, −1)**.



2 Finding the vertex

x = −b/2a | or the midpoint of the two x-intercepts

Both routes give the same x. Substitute it back to get the y. When a question gives you the factored form, the midpoint of the roots is faster than the formula.



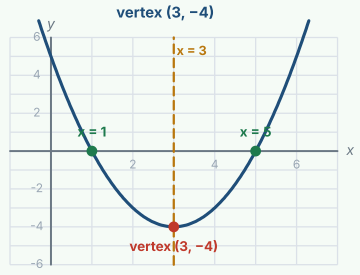
two routes, one answer

from standard form	x = −b/2a = 6/2 = 3
from the roots 1 and 5	midpoint (1 + 5)/2 = 3
then substitute	y = 9 − 18 + 5 = −4

The parabola is symmetric about its vertex, so the turning point sits exactly halfway between the two crossings — which is why both routes agree.

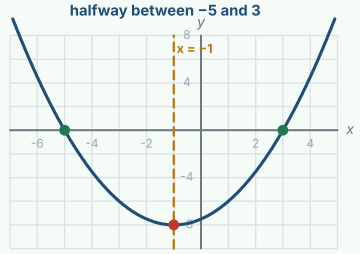
Example 1. **Find the vertex of $y = x^2 - 6x + 5$.**

Solution.
x = 6/2 = 3, and y = 9 − 18 + 5 = −4.
Vertex **(3, −4)**.



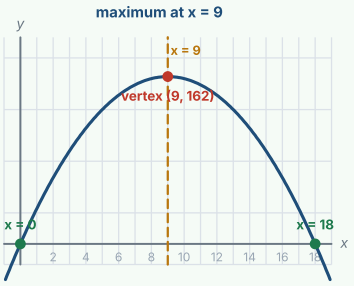
Example 2. **Find the vertex of $y = a(x - 3)(x + 5)$.**

Solution.
Midpoint of 3 and −5 is (3 − 5)/2 = **x = −1**.



Example 3. **R(x) = −2x² + 36x. Find the x that maximises R.**

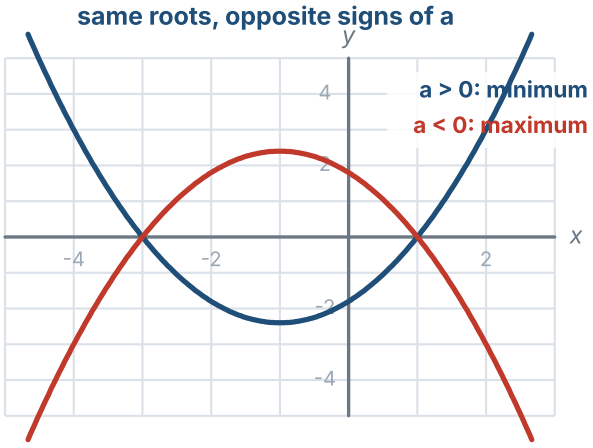
Solution.
x = −36/(2 × −2) = **9**.



3 Which way does it open, and what does that mean?

a > 0 → opens up, vertex is a minimum | a < 0 → opens down, vertex is a maximum

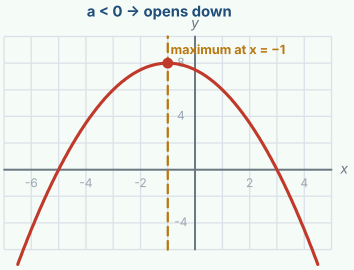
The sign of a settles maximum against minimum on its own. A negative a does not make everything else negative — the y-intercept, for instance, can still be positive.



Flipping the sign of a turns the curve over: the lowest point becomes the highest. The roots stay exactly where they were.

Example 1. **y = a(x − 3)(x + 5) with a < 0. Does the vertex give a maximum or a minimum?**

Solution.
a < 0 means the parabola opens downward, so the vertex is a **maximum**, at x = −1.



Example 2. **For that same function, is the y-intercept positive or negative?**

Solution.

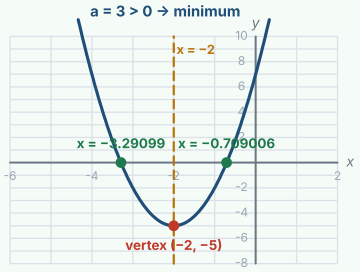
$y(0) = a(-3)(5) = -15a$. With $a < 0$ this is **positive**. A negative a does not make the intercept negative — seven of your class assumed it did.

$y(0) = a(-3)(5)$	$= -15a$
$a < 0$	so $-15a > 0$
the intercept	is positive

Example 3. **$y = 3x^2 + 12x + 7$ has what kind of turning point?**

Solution.

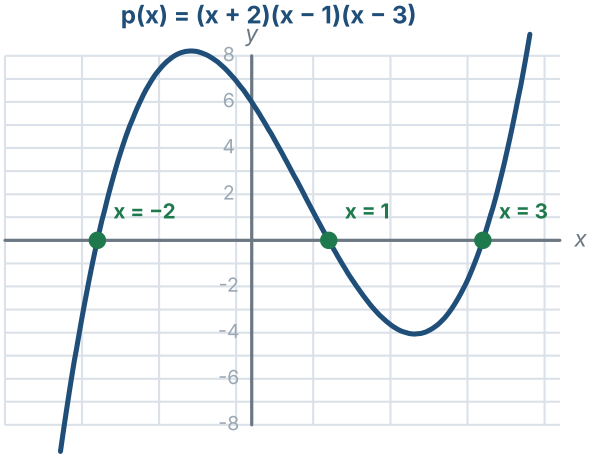
$a = 3 > 0$, so a **minimum**.



4 Reading intercepts off a graph

x-intercepts are where y = 0 | the y-intercept is where x = 0

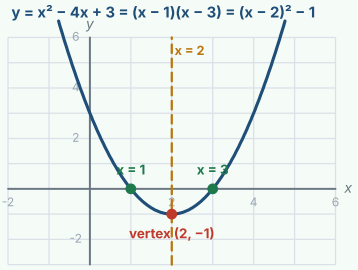
Substituting zero is the whole method. For the y-intercept of a product, multiply the constants: the y-intercept of (x + 2)(x − 1)(x − 3) is 2 × −1 × −3 = 6.



Where the curve meets the x-axis, y is zero; where it meets the y-axis, x is zero. Putting x = 0 into the three brackets gives the height 6.

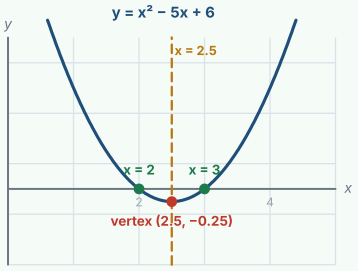
Example 1. **Find the y-intercept of y = (x − 1)(x − 3).**

Solution.
y(0) = (−1)(−3) = **3**.



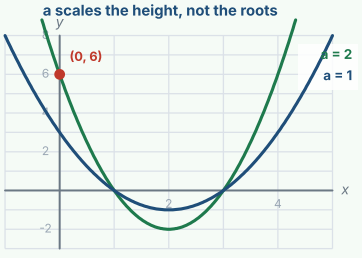
Example 2. **Find the x-intercepts of y = x² − 5x + 6.**

Solution.
(x − 2)(x − 3) = 0.
(2, 0) and (3, 0).



Example 3. **A parabola has x-intercepts 1 and 3 and passes through (0, 6). Find a in y = a(x−1)(x−3).**

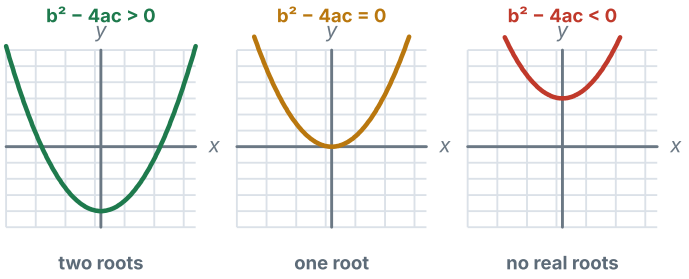
Solution.
6 = a(−1)(−3) = 3a, so a = **2**.



5 How many times does it cross?

the number of x-intercepts is decided by the discriminant $b^2 - 4ac$

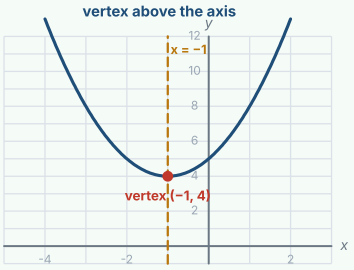
Two intercepts, one (a touch), or none. A question that says ‘touches the x-axis at exactly one point’ or ‘is tangent to the x-axis’ is asking you to set the discriminant to zero.



The same three cases as in 2.2, now read off the picture: cutting twice, grazing once, or missing the axis altogether.

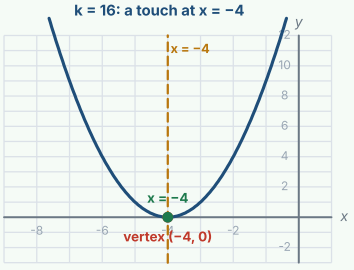
Example 1. **How many x-intercepts does $y = x^2 + 2x + 5$ have?**

Solution.
$4 - 20 = -16 < 0$.
None — the parabola sits entirely above the axis.



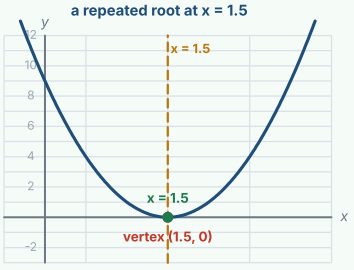
Example 2. **For what k does $y = x^2 + 8x + k$ touch the x-axis exactly once?**

Solution.
$64 - 4k = 0$.
$k = \mathbf{16}$.



Example 3. **How many x-intercepts does $y = 4x^2 - 12x + 9$ have?**

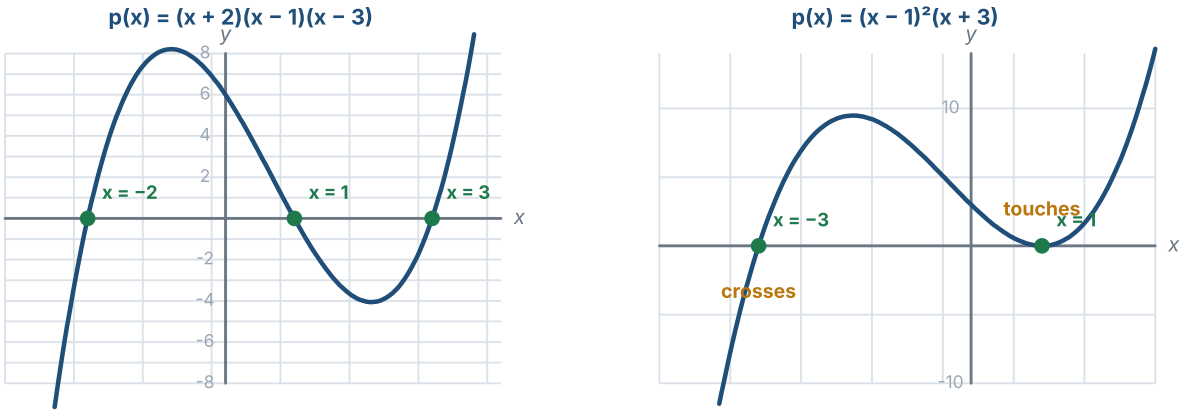
Solution.
$144 - 144 = 0$.
One, at the vertex $x = 1.5$.



6 Cubics and higher

the graph crosses at odd multiplicity and touches at even | degree n has at most n − 1 turning points

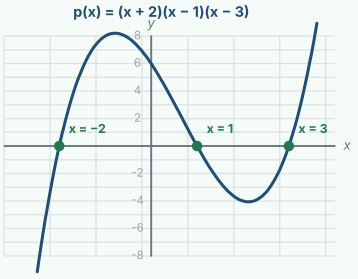
The factored form of a polynomial is a map of its graph. Each factor gives a zero, and the power on that factor says whether the curve passes through or bounces off.



Three single factors on the left: three crossings and two turning points. A squared factor on the right: the curve meets the axis at x = 1 and turns straight back.

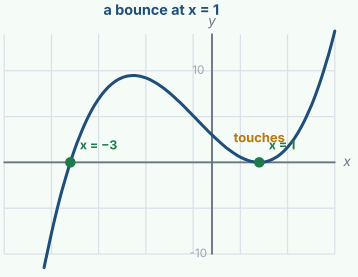
Example 1. **Describe the graph of y = (x + 2)(x − 1)(x − 3).**

Solution.
Three simple zeros, so it **crosses at −2, 1 and 3**, with at most 2 turning points.



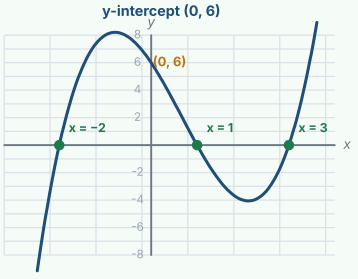
Example 2. **Describe the graph of y = (x − 1)²(x + 3) at x = 1.**

Solution.
Multiplicity 2, so the curve **touches the axis and turns back**.



Example 3. **What is the y-intercept of y = (x + 2)(x − 1)(x − 3)?**

Solution.
(2)(−1)(−3) = **6**.



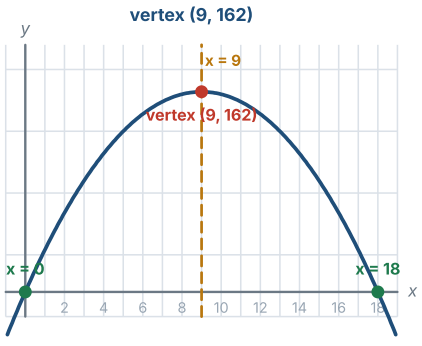
Part 1 — With Desmos (10 questions · every one is faster on the graph)

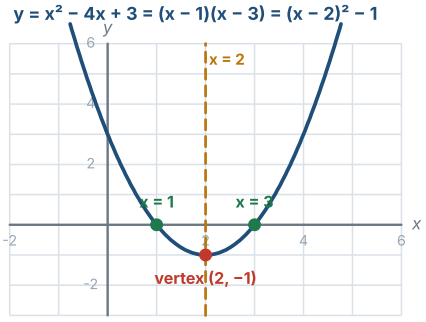
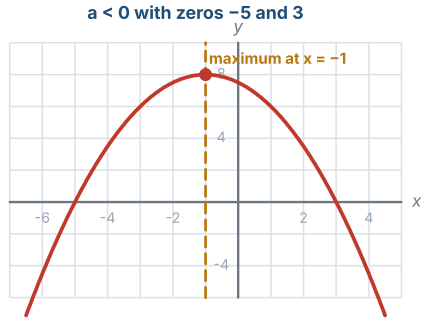
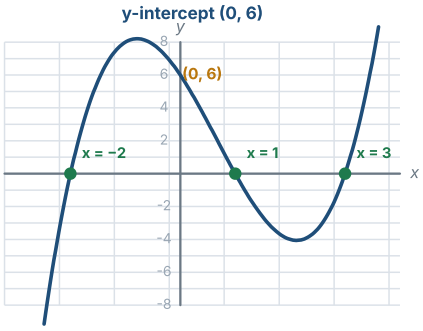
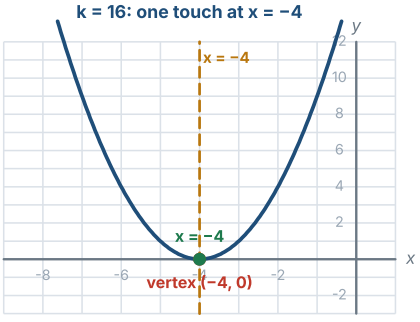
The SAT gives you Desmos on every maths question. This topic is **where Desmos earns its keep**. Graph the function, click the vertex and the intercepts, and the question usually answers itself.

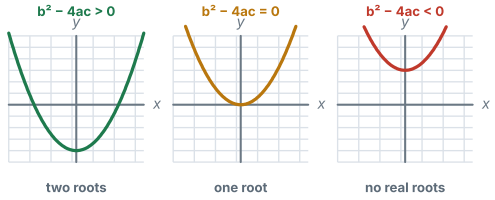
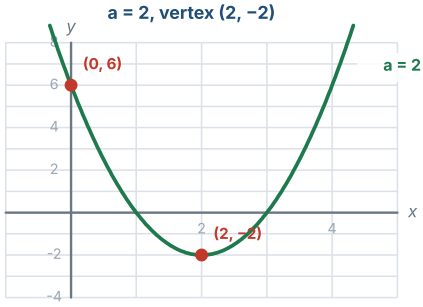
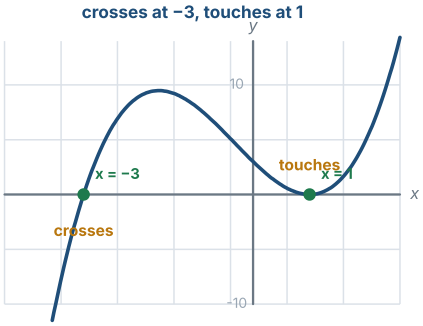
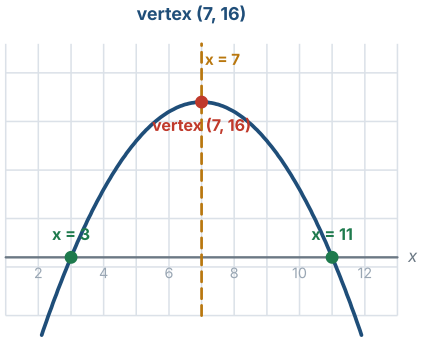
#	Question	Type into Desmos	Then	Answer
D1	Graph $y = x^2 - 4x + 3$. What is the vertex?	`y=x^2-4x+3`	Click the turning point; Desmos labels it.	(2, -1)
D2	For that graph, what are the x-intercepts?	`y=x^2-4x+3`	Click where the curve meets the axis.	1 and 3
D3	What is the minimum value of $y = x^2 - 6x + 5$?	`y=x^2-6x+5`	Read the y of the vertex.	-4
D4	$R(x) = -2x^2 + 36x$. What x gives the maximum?	`y=-2x^2+36x`	The x of the vertex, not the y.	9
D5	How many x-intercepts does $y = x^2 + 2x + 5$ have?	`y=x^2+2x+5`	The curve never reaches the axis.	0
D6	For what k does $x^2 + 8x + k = 0$ have exactly one solution?	`8^2/4`	Set the discriminant to zero.	16
D7	What is the y-intercept of $y = (x + 2)(x - 1)(x - 3)$?	`(0+2)*(0-1)*(0-3)`	Substitute x = 0.	6
D8	Graph $y = (x - 1)^2(x + 3)$. How many turning points does it have?	`y=(x-1)^2(x+3)`	Degree 3 allows at most 2.	2
D9	A parabola has x-intercepts 1 and 3 and passes through (0, 6). What is a in $y = a(x-1)(x-3)$?	`6/((0-1)*(0-3))`	Solve 3a = 6.	2
D10	What is the value of $-2x^2 + 36x$ at x = 9?	`-2*9^2+36*9`	The maximum revenue itself.	162

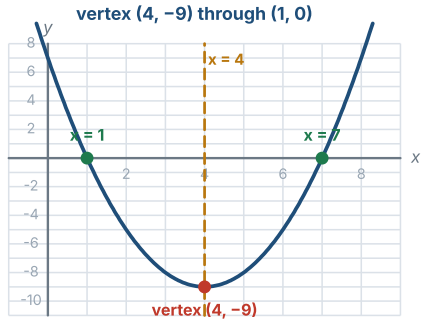
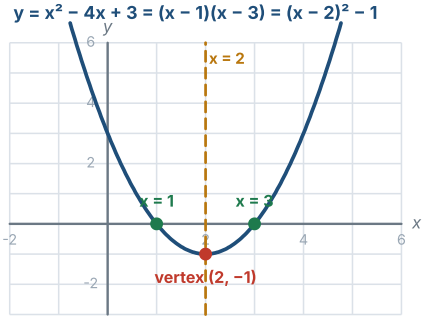
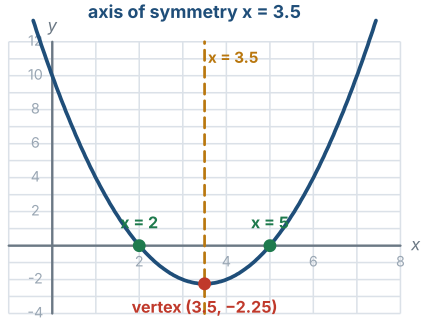
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Decide first whether the question wants **where** something happens (an x) or **how much** it is (a y). Most of the lost marks in this topic are the right number answering the wrong question.

#	Level	Question	Answer	Solution
P1	Hard	A candle shop's daily revenue, in dollars, is modelled by $R(x) = -2x^2 + 36x$, where x is the price of one candle in dollars. What price gives the maximum daily revenue? 	$9	The vertex sits at $x = -b/2a = -36/(-4) =$ **$9**. A common wrong answer is the maximum revenue R(9) = 162 — the answer to a different question. Five of your class chose C, reading 36 straight off the equation.

#	Level	Question	Answer	Solution
P2	Hard	The graph of $y = x^2 - 4x + 3$ has its vertex at which point? 	(2, −1)	$x = -b/2a = 4/2 = 2$, and $y = 4 - 8 + 3 = -1$. Vertex **(2, −1)**. A common wrong answer drops the sign on −b.
P3	Hard	A candle stall's weekly profit is modelled by $y = a(x - 3)(x + 5)$, where $a < 0$. Which must be true about the graph? 	The graph has a maximum at x = −1.	The zeros are 3 and −5, so the vertex is at their midpoint $x = -1$, and $a < 0$ makes it a **maximum**. The trap seven of your class took: the y-intercept is $a(-3)(5) = -15a$, which is *positive* when $a < 0$.
P4	Hard	What is the y-intercept of the graph of $y = (x + 2)(x - 1)(x - 3)$? 	6	Substitute $x = 0$: $(2)(-1)(-3) =$ **6**.
P5	Hard	For what value of k does the graph of $y = x^2 + 8x + k$ touch the x-axis at exactly one point? 	16	Touching once means the discriminant is zero: $64 - 4k = 0$. $k =$ **16**.

#	Level	Question	Answer	Solution
P6	Hard	How many x-intercepts does the graph of $y = x^2 + 2x + 5$ have? 	Zero	$b^2 - 4ac = 4 - 20 = -16 < 0$. The parabola opens upward with vertex above the axis, so it has **no** x-intercepts.
P7	Very hard	A parabola has x-intercepts at (1, 0) and (3, 0) and passes through (0, 6). What is the y-value of its vertex? 	−2	$y = a(x - 1)(x - 3)$, and $6 = a(-1)(-3) = 3a$ gives $a = 2$. The vertex sits at $x = 2$. $y = 2(1)(-1) = \mathbf{-2}$. A common wrong answer is the vertex y when $a = 1$.
P8	Very hard	The graph of a polynomial p crosses the x-axis at $x = -3$ and touches it at $x = 1$. Which could be p(x)? 	$(x + 3)(x - 1)^2$	Crossing needs odd multiplicity, touching needs even. $= \mathbf{(x + 3)(x - 1)^2}$. A common wrong answer swaps which zero is repeated; D has both signs wrong.
P9	Very hard	The function f is defined by $f(x) = -x^2 + 14x - 33$. What is the maximum value of f? 	16	The vertex is at $x = -14/(-2) = 7$. $f(7) = -49 + 98 - 33 = \mathbf{16}$. Answering 7 gives where the maximum occurs, not the maximum itself.

#	Level	Question	Answer	Solution
P10	Very hard	The graph of $y = a(x - h)^2 + k$ has its vertex at $(4, -9)$ and passes through $(1, 0)$. What is the value of a? 	1	$y = a(x - 4)^2 - 9$, and at $(1, 0)$: $0 = a(9) - 9$, so $9a = 9$. $a =$ **1**.
P11	Very hard	Two functions are given by $f(x) = x^2 - 4x + 3$ and $g(x) = (x - 2)^2 - 1$. State what is true here. 	f and g are the same function.	Expanding g: $x^2 - 4x + 4 - 1 = x^2 - 4x + 3 = f(x)$. They are **the same function written in two forms** — standard and vertex.
P12	Very hard	The graph of $y = x^2 + bx + 10$ has its axis of symmetry at $x = 3.5$. What is the value of b? 	−7	The axis of symmetry is $x = -b/2a = -b/2$. $-b/2 = 3.5$ gives $b =$ **−7**. A common wrong answer drops the minus sign.

The one habit to build. The x of the vertex is where the maximum occurs; the y is the maximum itself. On Test 5 five students answered 36 and one answered 162 when the question wanted 9 — all three numbers were in the problem, and only one answered what was asked.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

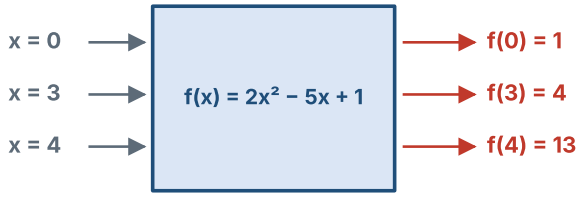
SAT Math 2.7 — Function Notation & Transformations

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Function notation

f(3) means put 3 in | f(x) = 3 means the output is 3 — solve for x

The brackets say substitute; they are not multiplication. f(a + b) is not f(a) + f(b), and the difference between f(3) = ? and f(x) = 3 is the difference between evaluating and solving.



two different jobs, one notation

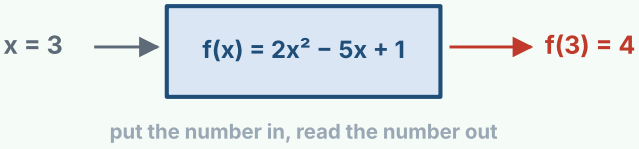
f(3) = ?	put 3 in, read the output
f(x) = 3	the output is 3, solve for x

The machine takes an input and returns an output. ‘Find f(3)’ runs it forwards; ‘solve f(x) = 3’ runs it backwards and can have two answers.

Example 1. **f(x) = 2x² − 5x + 1. Find f(3).**

Solution.

2(9) − 15 + 1 = 18 − 15 + 1 = **4**.

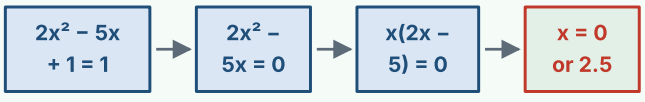


Example 2. **For that f, solve f(x) = 1.**

Solution.

2x² − 5x + 1 = 1 gives 2x² − 5x = 0, so x(2x − 5) = 0.

x = **0 or 2.5**.

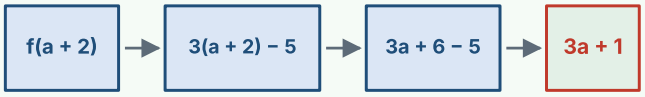


Example 3. **f(x) = 3x − 5. Find f(a + 2).**

Solution.

Substitute the whole expression: 3(a + 2) − 5.

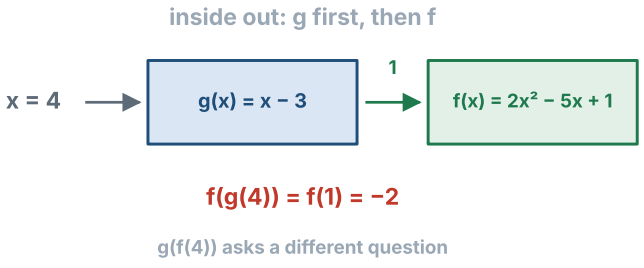
= **3a + 1**.



2 Composition

f(g(x)): work the inner function first, then feed that number outward

Composition is done from the inside out, and the order matters: f(g(2)) is usually not g(f(2)). Write the inner value down before touching the outer function.



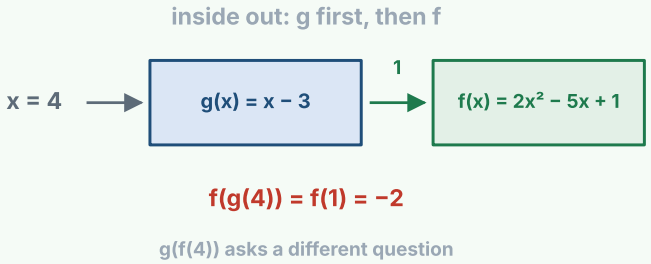
g(4) = 1 comes out of the first box and goes straight into the second, where f(1) = −2. Reversing the boxes gives a different number altogether.

Example 1. **f(x) = 2x² − 5x + 1 and g(x) = x − 3. Find f(g(4)).**

Solution.

g(4) = 1 first.

f(1) = 2 − 5 + 1 = **−2**.

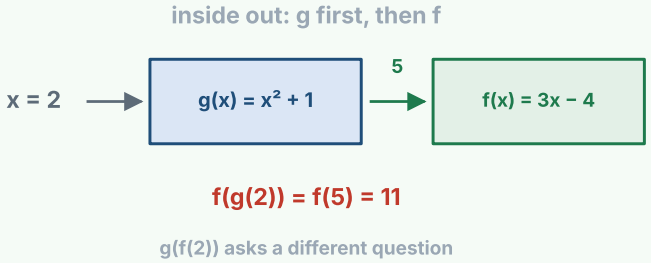


Example 2. **$f(x) = 3x - 4$ and $g(x) = x^2 + 1$. Find $f(g(2))$.**

Solution.

$g(2) = 5$.

$f(5) = 15 - 4 = \mathbf{11}$.

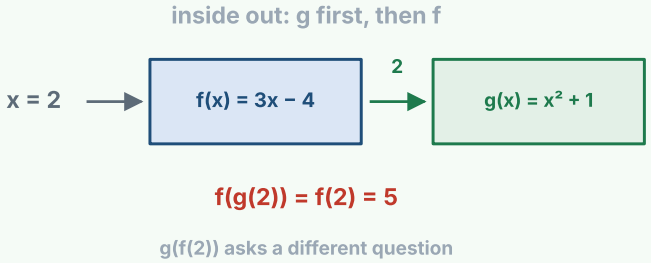


Example 3. **For those same functions, find $g(f(2))$.**

Solution.

$f(2) = 2$.

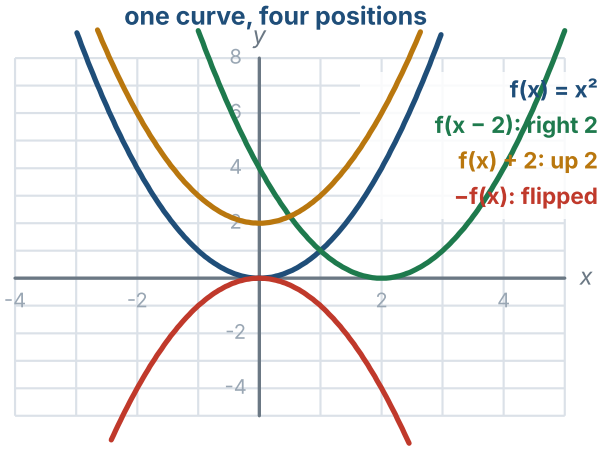
$g(2) = 5$. Different from $f(g(2)) = 11$ — **order matters**.



3 Vertical shifts and reflections

f(x) + k moves up k | f(x) − k moves down | −f(x) flips over the x-axis

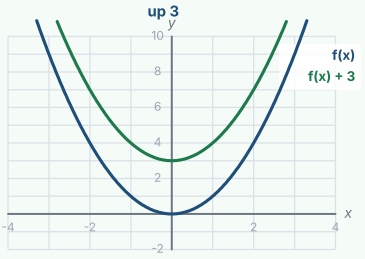
Changes made *outside* the function act on the output, so they behave exactly as you expect: add to go up, negate to flip vertically.



Everything outside the bracket does what it says: +2 lifts the curve, the minus sign turns it upside down. Only the green curve, whose change is inside the bracket, moves sideways.

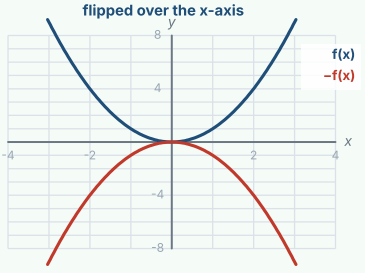
Example 1. **The graph of y = f(x) is shifted up 3. Write the new function.**

Solution.
y = f(x) + 3.



Example 2. **f(x) = x². Describe y = −f(x).**

Solution.
A reflection over the **x-axis**: the upward parabola becomes a downward one.



Example 3. **The point (2, 7) is on y = f(x). Where is it on y = f(x) − 4?**

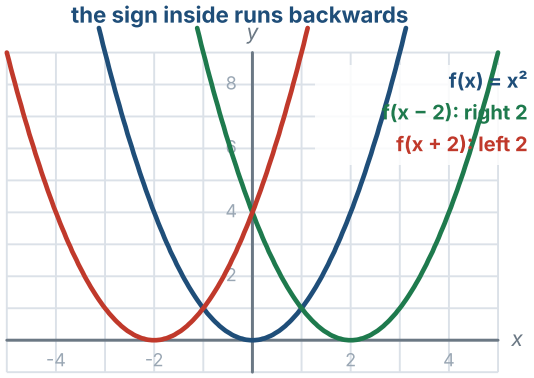
Solution.
Only the output changes.
(2, 3).

(2, 7) on f(x)	the input stays 2
f(x) − 4	the output drops by 4
new point	(2, 3)

4 Horizontal shifts — the opposite way

f(x − h) moves right h | f(x + h) moves left h

Changes made *inside* the brackets act on the input, and they run backwards. f(x − 2) shifts right, not left, because x must be 2 larger to produce the same output. This reversal is the single most-missed idea in transformations.



why inside runs backwards

f(x − 2)	x must be 2 bigger → right 2
f(x + 5)	x must be 5 smaller → left 5

To get the same output from f(x − 2) you must feed in an x that is 2 larger — so every point slides right. The minus sign moves the graph the opposite way to how it reads.

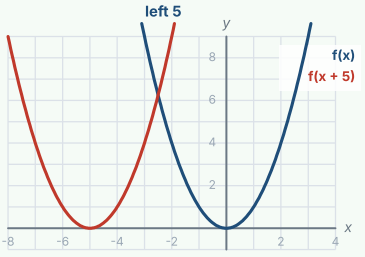
Example 1. **f(x) = x². Describe the graph of y = f(x − 2).**

Solution.
A shift **2 to the right**, giving y = (x − 2)² with vertex (2, 0).



Example 2. **Describe y = f(x + 5).**

Solution.
A shift **5 to the left** — the sign inside is the opposite of the direction.



Example 3. **The point (3, 8) is on y = f(x). Where is the matching point on y = f(x − 4)?**

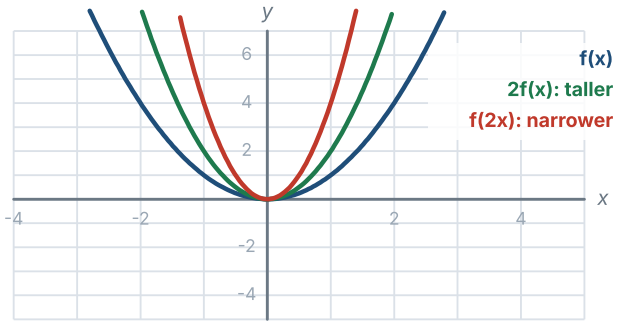
Solution.
The input must be 4 larger to give the same output.
(7, 8).

(3, 8) on f(x)	output 8 at input 3
f(x − 4)	needs x − 4 = 3
so x = 7	new point (7, 8)

5 Stretches

a·f(x) stretches vertically by a | f(bx) squeezes horizontally by 1/b

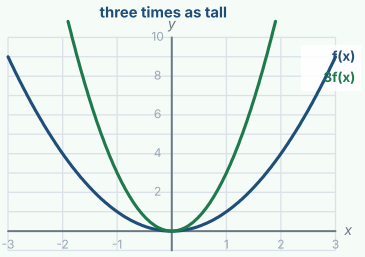
Outside multiplies the outputs, so heights change. Inside multiplies the inputs, so widths change — and again it runs backwards: f(2x) makes the graph half as wide, not twice.



Tripling outside makes every height three times as big. Doubling inside pulls the curve in to half its width, because x only has to reach half as far.

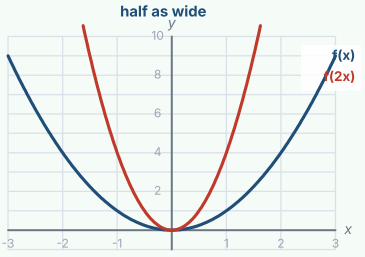
Example 1. **f(x) = x². Describe y = 3f(x).**

Solution.
Every y is tripled: a **vertical stretch by 3**, giving y = 3x².



Example 2. **Describe y = f(2x).**

Solution.
A **horizontal squeeze by a factor of 2** — the graph is half as wide.



Example 3. **The point (4, 6) is on y = f(x). Where is it on y = 2f(x)?**

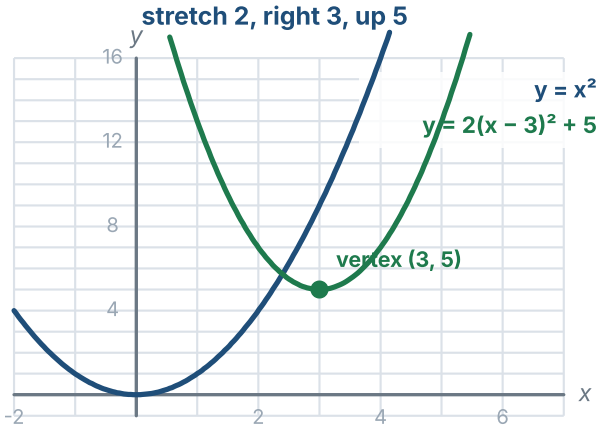
Solution.
The output doubles, the input does not.
(4, 12).

(4, 6) on f(x)	input 4, output 6
2f(x)	doubles the output only
new point	(4, 12)

6 Reading transformations from an equation

$y = a \cdot f(x - h) + k$ → stretch by a, right h, up k

Take the pieces in order and read the sign inside backwards. From vertex form $y = a(x - h)^2 + k$ you can state the whole transformation of $y = x^2$ in one sentence.

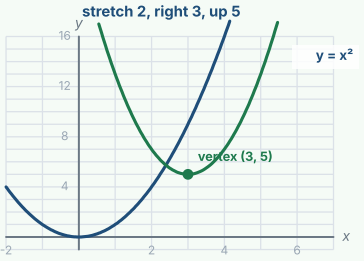


Read a, then h, then k: stretched by 2, moved 3 right (sign reversed) and 5 up. The vertex lands exactly at (h, k).

Example 1. **Describe $y = 2(x - 3)^2 + 5$ as a transformation of $y = x^2$.**

Solution.

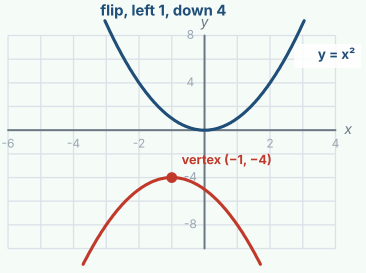
Stretched vertically by 2, moved 3 right and 5 up; vertex (3, 5).



Example 2. **Describe $y = -(x + 1)^2 - 4$.**

Solution.

Flipped over the x-axis, moved 1 left and 4 down; vertex (−1, −4).



Example 3. **$y = f(x)$ has a maximum at (2, 9). Where is the maximum of $y = f(x - 1) + 3$?**

Solution.

Right 1 and up 3.

(3, 12).

maximum (2, 9)	**on y = f(x)**
f(x − 1)	**moves it right 1 → x = 3**
+ 3	**moves it up 3 → y = 12**

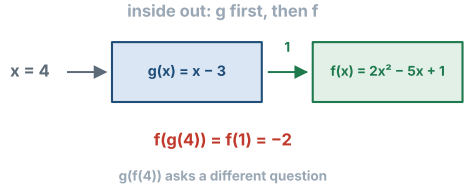
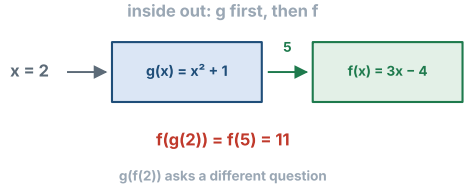
Part 1 — With Desmos (10 questions · every one is faster on the graph)

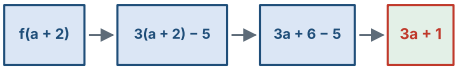
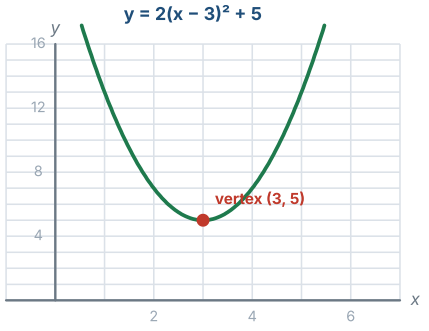
The SAT gives you Desmos on every maths question. Here it is **a transformation viewer**. Graph f and the transformed version together and watch which way the curve actually moves — it settles the inside-sign question for good.

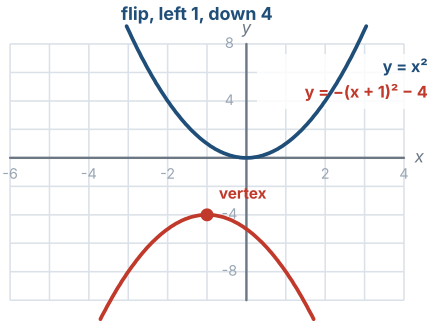
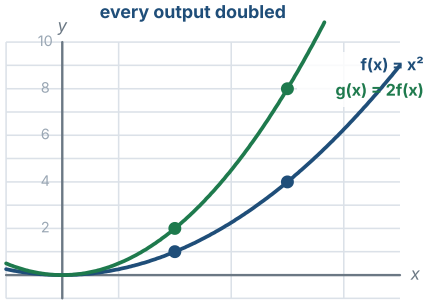
#	Question	Type into Desmos	Then	Answer
D1	f(x) = 2x² − 5x + 1. What is f(3)?	`2*3^2-5*3+1`	Substitute and evaluate in one line.	4
D2	f(x) = 2x² − 5x + 1 and g(x) = x − 3. What is f(g(4))?	`2*(4-3)^2-5*(4-3)+1`	Inner function first.	-2
D3	f(x) = 3x − 4 and g(x) = x² + 1. What is f(g(2))?	`3*(2^2+1)-4`	g(2) = 5, then f(5).	11
D4	For those same functions, what is g(f(2))?	`(3*2-4)^2+1`	Different from D3 — order matters.	5
D5	Graph y = x² and y = (x − 2)². Which way did it move?	`y=(x-2)^2`	The sign inside runs backwards.	right 2
D6	What is the vertex of y = 2(x − 3)² + 5?	`y=2(x-3)^2+5`	Read h and k straight off.	(3, 5)
D7	f(x) = 3x − 5. What is f(a + 2) when a = 4?	`3*(4+2)-5`	Substitute the whole expression.	13
D8	Solve 2x² − 5x + 1 = 1.	`2x^2-5x+1=1`	Two solutions; the question asks where the output is 1.	0 and 2.5
D9	The point (2, 7) is on y = f(x). What is the y-value on y = f(x) − 4?	`7-4`	Only the output shifts.	3
D10	What is the vertex of y = −(x + 1)² − 4?	`y=-(x+1)^2-4`	Inside sign flips; outside does not.	(-1, -4)

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Sort each change into **outside** (acts on the output, behaves as expected) or **inside** (acts on the input, runs backwards) before you answer.

#	Level	Question	Answer	Solution
P1	Hard	The functions f and g are defined by f(x) = 2x² − 5x + 1 and g(x) = x − 3. What is the value of f(g(4))? 	−2	g(4) = 1 first, then f(1) = 2 − 5 + 1 = **−2**. A common wrong answer is f(4) — the inner function g was never applied.
P2	Hard	For f(x) = 3x − 4 and g(x) = x² + 1, what is the value of f(g(2))? 	11	g(2) = 5, then f(5) = 15 − 4 = **11**. A common wrong answer works the functions in the wrong order: g(f(2)) = 5 is a different value again.

#	Level	Question	Answer	Solution
P3	Hard	The graph of y = f(x) is transformed to y = f(x − 2). Which describes the transformation? 	Shift 2 units right	A change inside the brackets acts on the input and runs *backwards*: x must be 2 larger to give the same output. The graph moves **2 right**. A common wrong answer is the instinctive one and the most common error in this topic.
P4	Hard	The point (2, 7) lies on the graph of y = f(x). Which point must lie on the graph of y = f(x) − 4? f(x) − 4 → outside → output drops 4 (2, 7) → becomes (2, 3) f(x − 4) → would give (6, 7) instead	(2, 3)	Subtracting outside the function changes only the output. **(2, 3).** A common wrong answer moves the input instead, which is what f(x − 4) would do.
P5	Hard	If f(x) = 3x − 5, which expression is equal to f(a + 2)? 	3a + 1	Substitute the whole expression: 3(a + 2) − 5 = 3a + 6 − 5. = **3a + 1**. A common wrong answer forgets to subtract the 5.
P6	Hard	What is the vertex of the graph of y = 2(x − 3)² + 5? 	(3, 5)	Vertex form y = a(x − h)² + k has vertex (h, k). **(3, 5)** — note h is +3 even though the bracket reads −3.
P7	Very hard	The graph of y = f(x) has a maximum at the point (2, 9). What are the coordinates of the maximum of y = f(x − 1) + 3? maximum (2, 9) → on y = f(x) f(x − 1) → right 1 → x = 3 + 3 → up 3 → y = 12	(3, 12)	Inside, −1 moves the graph right 1; outside, +3 moves it up 3. **(3, 12).** A common wrong answer moves left instead of right — the inside sign reverses.

#	Level	Question	Answer	Solution
P8	Very hard	Which describes the graph of $y = -(x + 1)^2 - 4$ as a transformation of $y = x^2$? 	**Reflected over the x-axis, 1 left, 4 down**	The minus outside reflects over the **x-axis**; (x + 1) moves **left** 1; −4 moves down 4. Vertex (−1, −4). A common wrong answer gets the direction of the shift backwards.
P9	Very hard	The point (4, 6) lies on the graph of y = f(x). Which point must lie on the graph of y = 2f(x)? 2f(x) — outside → output doubles (4, 6) — becomes (4, 12) f(2x) — would halve the input instead	**(4, 12)**	Multiplying outside the function doubles the **output** only. **(4, 12).** A common wrong answer doubles the input, which is what f(x/2) would do.
P10	Very hard	If $f(x) = 2x^2 - 5x + 1$, what is the greatest value of x for which f(x) = 1? f(x) = 1 → 2x² − 5x = 0 → x(2x − 5) = 0 → x = 0 or 2.5	**2.5**	$2x^2 - 5x + 1 = 1$ gives $2x^2 - 5x = 0$, so x(2x − 5) = 0. x = 0 or 2.5, and the greater is **2.5**. Note this question asks you to *solve*, not to evaluate.
P11	Very hard	The graph of y = f(x) passes through (0, 0), (1, 1) and (2, 4). The graph of y = g(x) passes through (0, 0), (1, 2) and (2, 8). Which relationship is consistent with these points? 	**g(x) = 2f(x)**	Each output of g is exactly twice the matching output of f: 0, 2, 8 against 0, 1, 4. **g(x) = 2f(x)**, a vertical stretch. A common wrong answer would double the inputs, giving g(1) = f(2) = 4, not 2.
P12	Very hard	The function h is defined by h(x) = f(x) + k, where k is a constant. If f(3) = 7 and h(3) = 2, what is the value of k? h(3) = f(3) + k — 2 = 7 + k k = −5 — the graph moved down 5	**−5**	h(3) = f(3) + k, so 2 = 7 + k. k = **−5**.

The one habit to build. Inside the brackets, everything is the opposite of what it looks like: f(x − 2) goes right, f(x + 5) goes left, f(2x) gets narrower. Your class scored 13% on transformations, the lowest figure on the whole syllabus, and this single rule is most of it.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

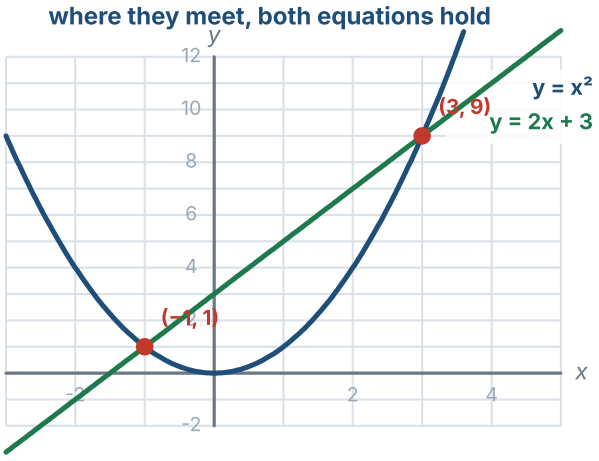
SAT Math 2.8 — Nonlinear Systems

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Solving by substitution

set the two expressions for y equal | solve the resulting equation | find each y

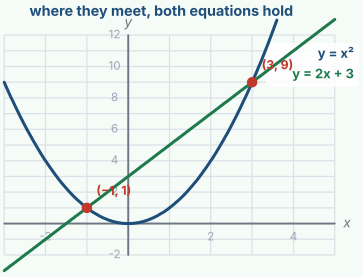
A nonlinear system is one curve and one line, or two curves. Where they meet, both equations hold, so equating them turns the pair into a single equation you already know how to solve. Do not stop at x — the solutions are points.



The two crossings are the two solutions. Equating $x^2 = 2x + 3$ finds their x-values; substituting back into either equation gives the matching y.

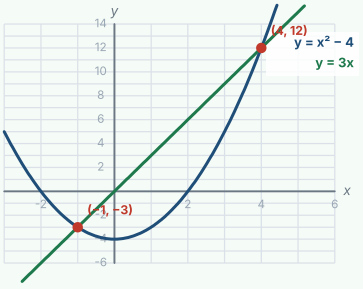
Example 1. **Solve $y = x^2$ and $y = 2x + 3$.**

Solution.
$x^2 = 2x + 3$, so $x^2 - 2x - 3 = 0$ and $(x - 3)(x + 1) = 0$.
$x = 3$ or -1, giving **(3, 9) and (−1, 1)**.



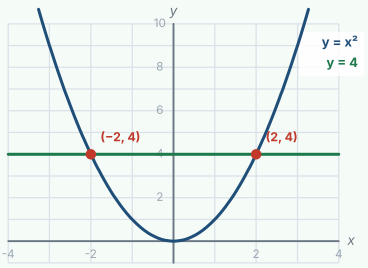
Example 2. **Solve $y = x^2 - 4$ and $y = 3x$.**

Solution.
$x^2 - 3x - 4 = 0$, so $(x - 4)(x + 1) = 0$.
(4, 12) and (−1, −3).



Example 3. **Solve $y = x^2$ and $y = 4$.**

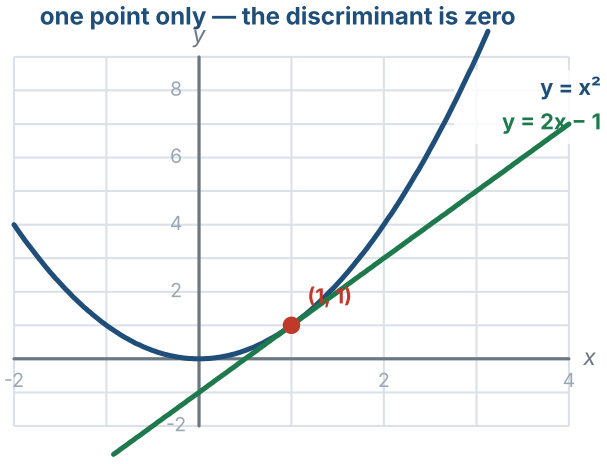
Solution.
$x^2 = 4$, so $x = \pm 2$.
(2, 4) and (−2, 4).



2 How many intersection points?

after substituting, the discriminant of the resulting quadratic counts the meetings

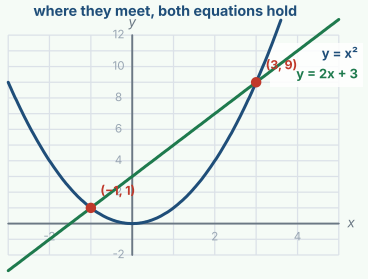
Two real solutions means the line cuts the curve twice; one means it is tangent; none means they miss entirely. This is the same discriminant test as always, wearing a different hat.



Substituting $y = 2x - 1$ into $y = x^2$ gives $x^2 - 2x + 1 = 0$ with discriminant 0 — and the picture shows the line just grazing the curve.

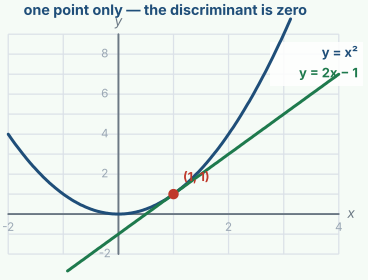
Example 1. **How many times does $y = 2x + 3$ meet $y = x^2$?**

Solution.
$x^2 - 2x - 3 = 0$ has discriminant $4 + 12 = 16 > 0$.
Twice.



Example 2. **How many times does $y = 2x - 1$ meet $y = x^2$?**

Solution.
$x^2 - 2x + 1 = 0$ has discriminant $4 - 4 = 0$.
Once — the line is tangent.

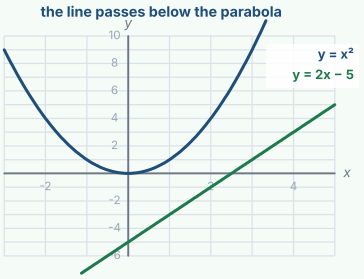


Example 3. **How many times does $y = 2x - 5$ meet $y = x^2$?**

Solution.

$x^2 - 2x + 5 = 0$ has discriminant $4 - 20 = -16 < 0$.

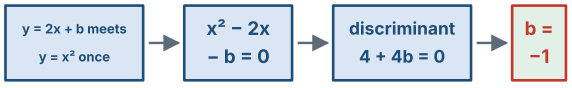
Never.



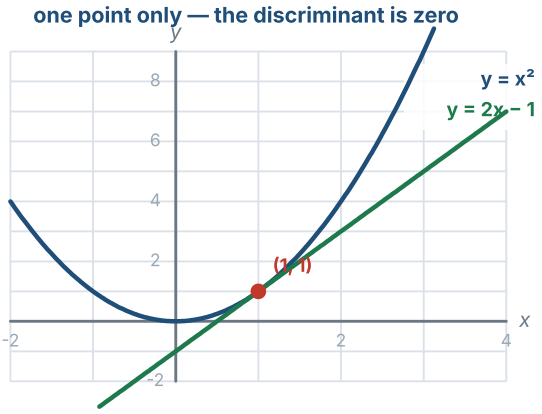
3 Tangency questions

‘meets at exactly one point’ ⇔ discriminant = 0

Whenever a question fixes a constant so that a line touches a curve once, substitute, collect into $ax^2 + bx + c = 0$, and set $b^2 - 4ac = 0$. Watch the sign at the last step — that is where the marks go.



the final sign is where five of your class lost the mark



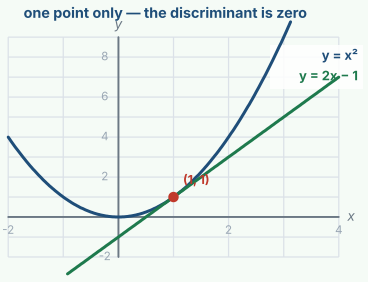
Set up, substitute, set the discriminant to zero, then solve for the constant — and check the sign of the answer against the picture.

Example 1. **For what b does $y = 2x + b$ meet $y = x^2$ exactly once?**

Solution.

$x^2 - 2x - b = 0$ with discriminant $4 + 4b = 0$.

$b = \mathbf{-1}$. Five of your class reached +1 here.

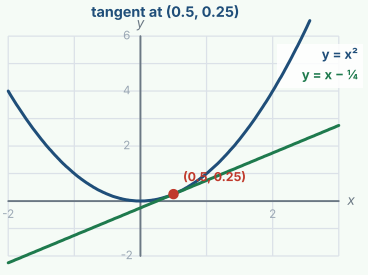


Example 2. **For what c does $y = x + c$ meet $y = x^2$ exactly once?**

Solution.

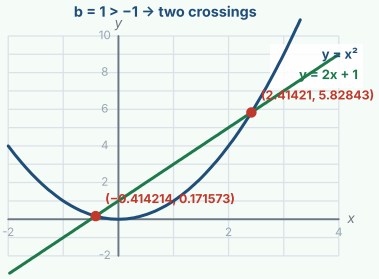
$x^2 - x - c = 0$, so $1 + 4c = 0$.

$c = \mathbf{-1/4}$.



Example 3. **Why does a positive b give two intersections in the first example?**

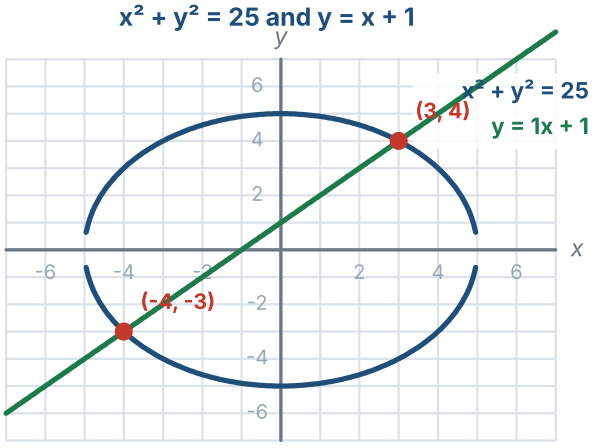
Solution.
$4 + 4b > 0$ whenever $b > -1$, so the discriminant is positive and the line cuts the parabola **twice**.



4 Circle and line systems

substitute the line into the circle equation, then solve the quadratic

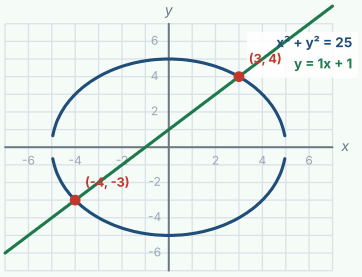
The method does not change when one equation is a circle. Replace y in $x^2 + y^2 = r^2$ using the line, expand carefully, and solve.



The line cuts the circle at two lattice points. Substituting $y = x + 1$ turns the system into $2x^2 + 2x - 24 = 0$, whose roots are those two x-values.

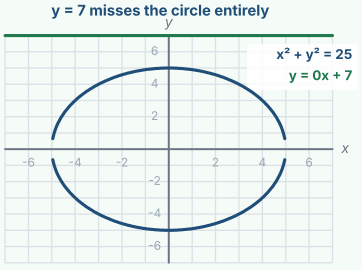
Example 1. **Solve $x^2 + y^2 = 25$ and $y = x + 1$.**

Solution.
$x^2 + (x + 1)^2 = 25$ gives $2x^2 + 2x - 24 = 0$, so $x^2 + x - 12 = 0$.
$x = 3$ or -4, giving **(3, 4) and (−4, −3)**.



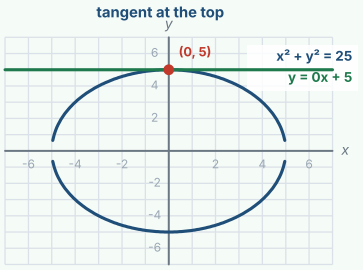
Example 2. **How many points does $y = 7$ share with $x^2 + y^2 = 25$?**

Solution.
$x^2 = 25 - 49 = -24$, which is impossible.
None — the line lies outside the circle.



Example 3. **How many points does y = 5 share with that circle?**

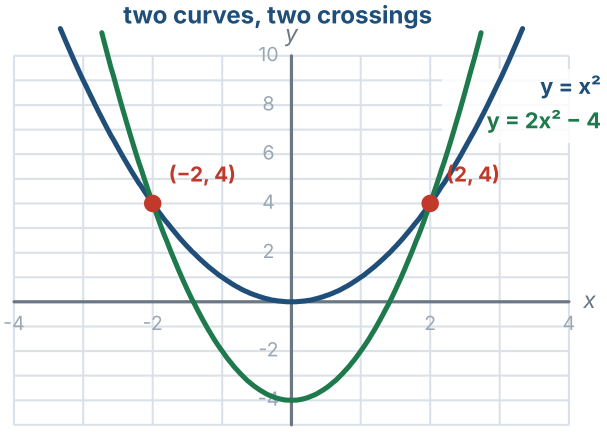
Solution.
$x^2 = 0$, so $x = 0$ only.
One: the line is tangent at (0, 5).



5 Systems with two curves

equate the two expressions | you may get a higher-degree equation

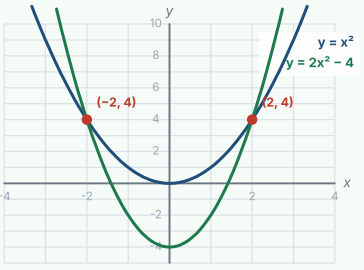
Two parabolas, or a parabola and a reciprocal, follow the same rule. The resulting equation may be cubic or quartic, so factor fully and keep every root.



Both curves are even, so the crossings come in a symmetric pair. Equating them gives $x^2 = 4$ and both roots count.

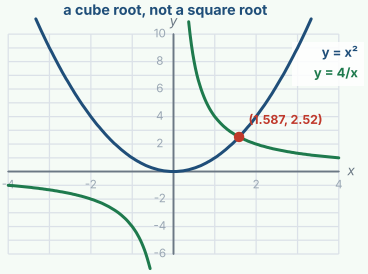
Example 1. **Solve $y = x^2$ and $y = 2x^2 - 4$.**

Solution.
$x^2 = 2x^2 - 4$ gives $x^2 = 4$.
(2, 4) and (−2, 4).



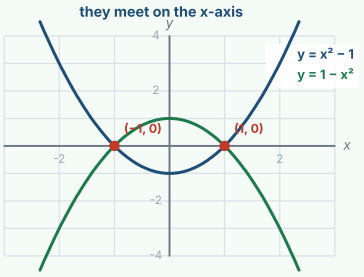
Example 2. **Solve $y = x^2$ and $y = 4/x$ for $x \neq 0$.**

Solution.
$x^2 = 4/x$, so $x^3 = 4$ and $x = \sqrt[3]{4} \approx 1.587$.
One real intersection, near **(1.587, 2.520)**.



Example 3. **Solve $y = x^2 - 1$ and $y = 1 - x^2$.**

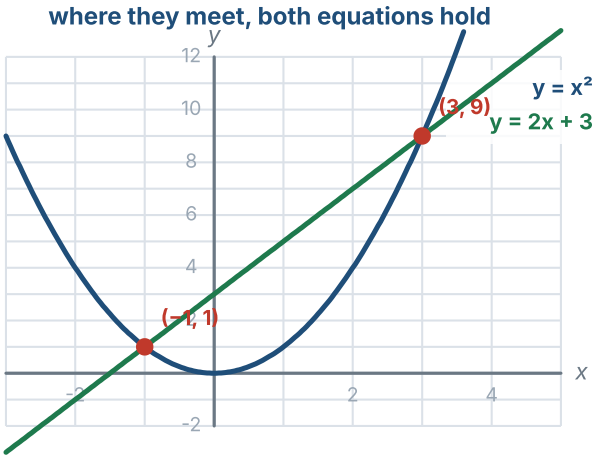
Solution.
$x^2 - 1 = 1 - x^2$ gives $2x^2 = 2$.
(1, 0) and (−1, 0).



6 Reading a system off a graph

a solution is a point the two graphs share | count intersections, not curves

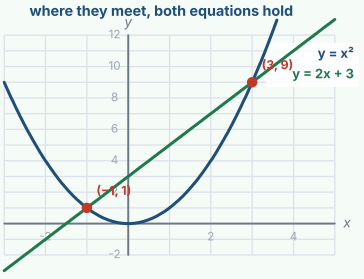
Desmos makes this almost free: enter both equations and click each crossing. The SAT often asks only for the number of solutions, or for one coordinate of one solution.



Two crossings means two solutions, and each one is a pair (x, y). Reading off only the x-values answers half the question.

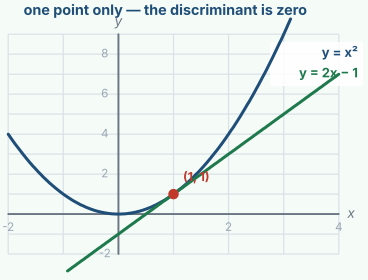
Example 1. **Two graphs cross at (3, 9) and (−1, 1). How many solutions does the system have?**

Solution.
Two — one for each shared point.

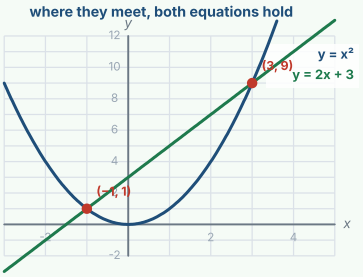


Example 2. **A line touches a parabola at one point. How many solutions does that system have?**

Solution.
One. A tangency is a single repeated solution.



Example 3. **A system's solutions are (3, 9) and (−1, 1). What is the sum of the x-coordinates?**



Solution.

3 + (−1) = **2**. This also equals −b/a for the quadratic you solved.

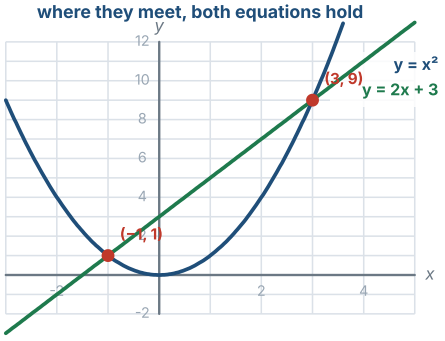
Part 1 — With Desmos (10 questions · every one is faster on the graph)

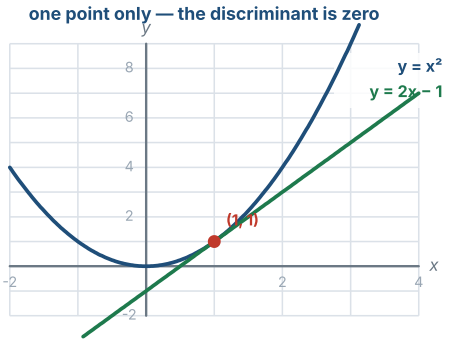
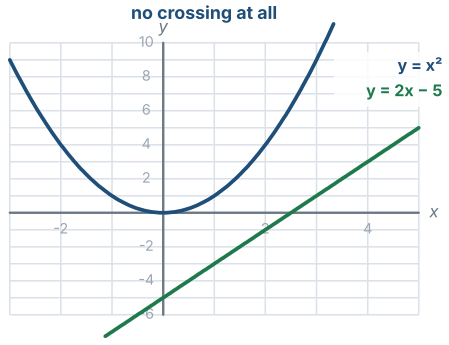
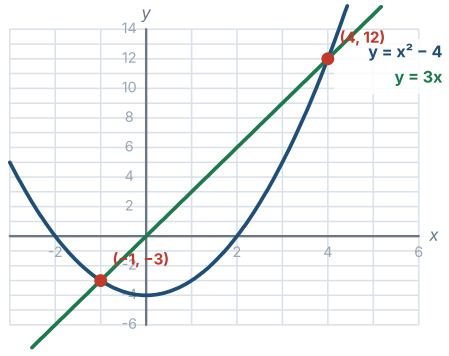
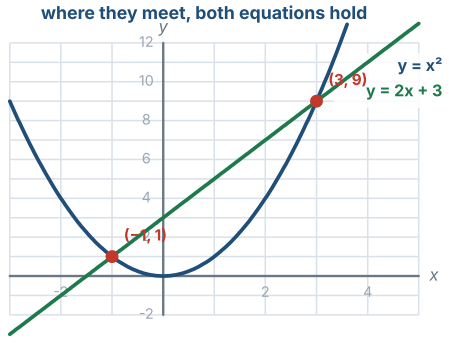
The SAT gives you Desmos on every maths question. Here it is **an intersection finder**. Enter both equations and click every crossing; Desmos gives you each point exactly.

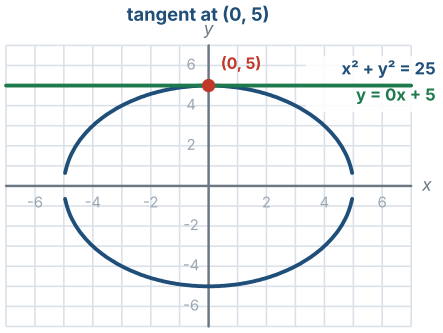
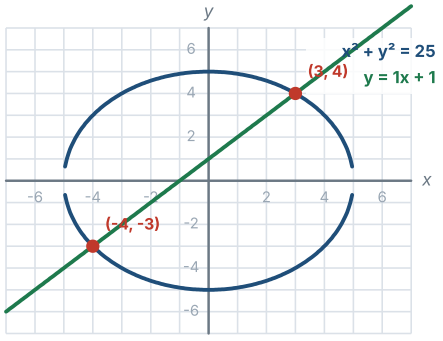
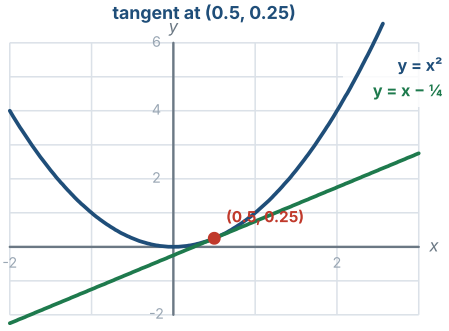
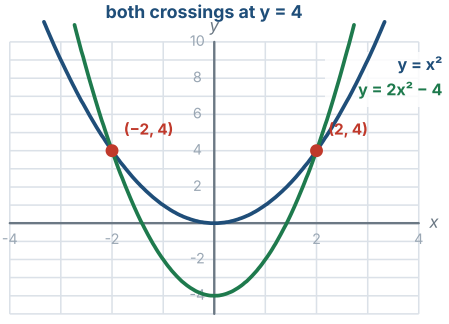
#	Question	Type into Desmos	Then	Answer
D1	Solve the system $y = x^2$ and $y = 2x + 3$. What is the larger x-value?	`y=x^2 and y=2x+3`	Graph both and click the intersections.	3
D2	For that system, what is the sum of the two x-values?	`3+(-1)`	Equals $-b/a$ for $x^2 - 2x - 3$.	2
D3	For what value of b does $y = 2x + b$ meet $y = x^2$ exactly once?	`-(2^2)/4`	Set $4 + 4b = 0$.	-1
D4	How many times does $y = 2x - 5$ meet $y = x^2$?	`y=x^2 and y=2x-5`	The line passes below the parabola.	0
D5	Solve $x^2 + y^2 = 25$ and $y = x + 1$. What is the larger x-value?	`x^2+(x+1)^2=25`	Substitute, then solve the quadratic.	3
D6	Solve $y = x^2$ and $y = 2x^2 - 4$. What are the x-values?	`x^2=2x^2-4`	Both curves are even, so the answers are symmetric.	2 and -2
D7	Solve $y = x^2$ and $y = 4/x$. What is x, to three decimal places?	`x^3=4`	A cube root, not a square root.	1.587
D8	How many points does the line $y = 5$ share with $x^2 + y^2 = 25$?	`25-5^2`	Zero under the root means tangency.	1
D9	Solve $y = x^2 - 4$ and $y = 3x$. What is the larger x-value?	`x^2-4=3x`	Factor $x^2 - 3x - 4$.	4
D10	For what value of c does $y = x + c$ meet $y = x^2$ exactly once?	`-1/4`	Set $1 + 4c = 0$.	-0.25

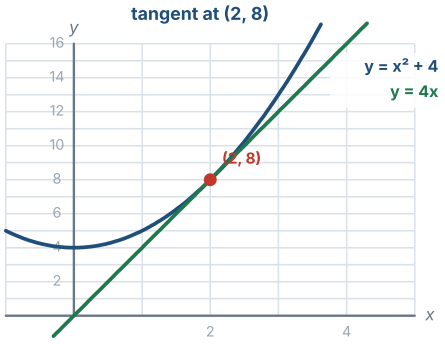
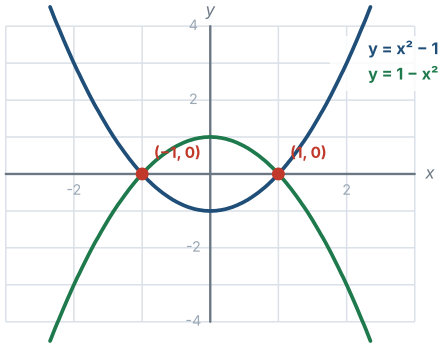
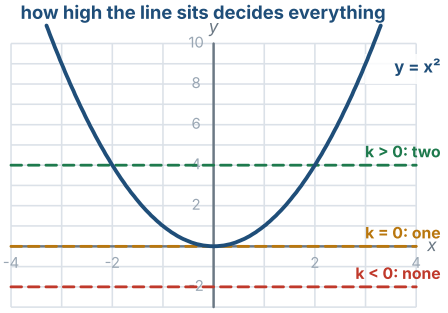
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. When a question fixes a constant so two graphs meet **exactly once**, it is a discriminant question wearing a disguise.

#	Level	Question	Answer	Solution
P1	Hard	In the xy-plane, the graphs of $y = x^2$ and $y = 2x + 3$ intersect at two points. What is the sum of the x-coordinates of those points? 	2	$x^2 = 2x + 3$ gives $x^2 - 2x - 3 = 0$, whose roots sum to $-b/a$ = **2**. The roots are 3 and −1. A common wrong answer is the larger root alone.

#	Level	Question	Answer	Solution
P2	Hard	In the xy-plane, the line $y = 2x + b$, where b is a constant, intersects the parabola $y = x^2$ at exactly one point. What is the value of b? 	−1	$x^2 - 2x - b = 0$ must have discriminant zero: $4 + 4b = 0$. $b = -1$. Five of your class answered +1 — the method was right and the sign was lost at the last step.
P3	Hard	How many solutions does the system $y = x^2$ and $y = 2x - 5$ have? 	Zero	$x^2 - 2x + 5 = 0$ has discriminant $4 - 20 = -16 < 0$. **No real solutions** — the line passes entirely below the parabola.
P4	Hard	The system $y = x^2 - 4$ and $y = 3x$ has two solutions. What is the greater of the two x-values? 	4	$x^2 - 3x - 4 = 0$ factors as $(x - 4)(x + 1) = 0$. $x = 4$ or -1, so the greater is **4**.
P5	Hard	Which ordered pair is a solution to the system $y = x^2$ and $y = 2x + 3$? 	(−1, 1)	Substituting $x = -1$: $y = 1$ from the parabola and $y = -2 + 3 = 1$ from the line, so both hold. **(−1, 1).** A common wrong answer uses the right y but the wrong sign of x.

#	Level	Question	Answer	Solution
P6	Hard	How many points of intersection do the line $y = 5$ and the circle $x^2 + y^2 = 25$ have? 	One	Substituting gives $x^2 + 25 = 25$, so $x^2 = 0$ and $x = 0$ only. **One** point, (0, 5): the line is tangent at the top of the circle.
P7	Very hard	The system $x^2 + y^2 = 25$ and $y = x + 1$ has two solutions. What is the greater of the two x-values? 	3	$x^2 + (x + 1)^2 = 25$ gives $2x^2 + 2x − 24 = 0$, so $x^2 + x − 12 = 0$ and $(x + 4)(x − 3) = 0$. $x = 3$ or $−4$, so the greater is **3**. A common wrong answer is the circle's radius, not a solution.
P8	Very hard	For what value of c does the line $y = x + c$ intersect the parabola $y = x^2$ at exactly one point? 	−1/4	$x^2 − x − c = 0$ must have discriminant zero: $1 + 4c = 0$. $c = \mathbf{-1/4}$. A common wrong answer drops the minus sign.
P9	Very hard	The graphs of $y = x^2$ and $y = 2x^2 − 4$ intersect at two points. What is the y-coordinate of both points? 	4	$x^2 = 2x^2 − 4$ gives $x^2 = 4$, so $x = ±2$. In either case $y = x^2 = \mathbf{4}$. Both points share the same height.

#	Level	Question	Answer	Solution
P10	Very hard	In the xy-plane, the line y = mx intersects the parabola y = x² + 4 at exactly one point, where m is a positive constant. What is the value of m? 	4	x² − mx + 4 = 0 needs m² − 16 = 0, so m = ±4. With m positive, m = **4**. A common wrong answer is m², stopped one step early.
P11	Very hard	The graphs of y = x² − 1 and y = 1 − x² intersect at how many points, and where? 	Two points, at (1, 0) and (−1, 0)	x² − 1 = 1 − x² gives 2x² = 2, so x = ±1, and then y = 1 − 1 = 0. **(1, 0) and (−1, 0)** — the two parabolas meet on the x-axis.
P12	Very hard	A system consists of y = x² and y = k, where k is a constant. For which values of k does the system have exactly two real solutions? 	k > 0	x² = k has two real solutions exactly when k is positive, one when k = 0, and none when k is negative. **k > 0.** The horizontal line must sit above the vertex.

The one habit to build. Set the discriminant to zero, then check the sign of your answer against the graph. On Test 5 ten students reached the right equation for b and five of them wrote +1 instead of −1.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

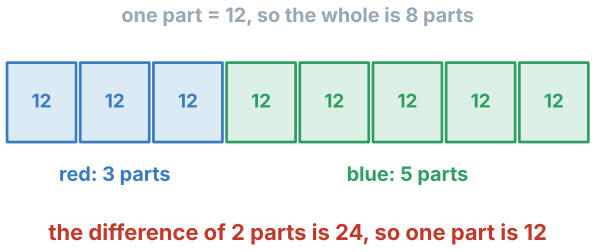
SAT Math 3.1 — Ratios, Rates, Proportion & Units

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Ratios — part to part, or part to whole?

a : b means a parts and b parts, so the whole is a + b parts

‘Boys to girls is 3 : 5’ means boys are **3/8** of the class, not 3/5. **Add the parts first**, then take the share you need. When a question gives a **difference** instead of a total, count how many parts that difference is worth.



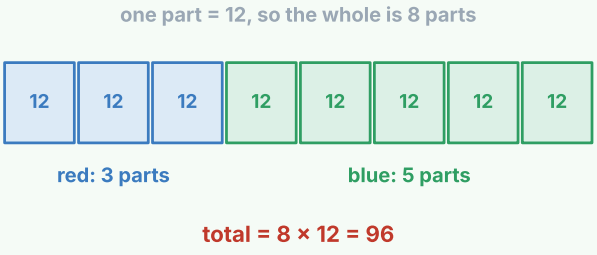
Eight equal parts in all: three red and five blue. The gap between them is two parts, which is what a ‘how many more’ question gives you.

Example 1. **Red and blue counters are in the ratio 3 : 5 and there are 24 more blue than red. How many counters altogether?**

Solution.

The difference is 5 − 3 = 2 parts, so 2 parts = 24 and one part = 12.

Total = 8 parts = **96**.

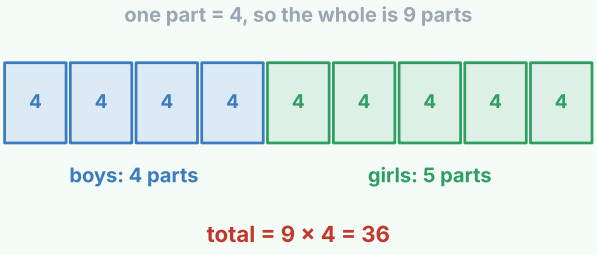


Example 2. **A class of 36 has boys and girls in the ratio 4 : 5. How many girls?**

Solution.

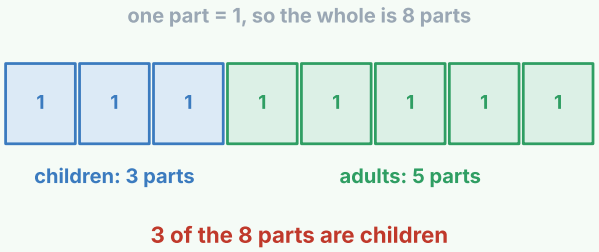
9 parts, so one part is 4.

Girls = 5 × 4 = **20**.



Example 3. **Three-eighths of a group are children. What is the ratio of children to adults?**

Solution.
Children 3 parts of 8, so adults are 8 − 3 = 5.
The ratio is **3 : 5**.



2 Proportion, and direct against inverse

direct: y = kx — double one, double the other | inverse: y = k/x — double one, halve the other

Decide which you have before calculating: more items cost more (**direct**), more workers take fewer days (**inverse**). Inverse questions are easiest as a fixed total of work — 5 painters × 12 days = 60 painter-days.

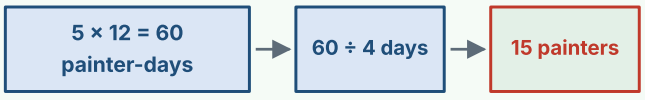
name the type before you calculate

direct: y = kx	twice the items, twice the cost
inverse: y = k/x	twice the workers, half the days
the test does more of one mean more, or less, of the other?	

The two behave in opposite directions, and the words in the question tell you which is which.

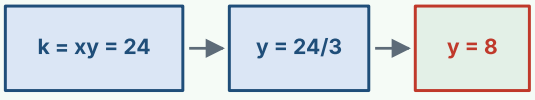
Example 1. **5 painters paint a house in 12 days. How many painters are needed to finish it in 4 days?**

Solution.
Total work = 5 × 12 = 60 painter-days.
60 ÷ 4 = **15 painters**. Fewer days needs more painters — inverse.



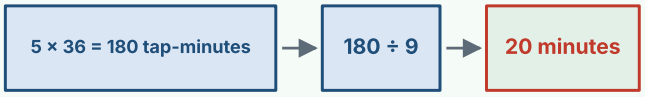
Example 2. **y is inversely proportional to x, and y = 6 when x = 4. Find y when x = 3.**

Solution.
k = xy = 24, so y = 24/3 = **8**. x went down, y went up.



Example 3. **The time to fill a tank is inversely proportional to the number of taps. 5 taps take 36 minutes. How long do 9 taps take?**

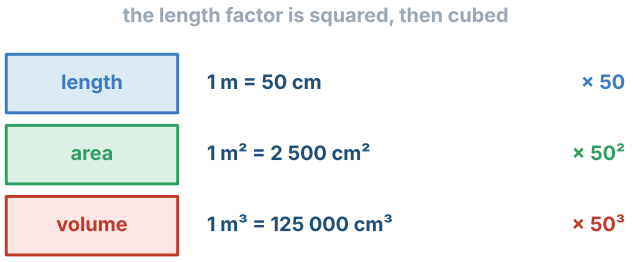
Solution.
Total work = 5 × 36 = 180 tap-minutes.
180 ÷ 9 = **20 minutes**.



3 Scale drawings and maps

a scale of 1 : n means 1 unit on the drawing is n of the same units in reality

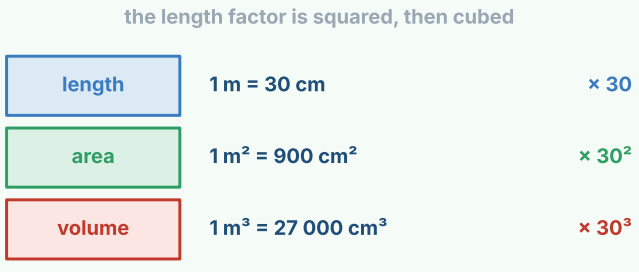
1 : 50,000 means one centimetre is 50,000 **centimetres**. Multiply first, **then** convert. And if lengths scale by k, **areas scale by k^2** and **volumes by k^3**.



The same factor applies once to a length, twice to an area and three times to a volume — which is why a 1 : 50 model's floor area is 2,500 times smaller, not 50.

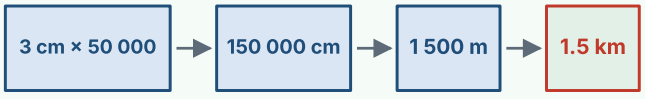
Example 1. **A model is built to scale 1 : 30 and holds 2 litres. How much does the real object hold?**

Solution.
Volumes scale by 30^3 = 27,000.
2 × 27,000 = **54,000 litres** — not 60.



Example 2. **A map has scale 1 : 50,000. A distance of 3 cm on the map is how many km?**

Solution.
3 × 50,000 = 150,000 cm = 1,500 m.
1.5 km. Multiply first, convert second.



Example 3. **On a 1 : 50 model the floor area is 1,200 cm². Find the real area in m².**

Solution.

Areas scale by $50^2 = 2{,}500$.

$1{,}200 \times 2{,}500 = 3{,}000{,}000$ cm² = **300 m²**.

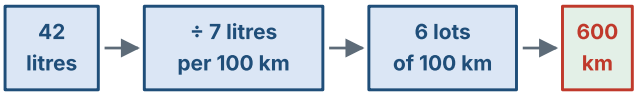
the length factor is squared, then cubed

length	1 m = 50 cm	× 50
area	1 m² = 2 500 cm²	× 50²
volume	1 m³ = 125 000 cm³	× 50³

4 Unit rates and unit conversion

a unit rate is the amount per one | convert by multiplying by a fraction equal to 1, arranged so the unwanted unit cancels

The word after 'per' goes on the bottom. Write a conversion as a fraction with the unit you want on top and the unit you are losing underneath, and **never stop at the number** — check you finished in the unit asked for.



the question asked for a distance, not a count of lots

Six is the trap: it is the number of 100-km lots, not the distance. One more step finishes the job.

Example 1. **A car uses 7 litres of fuel per 100 km. How far can it travel on 42 litres?**

Solution.

$42 \div 7 = 6$ lots of 100 km → **600 km**.

Stopping at 6 is the trap: the question asked for a distance, not a count.



Example 2. **Which is better value: 400 g for $3.20, or 650 g for $4.55?**

Solution.

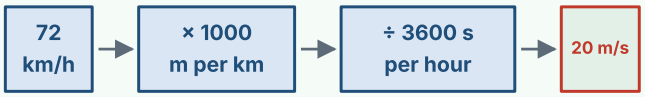
$8.00 per kg against **$7.00 per kg**, so the 650 g pack. Convert both to the same rate first.

400 g for $3.20	$8.00 per kg
650 g for $4.55	$7.00 per kg — better

Example 3. **Convert 72 km/h into metres per second.**

Solution.

72 × 1,000 ÷ 3,600 = **20 m/s**. Kilometres and hours both cancel.



5 Squared and cubed units, density and flow

1 m² = 10,000 cm² · 1 m³ = 1,000,000 cm³ · 1 m³ = 1,000 L | density = mass ÷ volume

Dividing an area by 100 instead of 10,000 is the error this box exists to prevent. Flow rates **add** when two taps run together; times never do.

the length factor is squared, then cubed

length	1 m = 100 cm	× 100
area	1 m² = 10 000 cm²	× 100²
volume	1 m³ = 1 000 000 cm³	× 100³

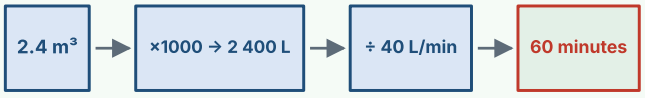
One hundred centimetres to the metre, so ten thousand square centimetres to the square metre and a million cubic ones. Each row is the one above, multiplied by 100 again.

Example 1. **A tank measures 2 m by 1.5 m by 0.8 m and fills at 40 litres per minute. How long does it take?**

Solution.

Volume = 2.4 m³ = **2,400 litres**.

2,400 ÷ 40 = **60 minutes**. Two unit steps, not one.



Example 2. **A material has density 2.7 g/cm³. Express this in kg/m³.**

Solution.

× 1,000,000 for the volume, ÷ 1,000 for the mass.

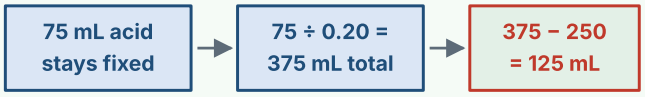
2,700 kg/m³.

the length factor is squared, then cubed

length	1 m = 100 cm	× 100
area	1 m² = 10 000 cm²	× 100²
volume	1 m³ = 1 000 000 cm³	× 100³

Example 3. **250 mL of a solution is 30% acid. How much water makes it 20% acid?**

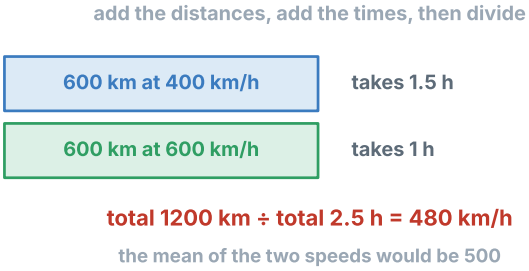
Solution.
Acid is 75 mL and does not change. 75 ÷ 0.20 = 375 mL in total.
Water added = **125 mL**.



6 Average speed — total over total

average speed = total distance ÷ total time (never the average of the speeds)

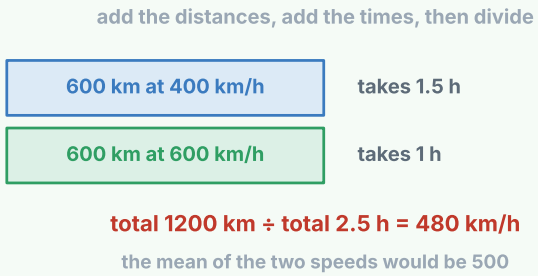
This was the worst-answered question on your class's mock: 2 of 15. Averaging the two speeds is only correct when the legs take the **same time**, which is almost never what the question describes.



The slow leg takes longer, so it carries more weight. That is why the answer sits below the halfway point between 400 and 600.

Example 1. **A plane flies 600 km at 400 km/h and returns at 600 km/h. Find the average speed.**

Solution.
Times 1.5 h and 1 h, so 2.5 h in total for 1,200 km.
1,200 ÷ 2.5 = **480 km/h** — not 500.

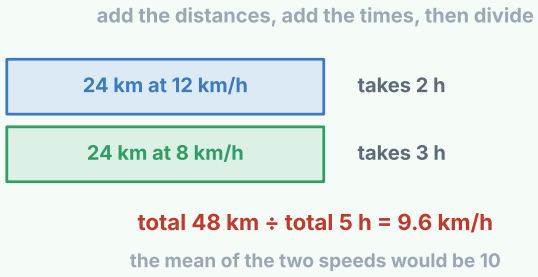


Example 2. **A cyclist rides 24 km at 12 km/h, then 24 km at 8 km/h. Find the average speed.**

Solution.

Times 2 h and 3 h, so 5 h for 48 km.

48 ÷ 5 = **9.6 km/h** — not 10.

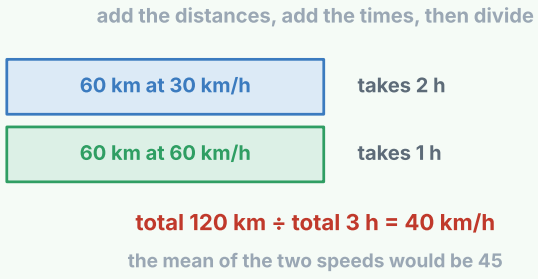


Example 3. **Half a journey is covered at 30 km/h and half at 60 km/h. Find the average speed.**

Solution.

For a distance 2d the time is d/30 + d/60 = d/20.

2d ÷ (d/20) = **40 km/h**, not 45.



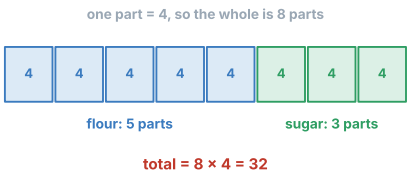
Part 1 — With Desmos (10 questions · every one is faster on the graph)

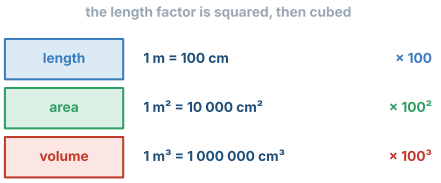
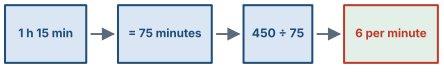
The SAT gives you Desmos on every maths question. For this topic it is **an equation solver and a unit chain**. Put the whole conversion on one line so no step is lost, and type a relationship with an x in it when the question hides one.

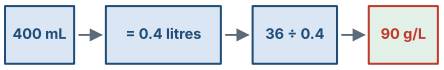
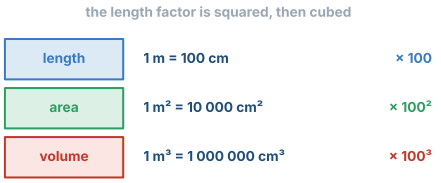
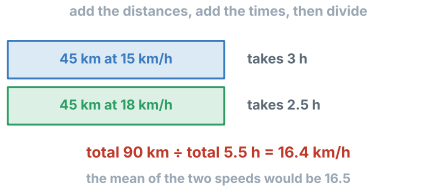
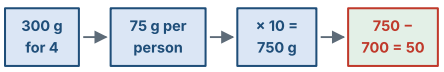
#	Question	Type into Desmos	Then	Answer
D1	A recipe uses flour and sugar in the ratio 5 : 3. If 20 cups of flour are used, how many cups of sugar are needed?	`20*3/5`	Scale the ratio in one line.	12
D2	A car travels 240 kilometres in 3 hours at a constant speed. What is its speed in kilometres per hour?	`240/3`	Distance over time.	80
D3	A printer produces 18 pages in 40 seconds. At this rate, how many pages does it produce in 5 minutes?	`18/40*300`	Convert minutes to seconds inside the line.	135
D4	A map uses a scale of 1 cm to 25 km. Two towns are 7.4 cm apart on the map. How many kilometres apart are they?	`7.4*25`	Multiply by the scale factor.	185
D5	A rectangular tank measures 40 cm by 25 cm by 30 cm. How many litres does it hold? (1 litre = 1000 cm^3)	`40*25*30/1000`	Divide by 1000, not by 1,000,000 — that factor converts to cubic metres.	30
D6	A solution contains 36 grams of salt in 400 millilitres. What is the concentration in grams per litre?	`36/0.4`	Convert millilitres to litres first.	90
D7	A machine fills 450 bottles in 1 hour 15 minutes. How many bottles per minute is that?	`450/75`	Work in a single unit before dividing.	6
D8	A metal block has a mass of 540 grams and a volume of 60 cubic centimetres. What is its density, in grams per cubic centimetre?	`540/60`	Density is mass over volume.	9
D9	A shop sells 3 notebooks for $7.50. At the same rate, what do 11 notebooks cost, in dollars?	`7.50/3*11`	Find the unit price, then scale.	27.5
D10	A runner covers 12 kilometres in 50 minutes. What is this speed in kilometres per hour?	`12/50*60`	Scale the time unit up to an hour.	14.4

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Two routes in: if the question hides a **relationship**, write an equation; if it hides a **unit**, write the conversion as a fraction and let the units cancel.

#	Level	Question	Answer	Solution
P1	Hard	A recipe calls for flour and sugar in the ratio 5 : 3. A baker uses 20 cups of flour. How many cups of sugar should the baker use? 	12	Each part is 20 ÷ 5 = 4 cups, and sugar is 3 parts. 3 × 4 = **12 cups**. A common wrong answer is the difference between the parts, not the sugar.

#	Level	Question	Answer	Solution
P2	Hard	A printer produces 18 pages in 40 seconds. At this rate, how many pages will it produce in 5 minutes? 	135	5 minutes is 300 seconds, and the rate is 18/40 = 0.45 pages per second. 0.45 × 300 = **135 pages**. A common wrong answer comes from using 5 minutes as 200 seconds.
P3	Hard	A rectangular oil tin measures 40 cm by 25 cm by 30 cm and is completely full. How many litres of oil does it hold? (1 litre = 1000 cubic centimetres) 	30	Volume = 40 × 25 × 30 = 30,000 cm³. 30,000 ÷ 1000 = **30 litres**. Dividing by 1,000,000 gives 0.03 — that factor converts cm³ to *cubic metres*, not litres.
P4	Hard	On a map, 1 centimetre represents 25 kilometres. Two towns are 7.4 centimetres apart. What is the actual distance between them, in kilometres? 	185	7.4 × 25 = **185 km**. A common wrong answer divides instead of multiplying; another adds the two numbers.
P5	Hard	A metal block has a mass of 540 grams and a volume of 60 cubic centimetres. What is its density, in grams per cubic centimetre? 	9	Density = mass ÷ volume = 540/60. = **9 g/cm³**. A common wrong answer inverts the fraction, giving volume per gram instead.
P6	Hard	A machine fills 450 bottles in 1 hour and 15 minutes. What is its rate, in bottles per minute? 	6	1 hour 15 minutes is 75 minutes. 450 ÷ 75 = **6 bottles per minute**. A common wrong answer divides by 60, forgetting the extra 15 minutes.
P7	Very hard	A runner covers 12 kilometres in 50 minutes. What is this speed in kilometres per hour? 	14.4	In one minute the runner covers 12/50 = 0.24 km, and an hour is 60 minutes. 0.24 × 60 = **14.4 km/h**. A common wrong answer divides 12 by 0.5, treating 50 minutes as half an hour.

#	Level	Question	Answer	Solution
P8	Very hard	A solution contains 36 grams of salt dissolved in 400 millilitres of water. What is the concentration, in grams per litre? 	90	400 mL = 0.4 L. 36 ÷ 0.4 = **90 g/L**. A common wrong answer divides by 4 instead of 0.4 — a factor-of-ten slip the units would have caught.
P9	Very hard	Three identical notebooks cost $7.50. A school buys 44 notebooks and receives a 10% discount on the total. How much does the school pay, in dollars? 	99	Unit price = 7.50/3 = $2.50, so 44 cost 44 × 2.50 = $110. A 10% discount means paying 90%: 0.90 × 110 = **$99**.
P10	Very hard	A cube has edges of 1 metre. A student says its volume is 100 cubic centimetres because 1 m = 100 cm. What is the correct volume, and what went wrong? 	1,000,000 cm³; the conversion factor must be cubed.	A length factor of 100 becomes 100² for area and 100³ for volume. 1 m³ = 100³ = **1,000,000 cm³**. This is the same slip that turns 30 litres into 0.03.
P11	Very hard	Two cyclists ride at constant speeds. The first covers 45 km in 3 hours; the second covers the same distance in 2 hours 30 minutes. How much faster is the second, in kilometres per hour? 	3	First: 45/3 = 15 km/h. Second: 2 h 30 min = 2.5 h, so 45/2.5 = 18 km/h. 18 − 15 = **3 km/h**. A common wrong answer gives the second speed itself, not the difference.
P12	Very hard	A recipe for 4 people uses 300 grams of rice. A cook prepares the same recipe for 10 people but has only 700 grams of rice. How many grams short is the cook? 	50	For 10 people: 300 ÷ 4 = 75 g per person, and 75 × 10 = 750 g. 750 − 700 = **50 grams short**.

The one habit to build. Eight of these twelve finish one step *before* the answer: 2.4 m³ is 2,400 litres, 6 lots of 100 km is 600 km, 48/5 is 9.6 and not 10. Every time you reach a number ask: *is this what they asked for, in the unit they asked for?*

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

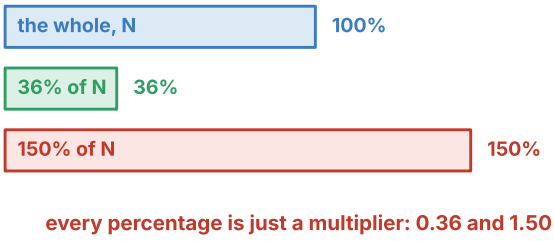
SAT Math 3.2 — Percentages

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Percent of a number

p% of N = (p/100) × N — the word *of* always means multiply

Turn the percentage into a decimal and multiply. Percentages above 100 are **not special**: 150% of 60 is 1.5 × 60 = 90. Run it backwards with one division — if 12% of x is 54, then x = 54 ÷ 0.12.



A percentage above 100 sits past the whole bar — nothing about the method changes, only the multiplier.

Example 1. **A shop has 1,250 items. 36% are books, and 20% of the books are hardback. How many hardback books are there?**

Solution.

0.36 × 1,250 = **450 books**.

0.20 × 450 = **90 hardback books**.



Example 2. **A shop has 1,250 items. 36% are books, and 20% of the books are hardback. How many hardbacks?**

Solution.

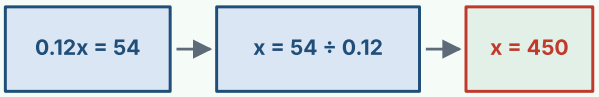
0.36 × 1,250 = 450 books; 0.20 × 450 = **90**.

the second percentage is taken of the books, not of 1,250

0.36 × 1,250	450 books
0.20 × 450	90 hardbacks

Example 3. **If 12% of x is 54, what is x?**

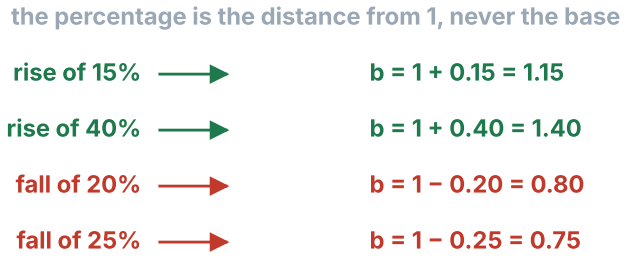
Solution.
0.12x = 54 → x = 54 ÷ 0.12 = **450**.



2 Increase and decrease — multiply by what is left

up p%: × (1 + p/100) | down p%: × (1 − p/100)

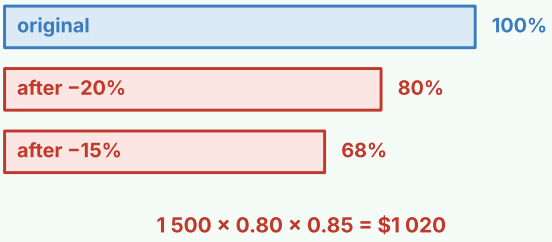
One multiplication, **not a subtraction**. To take 25% off, multiply by 0.75; to add 15%, multiply by 1.15. Writing the multiplier straight down is what makes every harder question in this topic a one-liner.



The percentage is the distance from 1. What you multiply by is what is left, or what you end up with — never the percentage itself.

Example 1. **A $1,500 laptop falls 20% in its first year and a further 15% in its second. Find the final price.**

Solution.
1,500 × 0.80 × 0.85 = **$1,020**.
Multiplying by 0.25 to take 25% off is the classic slip — the multiplier is what is *left*.



Example 2. **A $1,500 laptop falls 20% in year one and a further 15% in year two. Find the final price.**

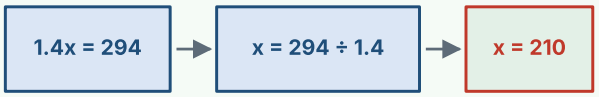
Solution.
1,500 × 0.80 × 0.85 = **$1,020**.



Example 3. **A number increased by 40% gives 294. Find the number.**

Solution.

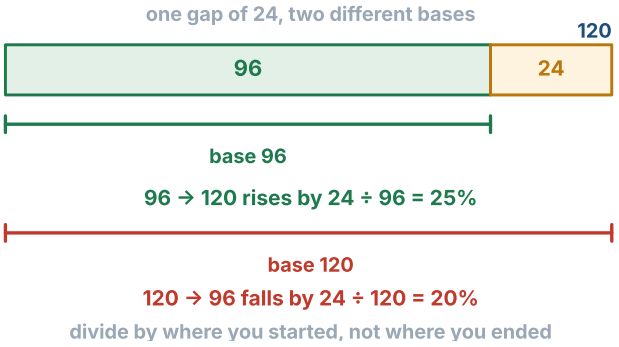
1.4x = 294 → x = **210**. Divide by the multiplier.



3 Percent change, and one quantity as a percent of another

percent change = (new − old) ÷ old × 100 | A as a percent of B = (A ÷ B) × 100

The denominator is always **what you started from**, or whatever follows the word *of*. The same gap is a different percentage in each direction: 96 → 120 is a 25% rise, but 120 → 96 is a 20% fall.



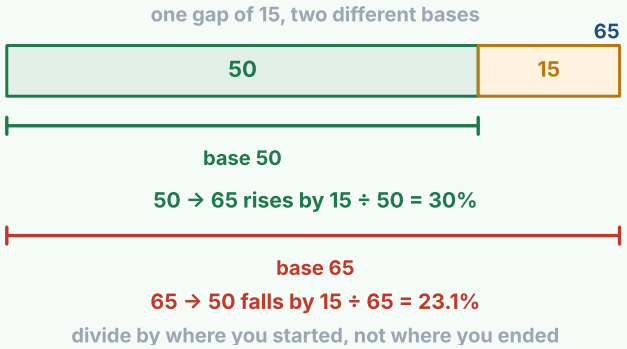
One gap of 24, two answers. Measured against 96 it is a quarter; measured against 120 it is a fifth. The base decides.

Example 1. **A ticket price rises from $50 to $65. What is the percent increase?**

Solution.

(65 − 50)/50 = 15/50 = **30%**.

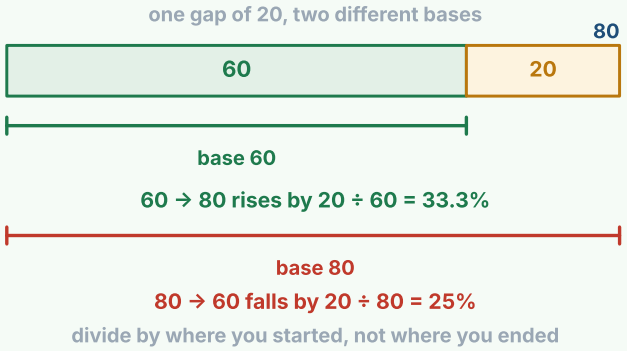
Dividing by 65 gives about 23% — the right subtraction and the wrong denominator.



Example 2. **A value falls from 80 to 60. Find the percent decrease.**

Solution.

20 ÷ 80 = **25%**. The denominator is the old value.

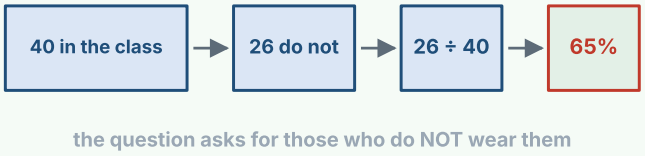


Example 3. **In a class of 40, 14 students wear glasses. What percent do not?**

Solution.

40 − 14 = 26 students.

26 ÷ 40 = **65%**.



4 Reverse percent — finding the original

if the final value is F after a change of p%, then original = F ÷ (1 ± p/100)

Given the price **after** a change, you divide. Adding the percentage back on is the trap: 15% off $120 is $102, but 15% more than $102 is only $117.30. Write 0.85x = 102 and solve it.



102 ÷ 0.85 = 120

one division by the whole chain — never add the percentage back on

The arrow runs forwards, so reading it backwards means dividing. Adding 15% of $102 back on lands at $117.30 — the wrong base.

Example 1. An item sells for $135 including 8% tax. What was the pre-tax price?

Solution.

1.08x = 135, so x = 135 ÷ 1.08 = **$125**.

Tax is *added*, so divide by 1.08 — not by 0.92.



135 ÷ 1.08 = 125

one division by the whole chain — never add the percentage back on

Example 2. A price is $288 after 20% off and then a further 10% off. Find the original.

Solution.

0.8 × 0.9 = 0.72, so 288 ÷ 0.72 = **$400**. Divide by the whole chain.



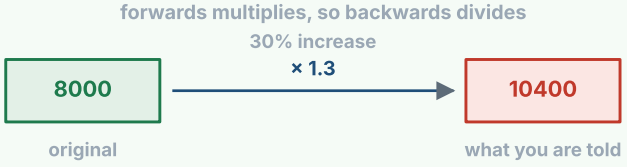
288 ÷ 0.72 = 400

one division by the whole chain — never add the percentage back on

Example 3. After a 30% increase a population is 10,400. What was it before?

Solution.

10,400 ÷ 1.3 = **8,000**.



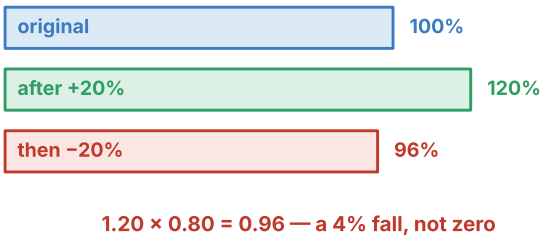
10400 ÷ 1.3 = 8000

one division by the whole chain — never add the percentage back on

5 Successive percent change — multiply, never add

up p% then down q% → multiply by (1 + p/100)(1 − q/100)

Two changes **do not add**, because the second is taken from a bigger or smaller number. The recovery is always larger than the fall: down 20% needs a 25% rise to get back, since 0.8 × 1.25 = 1.



The second 20% is taken from 120, not from 100, so it removes more than the first 20% added. The bar lands below where it started.

Example 1. **A price rises 20% and then falls 20%. What is the overall change?**

Solution.

1.20 × 0.80 = 0.96, so the price is 96% of the original — a **4% decrease**, not zero.

The order makes no difference: 0.80 × 1.20 is the same 0.96.



Example 2. **A price falls 20%. By what percent must it rise to restore the original?**

Solution.

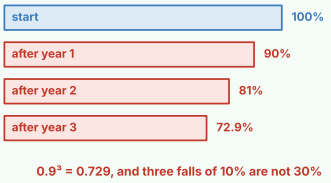
0.8 × m = 1 → m = 1.25, a **25% rise**. The recovery is always bigger than the fall.



Example 3. **A value falls 10% each year for three years. It ends at what percent of the original?**

Solution.

$0.9^3 = 0.729$, so **72.9%**. Repeated change is a power.



6 Discount, markup, tax and commission

markup is on the cost · discount is on the marked price · tax is on the price after the discount · percent profit is measured on the cost

The order is **not decoration** — doing tax before the discount gives a different, wrong answer. Commission is often tiered or applies only above a threshold, so read **which sales it covers**.

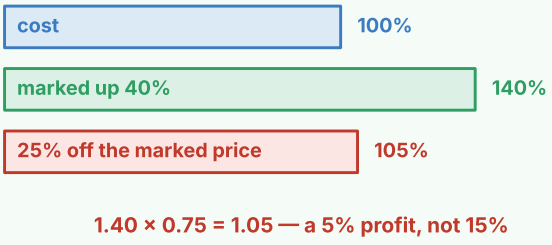
each percentage has its own base — work left to right

markup	taken on the cost
discount	taken on the marked price
tax	taken on the price after the discount
percent profit	measured against the cost

Four percentages, four different bases. Reading the chain in order is the whole method.

Example 1. A shop marks goods up 40% above cost and then offers 25% off. What percent profit is made on the cost?

Solution.
1.40 × 0.75 = 1.05, so the selling price is 105% of cost — a **5% profit**.
Not 15%. The 25% came off the *marked price*, not the cost.



Example 2. A $250 coat is reduced 30%, then 8% tax is added to the sale price. Find the total.

Solution.
250 × 0.7 = $175, then 175 × 1.08 = **$189**. Tax is on what you actually pay.

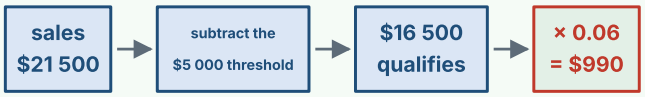


tax after the discount, never before

Example 3. **Commission is 6% on sales above $5,000 only. Find the commission on sales of $21,500.**

Solution.

Only the excess counts: 0.06 × 16,500 = **$990**.



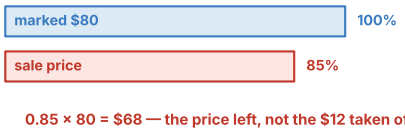
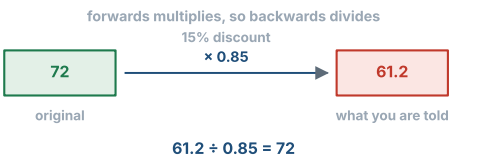
Part 1 — With Desmos (10 questions · every one is faster on the graph)

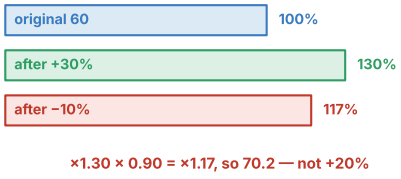
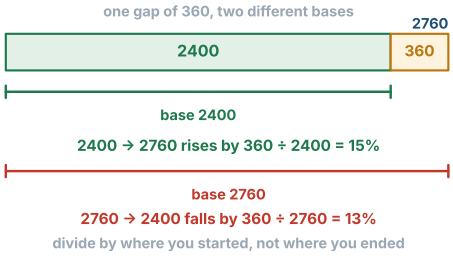
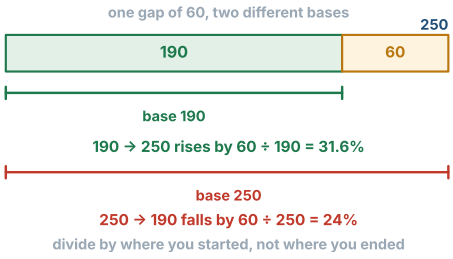
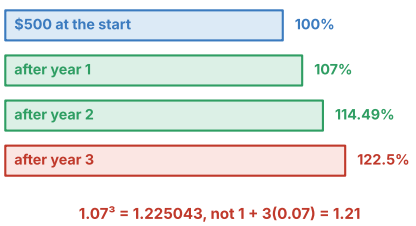
The SAT gives you Desmos on every maths question. For percentages it is **a multiplier chain**. Type every multiplier onto one line and read the final number; where the original is unknown, type the equation and let Desmos solve it.

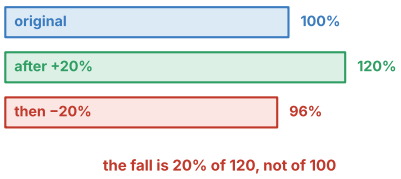
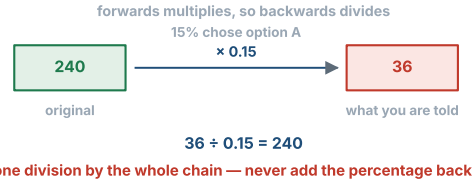
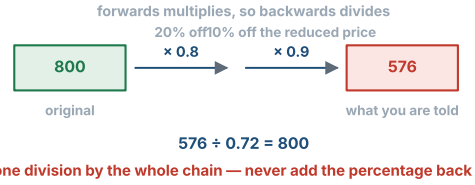
#	Question	Type into Desmos	Then	Answer
D1	A jacket priced at $80 is reduced by 15%. What is the sale price, in dollars?	`80*0.85`	A 15% cut multiplies by 0.85.	68
D2	A salary of $42,000 rises by 6%. What is the new salary, in dollars?	`42000*1.06`	A 6% rise multiplies by 1.06.	44520
D3	After a 15% discount an item sells for $61.20. What was the original price, in dollars?	`61.20/0.85`	Reversing a discount means dividing.	72
D4	A price of 60 is raised by 30% and the new price is then cut by 10%. What is the final price?	`60*1.3*0.9`	Chain the multipliers; the answer is not 60 × 1.20.	70.2
D5	A population grows from 2,400 to 2,760. What is the percentage increase?	`(2760-2400)/2400*100`	Change over the ORIGINAL, times 100.	15
D6	What is 18% of 350?	`0.18*350`	Write the percentage as a decimal.	63
D7	A $90 item is taxed at 8%. What is the total cost, in dollars?	`90*1.08`	Adding tax multiplies by 1 + rate.	97.2
D8	A number decreases from 250 to 190. What is the percentage decrease?	`(250-190)/250*100`	Still divide by the original value.	24
D9	A $500 investment grows by 7% a year for 3 years. What is it worth, in dollars?	`500*1.07^3`	Repeated multiplication, not 500 × 1.21.	612.52
D10	36 is what percentage of 240?	`36/240*100`	Part over whole, times 100.	15

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Two routes in: if a percentage is running **backwards**, divide; if two changes are **stacked**, multiply. And check which number the percentage is taken *of*.

#	Level	Question	Answer	Solution
P1	Hard	A jacket is priced at $80. During a sale the price is reduced by 15%. What is the sale price of the jacket? 	$68.00	A 15% reduction leaves 85% of the price. 0.85 × 80 = **$68**. A common wrong answer is the discount itself, not the new price.
P2	Hard	After a discount of 15%, a brass scale sells for 61.20 talens. What was its original price, in talens? 	72.00	The sale price is 85% of the original, so divide. 61.20 ÷ 0.85 = **72**. A common wrong answer adds 15% of 61.20 back on, which reverses the wrong base — the commonest error here.

#	Level	Question	Answer	Solution
P3	Hard	A copper kettle is priced at 60 talens. The shop raises the price by 30%, then lowers the new price by 10%. What is the final price, in talens? 	70.2	60 × 1.30 = 78, then 78 × 0.90 = **70.2**. Successive changes multiply: 1.30 × 0.90 = 1.17, a rise of 17% — not the 20% you get by adding 30 and −10.
P4	Hard	A town's population grows from 2,400 to 2,760. What is the percentage increase? 	15%	Change = 360, and the base is the original 2,400. 360/2400 = 0.15 = **15%**. A common wrong answer divides by the new value 2,760 instead.
P5	Hard	A $90 item is subject to 8% sales tax. What is the total cost, to the nearest cent? 	97.20	90 × 1.08 = **$97.20**.
P6	Hard	A number decreases from 250 to 190. What is the percentage decrease? 	24%	Change = 60, over the original 250. 60/250 = 0.24 = **24%**. A common wrong answer divides by 190; D is the proportion remaining, not the decrease.
P7	Very hard	A $500 investment grows by 7% each year. To the nearest cent, what is it worth after 3 years? 	$612.52	$500(1.07)^3 = 500 \times 1.225043 =$ **$612.52**. A common wrong answer is 500 × 1.21, which multiplies the rate by 3 instead of compounding it.

#	Level	Question	Answer	Solution
P8	Very hard	The price of a ticket rises by 20% and then falls by 20%. Compared with the original price, the final price is 	4% lower.	1.20 × 0.80 = 0.96, so the final price is 96% of the original — **4% lower**. The two 20%s are taken from different bases, which is why they do not cancel.
P9	Very hard	A shop marks up the cost of an item by 40% to set its selling price. During a sale it then takes 25% off the selling price. What is the overall percentage change from the original cost? 	5% increase	1.40 × 0.75 = 1.05, so the final price is 105% of the cost — a **5% increase**. A common wrong answer adds 40 and −25, which treats both percentages as if they had the same base.
P10	Very hard	In a survey, 36 of the respondents chose option A, which was 15% of all respondents. How many people responded in total? 	240	If 36 is 15% of the total, then total = 36 ÷ 0.15 = **240 people**.
P11	Very hard	A laptop's price is reduced by 20%, and a customer with a membership card receives a further 10% off the reduced price. The customer pays $576. What was the original price, in dollars? 	$800.00	The two reductions multiply: 0.80 × 0.90 = 0.72 of the original. 576 ÷ 0.72 = **$800**. A common wrong answer divides by 0.90 only; B treats the total discount as 20%.
P12	Very hard	A shop's revenue rose 25% in year 1 and fell 20% in year 2. A manager reports that revenue is now 5% above where it started. Is the manager right? 	No; revenue is back to exactly where it started.	1.25 × 0.80 = 1.00 exactly, so revenue is **unchanged**. Adding and subtracting the percentages (that answer) ignores that the 20% fall is taken from the larger, post-increase figure.

The one habit to build. Every percentage is a percentage **of something**, and the SAT's whole game is changing which something. Profit is on the cost, percent change is on the old value, tax is on the discounted price. Before you divide, say out loud: *of what?*

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

SAT Math 3.3 — One-Variable Data: Centre & Spread

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Mean, median and mode

mean = total ÷ how many | median = middle value once sorted | mode = most frequent

Sort the list before taking a median — that single step is the whole trick. With an even count the median is the average of the two middle values. The mean uses every value, so it moves when any one of them moves; the median usually does not.



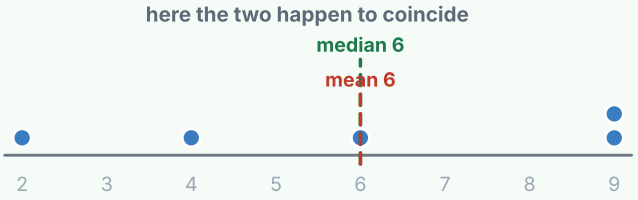
The median is a *position*, so the list has to be in order before you look at the middle. Here 6 sits third of five.

Example 1. **Find the mean and the median of 4, 9, 6, 2, 9.**

Solution.

Mean = 30 ÷ 5 = **6**.

Sorted: 2, 4, 6, 9, 9, so the median is **6**.

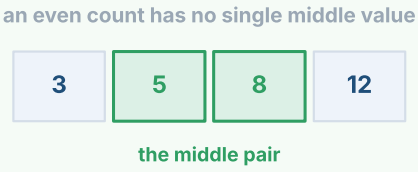


Example 2. **Find the median of 3, 8, 5, 12.**

Solution.

Sorted: 3, 5, 8, 12. Even count, so average the middle two.

(5 + 8)/2 = **6.5**.



Example 3. **The mean of 5 numbers is 14. What is their total?**

Solution.

Total = mean × how many = 14 × 5 = **70**.

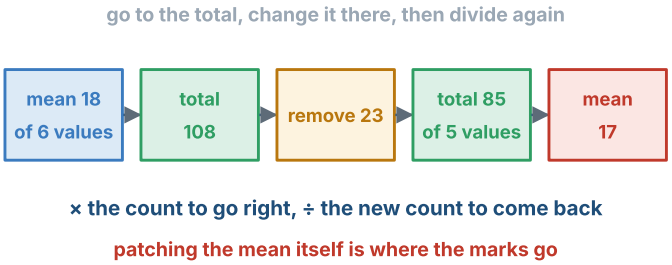
Turning a mean back into a total is the move behind most of these questions.



2 Changing the data set

work with the total, not the mean — total = mean × count

Whenever a value is added, removed or changed, **convert to the total first**, adjust it, then divide by the **new** count. Trying to patch the mean directly is where the errors come from.



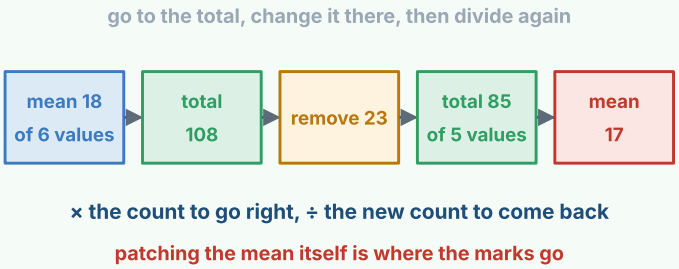
Four short steps, and the count changes in the middle of them. Dividing by the old count is the slip this picture is here to stop.

Example 1. **The mean of 6 numbers is 18. The number 23 is removed. Find the new mean.**

Solution.

Total = 108; removing 23 leaves 85 over 5 numbers.

85 ÷ 5 = **17**.

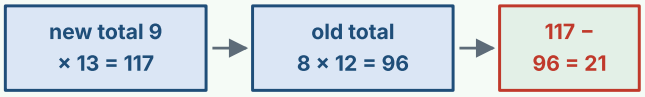


Example 2. **The mean of 8 numbers is 12. A ninth number is added and the mean becomes 13. Find the ninth.**

Solution.

New total = 9 × 13 = 117; old total = 96.

117 − 96 = **21**.



the new value is the difference between the two totals

Example 3. **A value of 20 in a list of 10 numbers with mean 15 is changed to 40. Find the new mean.**

Solution.

Total 150 rises by 20 to 170, still over 10 numbers.

170 ÷ 10 = **17**.

go to the total, change it there, then divide again



× the count to go right, ÷ the new count to come back

patching the mean itself is where the marks go

3 Weighted average

weighted mean = (sum of value × weight) ÷ (sum of weights)

When groups are different sizes you **cannot average the averages**. Multiply each group's mean by its size, add those up, then divide by the total number of items.

two classes, two different sizes

class mean	students	mean × students
75	20	1500
85	30	2550

mean = 4050 ÷ 50 = 81

The bigger class contributes more rows to the total, so the combined mean of 81 sits nearer 85 than the halfway 80.

Example 1. A class of 20 scores a mean of 75; another class of 30 scores a mean of 85. Find the mean of all 50.

Solution.
(20 × 75 + 30 × 85)/50 = (1,500 + 2,550)/50 = **81**.
Not 80 — the bigger class pulls the mean towards 85.

the 30-student class carries more weight

class mean	students	mean × students
75	20	1500
85	30	2550

mean = 4050 ÷ 50 = 81

Example 2. A student scores 80 on a test worth 40% and 90 on one worth 60%. Find the overall score.

Solution.
0.4(80) + 0.6(90) = 32 + 54 = **86**.

the weights are the percentages, and they add to 1

score	weight	score × weight
80	0.4	32
90	0.6	54

mean = 86 ÷ 1 = 86

Example 3. 5 items cost $4 each and 3 cost $12 each. Find the mean cost.

Solution.
(5 × 4 + 3 × 12)/8 = (20 + 36)/8 = **$7**.

total cost over total count

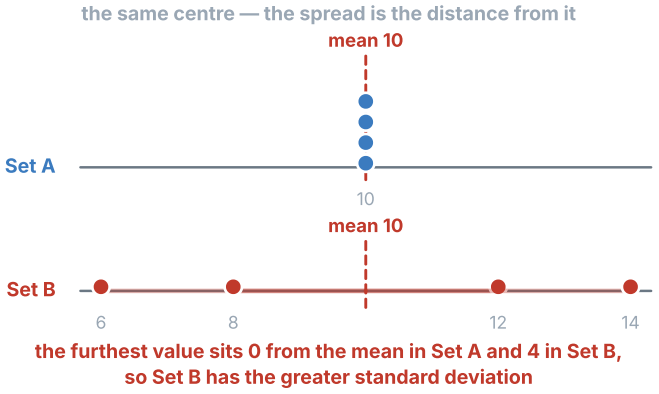
price	how many	price × how many
4	5	20
12	3	36

mean = 56 ÷ 8 = 7

4 Spread: range and standard deviation

range = largest − smallest | standard deviation measures how far values sit from the mean — compare it, never compute it

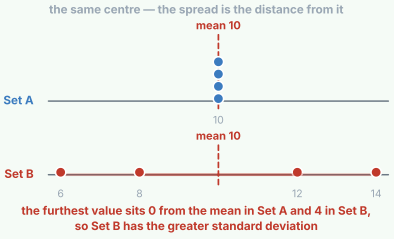
The SAT **never** asks you to calculate a standard deviation. It asks which of two data sets has the larger one, and the answer is whichever is **more spread out from its mean**. Values bunched together give a small standard deviation even if the mean is large.



Both sets centre on 10. Set A has no spread at all, so its standard deviation is zero — any spread beats none.

Example 1. **Set A is 10, 10, 10, 10 and Set B is 6, 8, 12, 14. Both have mean 10. Which has the greater standard deviation?**

Solution.
Set A has every value **at** the mean, so its standard deviation is 0.
Set B is greater — its values sit further from 10.



Example 2. **Find the range of 14, 9, 22, 9, 17.**

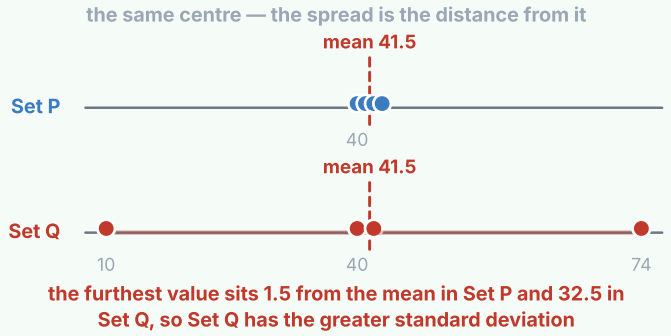
Solution.
22 − 9 = **13**.



the range uses only the two extremes

Example 3. **Set P is 40, 41, 42, 43 and Set Q is 10, 40, 42, 74. Which has the greater standard deviation, and do they have the same mean?**

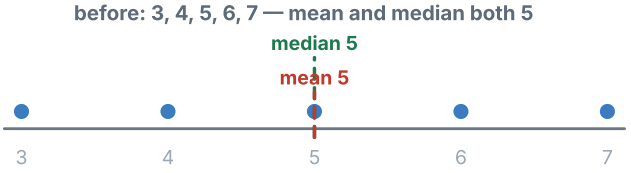
Solution.
Both have mean 41.5. **Set Q** has the greater standard deviation — the same centre, far wider spread.



5 Outliers: the mean moves, the median does not

an extreme value drags the mean towards it and leaves the median almost alone

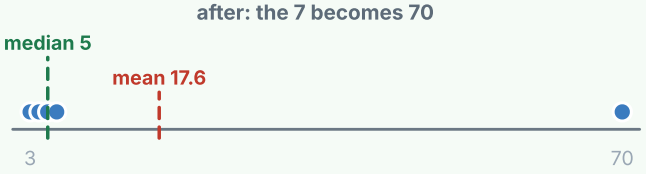
The median only cares about **position**, so one wild value changes it by at most one place. The mean adds every value into a total, so a single outlier can shift it a long way. That is why the median is the better summary of a skewed set.



This is the set before the outlier arrives. Compare it with the example below, where one value moves and only the mean follows.

Example 1. **For 3, 4, 5, 6, 7 the mean and median are both 5. The 7 is changed to 70. What happens to each?**

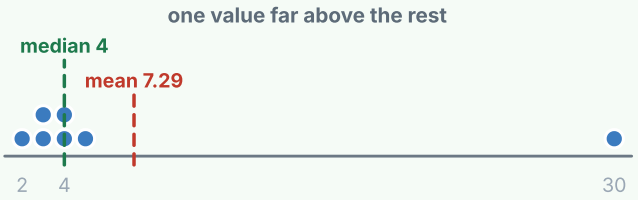
Solution.
New values 3, 4, 5, 6, 70: mean = 88/5 = **17.6**, median is still **5**.
The mean moved by 12.6; the median did not move at all.



Example 2. **A data set has an outlier far above the rest. Which is larger, the mean or the median?**

Solution.

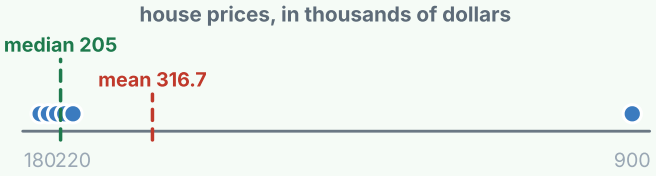
The mean — it is pulled up towards the outlier.



Example 3. **Which measure of centre should you report for house prices in a town with a few mansions?**

Solution.

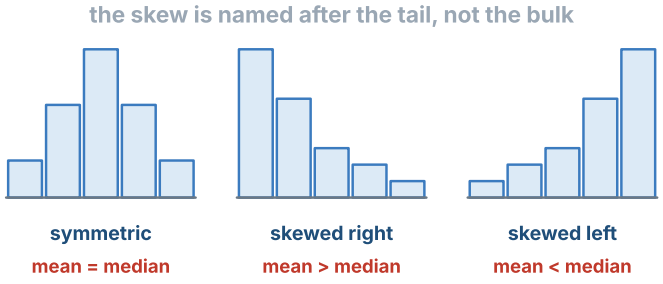
The median. A handful of very expensive houses would inflate the mean and misrepresent a typical house.



6 Shape: symmetric and skewed

symmetric → mean = median | skewed right → mean > median | skewed left → mean < median

The skew is named after **the tail**, not the bulk of the data. A long tail to the right drags the mean to the right of the median. Reading this off a histogram is a standard SAT question.

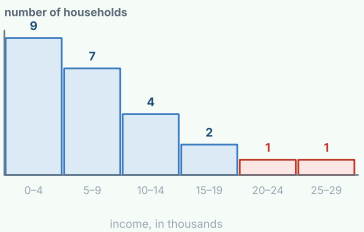


Three shapes, three verdicts. Find the tail first, then say which way the mean has been pulled.

Example 1. **A histogram has most of its bars on the left and a long tail stretching right. Compare the mean and the median.**

Solution.

Skewed right, so the tail pulls the mean up: **mean > median**.

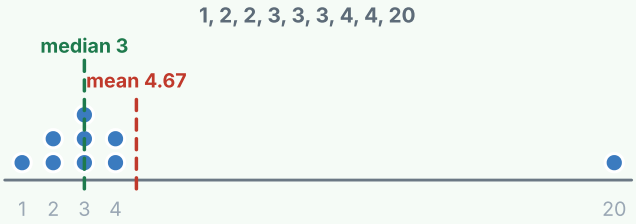


Example 2. **For the data 1, 2, 2, 3, 3, 3, 4, 4, 20, which is larger, the mean or the median?**

Solution.

Median is the 5th of 9 values = **3**. Mean = 42/9 ≈ **4.67**.

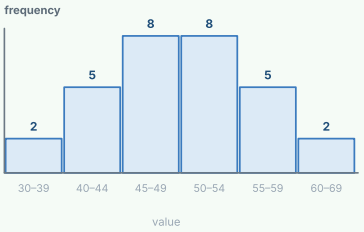
The 20 is a right tail, so mean > median. ✓



Example 3. **A distribution is roughly symmetric with mean 50. What is the approximate median?**

Solution.

About 50. In a symmetric distribution the two coincide.



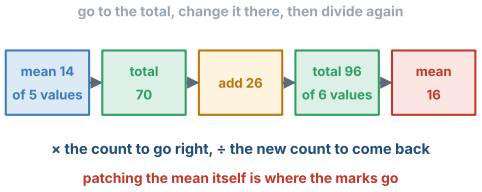
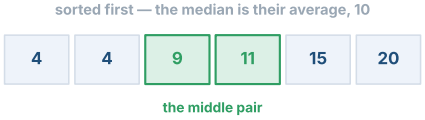
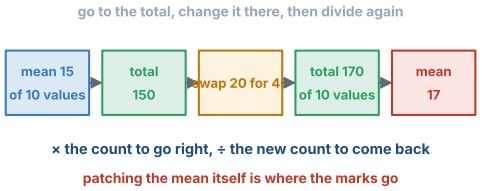
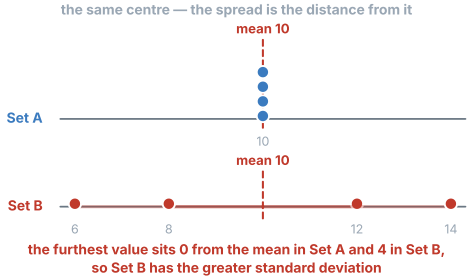
Part 1 — With Desmos (10 questions · every one is faster on the graph)

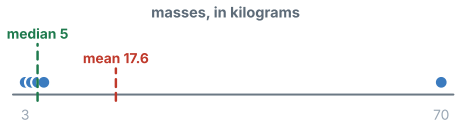
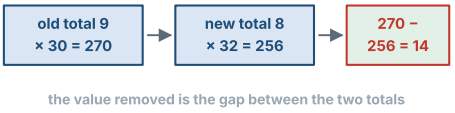
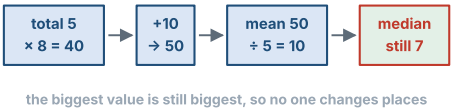
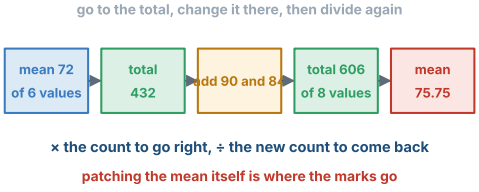
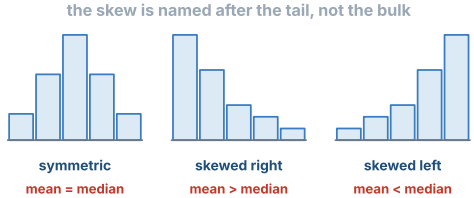
The SAT gives you Desmos on every maths question. Here it is **a calculator for totals**. The move that matters is turning a mean back into a total — total = mean × count — and you can build that straight into the line.

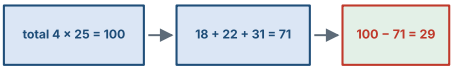
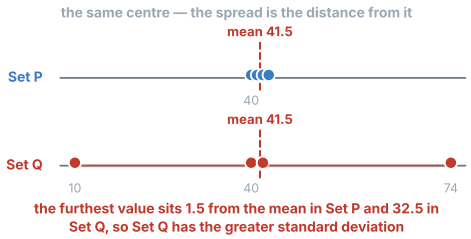
#	Question	Type into Desmos	Then	Answer
D1	A basketball player scored 12, 15, 19, 22 and 27 points in five games. What is the mean number of points per game?	`(12+15+19+22+27)/5`	Type the whole sum on one line.	19
D2	The mean mass of 6 parcels is 18 kilograms. One parcel with a mass of 23 kilograms is removed. What is the mean mass, in kilograms, of the remaining parcels?	`(6*18-23)/5`	Turn the mean into a total inside the same line.	17
D3	Eight students have a mean test score of 12. A ninth student joins and the mean becomes 13. What is the ninth student's score?	`9*13-8*12`	New total minus old total.	21
D4	A class of 20 students averaged 75 on a test and a class of 30 students averaged 85. What is the mean score of all 50 students?	`(20*75+30*85)/50`	Weighted — not the average of the averages.	81
D5	The daily high temperatures, in degrees Celsius, for five days were 14, 9, 22, 9 and 17. What is the range of these temperatures?	`22-9`	Largest minus smallest.	13
D6	Four runners finished a race in 3, 8, 5 and 12 minutes. What is the median finishing time, in minutes?	`(5+8)/2`	Sort first, then average the middle pair.	6.5
D7	A course grade is 40% midterm and 60% final. A student scored 80 on the midterm and 90 on the final. What is the student's course grade?	`0.4*80+0.6*90`	Weights as decimals on one line.	86
D8	The ages, in years, of five club members are 3, 4, 5, 6 and 70. What is the mean age, in years, of the members?	`(3+4+5+6+70)/5`	Compare with the median of 5 — the outlier moves only the mean.	17.6
D9	A librarian recorded the number of books read last month by nine students: 1, 2, 2, 3, 3, 3, 4, 4 and 20. What is the mean number of books read?	`(1+2+2+3+3+3+4+4+20)/9`	Desmos keeps the exact fraction 14/3 if you want it.	4.67 (42/9)
D10	A shop sold 5 notebooks at $4 each and 3 notebooks at $12 each. What is the mean price, in dollars, per notebook sold?	`(5*4+3*12)/8`	Total cost over total count.	7

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Almost every one is solved by converting to a **total** first, adjusting it, then dividing by the **new** count — decide what the count is before you divide.

#	Level	Question	Answer	Solution
P1	Hard	A list of 5 numbers has a mean of 14. A sixth number, 26, is added to the list. What is the mean of the 6 numbers? 	16	Total of the five is 5 × 14 = 70. New total 70 + 26 = 96 over 6 numbers. 96 ÷ 6 = **16**. The trap for forgetting the sixth value.
P2	Hard	The number of hours worked last week by each of 6 employees was 11, 4, 9, 4, 15 and 20. What is the median number of hours worked? 	10	Sorted: 4, 4, 9, 11, 15, 20. Even count, so average the middle pair. (9 + 11)/2 = **10**. A and D are the two middle values taken alone.
P3	Hard	In an experiment, 12 plants had a mean height of 150 centimetres and 8 plants had a mean height of 160 centimetres. What is the mean height, in centimetres, of all 20 plants? unequal groups, so weight each mean group mean / plants / mean × plants 150 / 12 / 1800 160 / 8 / 1280 mean = 3080 ÷ 20 = 154	154	(12 × 150 + 8 × 160)/20 = 3,080/20 = **154**. The trap: averaging 150 and 160 ignores the unequal group sizes.
P4	Hard	A list of 10 numbers has a mean of 15. One of the numbers, 20, is replaced by 40 and no other number is changed. What is the mean of the new list? 	17	The total rises by 40 − 20 = 20, from 150 to 170. 170 ÷ 10 = **17**.
P5	Hard	Data set A consists of the values 10, 10, 10 and 10. Data set B consists of the values 6, 8, 12 and 14. State what is true here. 	Data set B has the greater standard deviation.	Both sets have a mean of 10, but every value of A **is** the mean, so A has no spread at all and its standard deviation is 0. Any spread beats none.

#	Level	Question	Answer	Solution
P6	Hard	The masses, in kilograms, of 5 packages are 3, 4, 5, 6 and 70. Which correctly compares the mean and the median of these masses? 	The mean is greater than the median.	Mean = 88/5 = **17.6**; median = **5**. The single large value pulls the mean up but leaves the middle value where it was.
P7	Very hard	The mean of a list of 9 numbers is 30. When one number is removed from the list, the mean of the remaining 8 numbers is 32. What is the value of the number that was removed? 	14	Old total 9(30) = 270; the remaining 8 total 8(32) = 256. 270 − 256 = **14**. Removing a below-average value raises the mean.
P8	Very hard	A list of 5 numbers has a mean of 8 and a median of 7. The largest number in the list is increased by 10, and no other number is changed. What are the mean and the median of the new list? 	mean 10, median 7	Total 5(8) = 40 → 50, so the mean becomes 50/5 = **10**. The largest value is still the largest, so the sorted order is unchanged and the middle value — the median — stays **7**.
P9	Very hard	Six students in a class had a mean test score of 72. Two more students, with scores of 90 and 84, join the class. What is the mean score of all 8 students? 	75.75	Old total 6(72) = 432; new total 432 + 90 + 84 = 606 over 8 students. 606 ÷ 8 = **75.75**.
P10	Very hard	A histogram showing the distribution of household incomes in a city is skewed to the right. Which statements about the mean and the median income is true? 	The mean is greater than the median.	The long right tail contains a few very large incomes. They enter the mean at full weight but shift the middle value hardly at all, so the mean is dragged **above** the median.

#	Level	Question	Answer	Solution
P11	Very hard	The mean of 4 numbers is 25. Three of the numbers are 18, 22 and 31. What is the value of the fourth number? 	29	Total = 4 × 25 = 100. Known three sum to 71. 100 − 71 = **29**. The trap for stopping at the sum of the three.
P12	Very hard	Data set P consists of 40, 41, 42 and 43. Data set Q consists of 10, 40, 42 and 74. The two data sets have the same mean. State what is true here. 	Data set Q has the greater standard deviation.	Both have mean 166/4 = 41.5. P's values all sit within 1.5 of the centre; Q's reach 31.5 away. Standard deviation measures typical distance from the mean, so **Q's is far larger**.

The one habit to build. Never patch a mean directly. Turn it into a **total**, change the total, then divide by the **new** number of values. Half of these twelve change the count as well as the total, and that is where the marks go.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

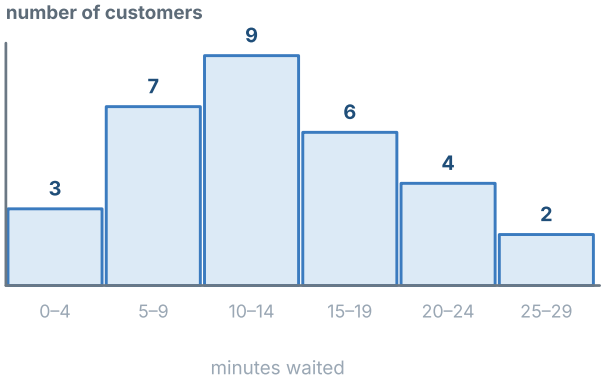
SAT Math 3.4 — Data Displays

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Reading a bar chart or histogram

a bar chart compares separate categories | a histogram shows how one number is distributed across intervals

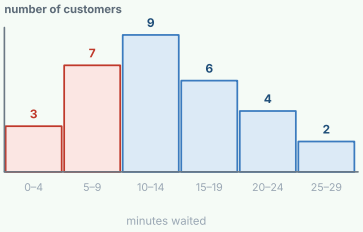
Read the scale before the bars. A bar chart has gaps between the bars because the categories are separate things; a histogram has none, because the intervals run into each other. The height is a **count**, so the total sample size is the sum of all the heights — not the number of bars.



Six touching intervals. The sample size is 3 + 7 + 9 + 6 + 4 + 2 = 31 customers, and six is only the number of bars.

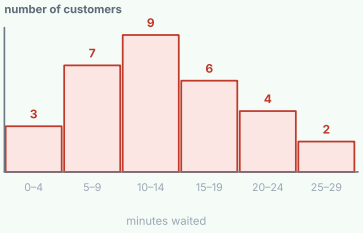
Example 1. **Using the histogram above, how many customers waited fewer than 10 minutes?**

Solution.
The first two intervals are 0–4 and 5–9, with heights 3 and 7.
3 + 7 = **10 customers**.



Example 2. **How many customers are in the sample altogether?**

Solution.
Add every bar: 3 + 7 + 9 + 6 + 4 + 2 = **31**. A common slip is to answer 6, the number of bars.



Example 3. **Which interval contains the median wait?**

Solution.
With 31 values the median is the 16th. Running totals: 3, 10, 19 — the 16th falls in the third interval, **10–14 minutes**.



2 Frequency tables and the mean

mean from a table = ∑(value × frequency) ÷ ∑frequency — divide by the total count, never by the number of rows

This is the **single most common error** on data-display questions. A table with five rows does not mean you divide by five. Multiply each value by how many times it occurred, add those products, then divide by the **total frequency**.

three rows, but twenty values

value	frequency	value × freq
1	4	4
2	6	12
3	10	30

mean = 46 ÷ 20 = 2.3

The right-hand column is what you add; the middle column is what you divide by. The number of rows plays no part at all.

Example 1. **Values 1, 2, 3 occur 4, 6 and 10 times. What is the mean?**

Solution.
∑(value × freq) = 1(4) + 2(6) + 3(10) = 4 + 12 + 30 = 46.
Total frequency = 20, so the mean is 46/20 = **2.3**.

46 over 20, not 46 over 3

value	frequency	value × freq
1	4	4
2	6	12
3	10	30

mean = 46 ÷ 20 = 2.3

Example 2. **A quiz is marked out of 4. Scores 2, 3 and 4 were earned by 5, 8 and 7 students. Find the mean.**

Solution.

2(5) + 3(8) + 4(7) = 10 + 24 + 28 = 62 over 20 students.

62/20 = **3.1**.

a quiz marked out of 4

score	students	score × students
2	5	10
3	8	24
4	7	28

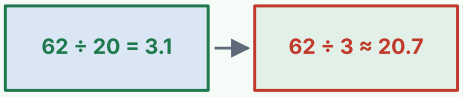
mean = 62 ÷ 20 = 3.1

Example 3. **Why is dividing by 3 wrong in the question above?**

Solution.

Three is the number of **distinct scores**, not the number of students.

62/3 ≈ 20.7, which is not even a possible score out of 4 — a quick sanity check that catches the error.

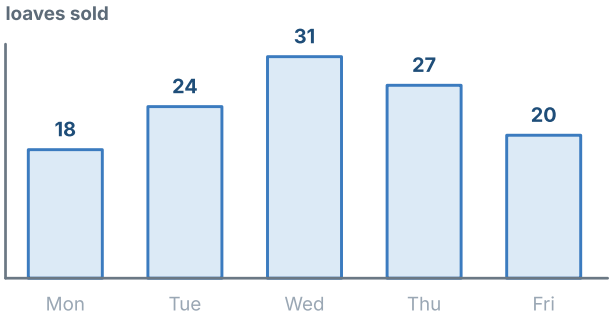


a score of 20.7 out of 4 is impossible — check the divisor

3 Bar charts: comparing categories

read the axis scale first | 'how many more' means subtract, 'how many times' means divide

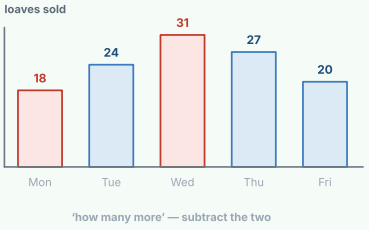
Most bar-chart questions are one of three things: **read one bar, compare two bars, or combine several.** The wording tells you which operation. Watch for an axis that does not start at zero — it makes differences look bigger than they are.



Five separate days. Nothing here is an interval, so nothing touches — that gap is what tells you it is a bar chart.

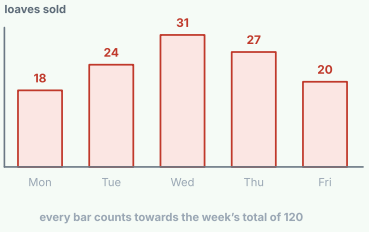
Example 1. **From the chart above, how many more loaves were sold on Wednesday than on Monday?**

Solution.
Wednesday 31, Monday 18.
31 − 18 = **13 loaves**.



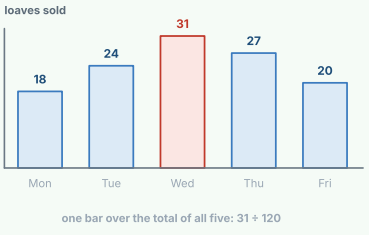
Example 2. **What was the total number of loaves sold over the five days?**

Solution.
18 + 24 + 31 + 27 + 20 = **120 loaves**.



Example 3. **What percentage of the week's sales happened on Wednesday?**

Solution.
31 out of 120.
31/120 = 0.2583... ≈ **25.8%**.



4 Two-way tables

each cell is a count | the row and column totals must add to the grand total

Fill in any missing totals **before** you answer anything. Then the only question that matters is **which number is the whole**: the grand total, one row total, or one column total. The wording picks it for you.

Survey of 180 market customers

	Cheese	No cheese	Total
Bought fruit	60	40	100
No fruit	20	60	80
Total	80	100	180

Three possible denominators sit in this one table: 180, a row total of 100 or 80, or a column total of 80 or 100.

Example 1. From the table above, how many customers bought cheese?

Solution.

The Cheese column totals 60 + 20 = **80 customers**.

Survey of 180 market customers

	Cheese	No cheese	Total
Bought fruit	60	40	100
No fruit	20	60	80
Total	80	100	180

Example 2. What fraction of all customers bought fruit but not cheese?

Solution.

That cell holds 40, and the grand total is 180.

40/180 = **2/9 ≈ 0.222**.

Survey of 180 market customers

	Cheese	No cheese	Total
Bought fruit	60	40	100
No fruit	20	60	80
Total	80	100	180

Example 3. **Of the customers who bought cheese, what fraction also bought fruit?**

Solution.

Now the whole is the Cheese column, **80** — not 180.

60/80 = **3/4**. Changing which total you divide by is the whole skill here.

Survey of 180 market customers

	Cheese	No cheese	Total
Bought fruit	60	40	100
No fruit	20	60	80
Total	80	100	180

5 Dot plots and the shape of data

every dot is one value | count dots for the sample size, not columns

A dot plot shows the **whole data set**, so you can read the mode straight off (the tallest stack), count to the median, and see the shape. Repeated values matter: three dots above 4 means the value 4 appears three times in the list.

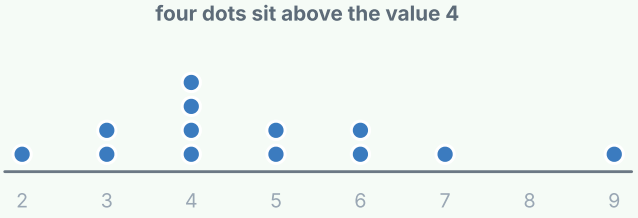


Thirteen dots, eight columns. The sample size is the number of dots, and the tallest stack is the mode.

Example 1. **Using the dot plot above, how many students read exactly 4 books?**

Solution.

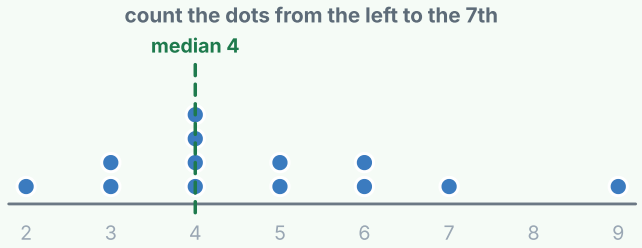
There are four dots above 4, so **4 students**.



Example 2. **What is the median number of books read?**

Solution.

13 values, so the median is the 7th. Counting up: 2 (1), 3 (3), 4 (7) — the 7th dot sits above **4 books**.



Example 3. **What is the range?**

Solution.

Largest 9, smallest 2.

9 − 2 = **7 books**.

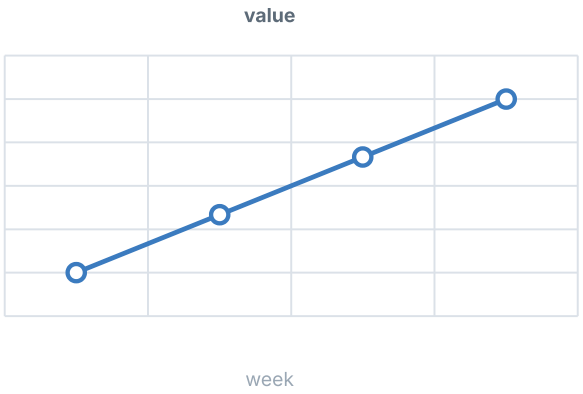


the range uses only the two ends of the plot

6 Line graphs and rates of change

steeper = faster | flat = no change | downward = decreasing

On a line graph the question is almost always about the **slope** between two points, not the heights themselves. ‘Greatest increase’ means the **steepest** upward segment, which is not necessarily the segment ending highest.



Equal steps of 4 a week give a constant slope. The rate of change is (32 − 20) ÷ (4 − 1) = 4 per week.

Example 1. **A graph shows 20 at week 1 and 32 at week 4. What is the average rate of change per week?**

Solution.
(32 − 20)/(4 − 1) = 12/3 = **4 per week**.



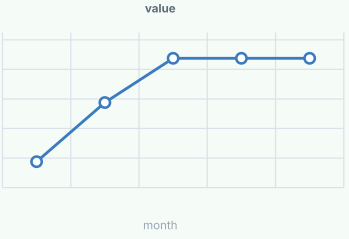
Example 2. **Between which pair of days did sales rise fastest: Mon→Tue (18 to 24) or Tue→Wed (24 to 31)?**

Solution.
Mon→Tue rises 6; Tue→Wed rises 7.
The faster rise is **Tuesday to Wednesday**.



Example 3. **A line graph is flat from month 3 to month 5. What does that mean?**

Solution.
The quantity **did not change** over those two months.
Flat does not mean zero — it means constant, and the constant could be any height.



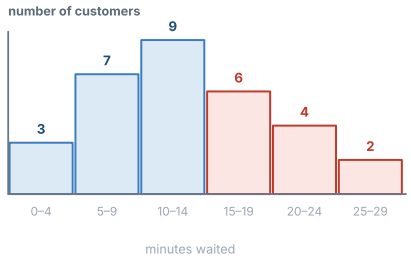
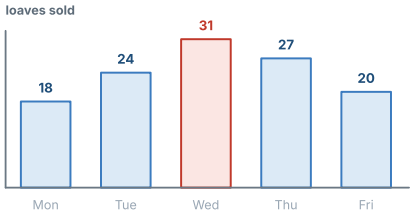
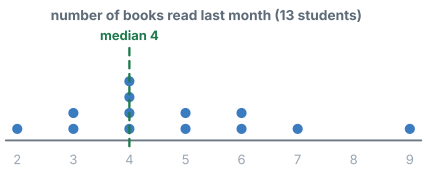
Part 1 — With Desmos (10 questions · every one is faster on the graph)

The SAT gives you Desmos on every maths question. Here it is **a calculator plus a careful denominator**. Build the frequency products and the correct total into a single line so you cannot divide by the wrong thing.

#	Question	Type into Desmos	Then	Answer
D1	A shop's daily sales were 18, 24, 31, 27 and 20 loaves. What is the mean number of loaves sold per day?	`(18+24+31+27+20)/5`	Type the whole sum on one line.	24
D2	In a survey the values 1, 2 and 3 occurred 4, 6 and 10 times. What is the mean value?	`(1*4+2*6+3*10)/(4+6+10)`	Products on top, total frequency underneath.	2.3
D3	A histogram has bars of height 3, 7, 9, 6, 4 and 2. How many values are in the data set?	`3+7+9+6+4+2`	Add the heights, not the bars.	31
D4	Of 180 customers surveyed, 100 bought fruit and 60 of those also bought cheese. What fraction of the fruit buyers bought cheese?	`60/100`	The fruit buyers are the whole here.	0.6
D5	A quiz out of 4 was scored 2, 3 and 4 by 5, 8 and 7 students. What is the mean score?	`(2*5+3*8+4*7)/20`	Weighted mean in one line.	3.1
D6	A bakery sold 31 loaves on Wednesday out of 120 for the week. What percentage of the week's loaves is that, to the nearest tenth?	`31/120*100`	Desmos gives the full decimal; round at the end.	25.8
D7	A line graph reads 20 at week 1 and 32 at week 4. What is the average rate of change per week?	`(32-20)/(4-1)`	Rise over run.	4
D8	A dot plot shows 2, 3, 3, 4, 4, 4, 4, 5, 5, 6, 6, 7 and 9. What is the mean, to two decimal places?	`(2+3+3+4+4+4+4+5+5+6+6+7+9)/13`	Desmos holds the exact fraction 62/13.	4.77
D9	A two-way table has row totals 100 and 80. What percentage of the whole sample is the first row?	`100/180*100`	Round only at the end.	55.6
D10	A histogram's intervals 0–4 and 5–9 hold 3 and 7 values out of 31. What percentage of the data is below 10?	`10/31*100`	Combine the two bars first.	32.3

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Before touching the numbers, decide what the display **shows** and what the question **wants**: a value, a difference, a total, or a fraction — and if a fraction, out of what.

#	Level	Question	Answer	Solution
P1	Hard	The histogram shows the time, in minutes, that each of 31 customers waited at a service desk. How many customers waited 15 minutes or longer? 	12	The last three intervals are 15–19, 20–24 and 25–29, with heights 6, 4 and 2. 6 + 4 + 2 = **12**. A common wrong answer is the first of those bars taken alone.
P2	Hard	In a frequency table, the value 1 occurs 4 times, the value 2 occurs 6 times and the value 3 occurs 10 times. What is the mean of the data? divide by 20, not by 3 and not by 7 value | frequency | value × freq 1 | 4 | 4 2 | 6 | 12 3 | 10 | 30 mean = 46 ÷ 20 = 2.3	2.3	∑(value × freq) = 4 + 12 + 30 = 46 over a total frequency of 20. 46/20 = **2.3**. A common wrong answer is the mean of 1, 2 and 3 with the frequencies ignored; D is 46 ÷ 7.
P3	Hard	The table shows the results of a survey of 180 market customers. Of the customers who bought cheese, what fraction also bought fruit? Survey of 180 market customers | | Cheese | No cheese | Total | Bought fruit | 60 | 40 | 100 No fruit | 20 | 60 | 80 Total | 80 | 100 | 180	3/4	The whole is the Cheese column, 60 + 20 = 80, not the grand total. 60/80 = **3/4**. A common wrong answer divides by the Fruit row (100) instead.
P4	Hard	The bar chart shows loaves sold on each of five days. What percentage of the week's total was sold on Wednesday, to the nearest tenth of a percent? 	25.8	Total = 18 + 24 + 31 + 27 + 20 = 120. 31/120 = 0.2583… = **25.8%**.
P5	Hard	The dot plot shows the number of books read last month by each of 13 students. What is the median number of books read? 	4	Thirteen values, so the median is the 7th in order. The running count 2 (1), 3 (3), 4 (7) puts the 7th value at **4**. A common wrong answer is the mean, which the question did not ask for.

#	Level	Question	Answer	Solution
P6	Hard	A bakery's sales rose from 18 loaves on Monday to 24 on Tuesday, then to 31 on Wednesday. Between which pair of consecutive days was the increase greatest, and by how much? 	**Tuesday to Wednesday, by 7**	Monday to Tuesday rises 24 − 18 = 6; Tuesday to Wednesday rises 31 − 24 = 7. The greater rise is **Tuesday to Wednesday, by 7**. B and C give day totals, not increases.
P7	Very hard	In the two-way table, what percentage of the customers who did not buy fruit also did not buy cheese? Survey of 180 market customers | | Cheese | No cheese | Total | |---|---|---|---| | Bought fruit | 60 | 40 | 100 | | No fruit | 20 | 60 | 80 | | Total | 80 | 100 | 180 |	**75**	The whole is the 'No fruit' row, which totals 20 + 60 = 80. 60/80 = **75%**. Dividing 60 by the grand total 180 gives 33.3%, which answers a different question.
P8	Very hard	A histogram of 31 wait times has bars of height 3, 7, 9, 6, 4 and 2 over the intervals 0–4, 5–9, 10–14, 15–19, 20–24 and 25–29 minutes. Which interval contains the median wait time? 	**10–14**	With 31 values the median is the 16th. Running totals are 3, 10, 19, so the 16th value sits in the third interval, **10–14**. A common wrong answer stops at the running total 10.
P9	Very hard	A dot plot shows the values 2, 3, 3, 4, 4, 4, 4, 5, 5, 6, 6, 7 and 9. One more student is added who read 9 books. What happens to the median? 	**It rises to 4.5.**	With 14 values the median is the average of the 7th and 8th. In order those are 4 and 5. (4 + 5)/2 = **4.5**. Adding a large value at the top shifts the middle by half a step, not a whole one.

#	Level	Question	Answer	Solution
P10	Very hard	A frequency table records a quiz marked out of 4: five students scored 2, eight scored 3 and seven scored 4. A student computes the mean as 62 ÷ 3 ≈ 20.7. What is the error, and what is the correct mean? twenty students, three rows score | students | score × students 2 | 5 | 10 3 | 8 | 24 4 | 7 | 28 mean = 62 ÷ 20 = 3.1	**Divided by the number of distinct scores instead of the number of students; the mean is 3.1.**	∑(value × freq) = 10 + 24 + 28 = 62 is right, but the divisor must be the total frequency, **20**, not the 3 rows. 62/20 = **3.1**. A mean of 20.7 on a test marked out of 4 is impossible, which is the quickest way to spot it.
P11	Very hard	Over five days a bakery sold 18, 24, 31, 27 and 20 loaves. On the sixth day it sold enough to raise the mean for all six days to 25. How many loaves were sold on the sixth day? five days total 120 → six-day total 6 × 25 = 150 → 150 − 120 = 30 work in totals, then take the difference	**30**	Five-day total = 120. For a six-day mean of 25 the total must be 6 × 25 = 150. 150 − 120 = **30**.
P12	Very hard	In a two-way table of 180 customers, 100 bought fruit and 80 did not; 60 fruit buyers and 20 non-buyers bought cheese. A report claims ‘most cheese buyers did not buy fruit’. Which statement best evaluates the claim? Survey of 180 market customers | | Cheese | No cheese | Total | Bought fruit | 60 | 40 | 100 | No fruit | 20 | 60 | 80 | Total | 80 | 100 | 180 |	**It is incorrect: 60 of the 80 cheese buyers also bought fruit.**	There are 80 cheese buyers in all, and 60 of them bought fruit — **three quarters**. The claim reverses the comparison. A common wrong answer quotes a true number that answers a different question.

The one habit to build. In this topic the answer is almost always right and **the denominator is almost always wrong**. Grand total, row total, column total, total frequency — name it out loud before you divide.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

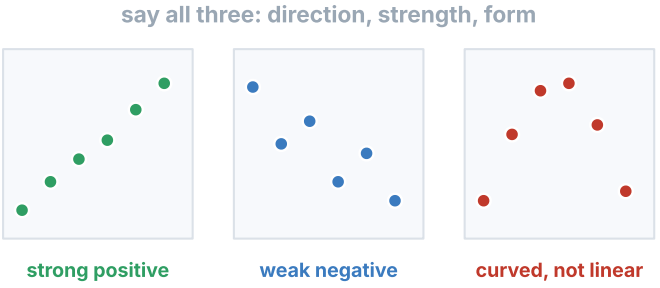
SAT Math 3.5 — Two-Variable Data & Models

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Describing a scatterplot

say three things: direction (up or down) | strength (tight or scattered) | form (straight or curved)

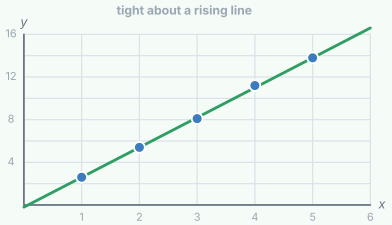
Direction is whether y rises or falls as x rises. **Strength** is how closely the points hug a line. **Form** is whether that pattern is a line at all — a tight parabola is a strong association but not a linear one. The SAT asks you to name all three, and a strong curve is the classic trap.



Three independent judgements. The third panel is strong *and* not linear, which is exactly the combination the test likes.

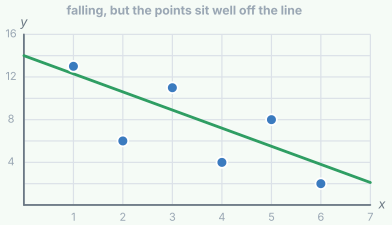
Example 1. **Points rise from bottom-left to top-right and lie very close to a line. Describe the association.**

Solution.
Strong positive linear association. Rising is positive, hugging the line is strong, straight is linear.



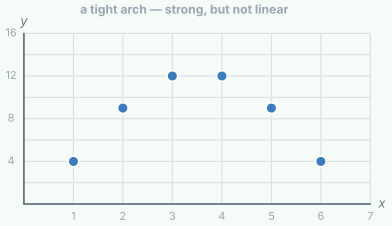
Example 2. **Points fall as x rises but are widely scattered. Describe it.**

Solution.
Weak negative linear association. Falling is negative; the scatter makes it weak.



Example 3. **Points rise, level off, then fall, forming a tight arch. Is this a strong linear association?**

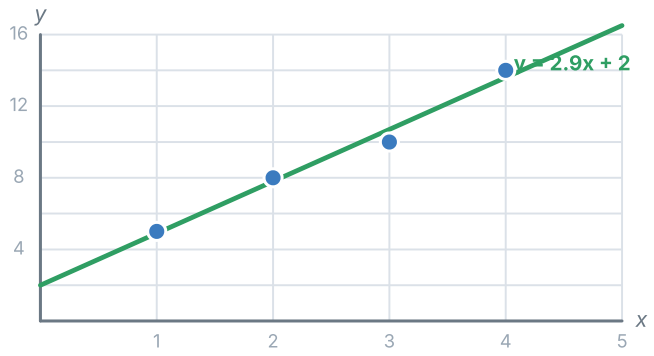
Solution.
No. It is a strong association but not a linear one.
Strength and form are separate judgements, and a line of best fit would describe this badly.



2 The line of best fit

the line of best fit minimises the total squared vertical distance | in Desmos type $y_1 \sim m x_1 + b$ under a table

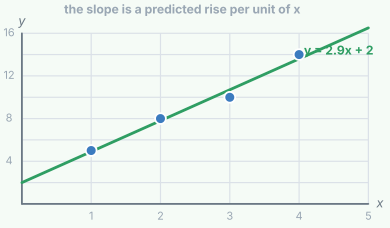
The line **does not have to pass through any data point**, and usually passes through none. Its **slope** is the predicted change in y for a one-unit rise in x, and its **y-intercept** is the predicted y when x is zero — which may be meaningless in context.



Four points and the line Desmos reports. It misses every one of them, which is normal and not an error.

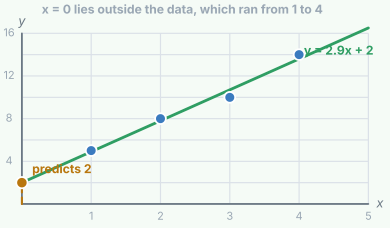
Example 1. **For the data (1, 5), (2, 8), (3, 10), (4, 14), Desmos reports y = 2.9x + 2. What does 2.9 mean?**

Solution.
For every increase of 1 in x, the model predicts y rises by about **2.9**.



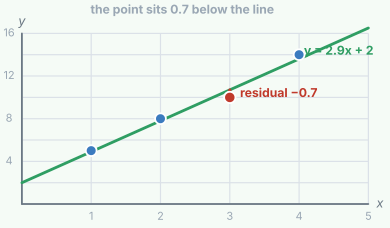
Example 2. **What does the intercept 2 mean in that model?**

Solution.
It is the predicted value of y when x = 0, so **about 2**.
If x = 0 is outside the data range, this is extrapolation and may have no sensible meaning.



Example 3. **Does the line pass through (3, 10)?**

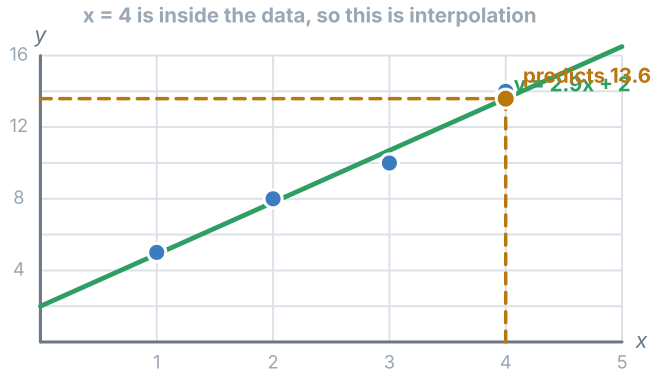
Solution.
At x = 3 the line gives 2.9(3) + 2 = **10.7**, not 10.
No — a line of best fit need not pass through any data point.



3 Predicting from a model

substitute the x you are given | inside the data range is interpolation, outside is extrapolation

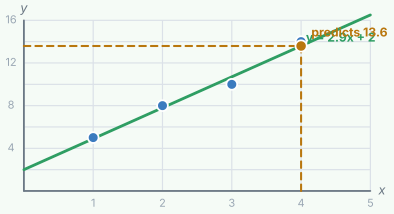
Prediction is just substitution. The judgement the SAT tests is whether the prediction is **trustworthy**: inside the range of the data it usually is; far outside it, the model has no evidence behind it and the prediction is unreliable however neat the arithmetic.



Read up from the x you are given, across to the y the line gives. Whether to trust it depends only on where x sits.

Example 1. **Using y = 2.9x + 2, predict y when x = 4.**

Solution.
2.9(4) + 2 = 11.6 + 2 = **13.6**.



Example 2. **The data covers x from 1 to 4. How much should you trust a prediction at x = 30?**

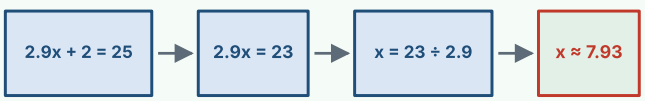
Solution.
Very little. x = 30 is far outside the data, so this is **extrapolation** — the model has no evidence that the pattern continues that far.

the arithmetic is fine; the evidence is not

x between 1 and 4	interpolation — supported by the data
x = 30	extrapolation — no evidence at all

Example 3. **Using y = 2.9x + 2, for what x does the model predict y = 25?**

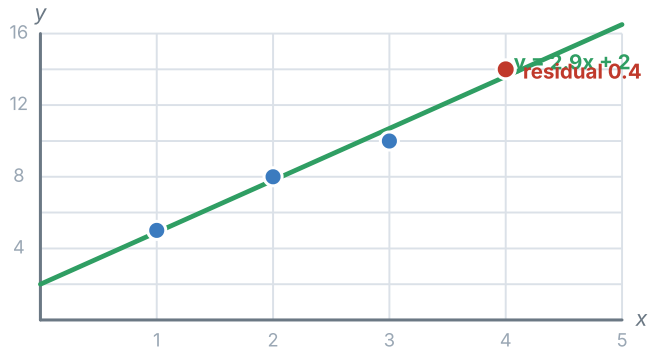
Solution.
2.9x + 2 = 25, so 2.9x = 23 and x = 23/2.9 ≈ **7.93**.



4 Residuals

residual = actual − predicted | positive means the point sits above the line

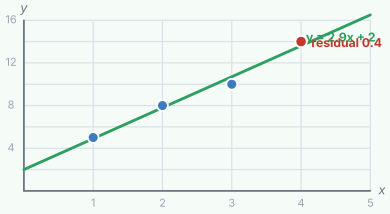
A residual measures **one point's miss**. Subtract in that order every time: **actual first**. A residual plot with no pattern says a line was the right model; a residual plot with a curve says it was not.



The red segment is the residual: the vertical gap from the line up to the actual point, +0.4 here.

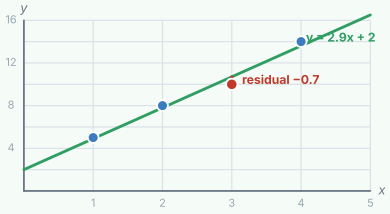
Example 1. **The model predicts 13.6 and the actual value is 14. Find the residual.**

Solution.
14 − 13.6 = **+0.4**. Positive, so the point lies just **above** the line.



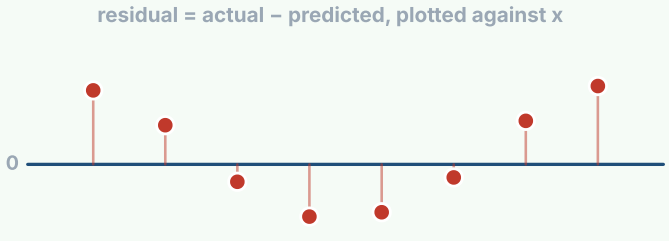
Example 2. **The model predicts 10.7 and the actual value is 10. Find the residual.**

Solution.
10 − 10.7 = **−0.7**. Negative, so this point lies **below** the line.



Example 3. **A residual plot shows a clear U shape. What does that tell you?**

Solution.
The straight-line model is **not appropriate** — the data curves, and a line systematically over- then under-predicts.



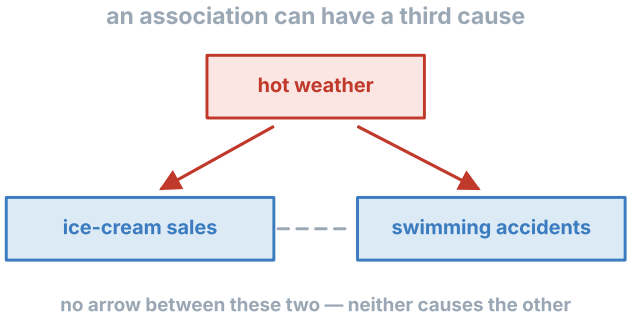
a clear U shape

the data curves — a straight line is the wrong model

5 Correlation is not causation

an association shows two things move together | only a randomised experiment shows one causes the other

This is tested **in words, not numbers**. However strong the association, an observational study cannot establish cause. Watch for a **lurking third variable** that drives both quantities.



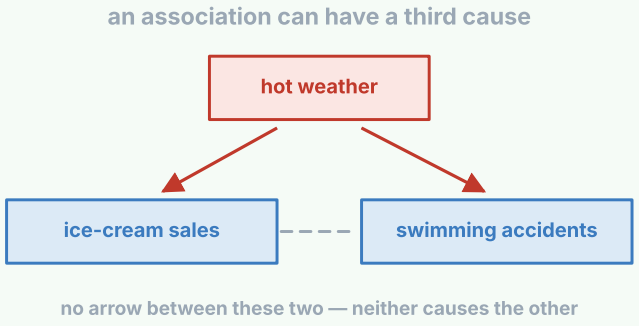
Two arrows out of one cause, and no arrow between the two effects. That missing arrow is the whole point.

Example 1. **Ice-cream sales and swimming accidents rise together. Does ice cream cause accidents?**

Solution.

No. Hot weather raises both — a lurking variable.

The association is real; the causal claim is not.

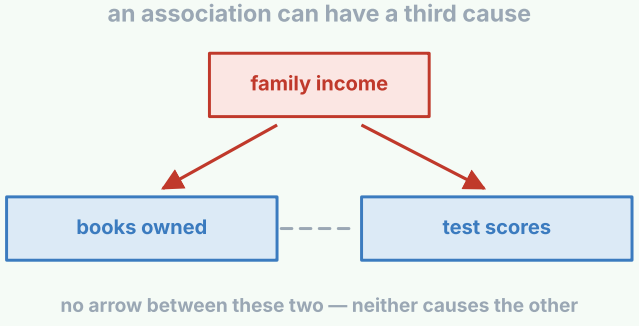


Example 2. **A study finds students who own more books score higher. Can the school conclude books raise scores?**

Solution.

No. This is observational, so it shows association only.

Family income could drive both.



Example 3. **What would be needed to support a causal claim?**

Solution.
A randomised controlled experiment: assign students to conditions at random, so the groups differ only in the treatment.

only one design licenses the word ‘causes’

observational study	association only
random assignment	cause can be claimed

6 Exponential against linear growth

constant difference → linear | constant ratio → exponential

Check the table before choosing a model. **Subtract** consecutive y-values: if those differences are about equal, the model is linear. **Divide** consecutive y-values instead: if those ratios are about equal, it is exponential. Exponential growth eventually overtakes any linear growth.

	x = 1	x = 2	x = 3	x = 4
	3	6	12	24
differences		+3	+6	+12
ratios		×2	×2	×2

equal ratios → exponential

Test the table both ways every time. Here the differences grow but the ratios hold at 2, so the model is exponential.

Example 1. **y goes 3, 7, 11, 15 as x goes 1, 2, 3, 4. Which model fits?**

Solution.
Differences 4, 4, 4 are constant, so the model is **linear, y = 4x − 1**.

	x = 1	x = 2	x = 3	x = 4
	3	7	11	15
differences		+4	+4	+4
ratios		×2.33	×1.57	×1.36

equal differences → linear

Example 2. **y goes 3, 6, 12, 24. Which model fits?**

Solution.

Ratios 2, 2, 2 are constant, so the model is **exponential, $y = 1.5(2)^x$**.

	x = 1	x = 2	x = 3	x = 4
	3	6	12	24
differences		+3	+6	+12
ratios		×2	×2	×2

equal ratios → exponential

Example 3. **y goes 100, 90, 81, 72.9. Linear or exponential?**

Solution.

Differences −10, −9, −8.1 are **not** constant, but ratios are all 0.9.

Exponential decay, falling 10% each step.

	x = 0	x = 1	x = 2	x = 3
	100	90	81	72.9
differences		−10	−9	−8.1
ratios		×0.9	×0.9	×0.9

equal ratios → exponential

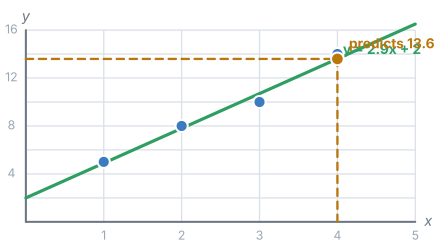
Part 1 — With Desmos (10 questions · every one is faster on the graph)

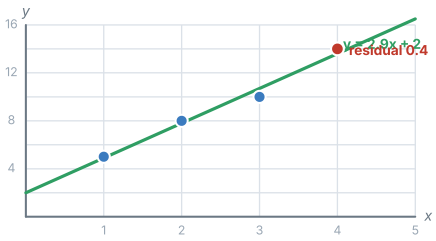
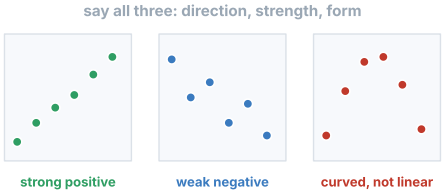
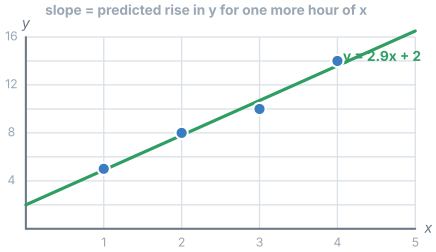
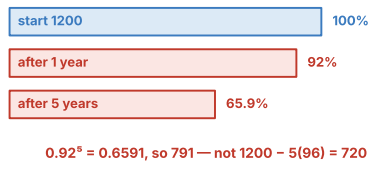
The SAT gives you Desmos on every maths question. This is the one data topic where Desmos really earns its place: **type a table and the regression line $y_1 \sim mx_1+b$**, and it reports the slope and intercept directly.

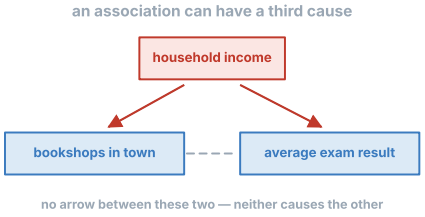
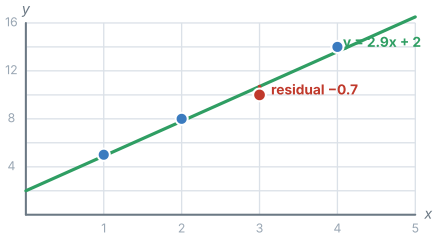
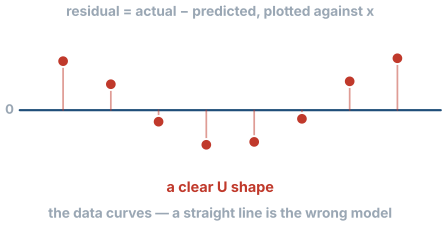
#	Question	Type into Desmos	Then	Answer
D1	A table lists (1, 5), (2, 8), (3, 10) and (4, 14). What is the slope of the line of best fit, to one decimal place?	`y_1~mx_1+b`	Enter the table, then type the regression under it.	2.9
D2	For that same data, what is the y-intercept of the line of best fit?	`y_1~mx_1+b`	Desmos reports m and b together.	2
D3	Using y = 2.9x + 2, what value does the model predict when x = 4?	`2.9*4+2`	Substitute and let Desmos evaluate.	13.6
D4	The actual value at x = 4 is 14 and the model predicts 13.6. What is the residual?	`14-(2.9*4+2)`	Actual minus predicted, in that order.	0.4
D5	Using y = 2.9x + 2, for what value of x does the model predict y = 25? Round to two decimal places.	`(25-2)/2.9`	Solve the linear equation in one line.	7.93
D6	A table shows y = 3, 6, 12, 24 for x = 1, 2, 3, 4. What is the common ratio?	`6/3`	Divide consecutive terms to test for an exponential model.	2
D7	A population of 1200 falls by 8% each year. What is it after 5 years, to the nearest whole number?	`1200*0.92^5`	A fall of 8% means a factor of 0.92.	791
D8	A line of best fit is y = 4x − 1. What does it predict at x = 9?	`4*9-1`	Interpolation or extrapolation depends on the data range, but the arithmetic is the same.	35
D9	Data points are (2, 11) and (6, 23). What is the slope of the line through them?	`(23-11)/(6-2)`	Rise over run.	3
D10	A model predicts 72.9 and the actual value is 70. What is the residual, to one decimal place?	`70-72.9`	A negative residual means the point is below the line.	−2.9

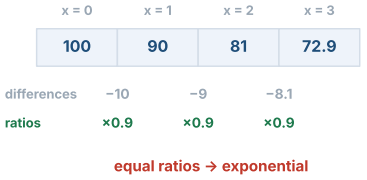
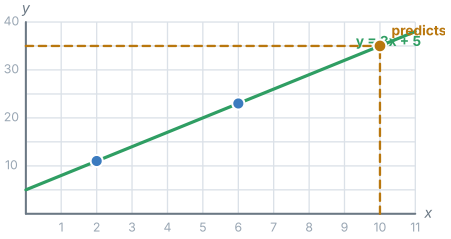
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Three things get tested: **what the slope and intercept mean**, **how far the model may be trusted**, and **what the data does and does not let you claim**.

#	Level	Question	Answer	Solution
P1	Hard	The scatterplot shows four data points and the line of best fit y = 2.9x + 2. What value does the model predict when x = 4? 	13.6	2.9(4) + 2 = 11.6 + 2 = **13.6**. A common wrong answer drops the intercept; C is the actual data value, not the prediction.

#	Level	Question	Answer	Solution
P2	Hard	For the data shown, the line of best fit is y = 2.9x + 2 and the actual value at x = 4 is 14. What is the residual at x = 4? 	0.4	Residual = actual − predicted = 14 − 13.6 = **0.4**. Positive, so the point sits **above** the line.
P3	Hard	A scatterplot of hours studied against test score shows points rising steeply and lying very close to a straight line. Which best describes the association? 	Strong positive linear	Rising is positive; points close to the line make it strong; straight makes it linear. **Strong positive linear.** A common wrong answer would need a visible curve.
P4	Hard	In the model y = 2.9x + 2 relating x hours of training to y units produced, what is the best interpretation of 2.9? 	The predicted rise in output for each extra hour of training.	A slope is a **rate of change**: per extra unit of x, y is predicted to rise by 2.9. A common wrong answer describes the intercept.
P5	Hard	A table shows y = 3, 6, 12, 24 as x = 1, 2, 3, 4. Which type of model best fits the data? 	Exponential, because the ratios are constant	Differences are 3, 6, 12 — not constant. Ratios are 2, 2, 2 — constant. A constant ratio means **exponential**. The pairing of reason to conclusion is what is being tested.
P6	Hard	A town's population of 1200 decreases by 8% each year. To the nearest whole number, what is the population after 5 years? 	791	A fall of 8% multiplies by 0.92 each year. $1200(0.92)^5 = 790.89... \approx$ **791**. Subtracting 8% of 1200 five times would give 720, which is the common error.

#	Level	Question	Answer	Solution
P7	Very hard	A study finds that towns with more bookshops have higher average exam results. State the best-supported conclusion. 	**There is an association, but a third factor such as household income may explain both.**	The study is **observational**, so it supports association only. A lurking variable such as income can raise both. A causal claim would need random assignment, which no town study can provide.
P8	Very hard	For a set of data the line of best fit is y = 2.9x + 2. At x = 3 the actual value is 10. What is the residual, and what does its sign mean? 	**−0.7; the point is below the line**	Predicted = 2.9(3) + 2 = 10.7. Residual = 10 − 10.7 = **−0.7**. Negative means the actual value is **below** the prediction. A common wrong answer reverses the subtraction.
P9	Very hard	The data used to build the model y = 2.9x + 2 covers x from 1 to 4. A manager uses it to predict the value at x = 30. Which is the best criticism? 	**x = 30 is far outside the data range, so the prediction is unreliable.**	2.9(30) + 2 = 89 is arithmetically fine, but x = 30 is well beyond the evidence. This is **extrapolation**, and nothing in the data shows the pattern continues.
P10	Very hard	A residual plot for a linear model shows the residuals forming a clear U shape rather than scattering randomly. What does this indicate? 	**A linear model is not appropriate for this data.**	A pattern in the residuals means the model is **missing structure** in the data. A line is the wrong model here — the relationship curves. A good residual plot looks like random scatter about zero.

#	Level	Question	Answer	Solution
P11	Very hard	Two quantities are recorded: y = 100, 90, 81, 72.9 as x = 0, 1, 2, 3. A student says the data is linear because y keeps falling by about the same amount. Why is the student wrong, and what model fits? 	**The differences are −10, −9, −8.1, which are not constant; the ratios are all 0.9, so the model is exponential decay.**	Differences −10, −9, −8.1 shrink; ratios 90/100 = 81/90 = 72.9/81 = 0.9 are constant. A constant ratio below 1 is **exponential decay**, a 10% fall each step. Over a short run a decay curve can look almost straight, which is exactly the trap.
P12	Very hard	Data points (2, 11) and (6, 23) lie on a line of best fit. What does the model predict at x = 10? 	**35**	Slope = (23 − 11)/(6 − 2) = 12/4 = 3, and 11 = 3(2) + b gives b = 5. At x = 10: 3(10) + 5 = **35**.

The one habit to build. A line of best fit is evidence **only inside the range it was built from**, and only ever about **association**. Two of these twelve are answered by refusing to overclaim — and that refusal is the correct answer, not a dodge.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

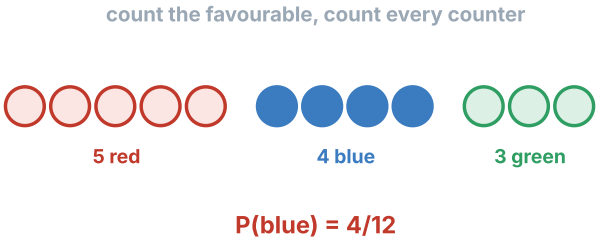
SAT Math 3.6 — Probability

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Basic probability

P(event) = favourable ÷ total | every probability lies between 0 and 1

Count what you want, count everything, divide. The commonest error is **counting the total wrong** — add all the categories, including the one you are not choosing. A probability above 1 always means the denominator is wrong.



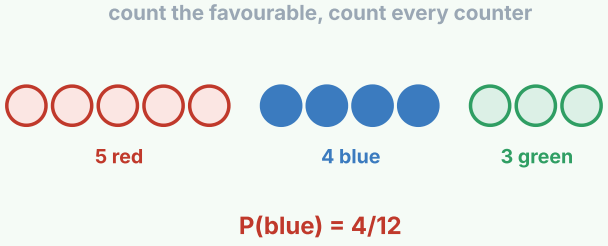
Twelve counters drawn out, so the denominator is counted rather than guessed. Four of them are blue.

Example 1. **A bag holds 5 red, 4 blue and 3 green counters. What is P(blue)?**

Solution.

Total = 5 + 4 + 3 = 12.

P(blue) = 4/12 = **1/3**.

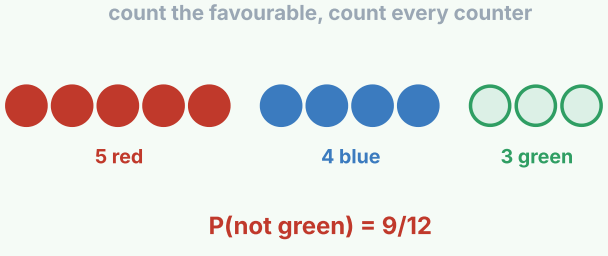


Example 2. **For the same bag, what is P(not green)?**

Solution.

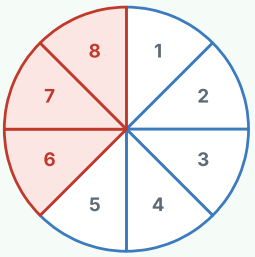
Not green means red or blue: 5 + 4 = 9 of 12.

9/12 = **3/4**. Equivalently 1 − 3/12.



Example 3. **A spinner has 8 equal sectors numbered 1 to 8. What is P(a number greater than 5)?**

Solution.
The values 6, 7, 8 qualify — three of them.
P = **3/8**. ‘Greater than 5’ excludes 5 itself.



P = 3/8

2 Probability from a two-way table

decide the whole first: grand total, one row, or one column

Almost every two-way-table question is easy arithmetic and **one hard decision** — which total goes underneath. Read the sentence for the group being described; that group is the denominator.

Survey of 180 market customers

	Cheese	No cheese	Total
Bought fruit	60	40	100
No fruit	20	60	80
Total	80	100	180

Everything you need is in the margins. The only question left is which of those margin numbers is the whole.

Example 1. **From the table above, what is the probability that a randomly chosen customer bought cheese?**

Solution.
The Cheese column totals 60 + 20 = 80, out of 180.
P = 80/180 = **4/9**.

Survey of 180 market customers

	Cheese	No cheese	Total
Bought fruit	60	40	100
No fruit	20	60	80
Total	80	100	180

Example 2. **What is the probability that a customer bought both fruit and cheese?**

Solution.

One cell, 60, out of the grand total 180.

P = 60/180 = **1/3**.

Survey of 180 market customers

	Cheese	No cheese	Total
Bought fruit	60	40	100
No fruit	20	60	80
Total	80	100	180

Example 3. **What is the probability that a customer bought fruit but not cheese?**

Solution.

That cell holds 40, out of 180.

P = 40/180 = **2/9**.

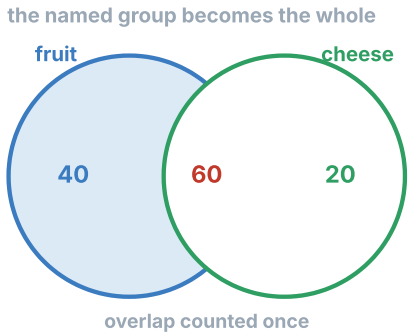
Survey of 180 market customers

	Cheese	No cheese	Total
Bought fruit	60	40	100
No fruit	20	60	80
Total	80	100	180

3 Conditional probability

‘given that’ or ‘of those who’ → the named group becomes the whole

This is the **highest-value idea in the topic** and the one your class loses marks on. The phrase ‘of the customers who bought fruit’ replaces the grand total with the fruit row. Nothing else changes — the numerator is still the overlap.



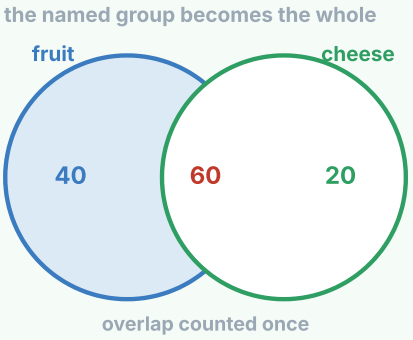
P(cheese | fruit) uses only the shaded left circle as the whole: 60 out of 100, not 60 out of 180.

Example 1. **Of the customers who bought fruit, what is the probability one also bought cheese?**

Solution.

The whole is now the Fruit row, 100.

P = 60/100 = **3/5**, not 60/180.

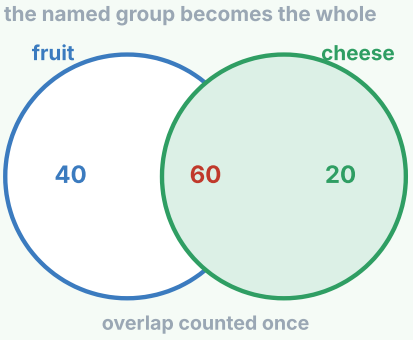


Example 2. **Given that a customer bought cheese, what is the probability they bought fruit?**

Solution.

Now the whole is the Cheese column, 80.

P = 60/80 = **3/4**. Notice this differs from the previous answer — order matters.



Example 3. **Of the customers who did not buy fruit, what fraction bought cheese?**

Solution.

The No-fruit row totals 20 + 60 = 80.

P = 20/80 = **1/4**.

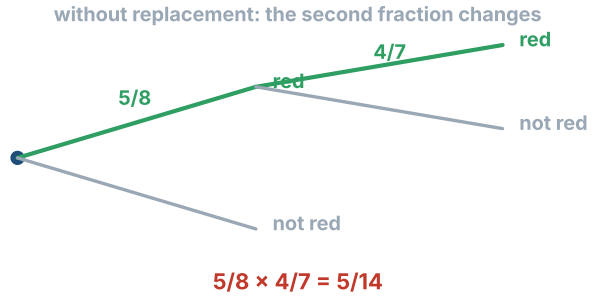
Survey of 180 market customers

	Cheese	No cheese	Total
Bought fruit	60	40	100
No fruit	20	60	80
Total	80	100	180

4 Independent events and ‘and’

independent events: P(A and B) = P(A) × P(B) | without replacement the second probability changes

Multiply along a sequence of events. The question to ask is whether the first event changes the second: drawing **with** replacement leaves the bag unchanged, drawing **without** replacement removes one item from both the numerator and the total.



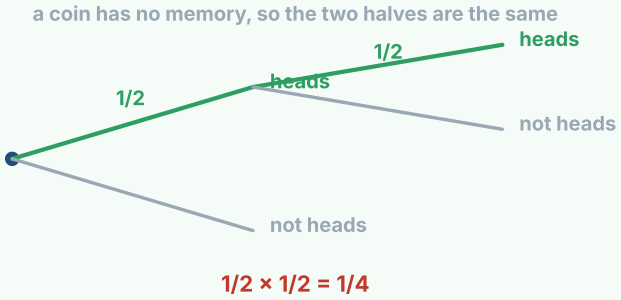
Follow one branch and multiply along it. The second fraction is 4/7, not 5/8, because a red counter has already gone.

Example 1. **A fair coin is tossed twice. What is P(two heads)?**

Solution.

The tosses are independent.

P = ½ × ½ = **1/4**.

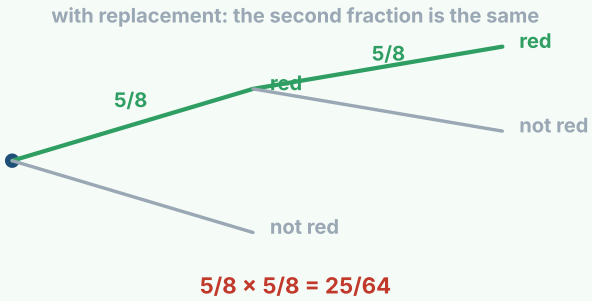


Example 2. **A bag has 5 red and 3 blue. Two are drawn with replacement. What is P(both red)?**

Solution.

The bag is unchanged between draws.

P = (5/8)(5/8) = **25/64**.

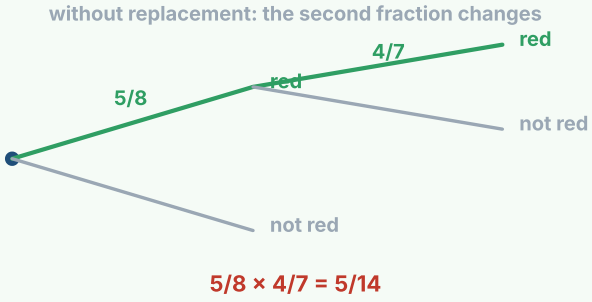


Example 3. **The same bag, but drawn without replacement. What is P(both red)?**

Solution.

After one red is taken, 4 reds remain out of 7.

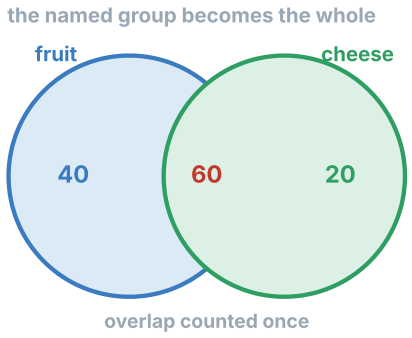
P = (5/8)(4/7) = 20/56 = **5/14**.



5 ‘Or’ and the overlap

P(A or B) = P(A) + P(B) − P(A and B)

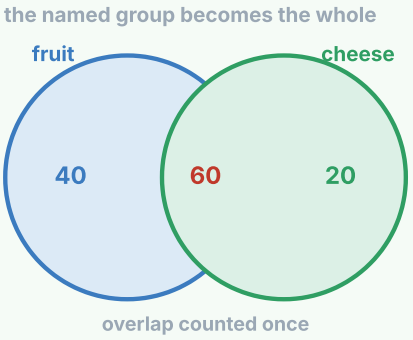
Adding two probabilities **double-counts** anything in both, so subtract the overlap once. If the events cannot happen together, the overlap is zero and the formula collapses to simple addition.



Shading both circles covers 40 + 60 + 20 = 120 people. Adding 100 and 80 would count the middle 60 twice.

Example 1. **From the table, what is P(bought fruit or bought cheese)?**

Solution.
100/180 + 80/180 − 60/180 = 120/180 = **2/3**. Without subtracting the 60, you would wrongly get 1.



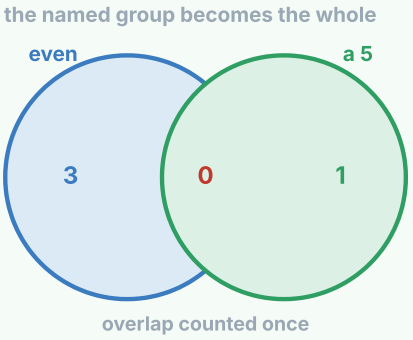
Example 2. **A card is drawn from 52. What is P(a heart or a king)?**

Solution.
13/52 + 4/52 − 1/52 (the king of hearts) = 16/52 = **4/13**.



Example 3. **A die is rolled. What is P(an even number or a 5)?**

Solution.
Even gives 3 outcomes, 5 gives 1, and they cannot both happen.
P = 3/6 + 1/6 = 4/6 = **2/3**.



6 Expected counts

expected number = probability × group size

When a question scales a proportion up to a larger population, **multiply**. This is the same arithmetic as a percentage of a total, and the answer should be rounded to a whole number of people or items when that is what is being counted.



a probability becomes a count the moment you multiply

The fraction is a rate; the group size turns it into people. Nothing more complicated than that.

Example 1. **The probability a customer buys cheese is 4/9. In a day with 450 customers, how many are expected to buy cheese?**

Solution.
(4/9)(450) = **200 customers**.



Example 2. **A machine produces faulty parts 3% of the time. How many faults are expected in 1,200 parts?**

Solution.
0.03 × 1200 = **36 parts**.



Example 3. **In a survey, 60 of 100 fruit buyers also bought cheese. In a town with 2,500 fruit buyers, how many are expected to buy cheese?**

Solution.

The rate is 60/100 = 0.6.

0.6 × 2500 = **1,500 people**.



Part 1 — With Desmos (10 questions · every one is faster on the graph)

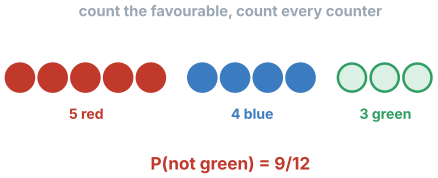
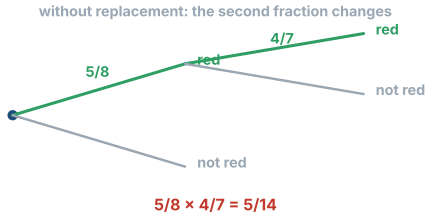
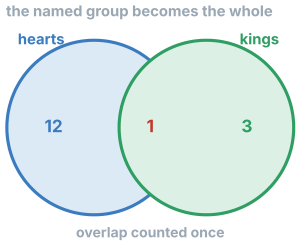
The SAT gives you Desmos on every maths question. Here it is **a fraction calculator**. Type the numerator and denominator you have chosen — the whole difficulty of this topic is choosing the denominator, and Desmos cannot do that.

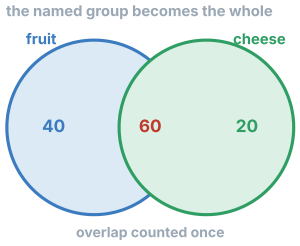
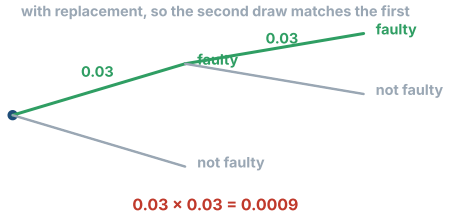
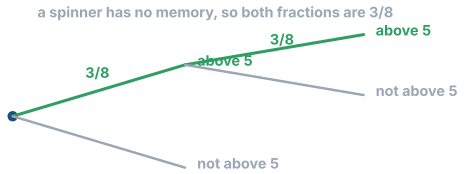
#	Question	Type into Desmos	Then	Answer
D1	A bag holds 5 red, 4 blue and 3 green counters. What is the probability of drawing a blue counter, as a decimal to three places?	`4/12`	Desmos keeps the exact fraction 1/3.	0.333
D2	In a survey of 180 customers, 80 bought cheese. What is the probability a randomly chosen customer bought cheese?	`80/180`	Desmos simplifies the fraction for you.	4/9
D3	Of 100 fruit buyers, 60 also bought cheese. What is this conditional probability?	`60/100`	The named group, 100, is the denominator.	0.6
D4	A fair coin is tossed twice. What is the probability of two heads?	`(1/2)*(1/2)`	Independent events multiply.	0.25
D5	A bag has 5 red and 3 blue. Two are drawn without replacement. What is P(both red)?	`(5/8)*(4/7)`	The second fraction changes because one red has gone.	5/14
D6	What is the probability that a card drawn from 52 is a heart or a king?	`13/52+4/52-1/52`	Subtract the king of hearts once.	4/13
D7	The probability a customer buys cheese is 4/9. How many of 450 customers are expected to?	`(4/9)*450`	Probability times group size.	200
D8	A machine is faulty 3% of the time. How many faults are expected in 1,200 parts?	`0.03*1200`	Write the percentage as a decimal.	36
D9	From a two-way table, 120 of 180 customers bought fruit or cheese. What is that probability?	`120/180`	Desmos reduces it.	2/3
D10	A spinner has 8 equal sectors numbered 1 to 8. What is P(greater than 5)?	`3/8`	Count 6, 7, 8 — not 5 itself.	0.375

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. One question decides almost all of these: **has the group been narrowed?** If the wording says ‘given that’, ‘of the’ or ‘among’, the denominator is a row or a column, not the whole table.

#	Level	Question	Answer	Solution
P1	Hard	The table shows the results of a survey of 180 market customers. If one customer is selected at random, what is the probability that the customer bought cheese? Survey of 180 market customers Cheese / No cheese / Total Bought fruit: 60 / 40 / 100 No fruit: 20 / 60 / 80 Total: 80 / 100 / 180	4/9	The Cheese column totals 60 + 20 = 80, out of the grand total 180. 80/180 = **4/9**. A common wrong answer is the single overlap cell 60/180.

#	Level	Question	Answer	Solution
P2	Hard	Using the same table: of the customers who bought fruit, what is the probability that the customer also bought cheese? Survey of 180 market customers | | Cheese | No cheese | Total | | Bought fruit | 60 | 40 | 100 | | No fruit | 20 | 60 | 80 | | Total | 80 | 100 | 180 |	3/5	'Of the customers who bought fruit' makes the Fruit row, 100, the whole. 60/100 = **3/5**. A common wrong answer divides by 180 instead — the single most common error on this topic.
P3	Hard	A bag contains 5 red, 4 blue and 3 green counters. One counter is drawn at random. What is the probability that it is not green? 	3/4	Not green means 5 + 4 = 9 of the 12 counters. 9/12 = **3/4**. A common wrong answer is P(green) itself.
P4	Hard	A bag contains 5 red and 3 blue counters. Two counters are drawn at random without replacement. What is the probability that both are red? 	5/14	First draw 5/8; after one red is removed, 4 reds remain of 7. (5/8)(4/7) = 20/56 = **5/14**. A common wrong answer is the answer if the counter is replaced.
P5	Hard	A card is drawn at random from a standard deck of 52. What is the probability that it is a heart or a king? 	4/13	13 hearts + 4 kings, minus the king of hearts counted twice. (13 + 4 − 1)/52 = 16/52 = **4/13**. Forgetting to subtract gives 17/52.
P6	Hard	The probability that a customer at a market buys cheese is 4/9. On a day when 450 customers visit, how many would be expected to buy cheese? 	200	Expected count = probability × group size. (4/9)(450) = **200 customers**.

#	Level	Question	Answer	Solution
P7	Very hard	Using the survey table, what is the probability that a randomly chosen customer bought fruit or cheese? 	2/3	100/180 + 80/180 would double-count the 60 who did both. (100 + 80 − 60)/180 = 120/180 = **2/3**. A common wrong answer is what you get by adding 100 and 80 and not subtracting the overlap.
P8	Very hard	In the same survey, let A be 'bought cheese given bought fruit' and B be 'bought fruit given bought cheese'. State what is true here. same numerator, different whole — so A ≠ B A = P(cheese | fruit) 60 ÷ 100 = 0.6 B = P(fruit | cheese) 60 ÷ 80 = 0.75	A < B, because 100 is larger than 80.	A = 60/100 = 0.6 and B = 60/80 = 0.75, so **A < B**. The numerator is the same 60 both times; only the denominator changes, and the larger denominator gives the smaller probability. A common wrong answer is the classic error of treating conditional probability as symmetric.
P9	Very hard	A quality check finds that 3% of parts are faulty. In a batch of 1,200 parts, two are drawn at random with replacement. What is the probability that both are faulty?  0.03 × 0.03 = 0.0009	0.0009	With replacement the draws are independent. (0.03)(0.03) = **0.0009**. A common wrong answer adds the two probabilities instead of multiplying; the batch size of 1,200 is not needed at all.
P10	Very hard	A spinner has 8 equal sectors numbered 1 to 8. It is spun twice. What is the probability that both spins give a number greater than 5?  3/8 × 3/8 = 9/64	9/64	Each spin: 6, 7, 8 qualify, so 3/8. The spins are independent. (3/8)(3/8) = **9/64**. A common wrong answer is one spin only.
P11	Very hard	In a survey of 180 customers, 100 bought fruit and 60 of those also bought cheese; 20 of the remaining 80 bought cheese. A manager claims that buying fruit and buying cheese are independent. Which calculation best tests the claim? independence is a comparison, not a formula P(cheese | fruit) 60 ÷ 100 = 0.6 P(cheese) overall 80 ÷ 180 ≈ 0.44 verdict different, so not independent	Compare 60/100 with 80/180.	Independence means P(cheese | fruit) equals P(cheese). P(cheese | fruit) = 60/100 = 0.6 and P(cheese) = 80/180 ≈ 0.44. They differ, so the events are **not independent** and the claim fails. A common wrong answer compares the two conditional probabilities, which tests something else.

#	Level	Question	Answer	Solution
P12	Very hard	Of 100 fruit buyers surveyed, 60 also bought cheese. A town has 2,500 fruit buyers. Assuming the same rate, how many are expected to buy cheese? 	1500	The rate is 60/100 = 0.6. 0.6 × 2500 = **1,500 people**.

The one habit to build. Your class scored 20% on conditional probability, and the cause is always the same: **the right numerator over the wrong denominator.** Before dividing, say which group you are now inside.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

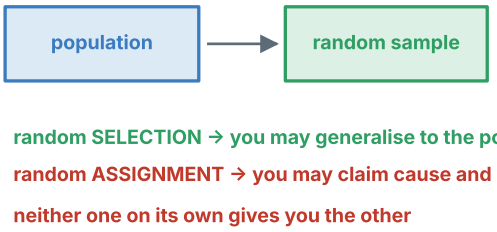
SAT Math 3.7 — Inference & Margin of Error

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Population, sample and parameter

the population is everyone you want to describe | the sample is who you actually asked

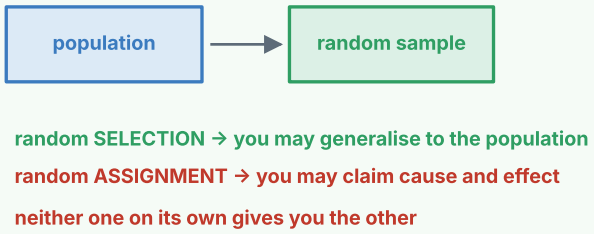
A **statistic** comes from the sample; a **parameter** describes the population. The whole point of sampling is to estimate the parameter you cannot measure. The sample must be drawn from the population you want to describe — surveying one school tells you nothing about a country.



One arrow, and everything downstream depends on it. What the sample licenses you to say is fixed by how it was drawn.

Example 1. **A school of 1,200 students is surveyed by asking 150 of them. Name the population and the sample.**

Solution.
Population = the 1,200 students; **sample** = the 150 asked.



Example 2. **In that survey 60% of those asked walk to school. Is 60% a statistic or a parameter?**

Solution.
It comes from the sample, so it is a **statistic**.
It estimates the unknown **parameter**, the true percentage among all 1,200.

which of the two did you actually measure?

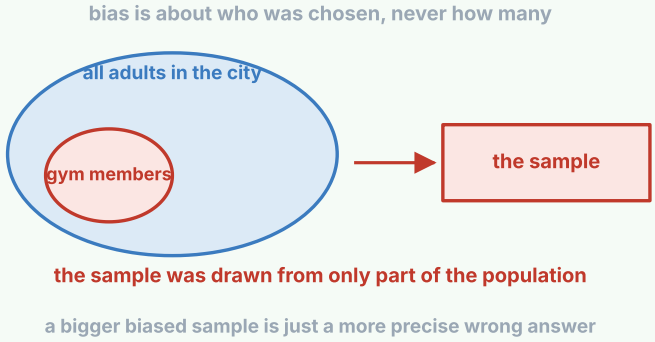
statistic	**measured on the sample — 60%**
parameter	**describes all 1,200 — unknown**

Example 3. **A researcher wants to describe all adults in a city but surveys only gym members. What is wrong?**

Solution.

The sample is **not drawn from the whole population**, so it is biased.

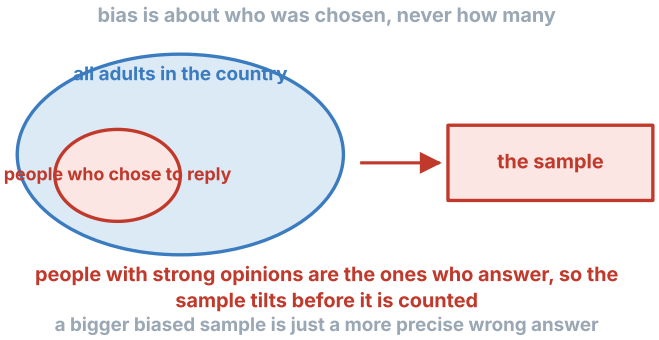
Gym members differ systematically from adults in general, and no sample size fixes that.



2 Random selection and bias

only a random sample supports generalising to the population

Bias is about **how** people were chosen, not **how many**. A voluntary online poll, a convenience sample at one location, or a survey of people who chose to reply are all biased no matter how large. Increasing a biased sample makes the wrong answer more precise, not more correct.



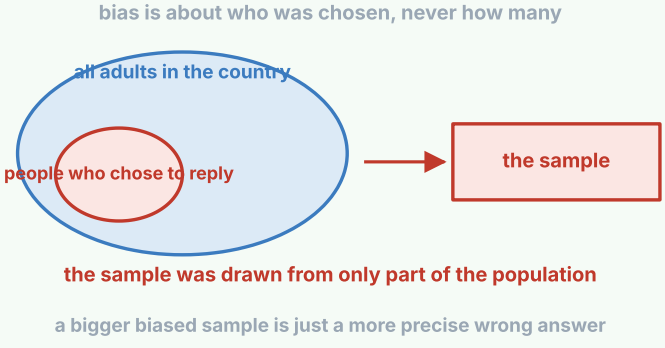
The red blob is who actually answered. It sits inside the population but is not representative of it, and 5,000 replies do not change its position.

Example 1. **A survey is posted online and 5,000 people reply voluntarily. Is it reliable for the population?**

Solution.

No. Voluntary response is biased — people with strong opinions answer.

5,000 replies do not repair that.



Example 2. **Which is better for estimating a school's average: 60 randomly chosen students, or 300 volunteers?**

Solution.

The 60 random students. Random selection removes systematic bias; volunteering does not.

size cannot buy what randomness gives

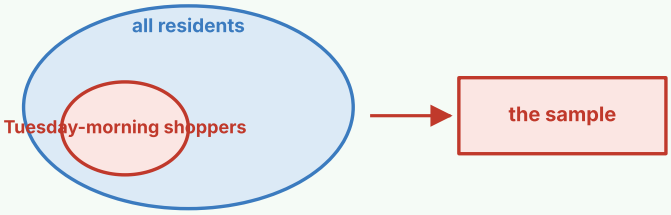
60 chosen at random	no systematic tilt — usable
300 volunteers	tilted, however many there are

Example 3. **A council surveys shoppers at one shopping centre on a Tuesday morning. Name the problem.**

Solution.

A **convenience sample**. Tuesday-morning shoppers are not representative of all residents, so the results cannot be generalised.

bias is about who was chosen, never how many



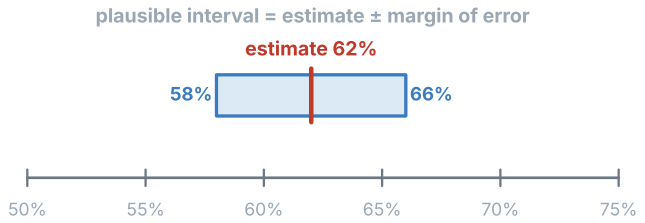
whoever happened to be there is not whoever lives in the town

a bigger biased sample is just a more precise wrong answer

3 Margin of error

plausible interval = estimate ± margin of error

The margin of error turns a single estimate into a **range of believable values** for the population. The SAT never asks you to compute one from scratch; it asks you to **build** the interval, **read** it, or say **what changes** it.



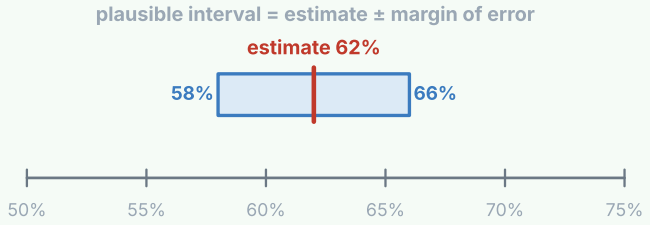
The estimate sits in the middle and the margin runs both ways. Every value inside the bar is a believable population value.

Example 1. **A poll estimates 62% with a margin of error of 4 percentage points. Give the plausible interval.**

Solution.

62 − 4 = 58 and 62 + 4 = 66.

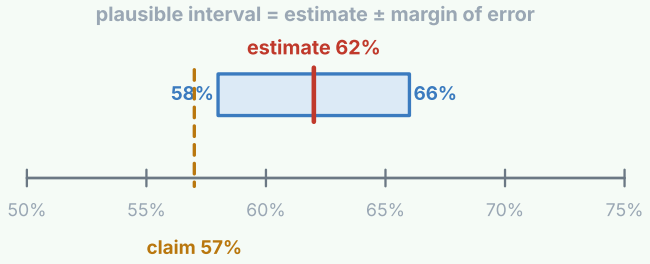
The interval is **58% to 66%**.



Example 2. **Is a true value of 57% consistent with that poll?**

Solution.

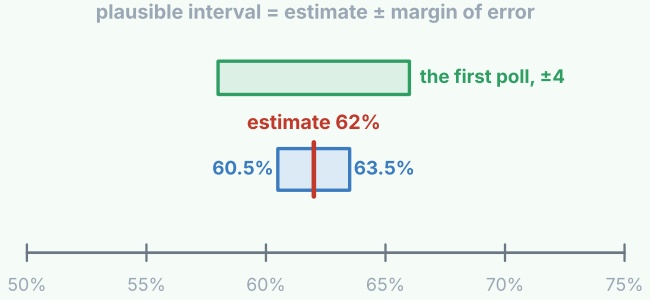
No. 57% lies outside 58%–66%, so it is not among the plausible values.



Example 3. **A second poll reports 62% with a margin of error of 1.5 points. What is its interval?**

Solution.

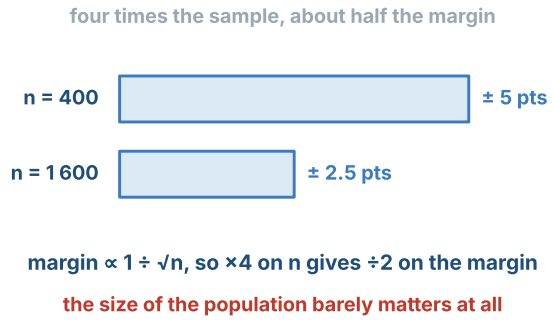
60.5% to 63.5% — a narrower, more precise estimate of the same quantity.



4 What makes the margin of error smaller

a larger sample narrows it | a higher confidence level widens it

Sample size is the lever the SAT tests. More data gives a tighter interval, but with diminishing returns — the margin shrinks with the **square root** of the sample size, so quadrupling the sample roughly halves it. Population size is almost irrelevant.



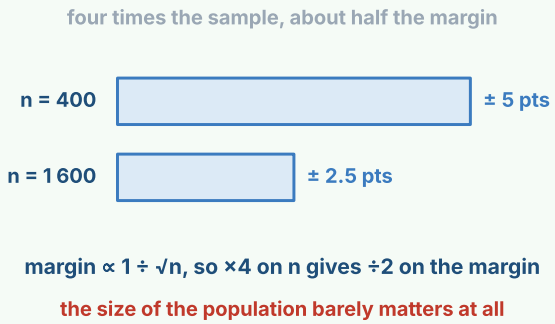
Four times as many people, about half the margin. Sixteen times as many would be needed to quarter it.

Example 1. **A survey of 400 gives a margin of error of 5 points. What would 1,600 give, roughly?**

Solution.

Four times the sample roughly halves the margin.

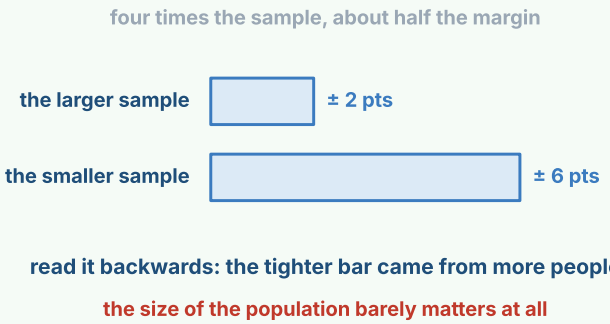
About 2.5 points.



Example 2. **Two polls of the same population report margins of 2 and 6 points. Which used a larger sample?**

Solution.

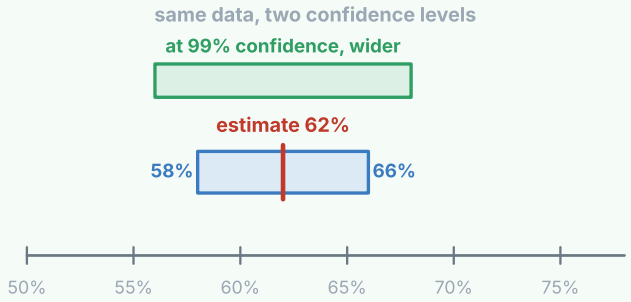
The one with the 2-point margin — smaller margin, larger sample.



Example 3. **Does raising the confidence level from 95% to 99% make the interval wider or narrower?**

Solution.

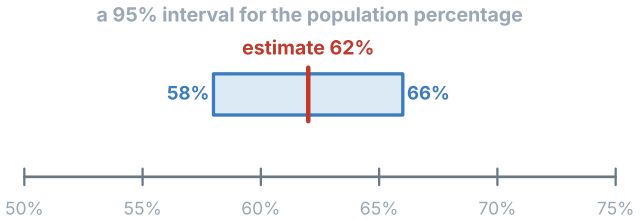
Wider. To be more confident of capturing the true value, you must allow a bigger range.



5 Reading a confidence statement

the interval is about the population value, not about individuals

The **language** matters more than the arithmetic. A 95% confidence interval of 58% to 66% says the plausible values **for the whole population** lie in that range. It does **not** say 95% of people, and it does not say any individual has a 95% chance of anything.

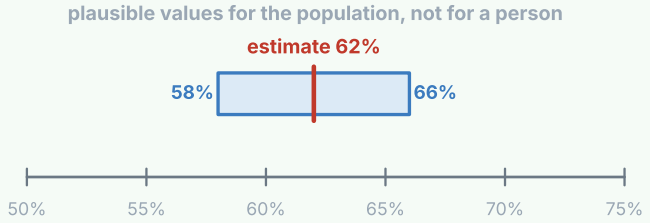


The 95% is a property of the method, not a number on this axis. What the axis shows is the share who walk to school.

Example 1. **A poll gives a 95% confidence interval of 58% to 66% for the share who walk to school. State what it means.**

Solution.

It is plausible that **the true percentage for the whole population** lies between 58% and 66%.



Example 2. **Does it mean 95% of students walk to school?**

Solution.

No. 95% is the confidence level, not a percentage of students.

The estimate of walkers is **62%**.

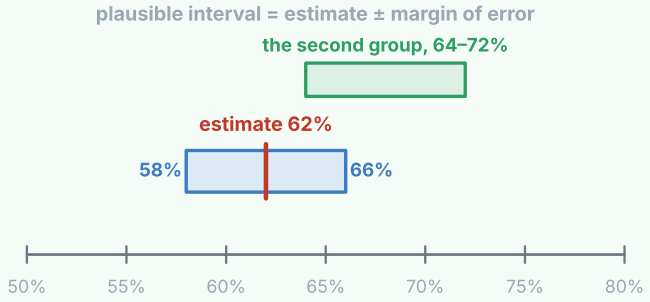
three numbers, three different jobs

95%	the confidence level — a property of the method
62%	the estimated share who walk
58% to 66%	the plausible values for that share

Example 3. **Two groups' intervals are 58–66% and 64–72%. Can you say one group is higher?**

Solution.

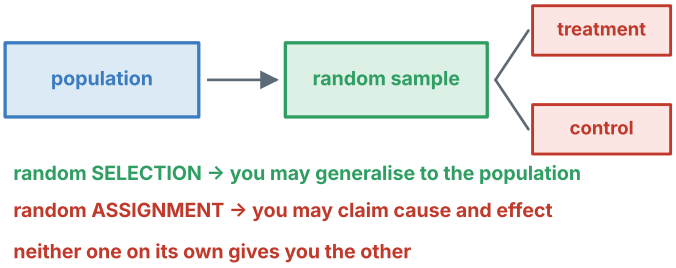
Not confidently — the intervals overlap between 64% and 66%, so the same true value is plausible for both.



6 Which conclusions a study supports

random selection → generalise | random assignment → cause

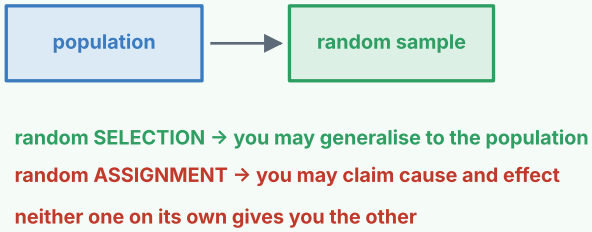
Two different randomisations license **two different conclusions**, and the SAT keeps them apart. Selecting participants at random lets you extend findings to the population. Assigning them to groups at random lets you claim the treatment caused the difference. **You need both for both.**



Two separate random steps. Each buys exactly one conclusion, and neither lends its conclusion to the other.

Example 1. **200 students are chosen at random from a school and surveyed. What can you conclude?**

Solution.
You may **generalise to that school**, but not to other schools and not about cause.



Example 2. **Volunteers are randomly assigned to two study methods and one scores higher. What follows?**

Solution.
You may claim the method **caused** the difference for people like these volunteers, but you **cannot generalise** beyond them.

random assignment without random selection

how they joined	they volunteered — not random
how they were grouped	at random
so you may say	cause, for these volunteers only

Example 3. **A study randomly selects participants from a city and randomly assigns them to treatments. What is supported?**

Solution.
Both a causal claim **and** generalisation to that city — the strongest design the SAT will show you.



random SELECTION → you may generalise to the population

random ASSIGNMENT → you may claim cause and effect

neither one on its own gives you the other

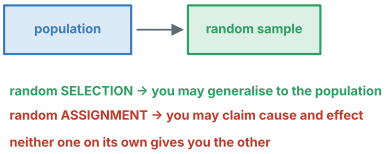
Part 1 — With Desmos (10 questions · calculator work only; the reasoning is in the wording)

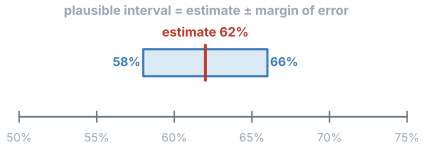
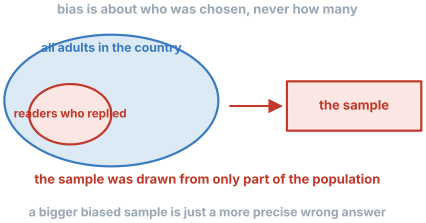
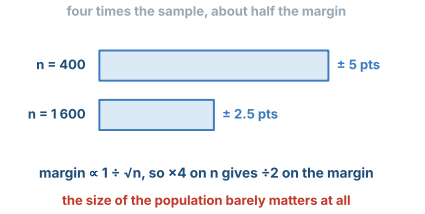
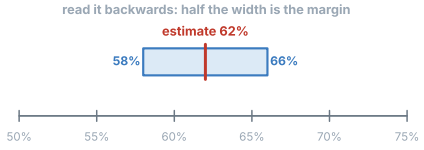
The SAT gives you Desmos on every maths question. Here it does almost nothing: **add and subtract the margin of error**, and scale a rate to a population. Every mark in this topic is in the wording, not the arithmetic.

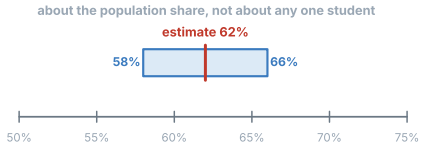
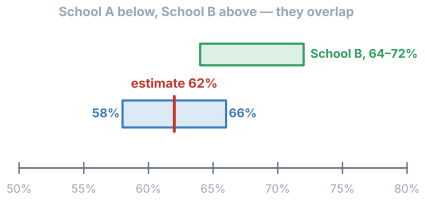
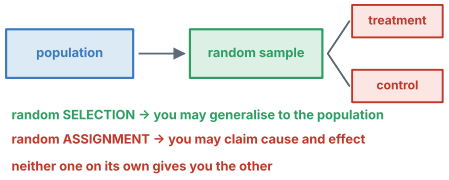
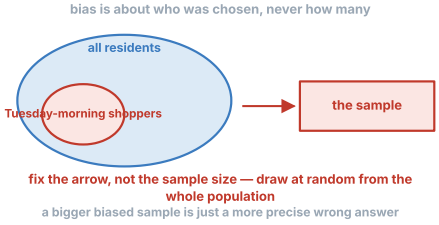
#	Question	Type into Desmos	Then	Answer
D1	A poll estimates 62% with a margin of error of 4 percentage points. What is the lower end of the plausible interval?	`62-4`	Subtract for the lower end.	58
D2	For the same poll, what is the upper end?	`62+4`	Add for the upper end.	66
D3	A sample of 150 out of 1,200 students is surveyed. What fraction of the population is sampled?	`150/1200`	A sampled fraction, not a probability.	0.125
D4	In a sample of 150, 90 students walk to school. What percentage is that?	`90/150*100`	Convert to a percentage at the end.	60
D5	A survey of 400 has a margin of error of 5 points. Four times the sample roughly halves it. What is the new margin?	`5/2`	Quadrupling the sample halves the margin.	2.5
D6	An interval runs from 60.5% to 63.5%. What is the margin of error?	`(63.5-60.5)/2`	Half the width of the interval.	1.5
D7	An interval runs from 58% to 66%. What is the point estimate at its centre?	`(58+66)/2`	The estimate sits in the middle.	62
D8	In a town of 24,000 adults, a poll estimates 38% support. How many adults is that?	`0.38*24000`	Scale the proportion up to the population.	9120
D9	A sample proportion is 0.44 in a survey of 180 people. How many people is that?	`0.44*180`	Desmos keeps the decimal; round to a whole person if the question asks.	79.2
D10	Two polls report margins of 2.0 and 6.0 points. How many times wider is the second interval?	`6/2`	Compare the margins directly.	3

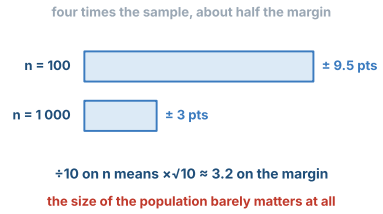
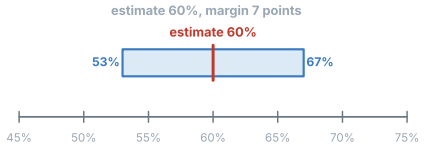
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Two questions answer nearly all of these: **who was sampled**, and **is the claimed value inside the interval**. Neither needs a calculation.

#	Level	Question	Answer	Solution
P1	Hard	A researcher surveys 150 students selected at random from the 1,200 students at a school and finds that 60% walk to school. Which is the parameter of interest?  random SELECTION → you may generalise to the population random ASSIGNMENT → you may claim cause and effect neither one on its own gives you the other	**The percentage of all 1,200 students who walk to school.**	A **parameter** describes the population. The 60% is a statistic that estimates it. The groups in B and D are the sample and the population themselves, not percentages.

#	Level	Question	Answer	Solution
P2	Hard	A poll of registered voters estimates that 62% support a proposal, with a margin of error of 4 percentage points. Which interval gives the plausible values for the whole population? 	**58% to 66%**	Estimate ± margin of error: 62 − 4 to 62 + 4. **58% to 66%.** Two common wrong answers use the margin on one side only.
P3	Hard	A website posts a survey and 5,000 readers choose to respond. Which is the strongest objection to generalising the results to all adults in the country? 	**Respondents chose themselves, so the sample is biased.**	This is **voluntary response bias**: people with strong views reply. 5,000 is a large sample, so A is wrong — and a large biased sample is still biased.
P4	Hard	A survey of 400 people produces a margin of error of 5 percentage points. Approximately what margin of error would a survey of 1,600 similar people produce? 	**2.5**	The margin shrinks with the square root of the sample size, so four times the sample roughly halves it. **About 2.5 points.** A common wrong answer divides by 4 instead of by 2.
P5	Hard	A 95% confidence interval for the percentage of residents who recycle runs from 58% to 66%. What is the margin of error, in percentage points? 	**4**	The interval is the estimate ± the margin, so the margin is half its width. (66 − 58)/2 = **4 percentage points**.
P6	Hard	In a town of 24,000 adults, a random sample estimates that 38% support a new library. Approximately how many adults in the town does this estimate represent? 	**9120**	0.38 × 24,000 = **9,120 adults**.

#	Level	Question	Answer	Solution
P7	Very hard	A study reports a 95% confidence interval of 58% to 66% for the share of students who walk to school. State the best-supported conclusion. 	**It is plausible that the true percentage for all students is between 58% and 66%.**	The interval describes plausible values for the **population percentage**. A common wrong answer confuses the confidence level with the quantity measured; B treats an estimate as exact.
P8	Very hard	Two schools are surveyed. School A's interval for the proportion who walk is 58% to 66%; School B's is 64% to 72%. What can be concluded? 	**The intervals overlap, so the data does not establish which school is higher.**	The intervals share the range 64%–66%, so a single true value could sit in both. No confident difference can be claimed. **Overlap is the thing to look for.**
P9	Very hard	Researchers randomly select 200 adults from a city and randomly assign them to one of two exercise programmes. Programme A's group shows greater improvement. State the best-supported conclusion. 	**Programme A causes greater improvement for adults in that city.**	Random **assignment** supports the causal claim; random **selection** from the city supports generalising to that city — but no further. A common wrong answer over-reaches to a population never sampled.
P10	Very hard	A council wants to estimate the average commute time of all its residents. It surveys shoppers at one shopping centre on a Tuesday morning. Which change would most improve the study? 	**Select residents at random from the council's records.**	The flaw is **how** people were chosen, not **how many**. Only random selection from the whole population removes it; A makes a biased estimate more precise, not more accurate.

#	Level	Question	Answer	Solution
P11	Very hard	A poll of 1,000 people reports 45% support with a margin of error of 3 points. The pollster then reports 45% with a margin of error of 3 points for a subgroup of only 100 of those people. What is wrong? 	**The subgroup is ten times smaller, so its margin of error must be larger — roughly 9.5 points.**	Margin of error depends on **sample size**. A tenth of the sample widens the margin by a factor of about √10 ≈ 3.2, so roughly 3 × 3.2 ≈ **9.5 points**. Reusing the full-sample margin for a subgroup overstates the precision badly.
P12	Very hard	A survey of a random sample of 150 of a school's 1,200 students finds 90 walk to school, with a margin of error of 7 percentage points. What is the lower end of the plausible interval for the whole school, as a percentage? 	**53**	The sample estimate is 90/150 = 60%. 60 − 7 = **53%**. The population size of 1,200 plays no part in the calculation.

The one habit to build. A margin of error covers **sampling variation and nothing else**. It does not cover a biased sample, a leading question or people answering dishonestly — and no sample size will fix those.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

SAT Math 3.8 — Evaluating Statistical Claims

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Observational study or experiment?

a study that only watches cannot show cause | an experiment imposes a treatment

Ask one question: **did the researchers decide who got what?** If they merely recorded what people already did, it is **observational** and the strongest claim available is association. If they assigned the treatment, it is an **experiment** and cause is on the table.



random SELECTION → you may generalise to the population

random ASSIGNMENT → you may claim cause and effect

neither one on its own gives you the other

The second random step is the one that creates an experiment. Without it there is no treatment, and so no cause to claim.

Example 1. **Researchers record the diets of 500 adults and their blood pressure. Which type of study is this?**

Solution.
Observational. Nobody was told what to eat, so no causal claim is supported.

observational: the researchers only watched

was a treatment imposed?	no — people ate as they already did	× no
so the claim available is	association only	✓ yes

Example 2. **Researchers randomly give 250 adults diet A and 250 diet B, then measure blood pressure. Which type?**

Solution.
An experiment — the treatment was imposed, and at random.



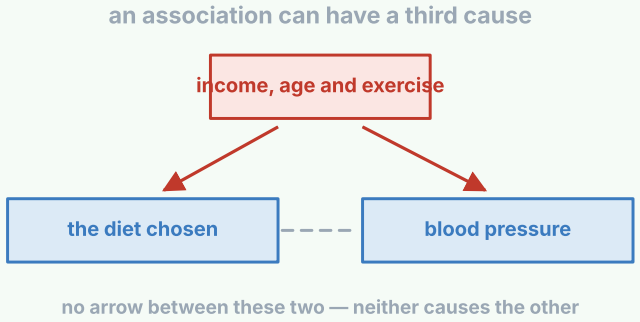
random SELECTION → you may generalise to the population

random ASSIGNMENT → you may claim cause and effect

neither one on its own gives you the other

Example 3. Why can the first study not show that diet causes high blood pressure?

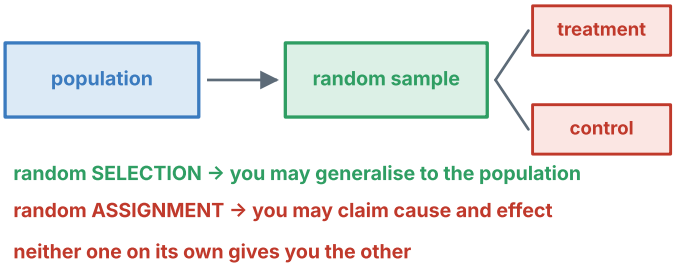
Solution.
People who choose a diet differ in other ways too — age, income, exercise.
Any of these **confounding variables** could explain the difference.



2 Random selection versus random assignment

random selection → generalise to the population | random assignment → claim cause

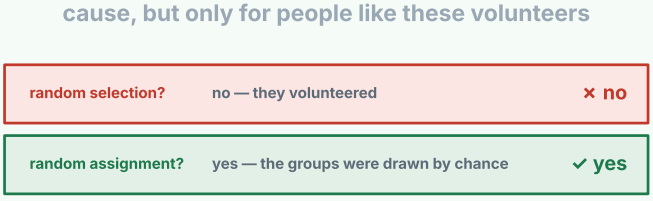
These two are **constantly swapped** on the SAT. **Selection** is how people entered the study; **assignment** is how they were split once inside. Each buys exactly one conclusion, and you need both to get both.



Two arrows, two licences. Cover either one and the conclusion it bought disappears with it.

Example 1. Volunteers are randomly assigned to two tutoring methods. What can and cannot be concluded?

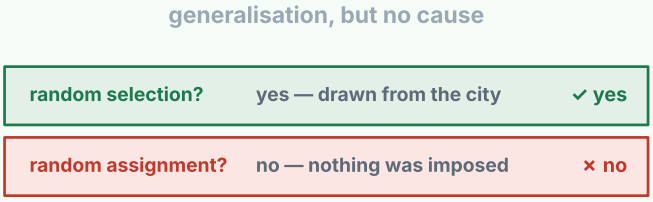
Solution.
Can: the method caused the difference, for people like these volunteers.
Cannot: generalise to all students — the volunteers were not randomly selected.



Example 2. **A random sample of a city's residents is surveyed about commuting. What is supported?**

Solution.

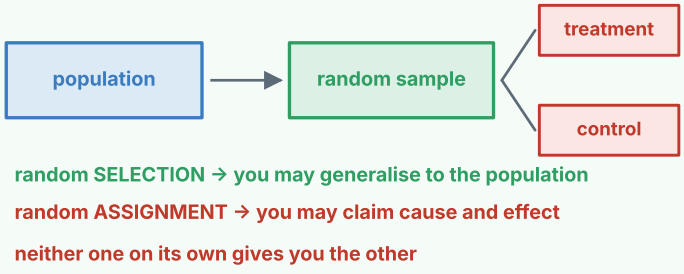
Generalisation to that city's residents. No treatment was imposed, so no causal claim.



Example 3. **Which design supports both conclusions?**

Solution.

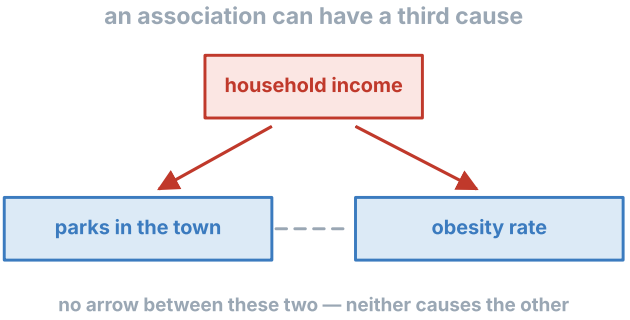
Random selection from the population and random assignment to groups — a randomised experiment on a random sample.



3 Confounding variables

a third variable linked to both the cause and the effect

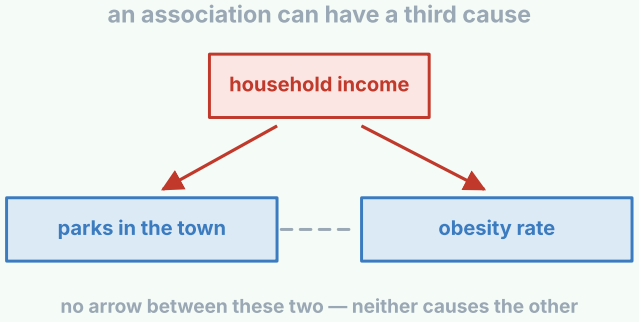
When a question asks why an observational result may mislead, the answer is usually a **named third variable**. A good answer identifies something plausibly connected to **both** quantities — not just any difference between the groups.



A confounder needs an arrow to each of the two quantities. One arrow is not enough, which is how the wrong options are built.

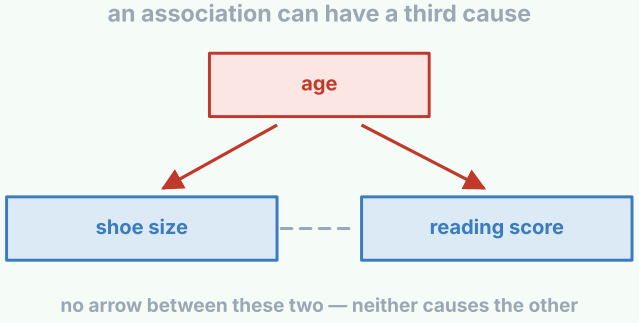
Example 1. **Towns with more parks have lower obesity rates. Suggest a confounding variable.**

Solution.
Household income. Wealthier areas tend to have both more parks and lower obesity, so the parks may not be the cause.



Example 2. **Children with larger shoe sizes read better. What explains it?**

Solution.
Age. Older children have bigger feet and read better; shoe size causes nothing.



Example 3. **Why does random assignment remove confounding?**

Solution.
It spreads the other variables **evenly across the groups on average**, so the groups differ only in the treatment.



random SELECTION → you may generalise to the population

random ASSIGNMENT → you may claim cause and effect

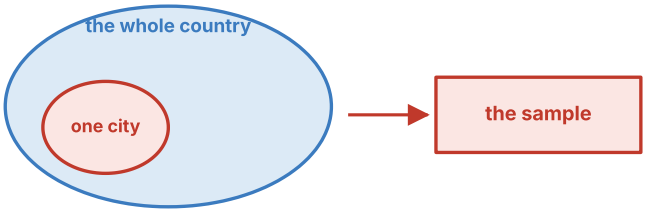
neither one on its own gives you the other

4 How the sample limits the conclusion

you may generalise only to the population the sample was drawn from

Match the conclusion to the **sampling frame**. A random sample of one school supports claims about **that school** and nothing wider. The commonest wrong answer on these questions is a true-sounding statement about a population that was never sampled.

bias is about who was chosen, never how many



the conclusion may reach the circle the sample came from, and no further

a bigger biased sample is just a more precise wrong answer

The sample defines its own ceiling. Everything outside the red blob is a population nobody asked.

Example 1. **200 students are chosen at random from one school. To whom do the results apply?**

Solution.
To that school's students only — not to the district, and not to students generally.



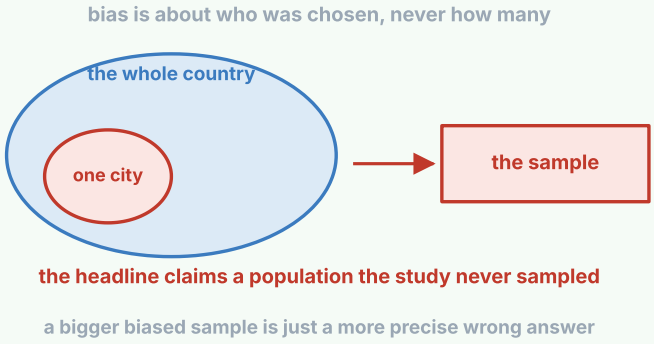
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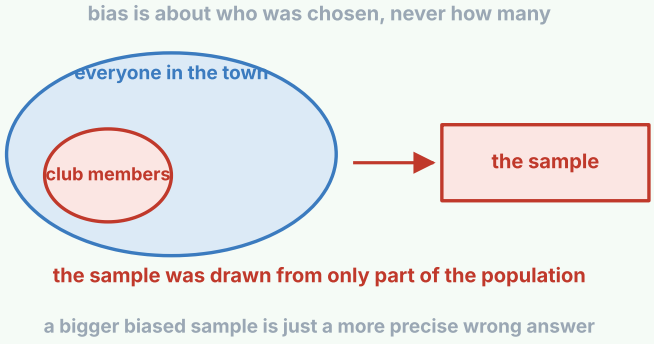
Example 2. **A study samples randomly from all adults in one city. A newspaper reports the result as a national finding. What is wrong?**

Solution.
The sampling frame was **one city**, so the conclusion cannot be stretched to the country.



Example 3. **A sample is drawn at random from members of a sports club. Can the result describe the town?**

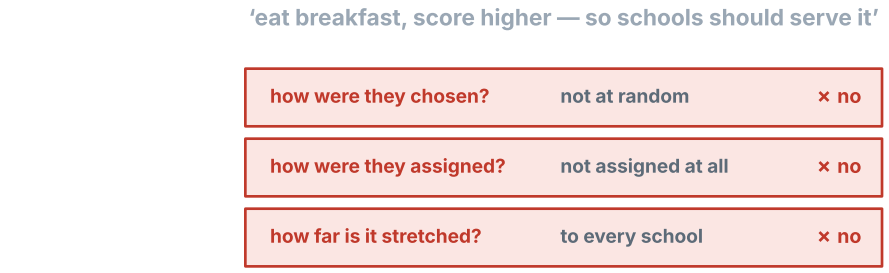
Solution.
No. The population is the club's members; club members differ systematically from the town as a whole.



5 Judging a reported claim

check three things: how chosen, how assigned, how far stretched

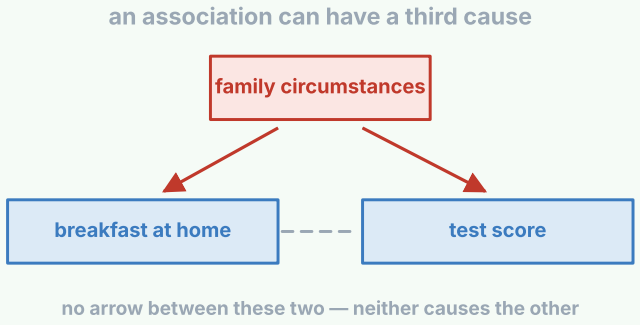
Most of these questions give you a headline and four conclusions, and **the right one is the cautious one**. Scan for the overreach: a causal verb on an observational study, or a population wider than the one sampled.



Three boxes, and this headline fails all three. The correct answer to such a question is always the one that claims least.

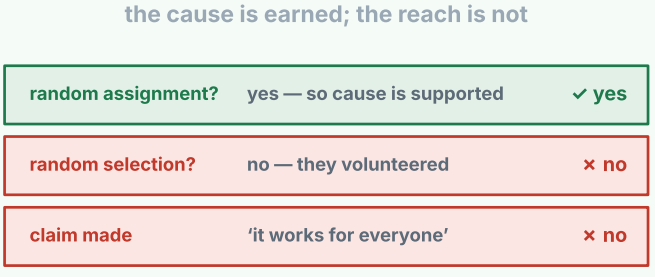
Example 1. **Headline: 'Students who eat breakfast score higher — so schools should serve breakfast.' Evaluate it.**

Solution.
The data is **observational**, so it shows association only.
Families who provide breakfast may differ in other ways. **The recommendation outruns the evidence.**



Example 2. **A randomised trial on 300 volunteers finds a drug works. A report says it works for everyone. What is the flaw?**

Solution.
Random **assignment** supports cause within the trial, but the volunteers were **not randomly selected**, so the result cannot be generalised to everyone.



Example 3. **What one change would let the breakfast study support its recommendation?**

Solution.

Randomly assign students to receive breakfast or not, then compare — turning the observational study into an experiment.



random SELECTION → you may generalise to the population

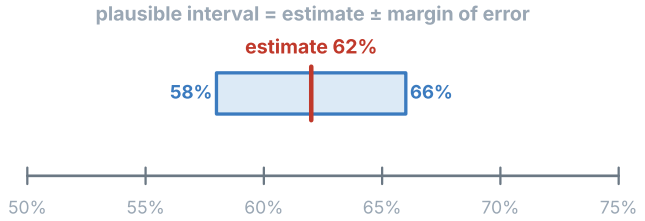
random ASSIGNMENT → you may claim cause and effect

neither one on its own gives you the other

6 Precision, bias and sample size

a bigger sample narrows the margin of error | only better selection removes bias

Sample size and bias solve **different problems** and the SAT tests whether you know which. A larger sample makes the estimate more **precise**; it does nothing about a sample drawn from the **wrong people**.

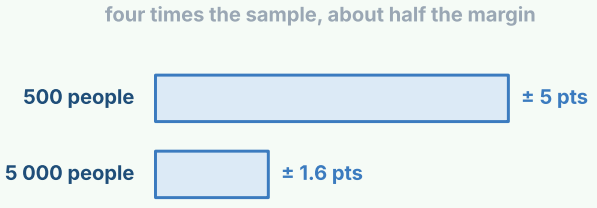


Sample size moves the ends of this bar inwards. Nothing about sample size moves the bar to where it should have been.

Example 1. **A biased survey of 500 is repeated with 5,000 people the same way. What improves?**

Solution.

Precision only. The margin of error narrows, but the estimate is still centred on the wrong value.



narrower, and still centred in the wrong place

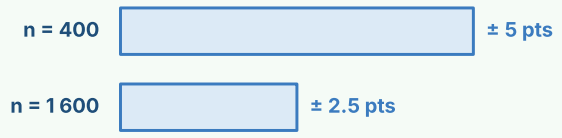
the size of the population barely matters at all

Example 2. **A survey of 400 gives a margin of 5 points. What does 1,600 give, roughly?**

Solution.

About 2.5 points — four times the sample roughly halves the margin.

four times the sample, about half the margin



margin ∝ 1 ÷ √n, so ×4 on n gives ÷2 on the margin

the size of the population barely matters at all

Example 3. **Which fixes bias: a larger sample, or random selection?**

Solution.

Random selection. Size addresses precision; selection addresses bias, and no amount of size repairs bad selection.

two problems, two different cures

a larger sample	narrows the margin of error
random selection	removes the bias

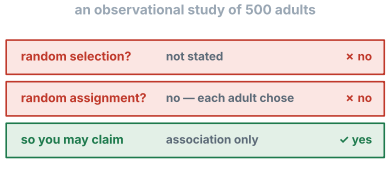
Part 1 — With Desmos (10 questions · calculator work only; the judgement is in the wording)

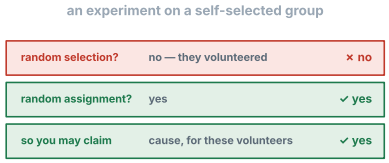
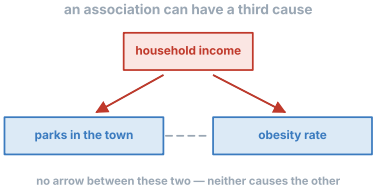
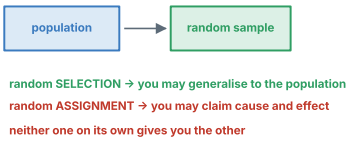
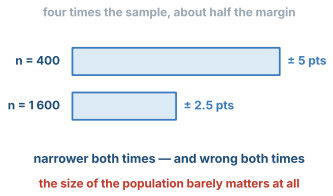
The SAT gives you Desmos on every maths question. Here it is **a calculator for the percentages inside the question**. The decision — what the study is allowed to claim — is entirely yours.

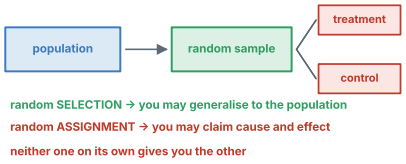
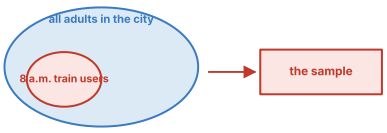
#	Question	Type into Desmos	Then	Answer
D1	A survey of 400 people has a margin of error of 5 percentage points. Four times the sample roughly halves it. What is the new margin?	`5/2`	Quadrupling the sample halves the margin.	2.5
D2	A random sample of 250 of a city's 40,000 adults is surveyed. What fraction of the population is sampled?	`250/40000`	A sampled fraction.	0.00625
D3	In an experiment, 250 adults receive diet A and 250 receive diet B. How many take part in all?	`250+250`	Group sizes add.	500
D4	A trial reports 38% improvement among 300 volunteers. How many people improved?	`0.38*300`	Proportion times group size.	114
D5	An estimate of 62% has a margin of error of 4 points. What is the width of the plausible interval, in percentage points?	`2*4`	The interval spans the margin on each side.	8
D6	Of 500 adults studied, 180 reported a high-salt diet. What percentage is that?	`180/500*100`	Convert at the end.	36
D7	A study of 1,200 students samples 150. By what factor would the sample need to grow to halve the margin of error?	`4`	Margin shrinks with the square root of the sample size.	4
D8	Two groups of 250 report mean improvements of 6.4 and 4.1 units. What is the difference?	`6.4-4.1`	A difference of means.	2.3
D9	A town of 24,000 adults is estimated at 38% support, with a margin of 3 points. How many adults does the lower end represent?	`0.35*24000`	Apply the lower percentage to the population.	8400
D10	A sample proportion is 90 out of 150. What percentage is that?	`90/150*100`	The sample statistic, before any margin is applied.	60

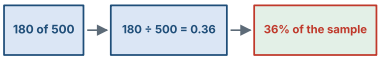
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Run the two-question test on every one: **was selection random?** (how far it generalises) and **was assignment random?** (whether cause may be claimed). They are independent.

#	Level	Question	Answer	Solution
P1	Hard	Researchers recorded the diets and blood pressure of 500 adults who had each chosen their own diet, and found that a high-salt diet was associated with higher blood pressure. State the best-supported conclusion. 	**There is an association between the two, but the study cannot establish cause.**	Nobody was assigned a diet, so this is an **observational study** and only association follows. Two common wrong answers assert a direction of cause the design cannot support.

#	Level	Question	Answer	Solution
P2	Hard	In a study, 300 volunteers are randomly assigned to either a new tutoring method or the usual one, and the new method's group scores higher. State the best-supported conclusion. 	**The new method causes higher scores for people like these volunteers.**	Random assignment supports a **causal claim**, but the volunteers were not randomly selected, so the finding cannot be stretched beyond people like them. A common wrong answer is the classic overreach.
P3	Hard	A study finds that towns with more public parks have lower obesity rates. Which is the most plausible confounding variable? 	**The average household income of the town.**	A confounder must plausibly affect **both** quantities. Wealthier towns tend to have more parks and lower obesity. Two common wrong answers name the two variables already in the study.
P4	Hard	A researcher randomly selects 200 students from one school and surveys them about study habits. To which group may the results be generalised? 	**All students at that school.**	The sample was drawn at random from that school, so the conclusion extends to **that school and no further**. A common wrong answer is too cautious: random sampling is exactly what licenses the step from sample to school.
P5	Hard	A survey of 400 people reports a margin of error of 5 percentage points. The researchers repeat it with 1,600 people, sampled the same biased way. Which improves? 	**Precision only; the estimate is still centred on the wrong value.**	A larger sample narrows the margin of error, so the estimate becomes more **precise**. But the same flawed selection means it is still **precisely wrong**. Size does not fix bias.

#	Level	Question	Answer	Solution
P6	Very hard	Two studies are described. Study 1 randomly selects 500 adults from a city and surveys their exercise habits. Study 2 takes 500 volunteers and randomly assigns them to two exercise programmes. State the best-supported conclusion.  random SELECTION → you may generalise to the population random ASSIGNMENT → you may claim cause and effect neither one on its own gives you the other	**Study 1 supports generalisation to the city; Study 2 supports a causal claim for its participants.**	Random **selection** (Study 1) buys generalisation; random **assignment** (Study 2) buys cause. A common wrong answer swaps them, which is the error the question is built around.
P7	Very hard	A newspaper reports: 'Students who eat breakfast score higher on tests, so schools should provide breakfast.' The data came from a survey of students' existing habits. What is the main flaw? the headline fails all three checks how were they chosen? not at random × no how were they assigned? not assigned at all × no how far is it stretched? to every school × no	**The recommendation assumes cause from an observational study.**	The survey recorded what students **already did**, so it shows association only. Families who provide breakfast may differ in other ways. The recommendation needs an experiment behind it.
P8	Very hard	A clinical trial randomly assigns 300 patients, all recruited from one hospital, to a drug or a placebo. The drug group improves significantly more. State the best-supported conclusion. a real experiment on a narrow group random selection? no — one hospital only × no random assignment? yes ✓ yes so you may claim cause, for these patients ✓ yes	**The drug caused the improvement in this group of patients.**	Random assignment establishes **cause within the trial**. Recruitment from a single hospital is not random selection from any wider population, so A over-reaches; C under-reads the assignment.
P9	Very hard	A researcher wants to estimate the average commute time of all adults in a city and surveys every adult leaving one train station at 8 a.m. Which single change would most improve the study? bias is about who was chosen, never how many  people who take the train commute differently from everyone else a bigger biased sample is just a more precise wrong answer	**Select adults at random from the city's electoral register.**	The flaw is the **sampling frame**: train users are not all adults, and no sample size fixes that. Only random selection from the whole population does. A common wrong answer makes a biased estimate more precise, not more accurate.

#	Level	Question	Answer	Solution
P10	Very hard	In an experiment, 250 adults on diet A improved by a mean of 6.4 units and 250 on diet B by a mean of 4.1 units. Assignment was random. What is the difference in mean improvement?  random assignment is what lets you blame the diets	2.3	6.4 − 4.1 = **2.3 units**. Because assignment was random, this difference can reasonably be attributed to the diets.
P11	Very hard	A town has 24,000 adults. A random sample estimates 38% support for a proposal, with a margin of error of 3 percentage points. Approximately how many adults does the lower end of the plausible interval represent?  take the lower end first, then scale it up	8400	Lower end = 38 − 3 = 35%. 0.35 × 24,000 = **8,400 adults**.
P12	Very hard	Of 500 adults in an observational study, 180 reported a high-salt diet. What percentage of the sample is this?  a statistic about the sample, not yet about the population	36	180/500 = 0.36 = **36%**. Note this is a statistic about **the sample**; extending it to the population needs the sample to have been randomly selected.

The one habit to build. Selection and assignment are **two different things** and answer two different questions. Most wrong answers on this topic are true statements about the wrong group, or causal claims from data where nobody assigned anything.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

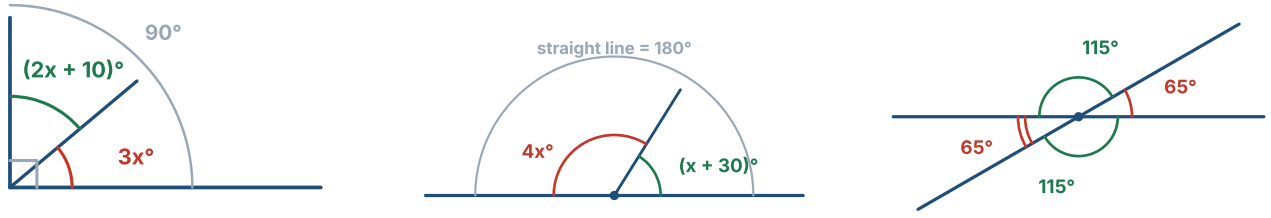
SAT Math 4.1 — Lines, Angles & Triangles

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Angles on a line and at a point

straight line = 180° | full turn = 360° | vertical angles are equal

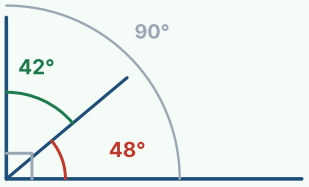
Complementary angles add to 90°; supplementary to 180°. Vertical (opposite) angles made by two crossing lines are always equal. Most angle-chasing questions are one of these three facts applied twice.



Three pictures, three facts: a right angle split in two parts that add to 90°; a straight line split in two that add to 180°; and two crossing lines, where opposite angles are equal and neighbours add to 180°.

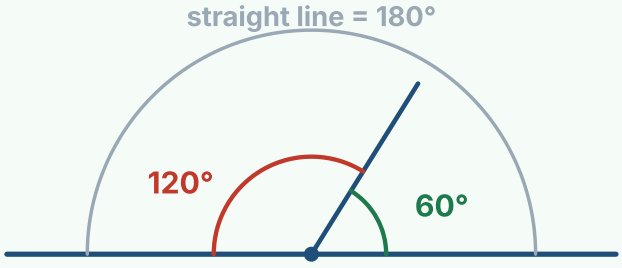
Example 1. **Two angles are complementary. One is 3x° and the other (2x + 10)°. Find x.**

Solution.
3x + 2x + 10 = 90, so 5x = 80.
x = **16**.



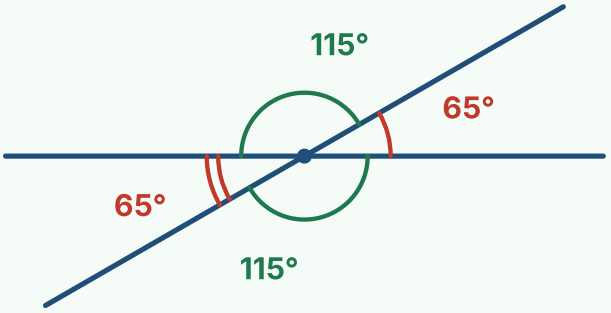
Example 2. **Two angles on a straight line are (4x)° and (x + 30)°. Find x.**

Solution.
4x + x + 30 = 180, so 5x = 150.
x = **30**.



Example 3. **Two lines cross. One angle is 65°. Find the other three.**

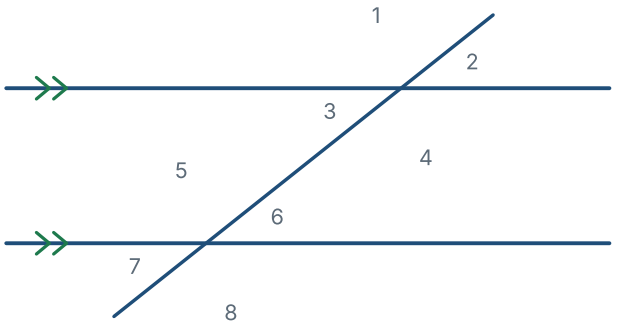
Solution.
The vertical angle is **65°**; the two adjacent ones are each 180 − 65 = **115°**.



2 Parallel lines and a transversal

corresponding and alternate angles are equal | co-interior angles add to 180°

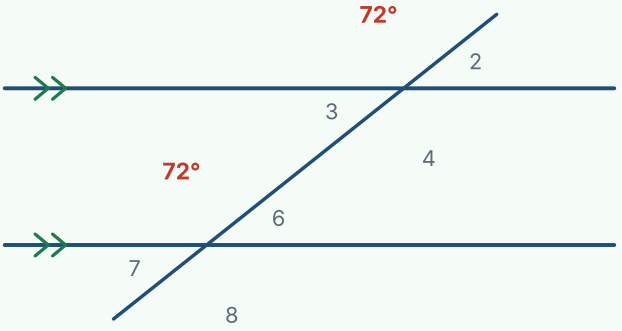
When a line crosses two parallels, every angle is either equal to a given one or supplementary to it. Find the equal pairs first; everything else follows from 180°.



The arrow marks say the two lines are parallel. Angles 1 and 5 are corresponding (equal), 4 and 5 are alternate interior (equal), and 3 and 5 are co-interior (they add to 180°).

Example 1. **Two parallel lines are cut by a transversal and one angle is 72°. What is its corresponding angle?**

Solution.
Corresponding angles are equal.
72°.

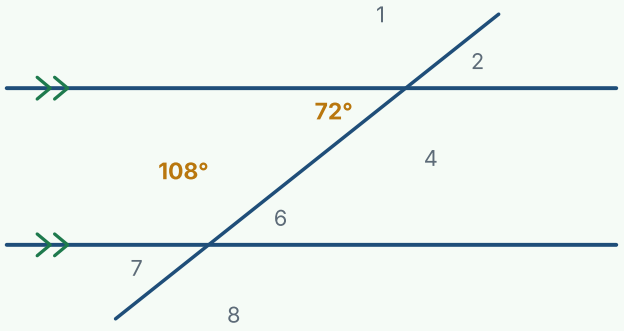


Example 2. **What is the co-interior angle to that 72°?**

Solution.

Co-interior angles add to 180°.

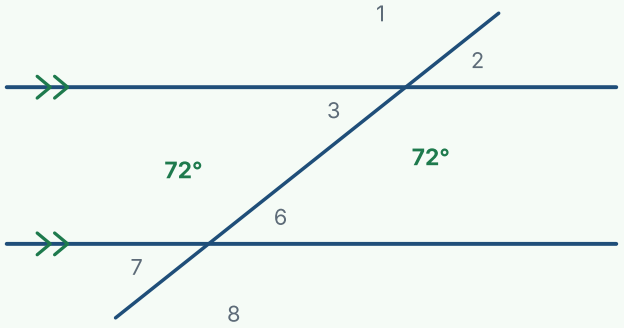
180 − 72 = **108°**.



Example 3. **In the figure above, angles 4 and 5 are what kind of pair?**

Solution.

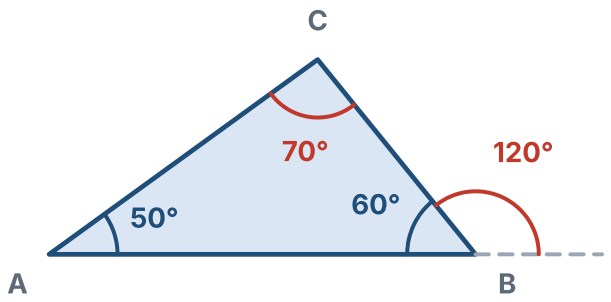
They sit between the parallels on opposite sides of the transversal: **alternate interior**, so they are equal.



3 Triangle angle facts

angles sum to 180° | an exterior angle equals the two remote interior angles

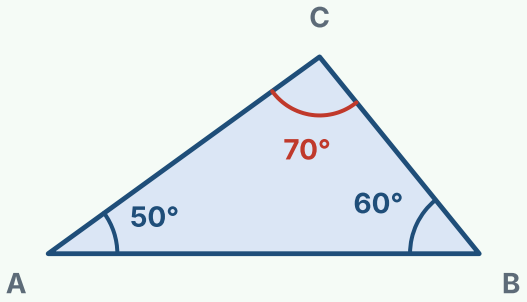
The exterior-angle shortcut saves a step: instead of finding the third angle and subtracting from 180, add the two far angles directly.



The three angles inside total 180°. The exterior angle at B is 120°, which is both 180 − 60 and 50 + 70 — the two angles far away from it.

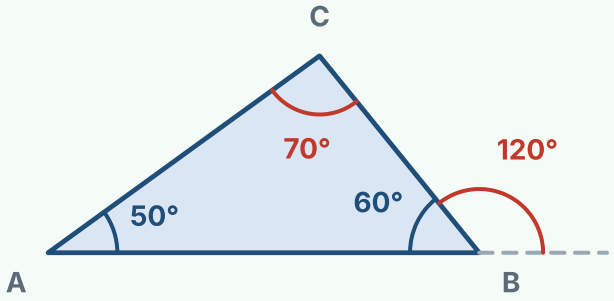
Example 1. **A triangle has angles 50° and 60°. Find the third.**

Solution.
180 − 50 − 60 = **70°**.



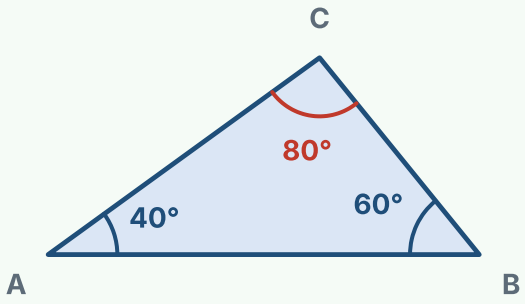
Example 2. **For that triangle, what is the exterior angle at the 60° vertex?**

Solution.
It equals the two remote interior angles: 50 + 70 = **120°**. Checking: 180 − 60 = 120.



Example 3. **A triangle's angles are in the ratio 2 : 3 : 4. Find the largest.**

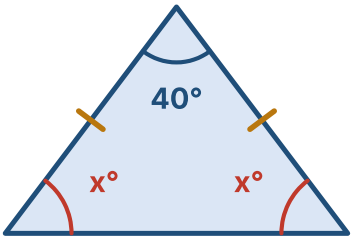
Solution.
The parts total 9, and 180/9 = 20° per part.
The largest is 4 × 20 = **80°**.

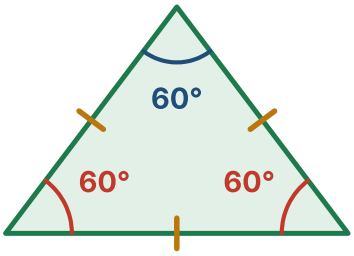


4 Isosceles and equilateral triangles

equal sides face equal angles | equilateral → every angle 60°

The tick marks on a diagram are doing real work: two marked sides mean two equal base angles. This turns most isosceles questions into a single equation.

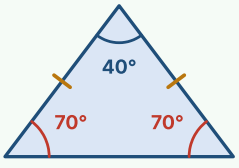




Two equal sides (the tick marks) force two equal base angles, so 2x + 40 = 180. When all three sides are marked, every angle is 60°.

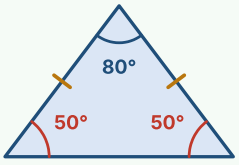
Example 1. **An isosceles triangle has apex angle 40°. Find each base angle.**

Solution.
The two base angles are equal: (180 − 40)/2.
70° each.



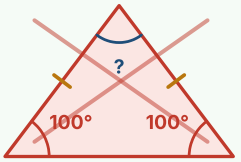
Example 2. **An isosceles triangle has a base angle of 50°. Find the apex angle.**

Solution.
Both base angles are 50°.
180 − 100 = **80°**.



Example 3. **Why can an isosceles triangle not have two angles of 100°?**

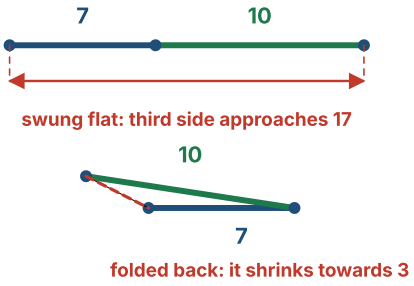
Solution.
They would total 200°, already above 180°. The equal angles in an isosceles triangle must each be **below 90°**.



5 The triangle inequality

any two sides must add to more than the third

Given two sides a and b, the third side x satisfies $|a - b| < x < a + b$. Questions ask for the possible values, or for the largest or smallest whole number that works.





Hinge the two known sides open and shut. Straightened out they reach 7 + 10 = 17; folded back they close to 10 − 7 = 3. The third side lives strictly between, which is the red stretch of the number line.

Example 1. **Two sides are 7 and 10. What values can the third side take?**

Solution.
$10 - 7 < x < 10 + 7$.
$3 < x < 17$.



Example 2. **For those sides, what is the largest possible whole-number third side?**

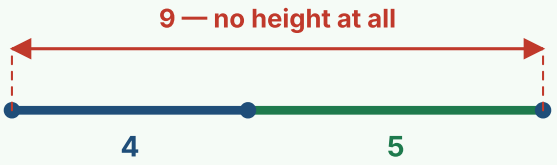
Solution.
x must be below 17.
16.



largest whole number: 16

Example 3. **Can a triangle have sides 4, 5 and 9?**

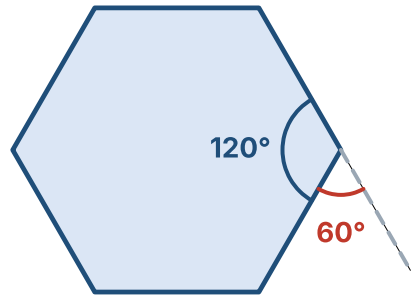
Solution.
4 + 5 = 9, which is not *more* than 9.
No — it collapses flat.



6 Polygons and quadrilaterals

interior angles of an n-gon sum to (n − 2) × 180° | exterior angles always sum to 360°

The exterior-angle fact is the quicker route for regular polygons: each exterior angle is 360/n, and the interior angle is its supplement. In a parallelogram opposite angles are equal and adjacent ones add to 180°.



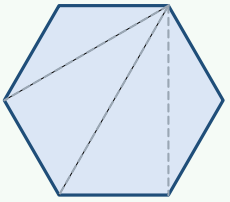
interior + exterior = 180°



Extend one side of the hexagon and the exterior angle appears: 60°, with 120° inside. Walk right round and the exterior angles total 360°. In the parallelogram, 65° sits opposite 65° and beside 115°.

Example 1. **Find the sum of the interior angles of a hexagon.**

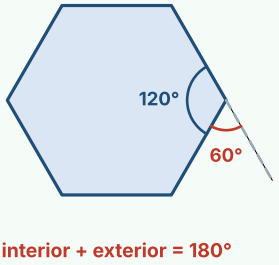
Solution.
(6 − 2) × 180 = **720°**.



4 triangles × 180° = 720°

Example 2. **Find each interior angle of a regular hexagon.**

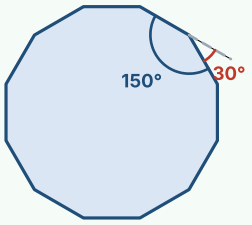
Solution.
Each exterior angle is 360/6 = 60°.
Interior = 180 − 60 = **120°**.



interior + exterior = 180°

Example 3. **A regular polygon has interior angles of 150°. How many sides?**

Solution.
Each exterior angle is 30°, and they total 360°.
360/30 = **12 sides**.



interior + exterior = 180°

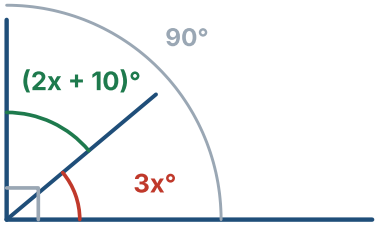
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

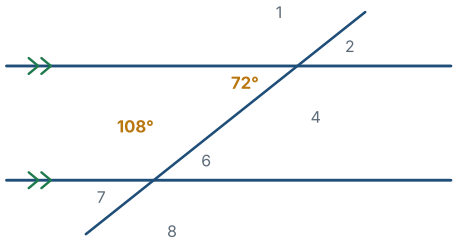
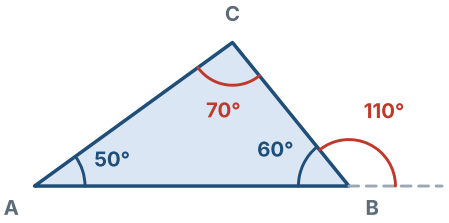
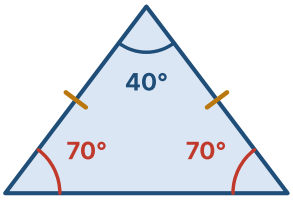
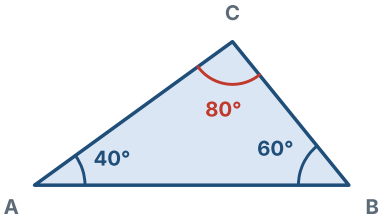
The SAT gives you Desmos on every maths question. For angle work it is **an equation solver**. Write the angle relationship as an equation and let Desmos finish it — the thinking is choosing which relationship applies.

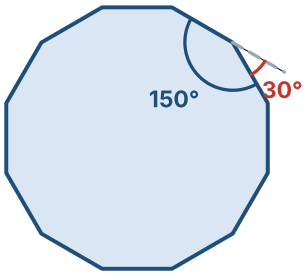
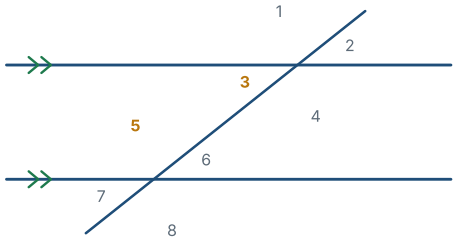
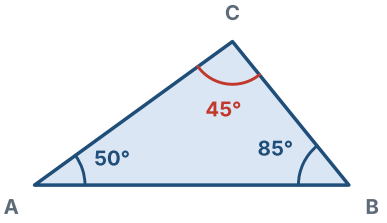
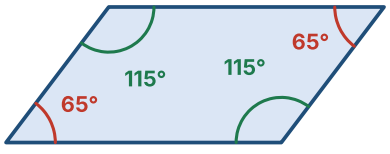
#	Question	Type into Desmos	Then	Answer
D1	Two complementary angles measure 3x° and (2x + 10)°. What is x?	`3x+2x+10=90`	Type the equation; Desmos solves it.	16
D2	Two angles on a straight line are 4x° and (x + 30)°. What is x?	`4x+x+30=180`	A straight line gives 180.	30
D3	A triangle has angles 50° and 60°. What is the third angle?	`180-50-60`	The three must total 180.	70
D4	A triangle's angles are in the ratio 2 : 3 : 4. What is the largest angle?	`180/9*4`	Nine parts in total.	80
D5	An isosceles triangle has apex angle 40°. What is each base angle?	`(180-40)/2`	The two base angles are equal.	70
D6	What is the sum of the interior angles of a hexagon?	`(6-2)*180`	The formula works for any polygon.	720
D7	What is each interior angle of a regular hexagon?	`180-360/6`	Exterior first, then subtract from 180.	120
D8	A regular polygon has interior angles of 150°. How many sides does it have?	`360/(180-150)`	Exterior angles always total 360.	12
D9	Two sides of a triangle are 7 and 10. What is the largest whole-number third side?	`7+10-1`	Strictly less than the sum.	16
D10	Two parallel lines are cut by a transversal and one angle is 72°. What is the co-interior angle?	`180-72`	Co-interior angles are supplementary.	108

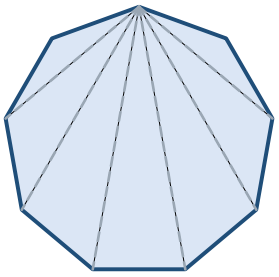
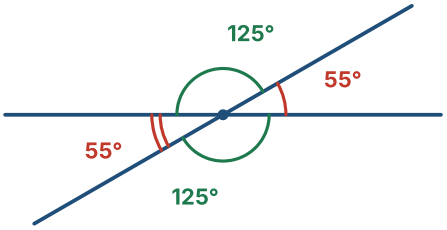
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Mark every angle you can deduce on the figure before you write an equation. Most of these fall out in two steps once the diagram is labelled.

#	Level	Question	Answer	Solution
P1	Hard	Two angles are complementary. One angle measures 3x° and the other measures (2x + 10)°. What is the value of x? 	16	Complementary means they total 90°: 3x + 2x + 10 = 90, so 5x = 80. x = **16**. Using 180 instead of 90 gives 34, which is not an option — a useful check.

#	Level	Question	Answer	Solution
P2	Hard	In the figure, two parallel lines are cut by a transversal. If angle 1 measures 72°, what is the measure of angle 3, in degrees? 	108	Angles 1 and 3 sit on a straight line, so they are supplementary. 180 − 72 = **108°**.
P3	Hard	In a triangle, two of the angles measure 50° and 60°. What is the measure of the exterior angle at the third vertex, in degrees? 	110	The third interior angle is 180 − 110 = 70°, and its exterior angle is 180 − 70 = **110°**. Equivalently, the exterior angle equals the two remote interior angles: 50 + 60 = 110.
P4	Hard	An isosceles triangle has an apex angle of 40°. What is the measure of each base angle? 	70°	The two base angles are equal and the three total 180°. (180 − 40)/2 = **70°**. A common wrong answer is the sum of the two base angles, not each one.
P5	Hard	The angles of a triangle are in the ratio 2 : 3 : 4. What is the measure of the largest angle, in degrees? 	80	The ratio has 2 + 3 + 4 = 9 parts, and 180/9 = 20° per part. 4 × 20 = **80°**.
P6	Hard	Two sides of a triangle measure 7 and 10. State the range of possible lengths for the third side. 	3 < x < 17	The third side x must satisfy **3 < x < 17**. Both boundaries are excluded, and each would give a flat, degenerate triangle.

#	Level	Question	Answer	Solution
P7	Very hard	In a regular polygon, each interior angle measures 150°. How many sides does the polygon have?  interior + exterior = 180°	12	Each exterior angle is 180 − 150 = 30°, and the exterior angles always total 360°. 360/30 = **12 sides**.
P8	Very hard	Two parallel lines are cut by a transversal. One of the eight angles formed measures (3x + 10)° and the co-interior angle on the same side measures (2x − 5)°. What is the value of x?  	35	Co-interior angles are supplementary: 3x + 10 + 2x − 5 = 180, so 5x + 5 = 180. x = **35**. Setting them equal instead gives x = −15, which is impossible — a sign you chose the wrong angle relationship.
P9	Very hard	In triangle ABC, angle A measures (2x)°, angle B measures (3x + 10)°, and angle C measures (x + 20)°. What is the measure of the largest angle, in degrees?  	85°	2x + 3x + 10 + x + 20 = 180 gives 6x + 30 = 180, so x = 25. The angles are 50°, 85° and 45°, so the largest is **85°**. A common wrong answer is x itself, which is not an angle of the triangle.
P10	Very hard	A parallelogram has one angle measuring 65°. What is the measure of the angle adjacent to it, in degrees?  	115	Adjacent angles of a parallelogram are supplementary. 180 − 65 = **115°**. The opposite angle, by contrast, is also 65°.

#	Level	Question	Answer	Solution
P11	Very hard	The sum of the interior angles of a polygon is 1,260°. How many sides does it have?  **7 triangles × 180° = 1260°**	9	(n − 2) × 180 = 1260, so n − 2 = 7. n = **9**. A common wrong answer is n − 2, stopped one step early.
P12	Very hard	In the figure, two lines intersect and one of the four angles formed measures (5x − 20)°. Its vertical angle measures (3x + 10)°. What is the measure of each of these angles, in degrees? 	55°	Vertical angles are equal: 5x − 20 = 3x + 10, so 2x = 30 and x = 15. The angle is 5(15) − 20 = **55°**. A common wrong answer is x itself, not the angle — read what the question asks for.

The one habit to build. Read what the question asks for. Several of these solve to a value of x that is *not* the answer — the angle is 5x − 20, not x.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

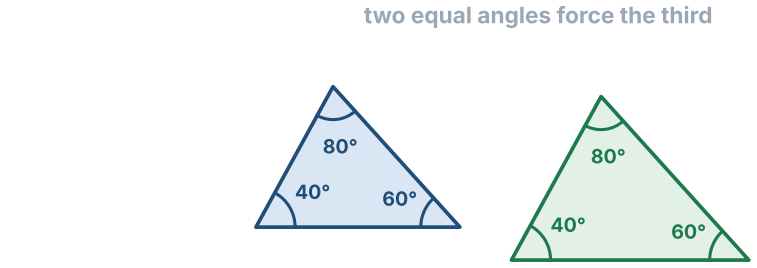
SAT Math 4.2 — Similarity & Congruence

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 When are two triangles similar?

AA — two pairs of equal angles is enough | similar means same shape, different size

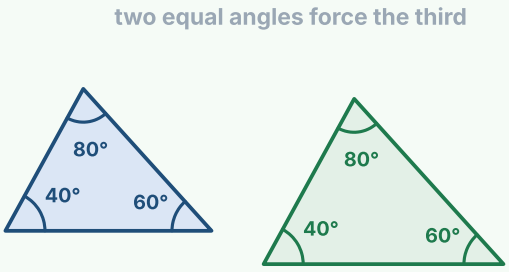
You never need all three angles: if two match, the third must. Similar triangles have equal angles and proportional sides, and the SAT almost always gives you the angles and asks for a side.



Same three angles, different sizes. Nothing here fixes the size, which is exactly why AA proves similarity but never congruence.

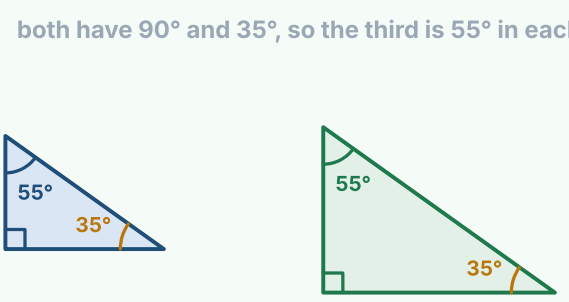
Example 1. **Two triangles have angles 40°, 60° and 40°, 80°. Are they similar?**

Solution.
The first has third angle 80°, the second 60°. Both are 40, 60, 80.
Yes, by AA.



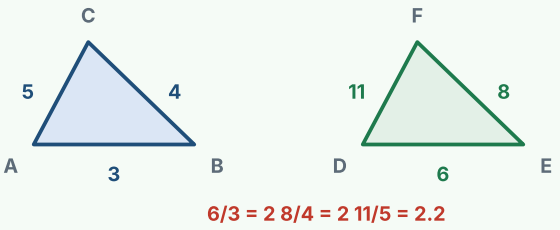
Example 2. **Two right triangles each have an angle of 35°. Are they similar?**

Solution.
Both have 90° and 35°, so the third is 55° in each.
Yes, by AA.



Example 3. **Two triangles have sides 3, 4, 5 and 6, 8, 11. Are they similar?**

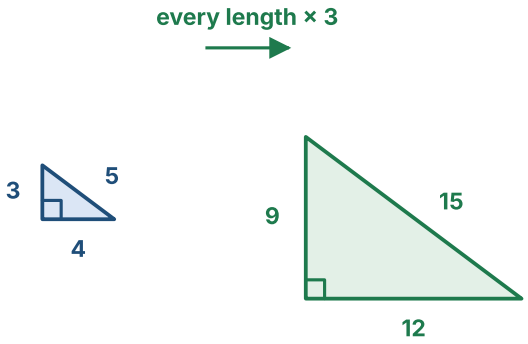
Solution.
6/3 = 2 and 8/4 = 2, but 11/5 = 2.2.
No — the ratios must all match.



2 The scale factor

scale factor k = new ÷ old | every matching length multiplies by k

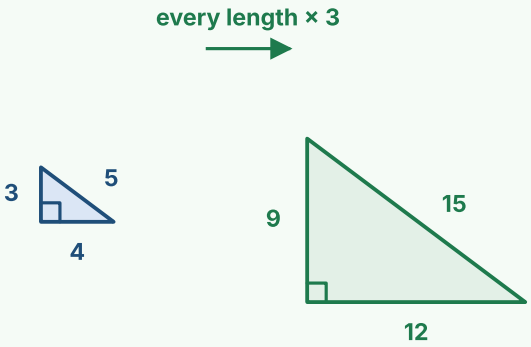
Find k from the one pair of sides you are given, then apply it to the side you want. Pair the sides correctly: the side opposite the 30° angle in one triangle matches the side opposite the 30° angle in the other.



One pair of matching sides gives k = 9/3 = 3. From there every other length follows: 4 → 12 and 5 → 15.

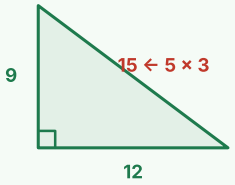
Example 1. **A triangle with sides 3, 4, 5 is enlarged so the shortest side becomes 9. Find the scale factor.**

Solution.
k = 9/3 = **3**.



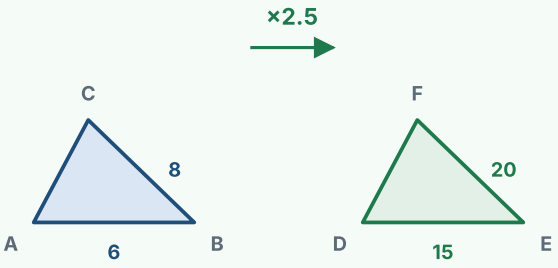
Example 2. **For that enlargement, what is the new longest side?**

Solution.
5 × 3 = **15**.



Example 3. **Triangle ABC has AB = 6; similar triangle DEF has DE = 15. If BC = 8, find EF.**

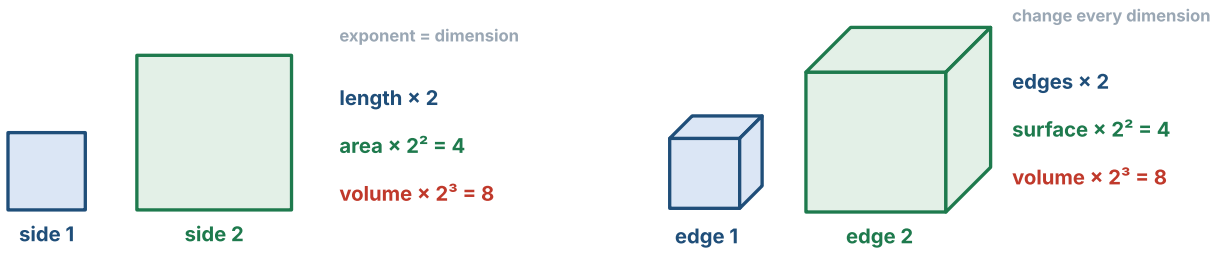
Solution.
k = 15/6 = 2.5.
EF = 8 × 2.5 = **20**.



3 Area and volume scale differently

lengths × k | areas × k^2 | volumes × k^3

This is the trap the SAT returns to. Doubling every length multiplies the area by 4 and the volume by 8. Going the other way, if areas are in the ratio 9 : 4, the lengths are in the ratio 3 : 2 — take the square root.



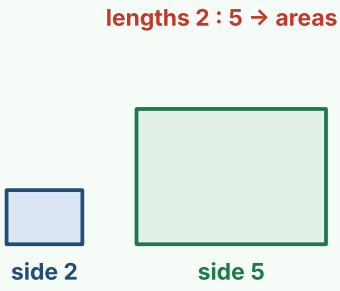
Double the side and four copies of the small square fit inside the big one; eight copies of the small cube fit inside the big cube. The exponent matches the dimension: 1 for length, 2 for area, 3 for volume.

Example 1. **Two similar rectangles have sides in the ratio 2 : 5. What is the ratio of their areas?**

Solution.

Square the factor.

4 : 25.

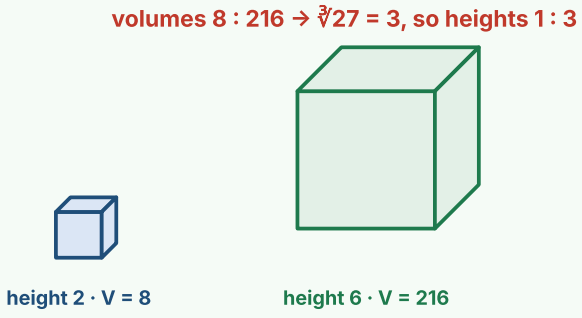


Example 2. **Two similar solids have volumes 8 cm³ and 216 cm³. Find the ratio of their heights.**

Solution.

216/8 = 27, and the cube root of 27 is 3.

1 : 3.

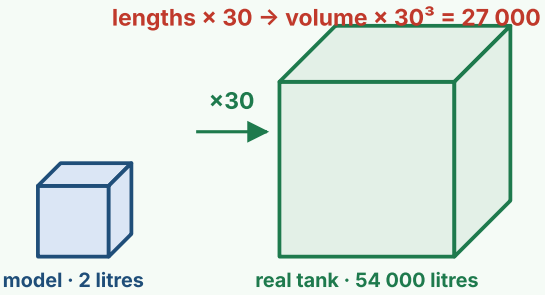


Example 3. **A model is built to scale 1 : 30. The model holds 2 litres. How much does the real tank hold?**

Solution.

Volume scales by $30^3 = 27{,}000$.

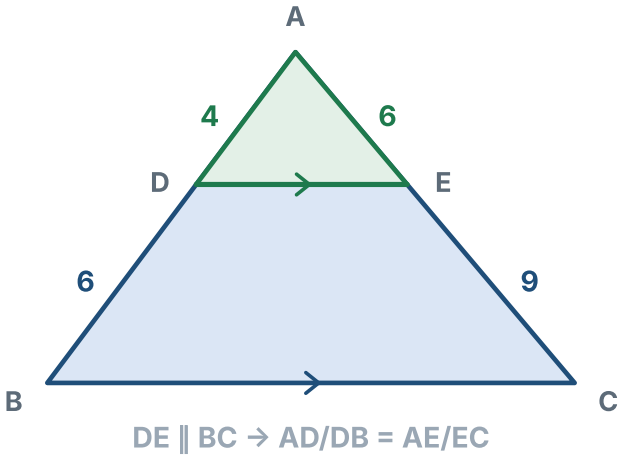
$2 \times 27{,}000 =$ **54,000 litres**.



4 Similar triangles hidden in one figure

a line parallel to one side cuts off a similar triangle

When a figure contains a line parallel to a side, two similar triangles are sitting inside it. Redraw them separately, the same way round, before writing any ratio.



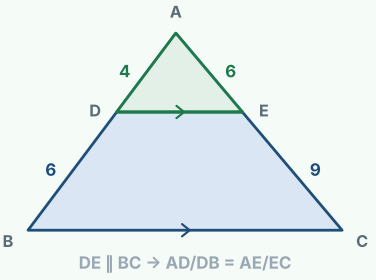
DE ∥ BC makes the angles at D and B equal, and the angle at A is shared, so triangle ADE is similar to triangle ABC. The matching sides give AD/DB = AE/EC.

Example 1. **In triangle ABC, DE is parallel to BC with AD = 4, DB = 6 and AE = 6. Find EC.**

Solution.

AD/DB = AE/EC, so 4/6 = 6/EC.

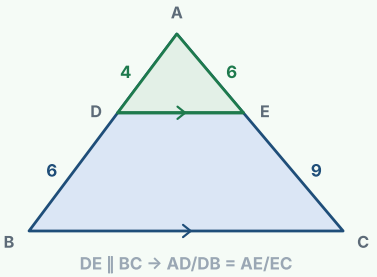
EC = **9**.



Example 2. **Why are the two triangles similar?**

Solution.

DE parallel to BC makes the corresponding angles equal, so **AA** applies.

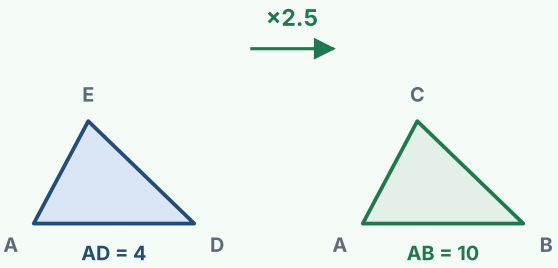


Example 3. **In that figure, AB = 10 and AC = 15. What is the scale factor from ADE to ABC?**

Solution.

AD/AB = 4/10.

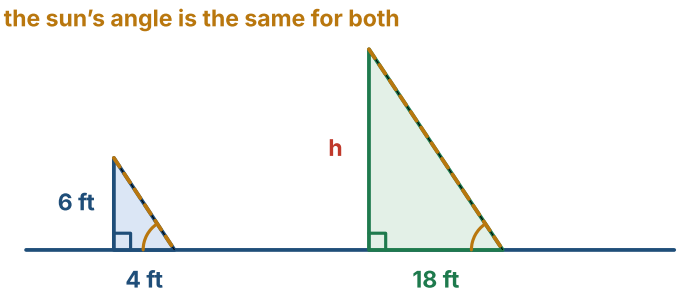
k = **2.5** going from the small triangle to the large one.



5 Shadows, mirrors and indirect measurement

same sun angle → similar triangles → height/shadow is the same ratio for both

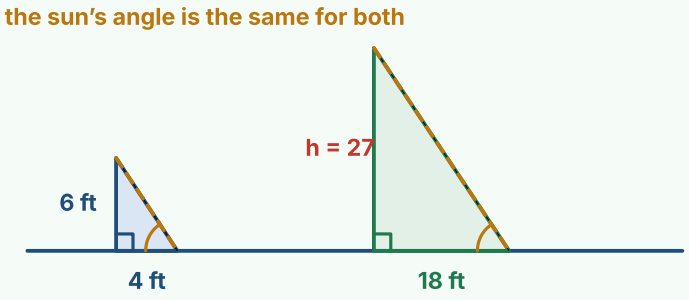
The classic SAT word problem. Write the ratio the same way round for both objects: height ÷ shadow for one equals height ÷ shadow for the other.



Both objects stand upright and both shadows run along the same ground, so the two triangles have the same three angles. 6/4 = h/18 gives h = 27.

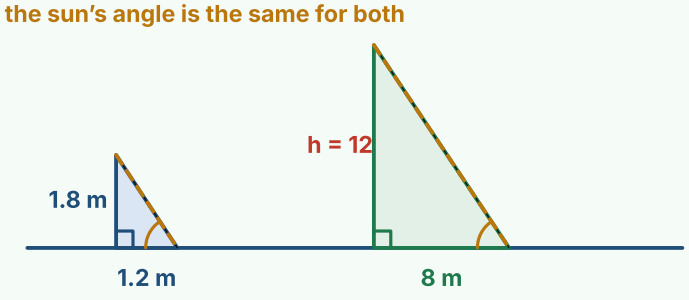
Example 1. **A 6-foot pole casts a 4-foot shadow. A signpost casts an 18-foot shadow. Find its height.**

Solution.
6/4 = h/18, so h = 18 × 1.5 = **27 feet**.



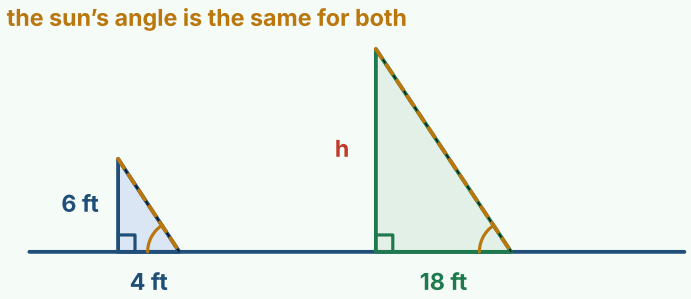
Example 2. **A 1.8 m person casts a 1.2 m shadow. A tree's shadow is 8 m. Find the tree's height.**

Solution.
1.8/1.2 = 1.5, and 8 × 1.5 = **12 m**.



Example 3. **Why must the two shadows be measured at the same time?**

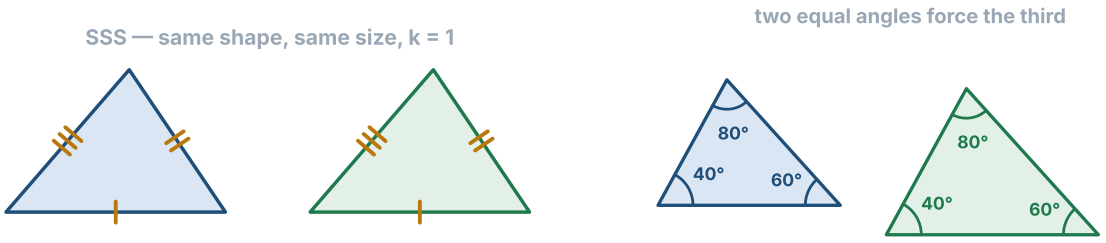
Solution.
The sun's angle changes through the day. Only at the same moment are the two triangles **similar**.



6 Congruence

congruent = same shape and same size, so the scale factor is 1

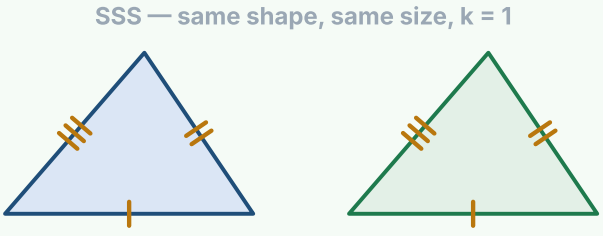
Congruent triangles match exactly: SSS, SAS, ASA or AAS. Note that AAA gives similarity only, never congruence, because the sizes can differ.



On the left the three pairs of sides match, so the triangles are congruent. On the right only the angles match, so they are similar — one is an enlargement of the other.

Example 1. **Two triangles have all three pairs of sides equal. Are they congruent?**

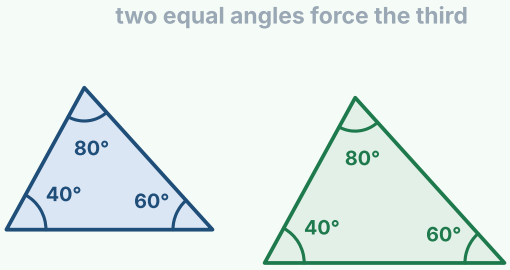
Solution.
Yes, by SSS.



Example 2. **Two triangles have all three pairs of angles equal. Are they congruent?**

Solution.

Not necessarily — they are similar, but one could be an enlargement of the other.

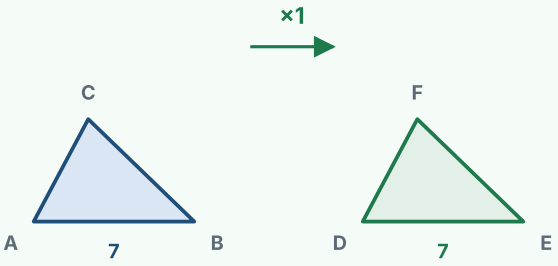


Example 3. **Triangles ABC and DEF are congruent with AB = 7. What is DE?**

Solution.

Congruent means the scale factor is 1.

DE = **7**.



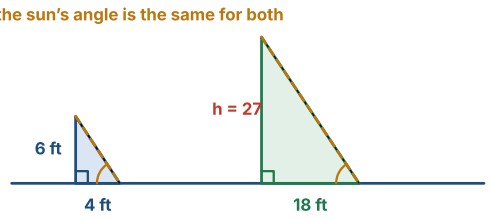
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

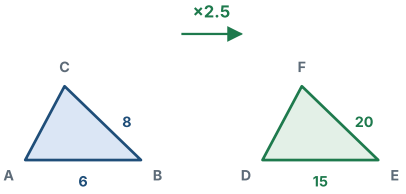
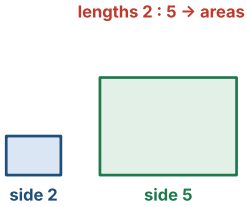
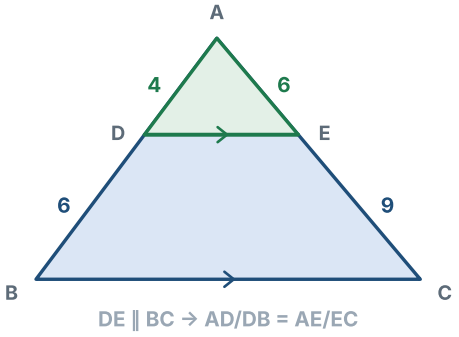
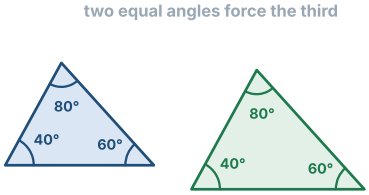
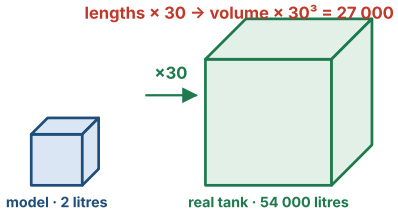
The SAT gives you Desmos on every maths question. Here it is **a proportion solver**. Type the ratio as an equation with the unknown in it and read the answer straight off.

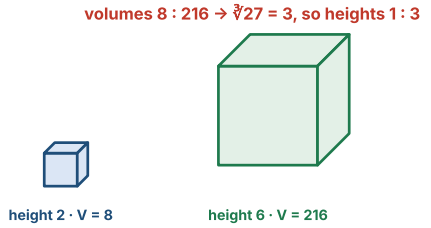
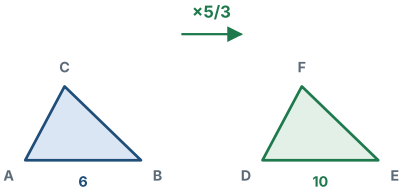
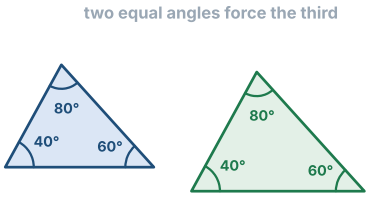
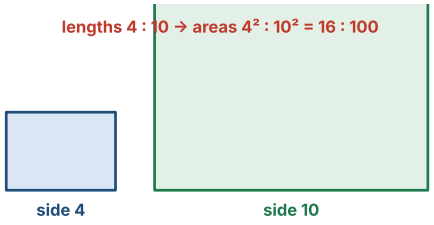
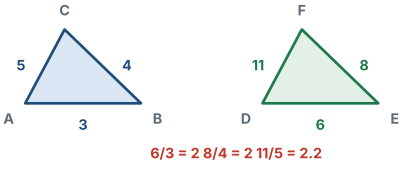
#	Question	Type into Desmos	Then	Answer
D1	A triangle with sides 3, 4, 5 is enlarged so the shortest side becomes 9. What is the new longest side?	`5*(9/3)`	Find the factor, then scale.	15
D2	Triangle ABC has AB = 6; similar triangle DEF has DE = 15. If BC = 8, what is EF?	`8*(15/6)`	The same factor applies to every side.	20
D3	Two similar rectangles have sides in the ratio 2 : 5. What is the ratio of their areas as a decimal?	`(2/5)^2`	Square the length ratio.	0.16
D4	Two similar solids have volumes 8 and 216. What is the ratio of their heights?	`(216/8)^(1/3)`	Cube root of the volume ratio.	3
D5	A model is built to scale 1 : 30 and holds 2 litres. How much does the real tank hold?	`2*30^3`	Volume scales by the cube.	54000
D6	A 6-foot pole casts a 4-foot shadow. A signpost casts an 18-foot shadow. How tall is it?	`18*(6/4)`	Same ratio for both objects.	27
D7	A 1.8 m person casts a 1.2 m shadow. A tree's shadow is 8 m. How tall is the tree?	`8*(1.8/1.2)`	Height over shadow is constant.	12
D8	In triangle ABC, DE is parallel to BC with AD = 4, DB = 6 and AE = 6. What is EC?	`6*(6/4)`	Set up AD/DB = AE/EC.	9
D9	Two similar triangles have areas 9 and 25. What is the ratio of their corresponding sides?	`sqrt(9/25)`	Take the square root of the area ratio.	0.6
D10	A photo 4 cm by 6 cm is enlarged by a factor of 2.5. What is the new area in square centimetres?	`4*6*2.5^2`	Area scales by the square of the factor.	150

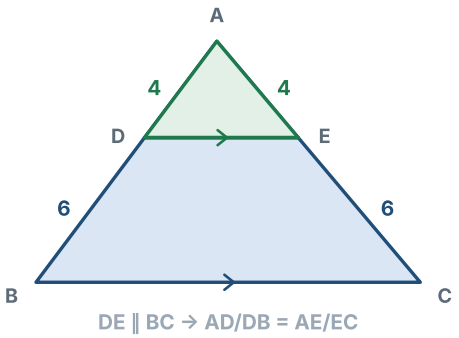
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. For every question, first ask whether the quantity being scaled is a **length**, an **area** or a **volume**. That one decision picks the exponent.

#	Level	Question	Answer	Solution
P1	Hard	Beside a market stall, a vertical pole 6 feet tall casts a shadow 4 feet long. At the same time of day, a signpost casts a shadow 18 feet long. What is the height of the signpost, in feet? 	27	The sun's angle is the same, so the triangles are similar: 6/4 = h/18. h = 18 × 1.5 = **27 feet**. A common wrong answer comes from 18 − 6, not a ratio at all.

#	Level	Question	Answer	Solution
P2	Hard	Triangle ABC is similar to triangle DEF, with AB = 6 and DE = 15. If BC = 8, what is the length of EF? 	20	The scale factor is 15/6 = 2.5, and it applies to every pair of matching sides. EF = 8 × 2.5 = **20**.
P3	Hard	Two similar rectangles have corresponding sides in the ratio 2 : 5. What is the ratio of their areas? 	4 : 25	Area scales by the **square** of the length ratio. $2^2 : 5^2$ = **4 : 25**. A common wrong answer forgets to square; D cubes it, which is the volume rule for solids.
P4	Hard	In triangle ABC, point D lies on AB and point E lies on AC so that DE is parallel to BC. If AD = 4, DB = 6 and AE = 6, what is the length of EC? 	9	DE parallel to BC makes the two triangles similar, so AD/DB = AE/EC. 4/6 = 6/EC gives EC = 36/4 = **9**.
P5	Hard	Two triangles have angles of 40° and 60° in the first, and 40° and 80° in the second. State what is true here. 	They are similar, because two angles of each match.	The first triangle's third angle is 80° and the second's is 60°, so both have angles 40, 60 and 80. **Similar by AA.** Congruence would need the sizes to match too, which nothing here establishes.
P6	Hard	A model of a water tank is built to a scale of 1 : 30. The model holds 2 litres. How many litres does the real tank hold? 	54,000	Volume scales by the **cube** of the length factor: 30^3 = 27,000. 2 × 27,000 = **54,000 litres**. A common wrong answer multiplies by 30 once; B squares it.

#	Level	Question	Answer	Solution
P7	Very hard	Two similar solid figures have volumes of 8 cubic centimetres and 216 cubic centimetres. What is the ratio of their corresponding heights? 	1 : 3	216/8 = 27, and the height ratio is the **cube root** of the volume ratio. ∛27 = 3, so the ratio is **1 : 3**. A common wrong answer uses the volume ratio itself.
P8	Very hard	Two similar triangles have areas of 9 and 25 square units. The shorter triangle has a side of length 6. What is the length of the corresponding side in the larger triangle? 	10	The side ratio is the square root of the area ratio: √(25/9) = 5/3. 6 × 5/3 = **10**.
P9	Very hard	Triangles ABC and DEF have all three pairs of angles equal. Which conclusion is guaranteed? 	The triangles are similar.	Equal angles guarantee the same **shape** but say nothing about size. **Similar**, and possibly congruent, but AAA alone never proves congruence — one could be an enlargement of the other.
P10	Very hard	A rectangular photograph measuring 4 cm by 6 cm is enlarged by a scale factor of 2.5. What is the area of the enlarged photograph, in square centimetres? 	150	Original area 24 cm²; area scales by 2.5² = 6.25. 24 × 6.25 = **150**. A common wrong answer multiplies the area by 2.5 once, the commonest error here.
P11	Very hard	Two triangles have sides 3, 4, 5 and 6, 8, 11 respectively. Are they similar, and why? 	No, because the three ratios are not all equal.	6/3 = 2 and 8/4 = 2, but 11/5 = 2.2. Similarity needs **all** matching ratios to agree, so they are not similar. A common wrong answer is true but irrelevant — similar triangles normally do have different perimeters.

#	Level	Question	Answer	Solution
P12	Very hard	In a figure, triangle ADE sits inside triangle ABC with DE parallel to BC. AD = 4, AB = 10 and the area of triangle ADE is 12 square units. What is the area of triangle ABC, in square units?  DE ∥ BC → AD/DB = AE/EC	75	The length factor is AB/AD = 10/4 = 2.5, so the area factor is $2.5^2 = 6.25$. 12 × 6.25 = **75**. A common wrong answer applies the length factor 2.5 to the area directly instead of squaring it.

The one habit to build. Areas scale by k squared and volumes by k cubed. Your class scored 47% here, and almost every lost mark was a length factor applied to an area or a volume.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

SAT Math 4.3 — Right Triangles & Trigonometry

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

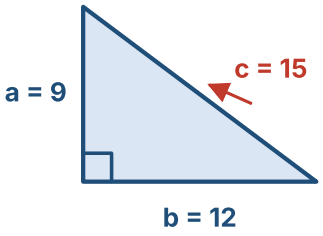
1 Pythagoras

$a^2 + b^2 = c^2$, where c is the hypotenuse — the side opposite the right angle

Identify the hypotenuse before substituting. It is always the longest side and always opposite the right angle. Putting a leg where c belongs is the error that costs this topic its marks.

$$a^2 + b^2 = c^2$$

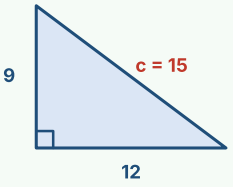
$$81 + 144 = 225$$



The arrow points from the right angle to the side facing it: that side is c. Here 81 + 144 = 225, so c = 15.

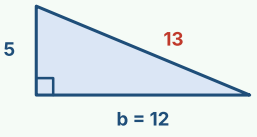
Example 1. **A right triangle has legs 9 and 12. Find the hypotenuse.**

Solution.
81 + 144 = 225.
c = √225 = **15**.



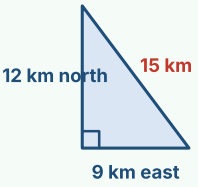
Example 2. **A right triangle has hypotenuse 13 and one leg 5. Find the other leg.**

Solution.
Here 13 is c, so 25 + b^2 = 169.
b = √144 = **12**.



Example 3. **A drone flies 9 km east then 12 km north. How far is it from the start?**

Solution.
The path makes a right angle.
√(81 + 144) = **15 km**.

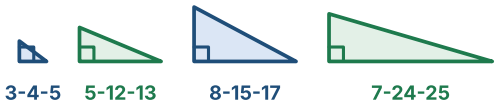


2 The triples worth memorising

3-4-5 | 5-12-13 | 8-15-17 | 7-24-25 — and all their multiples

Recognising a triple turns a calculation into a glance. 9-12-15 is 3-4-5 tripled; 10-24-26 is 5-12-13 doubled. If two sides fit a triple, the third is immediate.

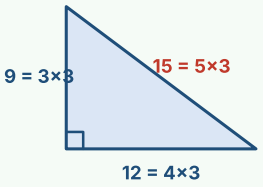
and every multiple: 6-8-10, 9-12-15, 10-24-26 ...



Four right triangles whose sides are whole numbers. Spotting one saves squaring, adding and taking a root.

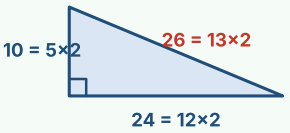
Example 1. **A right triangle has legs 9 and 12. Which triple is this?**

Solution.
3-4-5 multiplied by 3.
The hypotenuse is 5 × 3 = **15**.



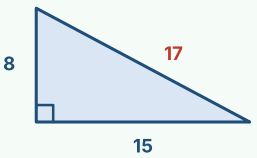
Example 2. **A right triangle has hypotenuse 26 and one leg 10. Find the other leg.**

Solution.
5-12-13 doubled.
The other leg is 12 × 2 = **24**.



Example 3. **Is a triangle with sides 8, 15, 17 right-angled?**

Solution.
$64 + 225 = 289 = 17^2$.
Yes — the converse of Pythagoras.

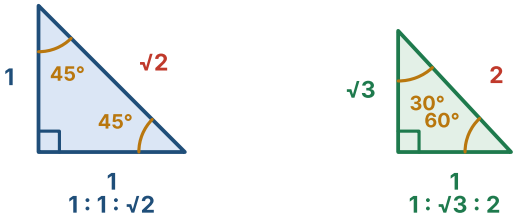


3 Special right triangles

45-45-90 → sides 1 : 1 : √2 | 30-60-90 → sides 1 : √3 : 2

These two appear constantly and are on the reference sheet. In a 30-60-90 the shortest side faces the 30° angle and is half the hypotenuse; the √3 side faces the 60°.

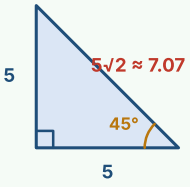
the short side faces 30° — half the hypotenuse



Memorise which side faces which angle. The hypotenuse is always the largest number in the ratio — √2 on the left, 2 on the right.

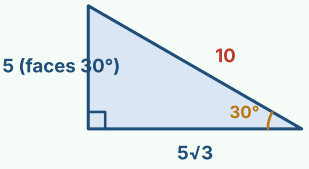
Example 1. **A 45-45-90 triangle has legs of 5. Find the hypotenuse.**

Solution.
5 × √2 = **5√2** ≈ 7.07.



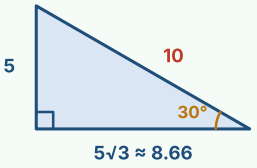
Example 2. **A 30-60-90 triangle has hypotenuse 10. Find the shortest side.**

Solution.
The side opposite 30° is half the hypotenuse.
5.



Example 3. **For that triangle, find the remaining side.**

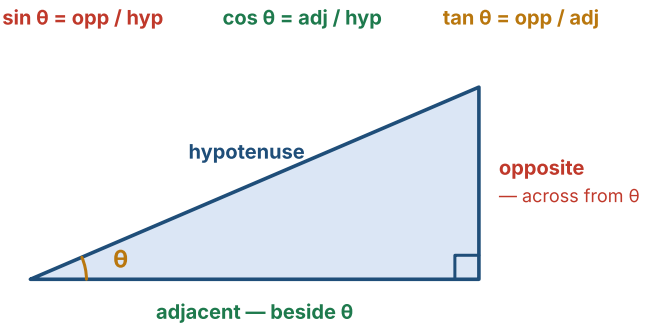
Solution.
It faces the 60° angle: 5√3 ≈ **8.66**.



4 SOH-CAH-TOA

sin = opp/hyp | cos = adj/hyp | tan = opp/adj

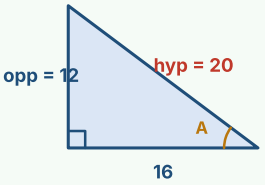
Label the three sides relative to the angle you are working with before choosing a ratio. Opposite and adjacent swap when you switch angles; the hypotenuse never moves.



Stand at θ: the side across the triangle is opposite, the side you are standing on is adjacent, and the long side facing the right angle is the hypotenuse.

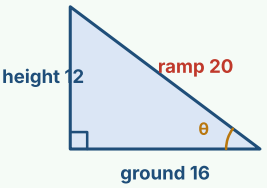
Example 1. **In a right triangle, sin A = 0.6 and the hypotenuse is 20. Find the side opposite A.**

Solution.
sin A = opp/hyp, so opp = 0.6 × 20 = **12**.



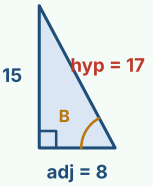
Example 2. **A ramp makes an angle whose sine is 0.6 and is 20 feet long. How high is the top?**

Solution.
The ramp is the hypotenuse.
0.6 × 20 = **12 feet**.



Example 3. **In a right triangle, the side adjacent to angle B is 8 and the hypotenuse is 17. Find cos B.**

Solution.
cos B = adj/hyp = **8/17** ≈ 0.471.

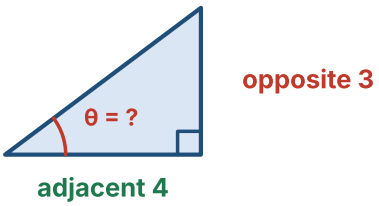


5 Finding an angle

use inverse trig: if $\tan\theta = 3/4$ then $\theta = \tan^{-1}(3/4)$

When the sides are known and the angle is not, the inverse functions give it. In Desmos type arctan, arcsin or arccos and make sure the calculator is in degrees.

$$\tan\theta = 3/4 \rightarrow \theta = \tan^{-1}(0.75) \approx 37°$$

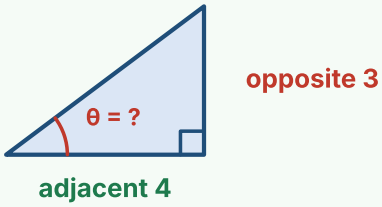


Two sides are known, the angle is not — so run the ratio backwards with the inverse function, and check the calculator says degrees.

Example 1. **A right triangle has opposite 3 and adjacent 4. Find the angle, to the nearest degree.**

Solution.
$\tan\theta = 0.75$, so $\theta = \tan^{-1}(0.75) \approx$ **37°**.

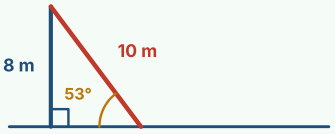
$$\tan\theta = 3/4 \rightarrow \theta = \tan^{-1}(0.75) \approx 37°$$



Example 2. **A ladder 10 m long reaches 8 m up a wall. What angle does it make with the ground?**

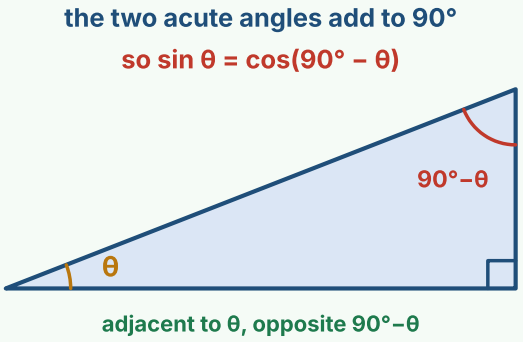
Solution.
$\sin\theta = 8/10 = 0.8$.
$\theta = \sin^{-1}(0.8) \approx$ **53°**.

$\sin\theta = 8/10 \rightarrow \theta \approx 53°$



Example 3. **Why do those two answers add to about 90°?**

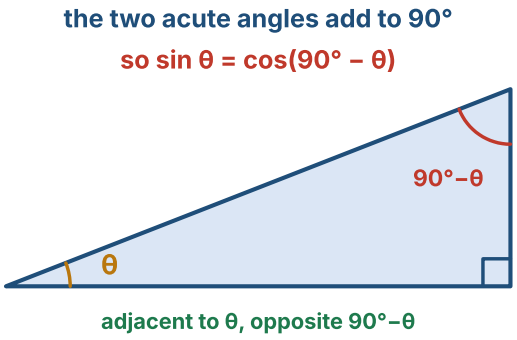
Solution.
They are the **two acute angles** of similar right triangles — 3-4-5 in both cases.



6 sin θ = cos(90° − θ)

the sine of an angle equals the cosine of its complement

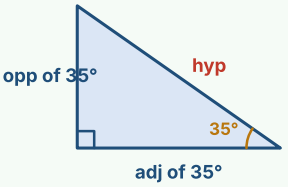
In a right triangle the two acute angles add to 90°, and what is opposite one is adjacent to the other. So sin and cos swap. Questions set sin(expression) = cos(expression) and ask you to make the two angles add to 90.



The side marked opposite for θ is the side adjacent for 90° − θ. Same two lengths, same fraction — only the name of the ratio changes.

Example 1. **If sin 35° = cos x° and x is acute, find x.**

Solution.
x = 90 − 35 = **55**.

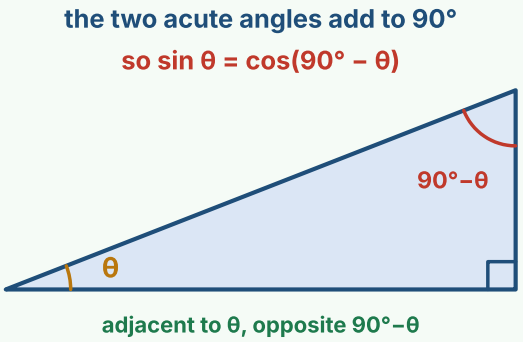


Example 2. **In a right triangle the acute angles are (2x + 10)° and (3x − 20)°. Find x.**

Solution.

They are complementary: 2x + 10 + 3x − 20 = 90, so 5x = 100.

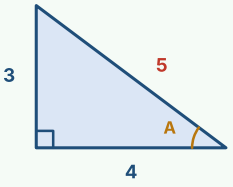
x = **20**.



Example 3. **If sin A = 3/5 for an acute angle A, what is cos(90° − A)?**

Solution.

3/5 — it is the same quantity, by the identity.



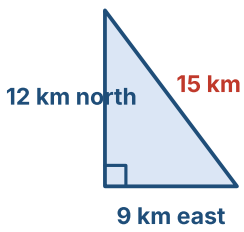
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

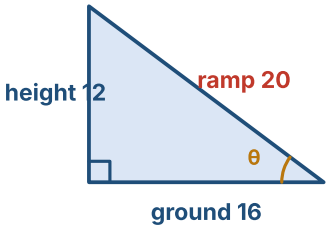
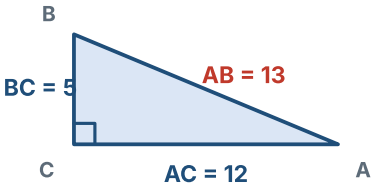
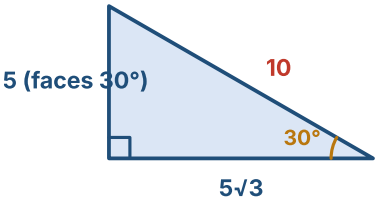
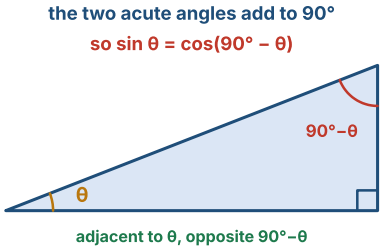
The SAT gives you Desmos on every maths question. Here it is **a trig calculator**. Use arcsin, arccos and arctan for angles, and check the setting is **degrees** before you trust anything.

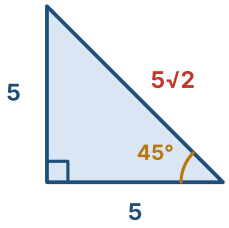
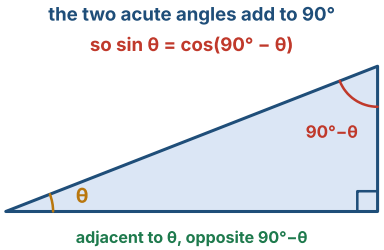
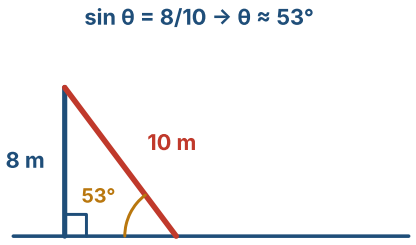
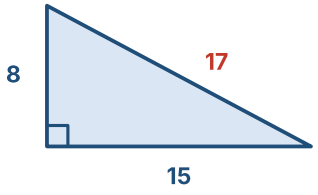
#	Question	Type into Desmos	Then	Answer
D1	A right triangle has legs 9 and 12. What is the hypotenuse?	`sqrt(9^2+12^2)`	Square, add, root — in one line.	15
D2	A right triangle has hypotenuse 13 and one leg 5. What is the other leg?	`sqrt(13^2-5^2)`	Subtract when you are given the hypotenuse.	12
D3	A 45-45-90 triangle has legs of 5. What is the hypotenuse, to two decimal places?	`5*sqrt(2)`	Desmos keeps the exact form too.	7.07
D4	A 30-60-90 triangle has hypotenuse 10. What is the side opposite 60°, to two decimals?	`5*sqrt(3)`	Half the hypotenuse, times √3.	8.66
D5	If sin A = 0.6 and the hypotenuse is 20, what is the opposite side?	`0.6*20`	sin = opp/hyp rearranged.	12
D6	A right triangle has opposite 3 and adjacent 4. What is the angle in degrees, to the nearest degree?	`arctan(3/4)`	Check Desmos is set to degrees.	37
D7	A ladder 10 m long reaches 8 m up a wall. What angle does it make with the ground?	`arcsin(8/10)`	Inverse sine.	53
D8	If sin 35° = cos x°, what is x?	`90-35`	Complementary angles.	55
D9	A right triangle has sides 8 and 17 as leg and hypotenuse. What is cos of that angle, to three decimals?	`8/17`	Adjacent over hypotenuse.	0.471
D10	In a right triangle the acute angles are (2x + 10)° and (3x − 20)°. What is x?	`2x+10+3x-20=90`	The two acute angles total 90.	20

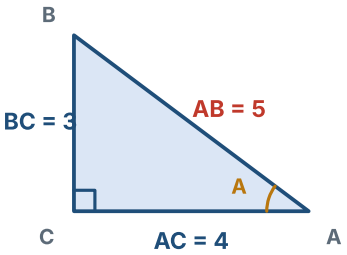
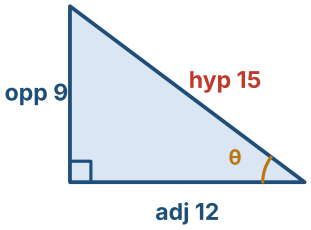
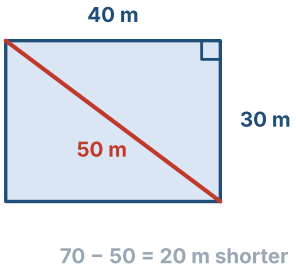
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Label opposite, adjacent and hypotenuse relative to the angle in the question *before* choosing a ratio. Opposite and adjacent swap when the angle does.

#	Level	Question	Answer	Solution
P1	Hard	A drone takes off from a stadium, flies 9 kilometres due east, then 12 kilometres due north. What is the straight-line distance, in kilometres, between the drone and the stadium?  	15	East and north are perpendicular, so this is a right triangle. √(81 + 144) = √225 = **15 km**. It is 3-4-5 tripled.

#	Level	Question	Answer	Solution
P2	Hard	A straight loading ramp forms a right triangle with the level ground. The sine of the angle between the ramp and the ground is 0.6, and the ramp is 20 feet long. How many feet above the ground is the top of the ramp? 	12	The ramp is the hypotenuse, and sin = opp/hyp. 0.6 × 20 = **12 feet**. A common wrong answer divides 20 by 0.6 × 2 or similar — the clue is that the opposite side must be shorter than the hypotenuse.
P3	Hard	In right triangle ABC, the right angle is at C and the hypotenuse AB = 13. If BC = 5, what is the length of AC? 	12	AB is the hypotenuse, so $5^2 + AC^2 = 13^2$. AC = √144 = **12**. This is the 5-12-13 triple.
P4	Hard	A 30-60-90 triangle has a hypotenuse of length 10. What is the length of the side opposite the 30° angle? 	5	In a 30-60-90 triangle the side opposite 30° is **half the hypotenuse**. 10/2 = **5**. A common wrong answer is the side opposite 60°.
P5	Hard	If sin 35° = cos x° and 0 < x < 90, what is the value of x? 	55	sin θ = cos(90° − θ). x = 90 − 35 = **55**.

#	Level	Question	Answer	Solution
P6	Hard	A 45-45-90 triangle has legs of length 5. What is the length of the hypotenuse? 	5√2	In a 45-45-90 triangle the sides are in the ratio 1 : 1 : √2. The hypotenuse is **5√2** ≈ 7.07. A common wrong answer doubles the leg instead.
P7	Very hard	In a right-triangle design on a banner, the two acute angles measure (2x + 10)° and (3x − 20)°, so sin((2x + 10)°) = cos((3x − 20)°). What is the value of x? 	20	The identity holds exactly when the two angles are complementary. 2x + 10 + 3x − 20 = 90 gives 5x = 100, so x = **20**. The angles are 50° and 40°.
P8	Very hard	A ladder 10 metres long leans against a wall and reaches 8 metres up it. What angle, to the nearest degree, does the ladder make with the ground? 	53°	The wall height is opposite the angle and the ladder is the hypotenuse: sin θ = 0.8. θ = $\sin^{-1}(0.8)$ ≈ **53°**. A common wrong answer is the angle at the *top* of the ladder, the complement.
P9	Very hard	A right triangle has legs of length 8 and 15. What is the length of the hypotenuse? 	17	64 + 225 = 289, and √289 = **17** — the 8-15-17 triple. A common wrong answer comes from subtracting instead of adding the squares.

#	Level	Question	Answer	Solution
P10	Very hard	In right triangle ABC the right angle is at C, and sin A = 3/5. What is the value of cos B? 	3/5	Angles A and B are complementary, and sin A = cos(90° − A) = cos B. cos B = **3/5**. A common wrong answer is cos A, which answers a different question.
P11	Very hard	A right triangle has an angle whose tangent is 3/4. If the side adjacent to that angle is 12, what is the length of the hypotenuse? 	15	tan = opp/adj = 3/4, so with adj = 12 the opposite is 9. This is 9-12-15, which is 3-4-5 tripled, so the hypotenuse is **15**. A common wrong answer is the opposite side, not the hypotenuse.
P12	Very hard	A rectangular field measures 40 metres by 30 metres. A path runs diagonally from one corner to the opposite corner. How much shorter, in metres, is the diagonal path than walking along two sides? 	20	The diagonal is √(1600 + 900) = √2500 = 50 m, and two sides total 70 m. 70 − 50 = **20 metres shorter**.

The one habit to build. Find the hypotenuse first and write it down. Your class scored 20% on Pythagoras, and the usual cause is putting a leg where c belongs — the hypotenuse is always opposite the right angle and always the longest side.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

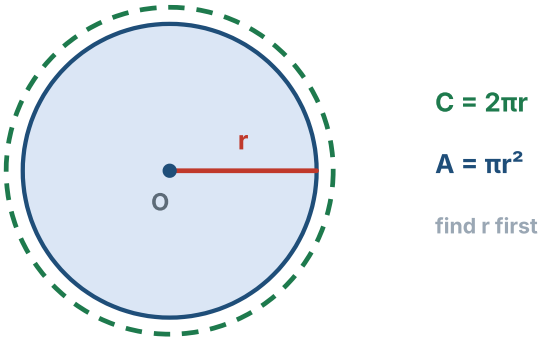
SAT Math 4.4 — Circles

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Circumference and area

$C = 2\pi r$ | $A = \pi r^2$ — both are on the reference sheet

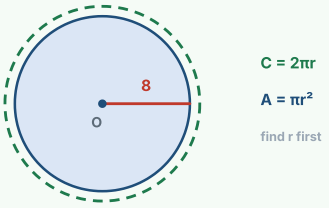
Almost every circle question starts by finding r. If you are given the area, take the square root; if the circumference, divide by 2π. Then answer what was asked.



The radius is the one number that unlocks both formulas: the green ring is the circumference and the shaded disc is the area.

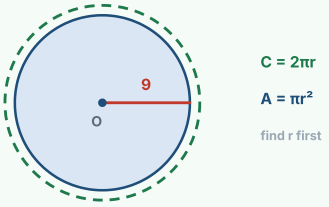
Example 1. **A circle has area 64π. Find its circumference.**

Solution.
$\pi r^2 = 64\pi$ gives $r = 8$.
$C = 2\pi(8) =$ **16π**.



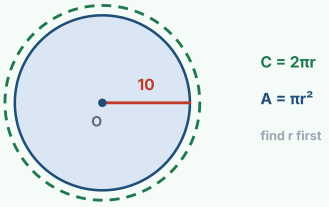
Example 2. **A circle has circumference 18π. Find its area.**

Solution.
$2\pi r = 18\pi$ gives $r = 9$.
$A = \pi(81) =$ **81π**.



Example 3. **A circular path encloses 100π square metres. How long is the path?**

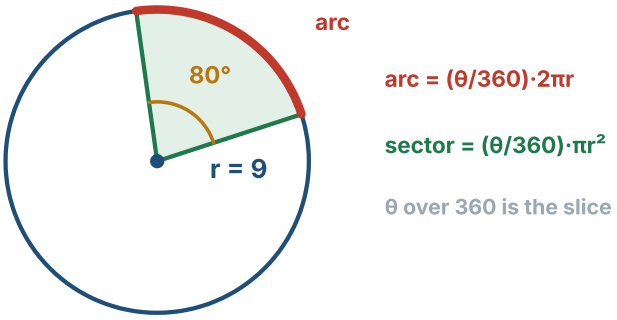
Solution.
$r = 10$, so the path is the circumference.
$2\pi(10) =$ **20π** metres.



2 Arcs and sectors

arc = (θ/360) × 2πr | sector area = (θ/360) × πr²

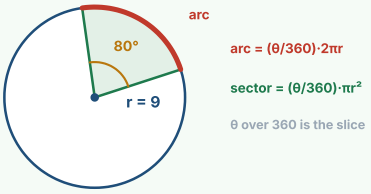
A sector is a fraction of the whole circle, and the fraction is the angle over 360. Work out the whole first, then take the slice.



The central angle picks the fraction: 80/360 = 2/9. Multiply the whole circumference by it for the arc, or the whole area for the sector.

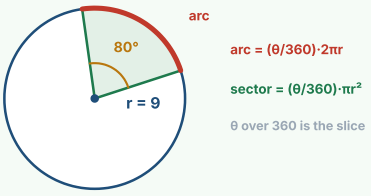
Example 1. **A circle of radius 9 has a sector with central angle 80°. Find its area.**

Solution.
Whole area = 81π, and 80/360 = 2/9.
(2/9)(81π) = **18π**.



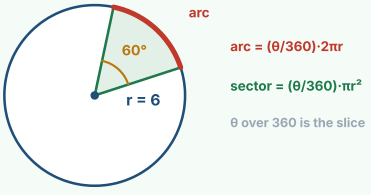
Example 2. **For that same circle, find the arc length of the sector.**

Solution.
Whole circumference = 18π.
(2/9)(18π) = **4π**.



Example 3. **A sector of a circle of radius 6 has area 6π. Find its central angle.**

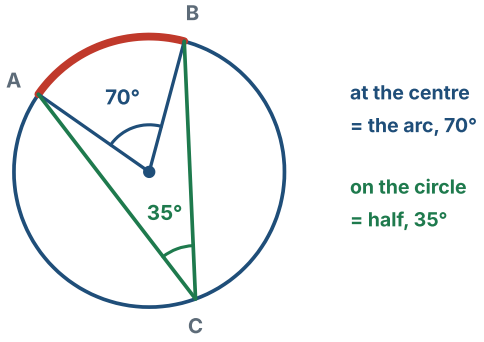
Solution.
Whole area 36π, and 6π/36π = 1/6.
(1/6)(360) = **60°**.

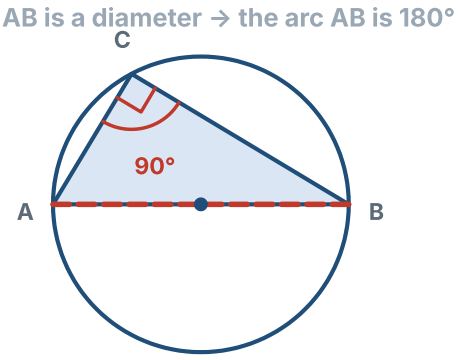


3 Central and inscribed angles

a central angle equals its arc | an inscribed angle is half its arc

The vertex position decides the rule: at the centre, the angle equals the arc; on the circle, it is half. The special case worth knowing cold is that an angle inscribed in a semicircle is a right angle.

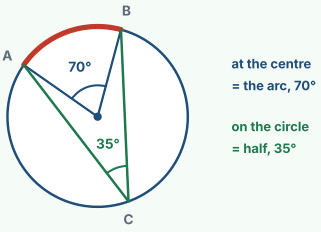




Both angles stand on the same red arc. The one at the centre is 70°; the one on the circle is 35°. When the arc is a semicircle (180°), half of it is a right angle.

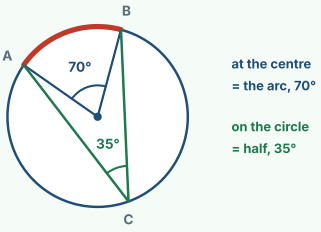
Example 1. **A central angle cuts an arc of 70°. What is the central angle?**

Solution.
70° — they are equal.



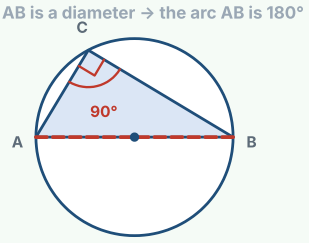
Example 2. **An inscribed angle cuts that same 70° arc. What is the inscribed angle?**

Solution.
Half the arc.
35°.



Example 3. **Triangle ABC is inscribed in a circle with AB a diameter. What is angle ACB?**

Solution.
The arc AB is 180°, and the inscribed angle is half.
90°.



4 Tangents and chords

a tangent is perpendicular to the radius at the point of contact

A tangent gives you a right angle for free, which usually means Pythagoras is next. A radius drawn perpendicular to a chord also bisects it.



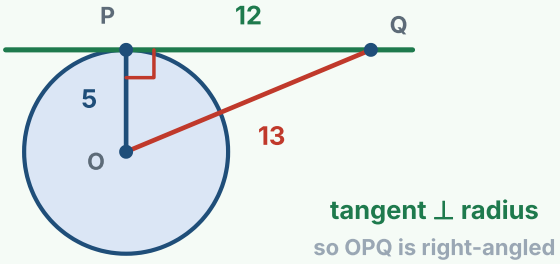
Left: the radius OP meets the tangent at 90°, so OPQ is a right triangle with legs 5 and 12. Right: the perpendicular from the centre cuts the chord in half, giving legs 8 and d with hypotenuse 10.

Example 1. **A tangent touches a circle of radius 5 at P. From an external point Q, PQ = 12. Find OQ.**

Solution.

Angle OPQ is 90°, so $OQ^2 = 25 + 144$.

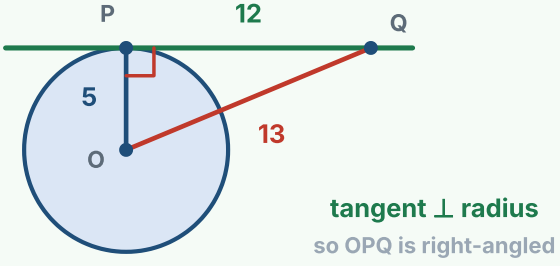
OQ = **13**.



Example 2. **Why is triangle OPQ right-angled?**

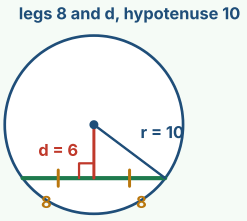
Solution.

OP is a radius to the point of contact, and a **tangent is perpendicular to the radius** there.



Example 3. **A chord of length 16 sits in a circle of radius 10. How far is it from the centre?**

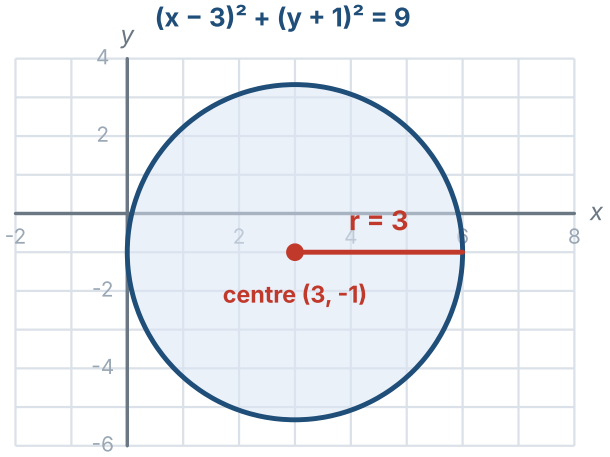
Solution.
The perpendicular from the centre bisects the chord, giving legs 8 and d with hypotenuse 10.
d = √(100 − 64) = **6**.



5 The equation of a circle

$(x - h)^2 + (y - k)^2 = r^2$, centre (h, k), radius r

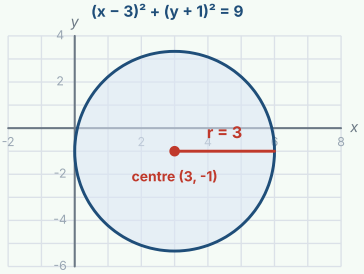
Read the centre with the signs flipped and remember the right-hand side is r squared, not r. This formula is *not* on the reference sheet.



$(x - 3)^2 + (y + 1)^2 = 9$ sits at (3, −1): the x term reads −3 so h = +3, the y term reads +1 so k = −1. And 9 is r^2, so r = 3.

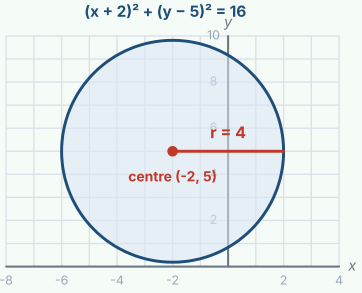
Example 1. **Find the centre and radius of $(x - 3)^2 + (y + 1)^2 = 9$.**

Solution.
Centre **(3, −1)**; radius **3**, because $9 = 3^2$.



Example 2. **Write the equation of a circle with centre (−2, 5) and radius 4.**

Solution.
$(x + 2)^2 + (y - 5)^2 = 16$.

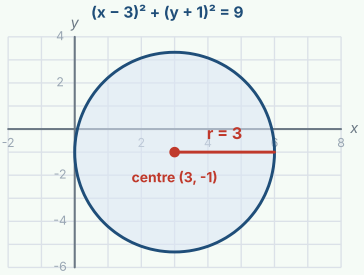


Example 3. **Does the point (6, −1) lie on the circle in the first example?**

Solution.

$(6 - 3)^2 + (-1 + 1)^2 = 9 + 0 = 9$.

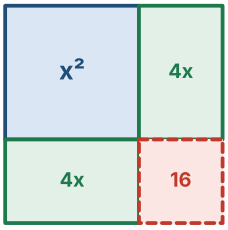
Yes, it is exactly on the circle.



6 Completing the square for a circle

group the x terms and the y terms, then halve and square each coefficient

An equation given as $x^2 + y^2 + Dx + Ey = F$ has to be rearranged before you can read anything off. Complete the square twice, once for x and once for y.



$x^2 + 8x$

$= (x + 4)^2 - 16$

halve 8, square it, then subtract it back

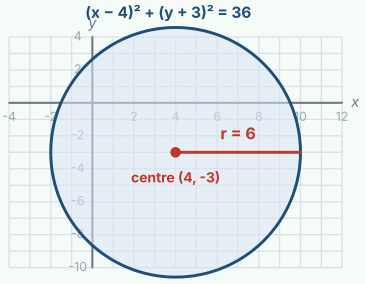
The square of side x plus the two strips of width 4 almost make a square of side x + 4. The dashed 16 is the missing corner — add it to both sides and the left-hand side becomes $(x + 4)^2$.

Example 1. **Find the radius of $x^2 + y^2 - 8x + 6y = 11$.**

Solution.

$(x - 4)^2 - 16 + (y + 3)^2 - 9 = 11$.

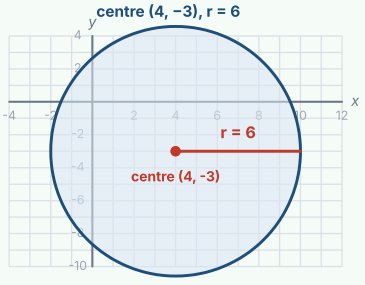
$(x - 4)^2 + (y + 3)^2 = 36$, so r = **6**.



Example 2. **Find its centre.**

Solution.

(4, −3) — the signs flip.

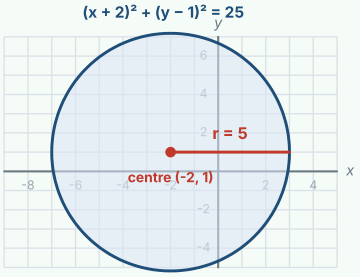


Example 3. **Find the radius of $x^2 + y^2 + 4x - 2y = 20$.**

Solution.

$(x + 2)^2 + (y - 1)^2 = 20 + 4 + 1 = 25$.

$r = \mathbf{5}$.



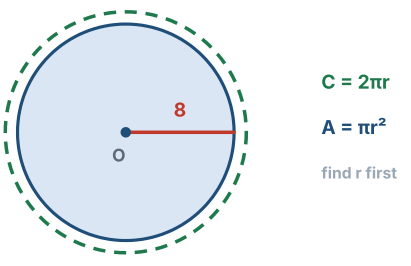
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

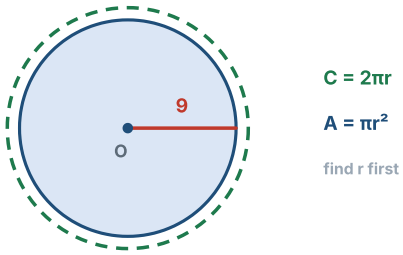
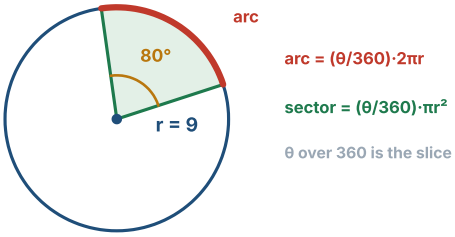
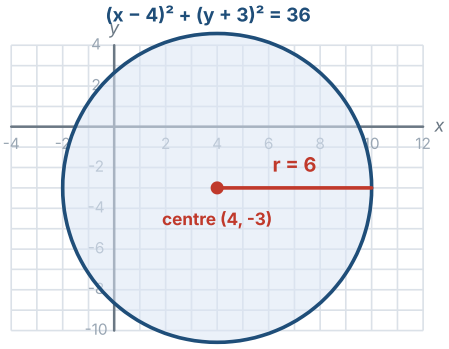
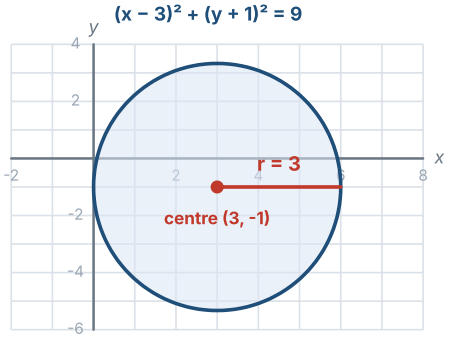
The SAT gives you Desmos on every maths question. Here it is **a circle grapher**. Type the equation in any form and Desmos draws it, centre and radius included — a fast check on your completing the square.

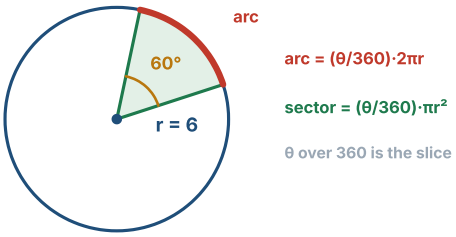
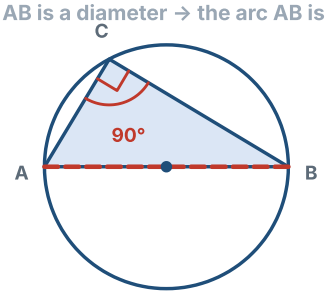
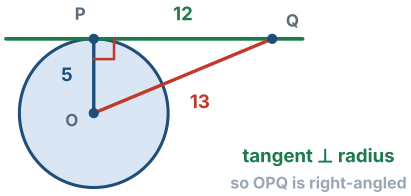
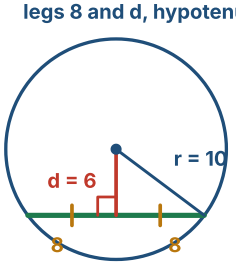
#	Question	Type into Desmos	Then	Answer
D1	A circle has area 64π. What is its circumference as a multiple of π?	`2*sqrt(64)`	Radius first, then the circumference.	16
D2	A circle has circumference 18π. What is its area as a multiple of π?	`(18/2)^2`	r = 9, then r squared.	81
D3	A sector of a circle of radius 9 has central angle 80°. What is its area as a multiple of π?	`80/360*81`	Fraction of the whole.	18
D4	For that same sector, what is the arc length as a multiple of π?	`80/360*18`	Same fraction, applied to the circumference.	4
D5	A sector of a circle of radius 6 has area 6π. What is its central angle in degrees?	`6/36*360`	The ratio of areas gives the ratio of angles.	60
D6	A tangent meets a circle of radius 5 at P, and PQ = 12. What is OQ?	`sqrt(5^2+12^2)`	The tangent makes a right angle with the radius.	13
D7	A chord of length 16 is in a circle of radius 10. How far is it from the centre?	`sqrt(10^2-8^2)`	Half the chord is one leg.	6
D8	What is the radius of $x^2 + y^2 - 8x + 6y = 11$?	`sqrt(11+16+9)`	Add the completing-the-square constants to both sides.	6
D9	What is the radius of $x^2 + y^2 + 4x - 2y = 20$?	`sqrt(20+4+1)`	Same method, different numbers.	5
D10	An inscribed angle cuts an arc of 70°. What is the inscribed angle?	`70/2`	Half the arc.	35

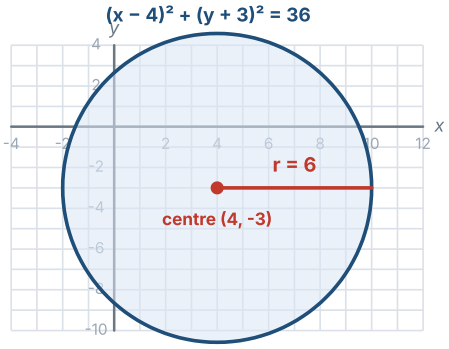
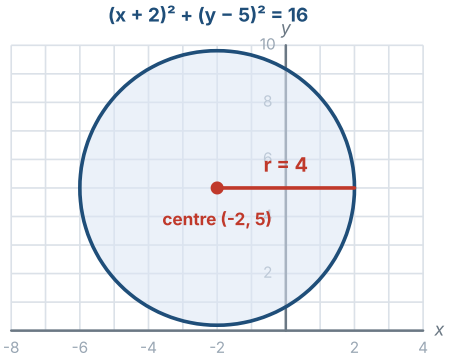
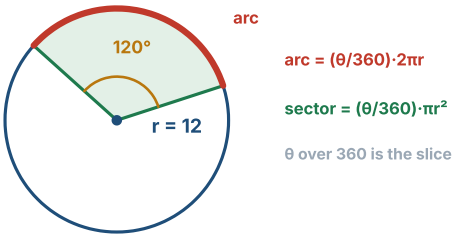
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Find the radius first in almost every question, then answer what was actually asked: circumference, area, arc or sector.

#	Level	Question	Answer	Solution
P1	Hard	The area enclosed by a circular jogging path is 64π square metres. What is the length of the path, in metres? 	16π	$\pi r^2 = 64\pi$ gives r = 8, and the path is the circumference. $2\pi(8) = \mathbf{16\pi}$. A common wrong answer is the radius times π, a common slip.

#	Level	Question	Answer	Solution
P2	Hard	A round cheese wheel has a circumference of 18π centimetres. What is the area, in square centimetres, of its circular top face? 	81π	2πr = 18π gives r = 9. A = π(9)² = **81π**. A common wrong answer squares the diameter instead of the radius.
P3	Hard	A wedge cut from a circular cheese of radius 9 inches has a central angle of 80°. What is the area, in square inches, of the top face of the wedge? 	18π	The whole area is 81π and the wedge is 80/360 = 2/9 of it. (2/9)(81π) = **18π**.
P4	Hard	What is the radius of the circle given by $x^2 + y^2 - 8x + 6y = 11$? 	6	Complete the square twice: $(x - 4)^2 + (y + 3)^2 = 11 + 16 + 9 = 36$. r = √36 = **6**. Answering 36 gives r^2, not r.
P5	Hard	A circle in the xy-plane has equation $(x - 3)^2 + (y + 1)^2 = 9$. What are the coordinates of its centre? 	(3, −1)	The form is $(x - h)^2 + (y - k)^2 = r^2$, so h = 3 and k = −1. **(3, −1)** — the signs flip. A common wrong answer flips them the wrong way.

#	Level	Question	Answer	Solution
P6	Hard	A sector of a circle of radius 6 has an area of 6π. What is its central angle, in degrees? 	60	The whole circle has area 36π, and 6π/36π = 1/6. (1/6)(360) = **60°**.
P7	Very hard	In a circle with centre O, AB is a diameter and C is a point on the circle distinct from A and B. What is the measure of angle ACB, in degrees? 	90	Angle ACB is inscribed in a semicircle, so it cuts an arc of 180° and is half of it. **90°** wherever C sits — which is exactly why any other answer is wrong.
P8	Very hard	A tangent touches a circle with centre O and radius 5 at point P. An external point Q lies on the tangent with PQ = 12. What is the length of OQ? 	13	The tangent is perpendicular to the radius OP, so triangle OPQ has a right angle at P. OQ = √(25 + 144) = **13**, the 5-12-13 triple.
P9	Very hard	A chord of length 16 is drawn in a circle of radius 10. What is the distance from the centre of the circle to the chord? 	6	The perpendicular from the centre bisects the chord, giving a right triangle with hypotenuse 10 and one leg 8. d = √(100 − 64) = **6**. A common wrong answer is half the chord itself.

#	Level	Question	Answer	Solution
P10	Very hard	The circular fountain plaza at the centre of a market square is modelled by $x^2 + y^2 - 8x + 6y = 11$, with units in metres. What is the radius, in metres? 	6	Completing the square gives $(x - 4)^2 + (y + 3)^2 = 36$. r = **6**. A common wrong answer is r^2, and C is the number on the right of the original equation — both are answers to steps along the way, not the question.
P11	Very hard	A circle has centre (−2, 5) and radius 4. Write the equation of this circle. 	$(x + 2)^2 + (y - 5)^2 = 16$	Signs flip inside the brackets and the right-hand side is r^2. **$(x + 2)^2 + (y - 5)^2 = 16$.** A common wrong answer uses r instead of r^2; another gets both signs wrong.
P12	Very hard	A circle of radius 12 centimetres has a sector whose arc length is 8π centimetres. What is the area of that sector, in square centimetres? 	48π	The full circumference is 24π, so the sector is 8π/24π = 1/3 of the circle. The full area is 144π, and (1/3)(144π) = **48π**.

The one habit to build. The right-hand side of a circle equation is r^2, not r. Your class scored 47% on the circle equation, and answering 36 instead of 6 was the single most common slip.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

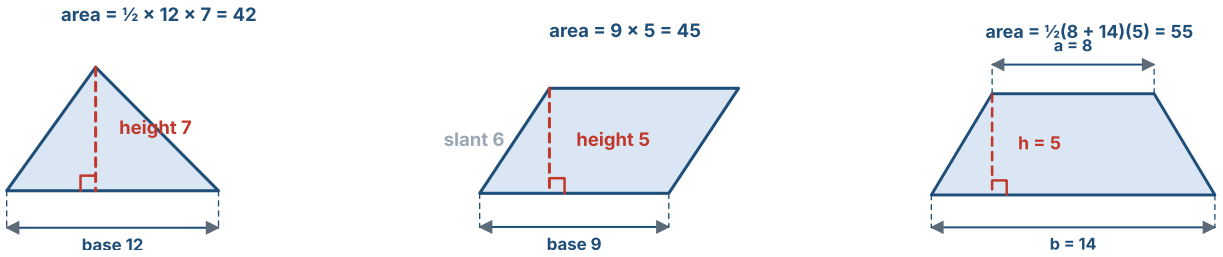
SAT Math 4.5 — Area & Volume

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Areas of plane figures

triangle ½bh | parallelogram bh | trapezoid ½(a + b)h | circle πr²

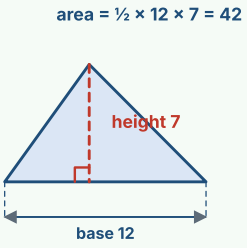
The height is always the *perpendicular* distance, not the slanted side. A triangle drawn leaning over still uses the vertical height.



In all three the red dashed line is the height: it meets the base at a right angle. The slanted side of 6 in the parallelogram is a distraction — the height is 5.

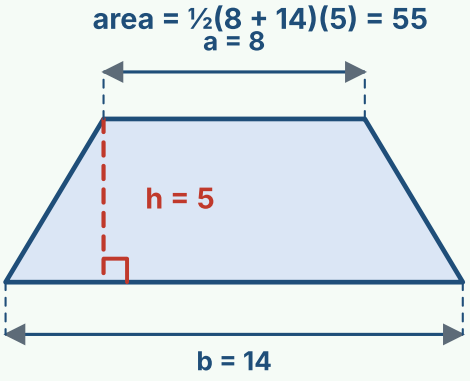
Example 1. **A triangle has base 12 and perpendicular height 7. Find its area.**

Solution.
½(12)(7) = **42**.



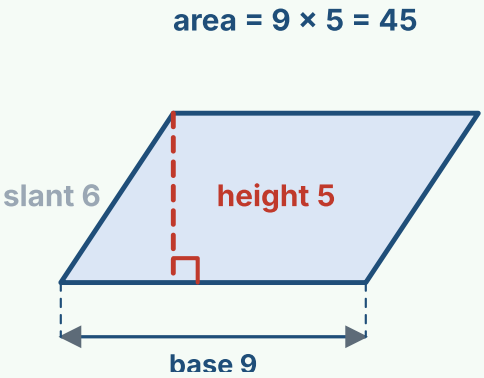
Example 2. **A trapezoid has parallel sides 8 and 14 and height 5. Find its area.**

Solution.
½(8 + 14)(5) = ½(22)(5) = **55**.



Example 3. **A parallelogram has base 9 and slant side 6, with perpendicular height 5. Find its area.**

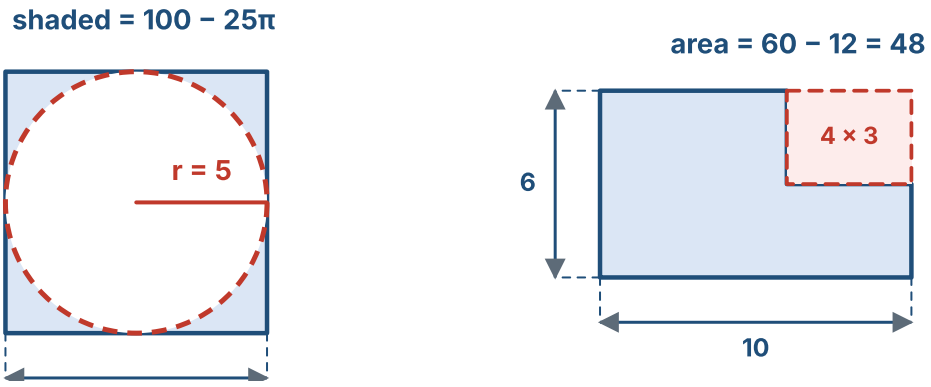
Solution.
Use the perpendicular height, not the slant.
9 × 5 = **45**.



2 Composite figures

split the shape into pieces you know, or subtract a hole from a whole

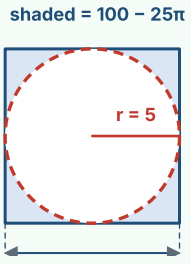
Decide whether adding or subtracting is quicker, then label every piece before calculating. A shaded region is nearly always one area minus another.



Left: the shaded part is the square minus the circular hole. Right: the L-shape is the full rectangle minus the dashed corner that was removed.

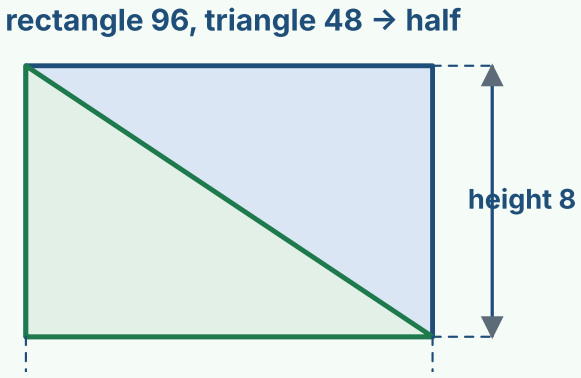
Example 1. **A square of side 10 has a circle of radius 5 cut from it. Find the shaded area left over.**

Solution.
100 − 25π ≈ 100 − 78.54 = **21.46**.



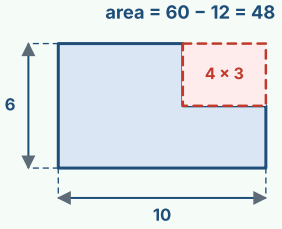
Example 2. **A rectangle 12 by 8 has a triangle of base 12 and height 8 removed. What fraction remains?**

Solution.
Rectangle 96, triangle 48.
Half remains.



Example 3. **An L-shape is a 10 by 6 rectangle with a 4 by 3 rectangle removed from one corner. Find its area.**

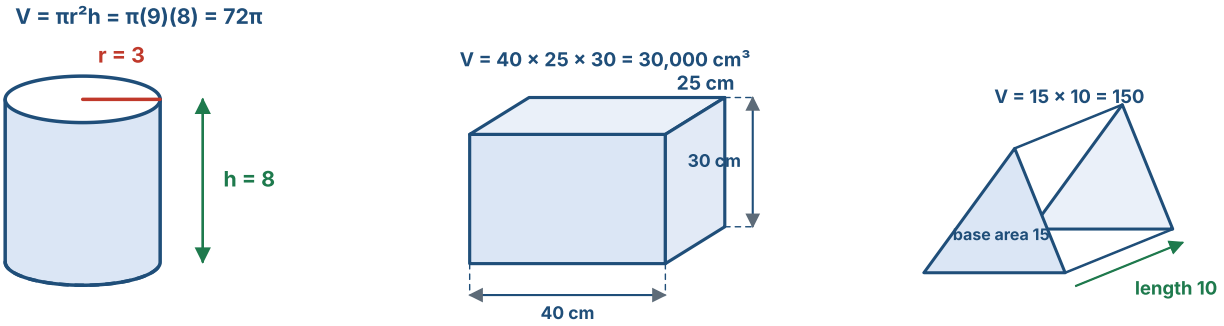
Solution.
60 − 12 = **48**.



3 Volume of prisms and cylinders

V = (area of the base) × height | cylinder V = πr²h

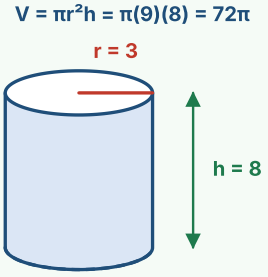
Any solid with the same cross-section all the way up follows one rule: find the base area, multiply by the height. The reference sheet gives these, so the work is choosing the base.



Each solid has one cross-section repeated along its length. Find that face's area — a circle, a rectangle, a triangle — then multiply by how far it travels.

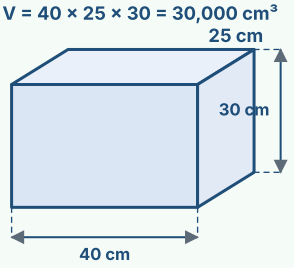
Example 1. **A cylinder has radius 3 and height 8. Find its volume.**

Solution.
π(9)(8) = **72π** ≈ 226.19.



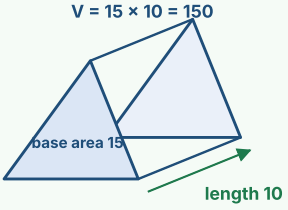
Example 2. **A rectangular tank measures 40 by 25 by 30 centimetres. Find its volume.**

Solution.
40 × 25 × 30 = **30,000 cm³**, which is 30 litres.



Example 3. **A triangular prism has a base triangle of area 15 and length 10. Find its volume.**

Solution.
15 × 10 = **150**.

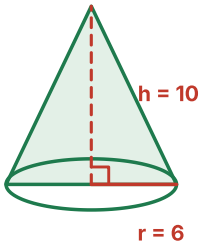


4 Cones, pyramids and spheres

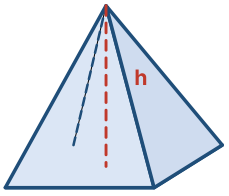
cone $\frac{1}{3}\pi r^2 h$ | pyramid $\frac{1}{3}Bh$ | sphere $\frac{4}{3}\pi r^3$

The one-third is the whole difficulty: a cone is a third of the cylinder that would contain it. All three formulas are on the reference sheet, so check you have copied the fraction.

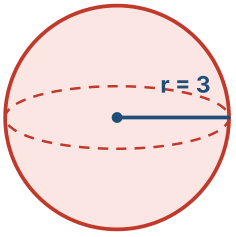
$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(10) = 120\pi$



$V = \frac{1}{3} \times$ base area $\times h$



$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(27) = 36\pi$

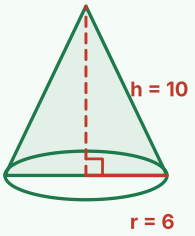


The cone and the pyramid both carry the one third; the sphere carries four thirds and cubes the radius. Copy the fraction before you substitute.

Example 1. **A cone has radius 6 and height 10. Find its volume.**

Solution.
$(1/3)\pi(36)(10) = \mathbf{120\pi}$.

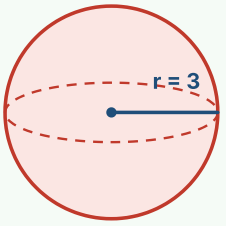
$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(10) = 120\pi$



Example 2. **A sphere has radius 3. Find its volume.**

Solution.
$(4/3)\pi(27) = \mathbf{36\pi}$.

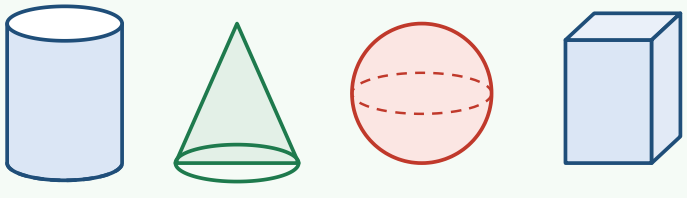
$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(27) = 36\pi$



Example 3. **A cylinder and a cone share the same radius and height. How do their volumes compare?**

Solution.
The cone is exactly **one third** of the cylinder.

all four are on the reference sheet

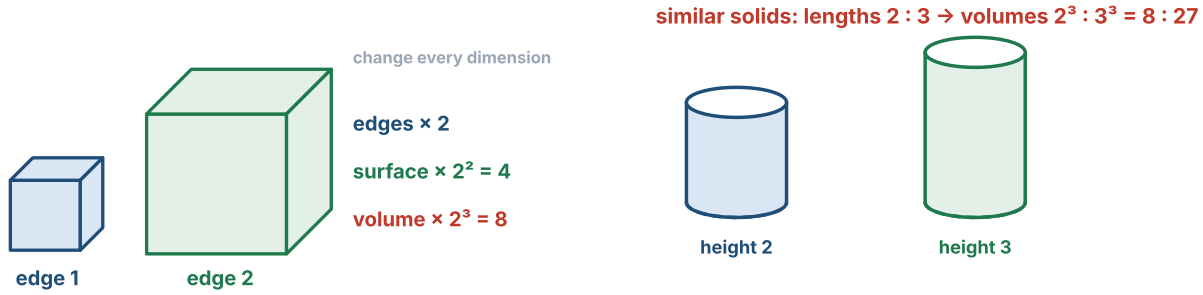


$V = \pi r^2 h$ $V = \frac{1}{3}\pi r^2 h$ $V = \frac{4}{3}\pi r^3$ $V = Bh$

5 Scaling a solid

lengths × k | surface area × k^2 | volume × k^3

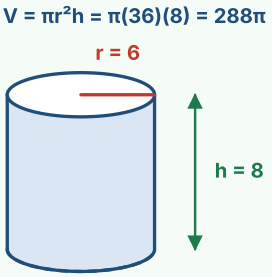
Changing one dimension multiplies the volume by that factor alone. Changing every dimension by k multiplies the volume by k cubed. Read carefully which the question means.



Left: every edge doubles, so the volume is eight times as large. Right: two similar cans in the ratio 2 : 3 have volumes in the ratio 8 : 27.

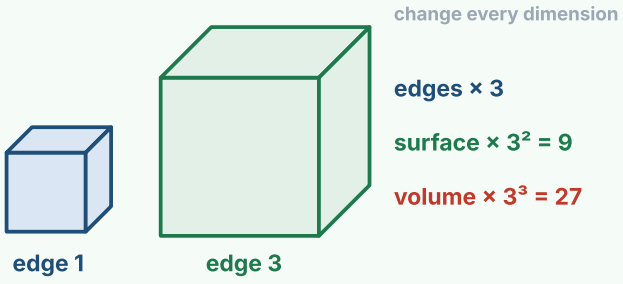
Example 1. **A cylinder's radius is doubled and its height is unchanged. What happens to the volume?**

Solution.
V depends on r^2.
The volume is multiplied by **4**.



Example 2. **A cube's edges are all tripled. What happens to its volume?**

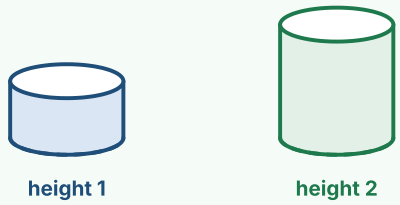
Solution.
Every dimension triples.
Volume × 3^3 = **27**.



Example 3. **A cylinder's height alone is doubled. What happens to the volume?**

Solution.
Only one dimension changes.
Volume **doubles**.

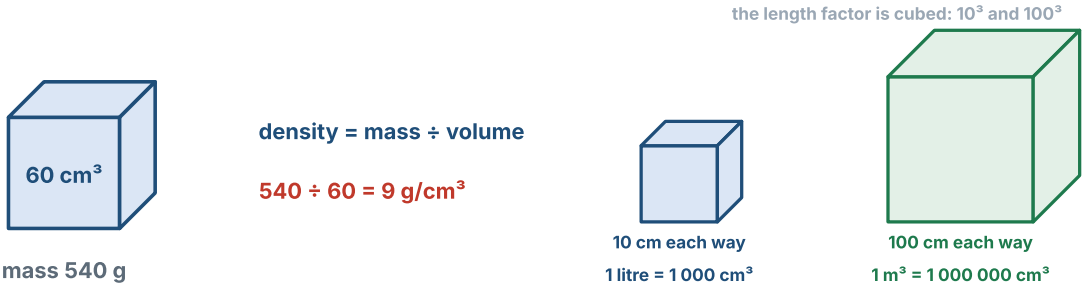
similar solids: lengths $1:2 \rightarrow$ volumes $1^3:2^3 = 1:8$



6 Density and unit conversion

density = mass ÷ volume | for volume the conversion factor is cubed

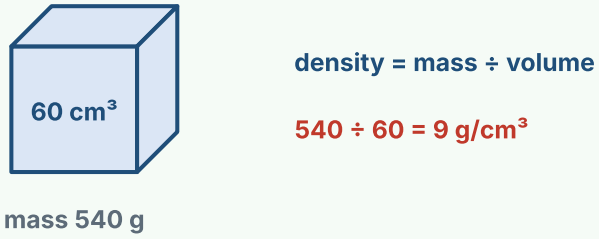
Density questions are a rate in disguise. For units, remember 1 litre = 1000 cm^3 while 1 m^3 = 1,000,000 cm^3 — the two are not the same factor.



A litre is a 10 cm cube, so 10^3 = 1 000 cm^3. A cubic metre is a 100 cm cube, so 100^3 = 1 000 000 cm^3. Using one factor for the other is the classic slip.

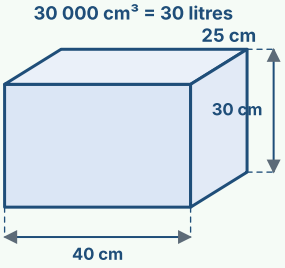
Example 1. **A block has mass 540 g and volume 60 cm^3. Find its density.**

Solution.
540/60 = **9 g/cm^3**.



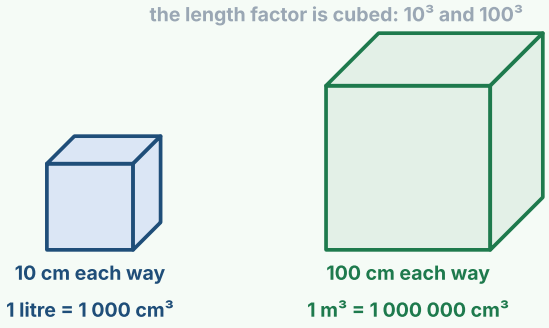
Example 2. **A tank holds 30,000 cm³. How many litres is that?**

Solution.
30,000 ÷ 1000 = **30 litres**.



Example 3. **How many cubic centimetres are in 1 cubic metre?**

Solution.
100^3 = **1,000,000**. The length factor 100 gets cubed.



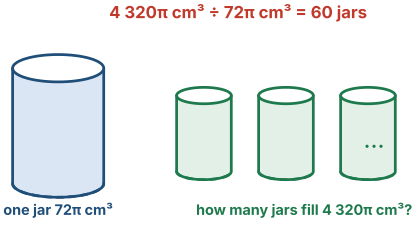
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

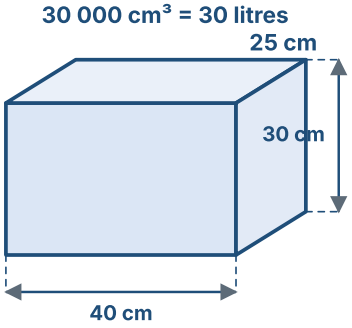
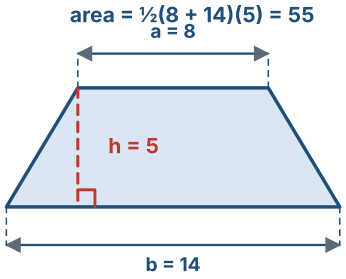
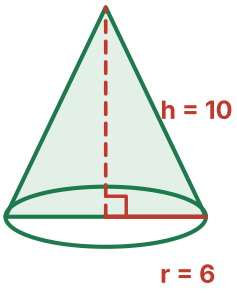
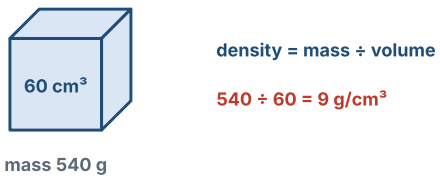
The SAT gives you Desmos on every maths question. Here it is **an arithmetic engine for formulas**. Type the whole formula with its numbers in one line so the 1/3 or the ½ cannot go missing.

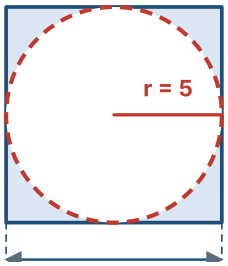
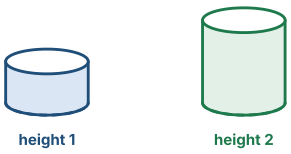
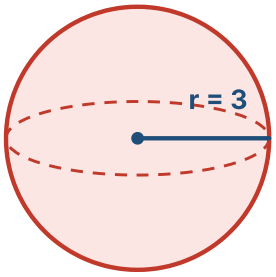
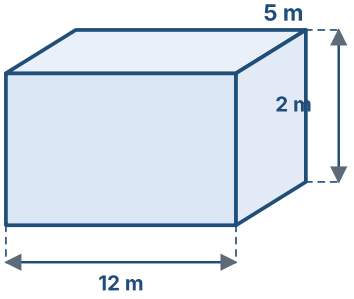
#	Question	Type into Desmos	Then	Answer
D1	A triangle has base 12 and height 7. What is its area?	`0.5*12*7`	Half base times height.	42
D2	A trapezoid has parallel sides 8 and 14 and height 5. What is its area?	`0.5*(8+14)*5`	Average the parallel sides, then multiply by the height.	55
D3	A cylinder has radius 3 and height 8. What is its volume as a multiple of π?	`3^2*8`	r squared times h.	72
D4	A cone has radius 6 and height 10. What is its volume as a multiple of π?	`(1/3)*6^2*10`	Do not forget the one third.	120
D5	A sphere has radius 3. What is its volume as a multiple of π?	`(4/3)*3^3`	Four thirds times r cubed.	36
D6	A tank measures 40 by 25 by 30 cm. How many litres does it hold?	`40*25*30/1000`	Divide by 1000, not by a million.	30
D7	A block has mass 540 g and volume 60 cm³. What is its density in g/cm³?	`540/60`	Mass over volume.	9
D8	A square of side 10 has a circle of radius 5 removed. What area remains, to two decimals?	`100-pi*25`	Subtract the hole from the whole.	21.46
D9	A cylinder's radius is doubled and its height is unchanged. By what factor does the volume grow?	`2^2`	Volume depends on r squared.	4
D10	How many cubic centimetres are in one cubic metre?	`100^3`	The length factor is cubed.	1000000

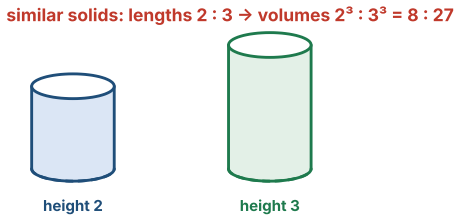
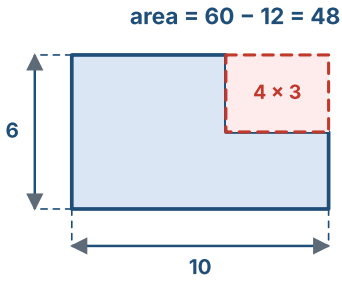
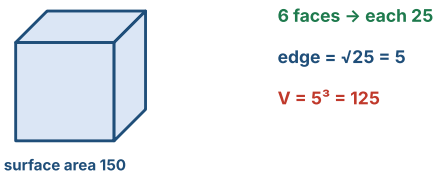
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Write the formula down before substituting. Most lost marks in this topic are a correct calculation of the wrong formula.

#	Level	Question	Answer	Solution
P1	Hard	A market supplier pours 4,320π cubic centimetres of olive oil into identical cylindrical jars. Each jar has an inside radius of 3 centimetres and is filled to a height of 8 centimetres. How many jars does the supplier fill? 	60	Each jar holds π(9)(8) = 72π cm³. 4320π ÷ 72π = **60 jars**. A common wrong answer is the volume of one jar in multiples of π, not a count.

#	Level	Question	Answer	Solution
P2	Hard	An oil tin is a rectangular prism measuring 40 cm by 25 cm by 30 cm and is completely full. How many litres of oil does it hold? (1 litre = 1000 cubic centimetres) 	30	40 × 25 × 30 = 30,000 cm³. 30,000 ÷ 1000 = **30 litres**. Dividing by 1,000,000 gives 0.03 — that factor converts to cubic *metres*, and eight of your class used it on Test 5.
P3	Hard	A trapezoid has parallel sides of 8 and 14 and a perpendicular height of 5. What is its area? 	55	Area = ½(a + b)h = ½(22)(5) = **55**. A common wrong answer forgets the one half; another averages the sides but omits the height.
P4	Hard	A cone has a radius of 6 and a height of 10. What is its volume? $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(10) = 120\pi$ 	120π	V = (1/3)πr²h = (1/3)π(36)(10). = **120π**. A common wrong answer is the cylinder's volume — the one third was dropped.
P5	Hard	A metal block has a mass of 540 grams and a volume of 60 cubic centimetres. What is its density, in grams per cubic centimetre? 	9	Density = mass ÷ volume = 540/60 = **9 g/cm³**. A common wrong answer inverts the fraction; another multiplies instead of dividing.

#	Level	Question	Answer	Solution
P6	Hard	A square of side 10 centimetres has a circle of radius 5 centimetres cut out of it. What is the area of the remaining region, to the nearest hundredth of a square centimetre? **shaded = 100 − 25π** 	21.46	100 − π(25) = 100 − 78.5398… = **21.46 cm²**.
P7	Very hard	A cylinder's radius is doubled while its height stays the same. By what factor is its volume multiplied? similar solids: lengths 1 : 2 → volumes $1^3 : 2^3 = 1 : 8$ 	4	V = πr²h, and only r changes, so the factor is 2^2. The volume is **4 times** as large. A common wrong answer would be right if *every* dimension doubled.
P8	Very hard	A sphere and a cylinder both have radius 3. The cylinder has height 3. What is the ratio of the sphere's volume to the cylinder's? $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(27) = 36\pi$ 	4 : 3	Sphere = (4/3)π(27) = 36π; cylinder = π(9)(3) = 27π. 36 : 27 = **4 : 3**. A common wrong answer inverts the ratio.
P9	Very hard	A rectangular swimming pool is 12 metres long, 5 metres wide and 2 metres deep. Water is pumped in at 40 litres per minute. How many minutes does it take to fill? **120 m³ = 120 000 litres** 	3000	Volume = 120 m³, and 1 m³ = 1000 litres, so 120,000 litres. 120,000 ÷ 40 = **3,000 minutes**. Using 1 m³ = 1,000,000 cm³ and then forgetting to convert to litres is the usual slip here.

#	Level	Question	Answer	Solution
P10	Very hard	Two similar cylindrical cans have heights in the ratio 2 : 3. What is the ratio of their volumes? 	8 : 27	Similar solids scale by k in length, so volumes scale by k^3. $2^3 : 3^3$ = **8 : 27**. A common wrong answer squares instead of cubing, which is the surface-area rule.
P11	Very hard	An L-shaped floor is formed by removing a 4 m by 3 m rectangle from one corner of a 10 m by 6 m rectangle. Tiles cost $12 per square metre. What is the total cost of tiling the floor? 	$576	Area = 60 − 12 = 48 m². 48 × 12 = **$576**. A common wrong answer tiles only the 12 m² corner that was removed.
P12	Very hard	A cube has a surface area of 150 square centimetres. What is its volume, in cubic centimetres? 	125	A cube has 6 faces, so each face has area 150/6 = 25 and the edge is 5. $V = 5^3$ = **125**. A common wrong answer is the area of one face, not the volume.

The one habit to build. Litres and cubic metres use different conversion factors: 1 litre = 1,000 cm^3 but 1 m^3 = 1,000,000 cm^3. Eight of thirteen of your class used the wrong one on Test 5 and turned 30 litres into 0.03.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

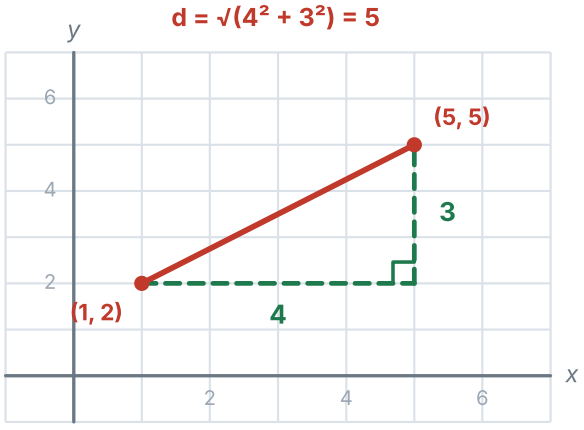
SAT Math 4.6 — Coordinate Geometry

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Distance between two points

$$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]} \text{ — it is Pythagoras in disguise}$$

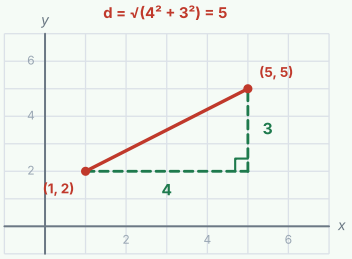
Draw the horizontal and vertical legs and the formula becomes obvious. Because both differences are squared, the order of subtraction does not matter.



The green legs are just the differences in x and in y: 4 across and 3 up. The red segment is the hypotenuse of that right triangle, so d = 5.

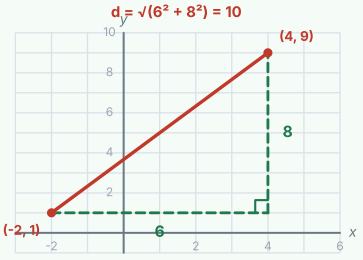
Example 1. **Find the distance between (1, 2) and (5, 5).**

Solution.
Legs 4 and 3.
√(16 + 9) = **5**.



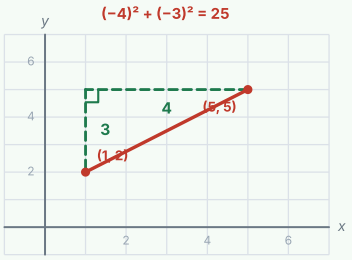
Example 2. **Find the distance between (−2, 1) and (4, 9).**

Solution.
Legs 6 and 8.
√(36 + 64) = **10**.



Example 3. **Why does subtracting the other way round give the same answer?**

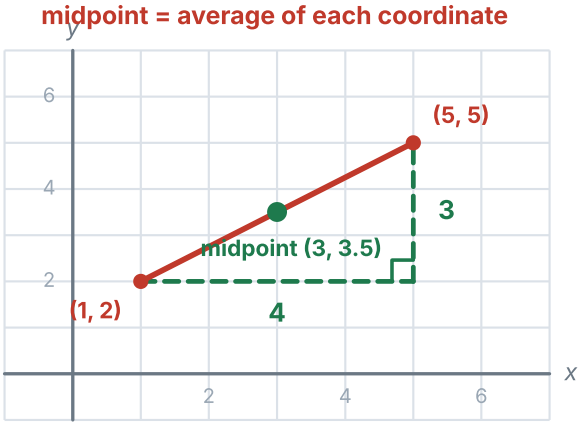
Solution.
Squaring removes the sign: $(-4)^2 = 4^2$.



2 Midpoint

midpoint = $((x_1 + x_2)/2, (y_1 + y_2)/2)$ — average each coordinate

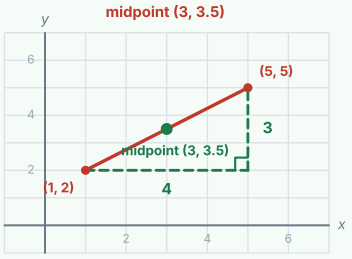
Average the x's and average the y's, separately. If you are given one endpoint and the midpoint, work backwards: the midpoint is as far past itself again.



The midpoint sits halfway along in both directions: x goes 1 → 3 → 5 and y goes 2 → 3.5 → 5.

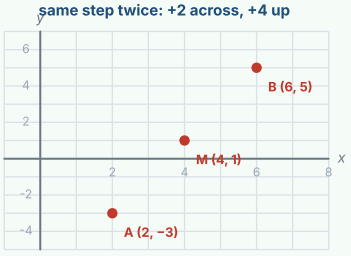
Example 1. **Find the midpoint of (1, 2) and (5, 5).**

Solution.
((1+5)/2, (2+5)/2) = **(3, 3.5)**.



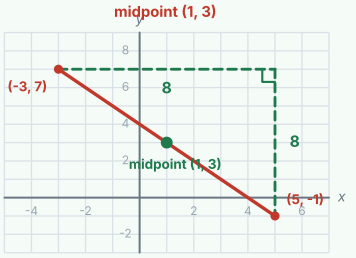
Example 2. **The midpoint of AB is (4, 1) and A is (2, −3). Find B.**

Solution.
B is the same step again: (4 + 2, 1 + 4) = **(6, 5)**.



Example 3. **Find the midpoint of (−3, 7) and (5, −1).**

Solution.
((−3+5)/2, (7−1)/2) = **(1, 3)**.

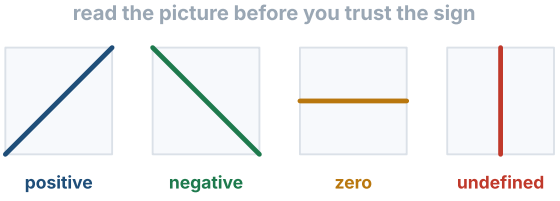


3 Slope

$m = (y_2 - y_1)/(x_2 - x_1)$ — rise over run

Keep the two points in the same order on top and bottom. A positive slope rises left to right; a horizontal line has slope 0 and a vertical line has no slope at all.

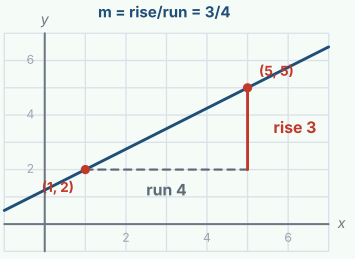




Rise 3 over run 4 gives m = 3/4. The four small panels show the only four possibilities: up, down, flat (0) and vertical (undefined).

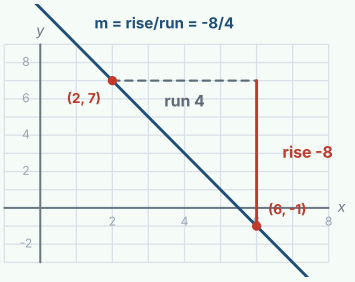
Example 1. **Find the slope through (1, 2) and (5, 5).**

Solution.
(5 − 2)/(5 − 1) = **3/4**.



Example 2. **Find the slope through (2, 7) and (6, −1).**

Solution.
(−1 − 7)/(6 − 2) = −8/4 = **−2**.



Example 3. **What is the slope of the line through (3, 4) and (3, 9)?**

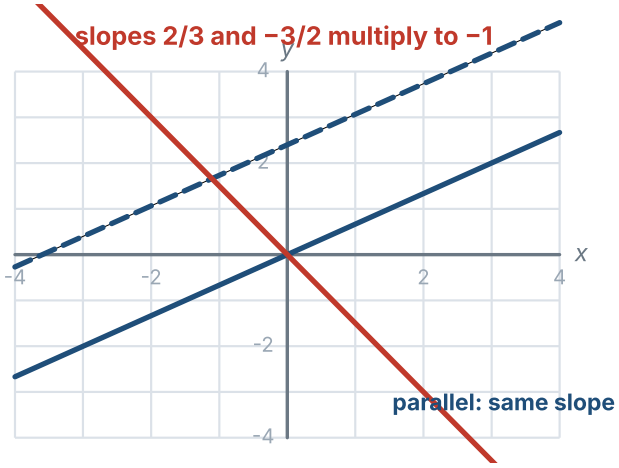
Solution.
The run is 0, so the slope is **undefined** — the line is vertical.



4 Parallel and perpendicular

parallel → equal slopes | perpendicular → slopes multiply to −1 (negative reciprocals)

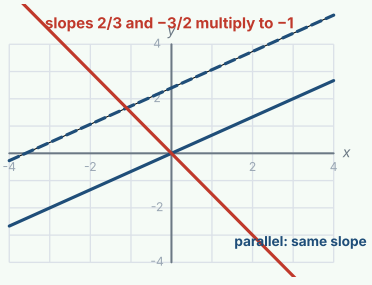
To get a perpendicular slope, flip the fraction and change the sign. The perpendicular to a slope of 2/3 is −3/2.



The two blue lines never meet — same slope 2/3. The red line crosses them at a right angle, and its slope −3/2 is the fraction flipped and negated.

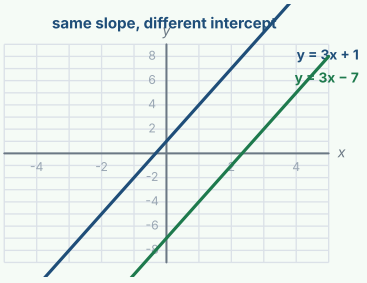
Example 1. **A line has slope 2/3. What is the slope of a line perpendicular to it?**

Solution.
Flip and negate.
−3/2.



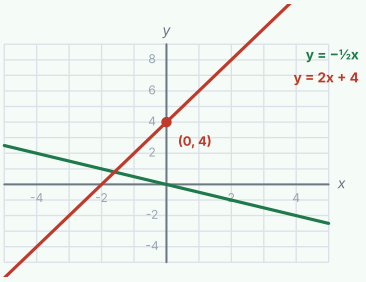
Example 2. **Is y = 3x + 1 parallel to y = 3x − 7?**

Solution.
Both have slope 3.
Yes — parallel, and never meeting.



Example 3. **A line through (0, 4) is perpendicular to $y = -\frac{1}{2}x$. Write its equation.**

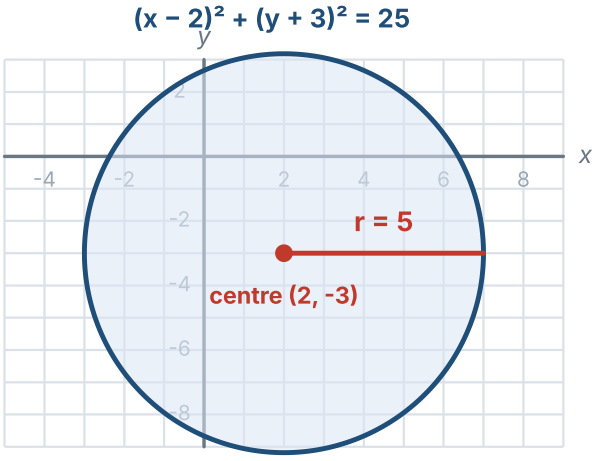
Solution.
The perpendicular slope is 2.
$y = 2x + 4$.



5 The equation of a circle in the plane

$$(x - h)^2 + (y - k)^2 = r^2$$

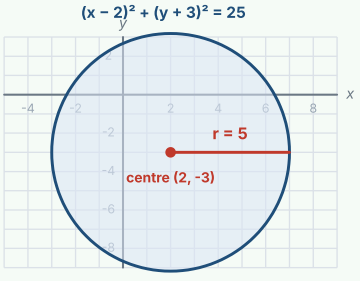
This is coordinate geometry's one memorised formula, and it is not on the reference sheet. The centre's coordinates appear with their signs flipped.



Centre (2, −3) with radius 5. In the equation the 2 appears as − 2 and the −3 appears as + 3, and the right-hand side is $5^2 = 25$.

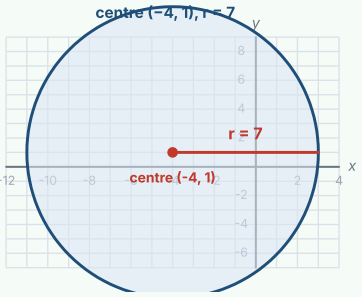
Example 1. **A circle has centre (2, −3) and radius 5. Write its equation.**

Solution.
$(x - 2)^2 + (y + 3)^2 = 25$.



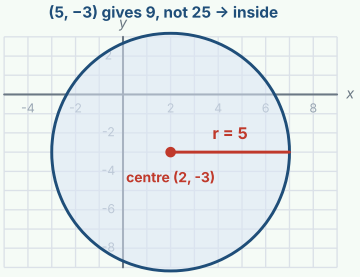
Example 2. **Find the centre and radius of $(x + 4)^2 + (y - 1)^2 = 49$.**

Solution.
Centre **(−4, 1)**, radius **7**.



Example 3. **Is (5, −3) on the circle with centre (2, −3) and radius 5?**

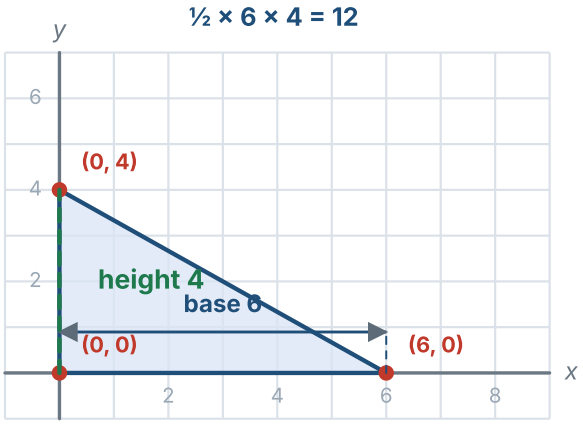
Solution.
$(5 - 2)^2 + 0 = 9$, not 25.
No — it is inside.



6 Area on a grid

for a right-angled or axis-aligned figure, read the legs straight off the grid

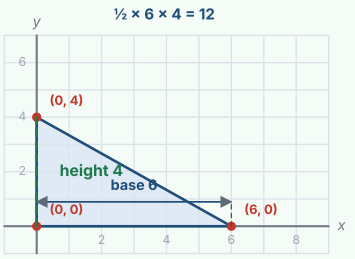
When a figure sits on the coordinate axes, you rarely need a formula beyond the basics: count the horizontal and vertical lengths and apply the ordinary area rule.



Both legs lie along the axes, so the base is 6 and the height is 4 — no distance formula needed.

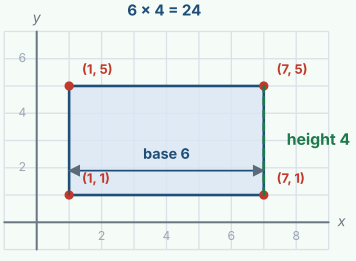
Example 1. **Find the area of a triangle with vertices (0, 0), (6, 0) and (0, 4).**

Solution.
Legs 6 and 4 along the axes.
½(6)(4) = **12**.



Example 2. **Find the area of a rectangle with vertices (1, 1), (7, 1), (7, 5) and (1, 5).**

Solution.
Width 6, height 4.
24.

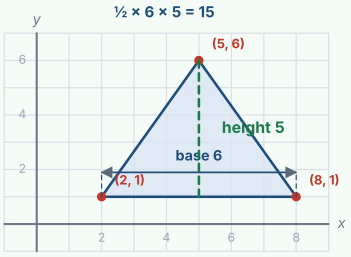


Example 3. **Find the area of a triangle with vertices (2, 1), (8, 1) and (5, 6).**

Solution.

Base 6 along y = 1, height 5.

½(6)(5) = **15**.



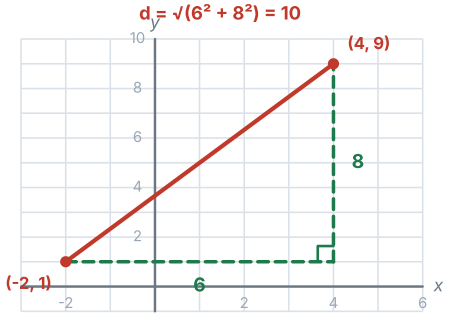
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

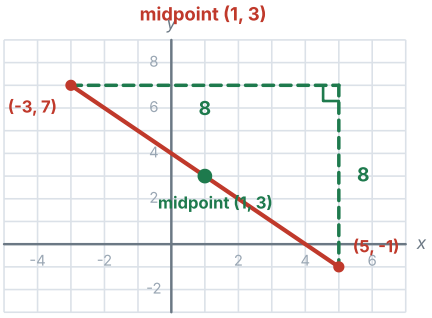
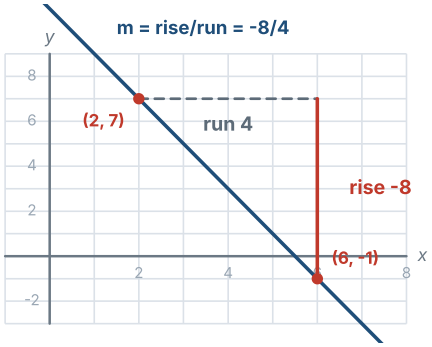
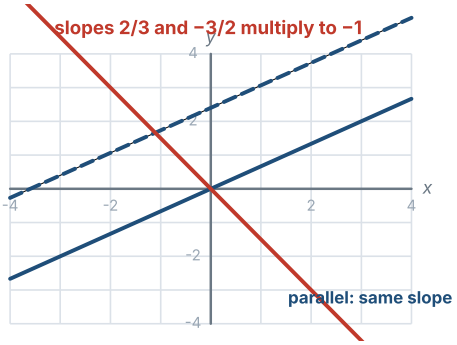
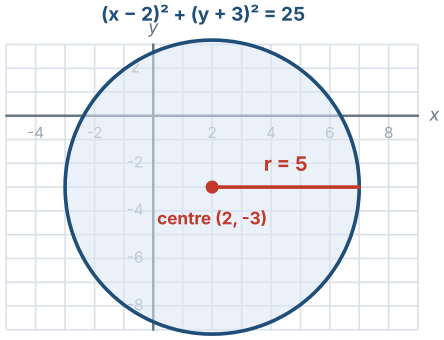
The SAT gives you Desmos on every maths question. Here it is **a plotter**. Enter the points and the line; seeing the picture settles slope signs and perpendicularity faster than any rule.

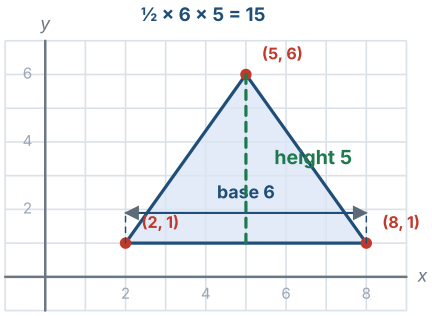
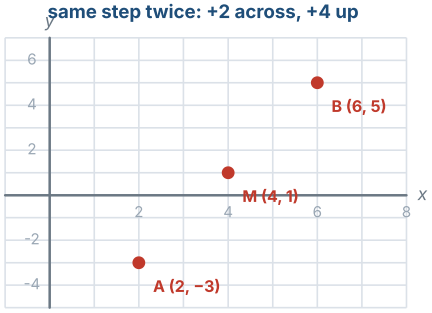
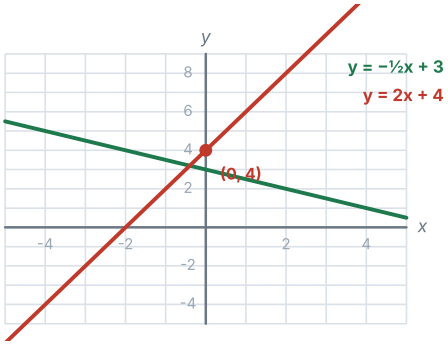
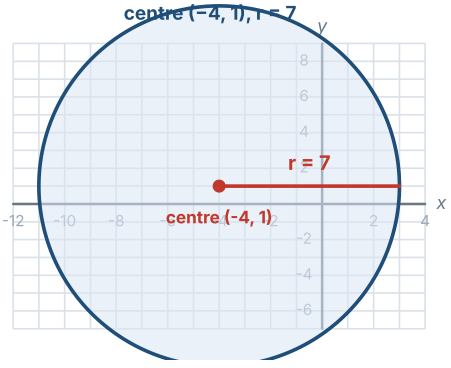
#	Question	Type into Desmos	Then	Answer
D1	What is the distance between (1, 2) and (5, 5)?	`sqrt((5-1)^2+(5-2)^2)`	Pythagoras on the two differences.	5
D2	What is the distance between (−2, 1) and (4, 9)?	`sqrt(6^2+8^2)`	A 6-8-10 triangle.	10
D3	What is the midpoint of (1, 2) and (5, 5)?	`((1+5)/2,(2+5)/2)`	Average each coordinate.	(3, 3.5)
D4	What is the slope through (2, 7) and (6, −1)?	`(-1-7)/(6-2)`	Keep the order consistent.	−2
D5	A line has slope 2/3. What is the perpendicular slope?	`-1/(2/3)`	Flip and negate.	−1.5
D6	The midpoint of AB is (4, 1) and A is (2, −3). What is the x-coordinate of B?	`2*4-2`	Twice the midpoint minus the known endpoint.	6
D7	What is the radius of $(x + 4)^2 + (y − 1)^2 = 49$?	`sqrt(49)`	The right-hand side is r squared.	7
D8	What is the area of a triangle with vertices (0, 0), (6, 0) and (0, 4)?	`0.5*6*4`	The legs lie along the axes.	12
D9	What is the area of a triangle with vertices (2, 1), (8, 1) and (5, 6)?	`0.5*6*5`	Base along y = 1, height from y = 1 to y = 6.	15
D10	Is (5, −3) on the circle centred (2, −3) with radius 5?	`(5-2)^2+(-3+3)^2`	Compare the result with r squared.	no, 9 is not 25

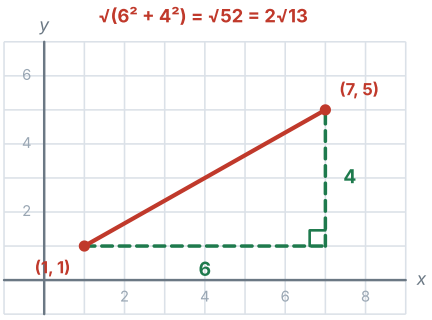
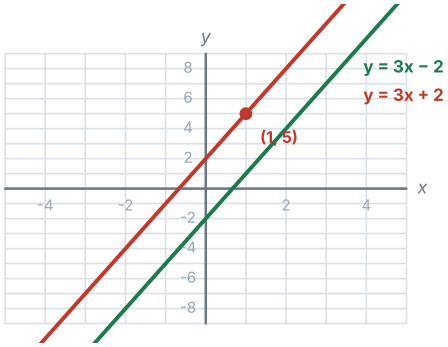
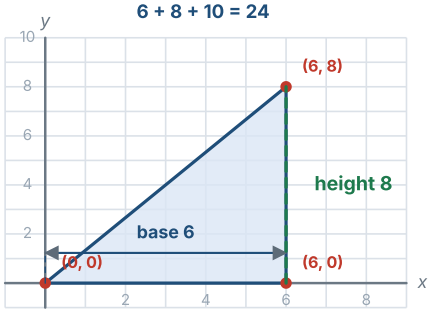
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Sketch the points on a quick set of axes first. Nearly every question in this topic is Pythagoras, an average, or a slope — and the sketch tells you which.

#	Level	Question	Answer	Solution
P1	Hard	What is the distance between the points (−2, 1) and (4, 9) in the xy-plane? 	10	The horizontal difference is 6 and the vertical difference is 8. √(36 + 64) = √100 = **10**.

#	Level	Question	Answer	Solution
P2	Hard	What is the midpoint of the segment joining (−3, 7) and (5, −1)? 	(1, 3)	Average each coordinate: ((−3 + 5)/2, (7 − 1)/2). **(1, 3)**. A common wrong answer subtracts the coordinates instead of averaging them.
P3	Hard	What is the slope of the line passing through (2, 7) and (6, −1)? 	−2	(−1 − 7)/(6 − 2) = −8/4 = **−2**. A common wrong answer inverts the fraction, giving run over rise.
P4	Hard	A line has a slope of 2/3. What is the slope of any line perpendicular to it? 	−3/2	Perpendicular slopes are negative reciprocals: flip the fraction and change the sign. **−3/2**. A common wrong answer flips without negating; another negates without flipping.
P5	Hard	A circle has centre (2, −3) and radius 5. Write the equation of that circle. 	(x − 2)² + (y + 3)² = 25	Signs flip inside the brackets and the right side is r². **(x − 2)² + (y + 3)² = 25.** A common wrong answer uses r instead of r².

#	Level	Question	Answer	Solution
P6	Hard	A triangle has vertices (2, 1), (8, 1) and (5, 6). What is its area, in square units? 	15	The base lies along y = 1 from x = 2 to x = 8, so it is 6 long, and the height is 6 − 1 = 5. ½(6)(5) = **15**.
P7	Very hard	The midpoint of segment AB is (4, 1). If A has coordinates (2, −3), what are the coordinates of B? 	(6, 5)	The midpoint is the average, so B is the same step again: (4 + 2, 1 + 4). **(6, 5)**. A common wrong answer is the midpoint of A and the given midpoint.
P8	Very hard	Line k passes through (0, 4) and is perpendicular to the line y = −½x + 3. What is the equation of line k? 	y = 2x + 4	The perpendicular slope to −½ is **2**, and the y-intercept is given as 4. **y = 2x + 4.** A common wrong answer flips the fraction without changing the sign.
P9	Very hard	In the xy-plane, a circle has equation $(x + 4)^2 + (y - 1)^2 = 49$. What is the sum of the coordinates of its centre and its radius? 	4	Centre (−4, 1) and radius 7. −4 + 1 + 7 = **4**.

#	Level	Question	Answer	Solution
P10	Very hard	A square has two opposite vertices at (1, 1) and (7, 5). What is the length of its diagonal? 	2√13	The diagonal joins the two given points: √(36 + 16) = √52. √52 = 2√13 ≈ 7.21, so **2√13**. A common wrong answer is the same value unsimplified — the SAT expects the simplified radical.
P11	Very hard	Line m has equation y = 3x − 2. Which line is parallel to m and passes through (1, 5)? 	y = 3x + 2	Parallel means the same slope, 3, and 5 = 3(1) + b gives b = 2. **y = 3x + 2.** A common wrong answer is the original line; another does not pass through (1, 5).
P12	Very hard	Points A(0, 0), B(6, 0) and C(6, 8) form a triangle. What is its perimeter? 	24	AB = 6 and BC = 8, and the right angle at B makes AC = 10 by the 6-8-10 triple. 6 + 8 + 10 = **24**. A common wrong answer omits the hypotenuse; another is twice the area.

The one habit to build. Distance is Pythagoras; midpoint is an average. Confusing the two is the commonest error here, and a thirty-second sketch prevents it.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.